Quantum optics with atoms and molecules

Quantenoptik mit Atomen und Molekülen

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Abstract

In this thesis, we investigate cooperative phenomena in light-matter platforms, focusing on superradiance, subradiance, dipole-dipole interactions, and their technological applications. We utilize Lindblad master equations for open quantum systems and coupled dipole models to analyze these interactions.

We begin by examining the properties of light scattering from cold atomic clouds, particularly the effects of static magnetic dipole moments on superradiance, subradiance, and the vacuum Rabi splitting. To study this system, we extend the commonly used coupled dipole model to include static magnetic dipole interactions.

Next, we explore regular two-dimensional arrangements of atoms, emphasizing technological applications. By leveraging atomic level structures, we design linear optical elements using a single array of atoms. We also address the challenge of chiral sensing of molecules, utilizing a cavity formed by helicity-preserving mirrors, where each mirror consists of two layers of atomic arrays.

Furthermore, we investigate molecular systems, deriving analytical rates for Kasha's rule in H-aggregates and illustrating the role of vibrational degrees of freedom in these molecular systems. Additionally, we study the Doppler cooling of molecules in optical cavities, aiming to leverage the Purcell effect to enhance cooling by suppressing the trapping of populations in rovibrational manifolds. We find improvements for single molecules, however, there is no collective enhancement and the final velocity shows no dependence on particle number.

Overall, these investigations contribute to a deeper understanding of cooperative effects in atomic and molecular platforms, which hold promise for various technological applications.

Deutsche Zusammenfassung

In dieser Arbeit untersuchen wir kooperative Phänomene in Licht-Materie-Plattformen, mit einem Schwerpunkt auf Superradianz, Subradianz, Dipol-Dipol-Wechselwirkungen und deren technologischen Anwendungen. Wir verwenden Lindblad-Master-Gleichungen für offene Quantensysteme und gekoppelte Dipolmodelle, um diese Wechselwirkungen zu analysieren.

Wir beginnen mit der Untersuchung der Eigenschaften der Streuung von Licht von kalten Atomwolken, insbesondere der Auswirkungen statischer magnetischer Dipolmomente auf Superradianz, Subradianz und die Vakuum-Rabi-Aufspaltung. Um dieses System zu untersuchen, erweitern wir das häufig verwendete gekoppelte Dipolmodell, um statische magnetische Dipolbeiträge zu berücksichtigen.

Anschließend untersuchen wir reguläre zweidimensionale Anordnungen von Atomen und legen dabei einen Schwerpunkt auf technologische Anwendungen. Durch die Nutzung der atomaren Energielevelstrukturen entwerfen wir lineare optische Elemente mithilfe eines einzelnen Atomarrays. Wir widmen uns auch der Herausforderung der chiralen Detektion von Molekülen, indem wir einen optischen Resonator nutzen, der aus Helizität-erhaltenden Spiegeln besteht, die aus zwei Schichten von Atomarrays gebildet werden.

Darüber hinaus untersuchen wir molekulare Systeme, leiten analytische Raten für Kasha's Regel in H-Aggregaten ab und veranschaulichen die Rolle der Vibrationsfreiheitsgrade in diesen molekularen Systemen. Zusätzlich untersuchen wir die Doppler-Kühlung von Molekülen in optischen Resonatoren, mit dem Ziel, den Purcell-Effekt zu nutzen, um die Kühlung durch die Unterdrückung des Verlusts von Populationen in rovibrationalen Zuständen zu verbessern. Wir finden Verbesserungen für einzelne Moleküle, jedoch keine kollektive Verbesserung, und die Endgeschwindigkeit zeigt keine Abhängigkeit von der Teilchenanzahl.

Insgesamt tragen diese Untersuchungen zu einem tieferen Verständnis kooperativer Effekte in atomaren und molekularen Plattformen bei, die vielversprechend für verschiedene technologische Anwendungen sind.

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1 Scientific context

In recent years, advances in experimental techniques have enabled the investigation of cooperative and collective effects in atomic and molecular systems with unprecedented precision. The overarching theme of this thesis is the theoretical investigation of collective and cooperative phenomena, such as superradiance, subradiance, dipole-dipole interactions, and interference. These effects manifest across various platforms, including cold atomic gases, quantum emitter arrays, metasurfaces, molecular aggregates, and molecules in cavities. The primary goal is to understand the physics of the aforementioned effects in these platforms, but when possible, also to consider technological applications such as sensing and the design of performant optical elements.

In 1954, the idea of superradiance was introduced in the seminal article by Dicke, which is now referred to as Dicke superradiance [1]. The system he investigated was a dense cloud of fully inverted atoms where each emitter is coupled to the electromagnetic vacuum identically, leading to a burst of radiation that is more than simply the radiation of each independent emitter [2]. Many experimental [3] and theoretical [4–8] studies on more realistic systems as sketched in Fig. 1(a) have been performed to see the emergence and modification of superradiance for dense cold atomic gases. More generally, light scattering from cold atomic gases has been an active field for a long time with rich physical phenomena that can occur [9–12]. An accompanying effect to superradiance is subradiance, where radiation from an ensemble is suppressed [13, 14]. It is the dipole-dipole interaction between the transition dipole moments of the atoms that leads to super and subradiance as well as many other interesting effects (collective lineshifts, synchronization, localization) [15–18]. Much of the literature is also concerned with the associated collective motional effects in the cloud due to radiation pressure [19–21]. Another avenue of fundamental research is the limit of extremely dense cold (quantum degenerate) gases where the fermionic and bosonic nature of the motional degrees of freedom become relevant and how the scattering of light is affected by the quantum statistics of the constituents leading to effects like bosonic superradiance and fermionic Pauli blocking [22–27].

The platform of cold atomic clouds is very complex, and the control over the system

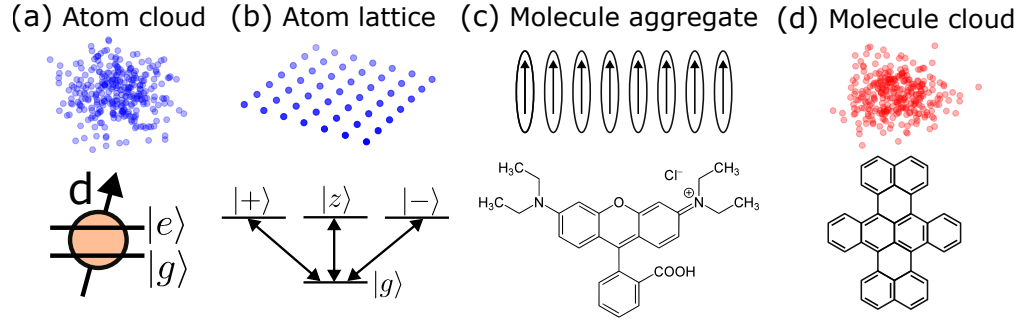


Figure 1: Here, the different types of systems that are investigated in this thesis are shown. In (a), a cold atomic cloud of two-level atoms is sketched. In (b), a lattice of atoms is illustrated. A Λ type system is displayed as an example of the type of atom in the lattice. An H-aggregate of molecules is sketched in (c) with all dipole moments aligned perpendicular to the aggregation axis. Below is the chemical formula of Rhodamine B, which is a typical dye molecule that aggregates in solution. In (d) a cloud of molecules in the gaseous phase is shown with the chemical structure of Dibenzoterrylene.

is limited. A system where greater control over the emitters is possible and in which emitters achieve distances where dipolar interactions become relevant (subwavelength distance) is the platform of neutral atoms trapped in an optical lattice [28] illustrated in Fig. 1(b). The periodic arrangement of atoms greatly simplifies the theoretical treatment of such arrays [29] compared to disordered clouds. A recent perspective reviewing progress in this field can be found in Ref. [30]. Much of the resurgence of this field is related to the high optical cross-section of such an array, which enables, in principle, perfect reflection of an incoming field due to destructive interference of the scattered and incoming radiation behind the array [31–33]. Motivated by the established field of metasurfaces [34, 35], there has been theoretical progress on the tailoring of the linear optical properties of such emitter arrays [36, 37]. In this linear regime, topological band structure effects, just as in condensed matter hopping models, can be investigated [38–40]. However, these surfaces consist of quantum emitters, which distinguishes this platform from conventional metasurfaces with dielectric or metallic scatterers. There have been theoretical suggestions to leverage the subradiance due to the dipolar interactions to store photons [37, 41–44] and to engineer quantum states of light using such arrays [45–47].

Capturing a sufficient number of neutral atoms in an atomic lattice is technologically challenging and requires a sophisticated experimental setup. A platform where closely spaced emitters in a regular arrangement occur naturally without any engineering is dye molecules in solution as shown in Fig. 1(c). Many such dye molecules, e.g. Rhodamine, aggregate to form linear chains which are called H or J aggregates depending on the

orientation of the dipoles with respect to the aggregation axis [48, 49]. However, molecular vibrations and solvent-induced thermalization strongly influence the physics of these aggregates. The physical consequence of the interplay between dipolar interactions and molecular vibrations is called Kasha's rule [50], where a laser excites a blue-shifted symmetric state in an H aggregate, which then relaxes to a subradiant state due to the vibrational degrees of freedom [51]. For emitters embedded in a solid-state matrix, on the other hand, there is an intricate interplay between the photonic, electronic, vibrational, and phononic degrees of freedom [52–54].

Molecular physics is naturally combined with the physics of atomic arrays and metasurfaces when considering the problem of chiral sensing [55–58]. Indeed, helicity-preserving metasurface structures have been proposed to be a good candidate to sense the optical activity of an ensemble of molecules [59–61]. Here, a cavity is designed with helicity-preserving mirrors (where atomic lattices Fig. 1(b) are proposed as a platform) to enhance the light-matter interaction between a molecule and a mode with fixed circular polarization. This allows the characterization of the optical activity of an ensemble of molecules as sketched in Fig. 1(d). Many designs for metasurfaces that also exhibit chiral dichroism exist [62–64], and optical chirality, in general, is a fascinating research subject [65, 66]. Beyond sensing, the subject of material engineering using chiral vacuum modes to break time-reversal symmetry in a material is of interest [67–69].

Going beyond simply the observation of effects in single molecules and aggregates, molecules are also a promising platform for quantum applications since they provide many more degrees of freedom than atomic systems and larger transition dipole moments as well as electrostatic interactions [70]. Cold molecules as sketched in Fig. 1(d) are also a relevant platform for the study of more fundamental physics [71], such as the search for the permanent dipole moment of the electron. One of the main caveats is that the cooling of molecular systems is much more complex than the cooling of atomic systems since the usual laser cooling techniques generally fail [72, 73]. However, diatomic molecules have been successfully cooled [74–77], and even Bose-Einstein condensates of diatomic molecules have been reported [78, 79]. If the molecules have special symmetries, laser cooling is also possible [80]. Furthermore, cavity cooling is a promising avenue for the cooling of molecules [81] since conceptually, it works for any polarizable object [82–85], but problems due to dark state trapping may persist. Recent experiments on molecules in cavities [86] suggest that the Purcell enhancement may be leveraged to cool single or few molecules.

In this introduction, we will mostly review content that is either part of the main text or the appendices of the publications which are also part of this thesis. Since the

theoretical description in these papers is fairly complete, this is however inevitable. The task is then to put the theory presented in these calculations into proper context and convey a coherent story so that the connection and context of the publications can be better understood.

First, we derive the starting point, the coupled dipole model with the minimal coupling Hamiltonian in Sect. 1.1. We start from the simple case of an ensemble of two-level emitters coupled to the electromagnetic vacuum in Sect. 1.2. We illustrate the emergence of super-and subradiance in this system. Then, we take several steps to add more complexity to this setup and discuss the relevant publications.

We explicitly consider magnetic structures in Sect. 1.3. This means Zeeman sublevels, permanent magnetic dipole moments, and magnetic transition dipole moments. Then we discuss the relevance of two of these three types of emitters when considering their regular arrangement in an array in Sect. 1.4. Departing from "simple" atomic emitters, we introduce molecular scatterers in Sect. 1.5 and discuss a chain of molecules. Then, we give a brief introduction to the cooling of molecules approximated by a toy model in Sect. 1.6. We conclude by summarizing this thesis and its results in Sect. 1.7 and outlining plans in Sect. 1.8. We take units of $\hbar = 1$ in this thesis, but keep c in all expressions.

1.1 Derivation of the dipolar coupling Hamiltonian

We give a brief introduction as to how the dipolar coupling Hamiltonian can be derived and approximated from the minimal coupling Hamiltonian, and which approximations are involved. We roughly follow the discussion in Ref. [87] for this derivation. We choose SI units for the electromagnetic degrees of freedom. The full minimal coupling Hamiltonian for a single electron per atom with momentum $\boldsymbol{p}_j$ at position $\boldsymbol{r}_j$ and mass m and a single nuclear momentum $\boldsymbol{P}_j$ (at position $\boldsymbol{R}_j$) per atom and quantized electromagnetic vector

potential $\hat{\boldsymbol{A}}(\boldsymbol{r})$ reads

$$\begin{aligned}
\mathcal{H} &= H_a + \mathcal{H}_{aa} + \mathcal{H}_{\text{field}} \\
\mathcal{H}_a &= \sum_{j=1}^{N} \left[\frac{\left(\boldsymbol{P}_j - e\hat{\boldsymbol{A}}(\boldsymbol{R}_j)\right)^2}{2M} + \frac{\left(\boldsymbol{p}_j + e\hat{\boldsymbol{A}}(\boldsymbol{r}_j)\right)^2}{2m} + V_{ec}(\boldsymbol{R}_j, \boldsymbol{r}_j) \right] \\
\mathcal{H}_{aa} &= \frac{1}{2} \sum_{j \neq j'} \left[V_{cc}(\boldsymbol{R}_j, \boldsymbol{R}_{j'}) + V_{ec}(\boldsymbol{R}_j, \boldsymbol{r}_{j'}) + V_{ec}(\boldsymbol{R}_{j'}, \boldsymbol{r}_j) + V_{ee}(\boldsymbol{r}_j, \boldsymbol{r}_{j'}) \right] \\
\mathcal{H}_{\text{field}} &= \sum_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})} \omega_k \, \hat{a}^\dagger_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})} \hat{a}_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})},
\end{aligned} \tag{1}$$

where V_{ec} is the electron-core interaction, V_{ee} the electron-electron interaction and V_{cc} the core-core interaction. In the simplest case, V_{ec}, V_{ee} and V_{cc} are all Coulomb interactions but in general they are shielded by the electronic cloud of the nuclear and electronic degrees of freedom which will not be considered in the single-electron picture used here. The Hamiltonian is split into an atomic part $\mathcal{H}_a$, an atom-atom interaction part $\mathcal{H}_{aa}$, and a free electromagnetic field energy term $\mathcal{H}_{\text{field}}$. This contains the bosonic creation $\hat{a}^\dagger_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})}$ and annihilation operators $\hat{a}_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})}$ with canonical commutation relation $\left[\hat{a}_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})}, \hat{a}^\dagger_{\boldsymbol{k}', \boldsymbol{\epsilon}'(\boldsymbol{k}')}\right] = \delta_{\boldsymbol{k}, \boldsymbol{k}'} \delta_{\boldsymbol{\epsilon}(\boldsymbol{k}), \boldsymbol{\epsilon}'(\boldsymbol{k}')}$. Here $\boldsymbol{k}$ is the wavevector and $\boldsymbol{\epsilon}(\boldsymbol{k})$ is the polarization vector perpendicular to the wavevector, which is a complete set of quantum numbers for the electromagnetic field. The frequency $\omega_k = ck$ follows the linear dispersion relation of light in free space. We have used the quantities e, representing the electron charge, and c for the speed of light in our definitions. This Hamiltonian is already approximate, due to the underlying assumption that the core and electronic degree of freedom are hydrogen-like and can be described with single core and electron momenta as well as Coulomb-like interactions. The quantized electromagnetic vector potential is defined as

$$\hat{\boldsymbol{A}}(\boldsymbol{r}) = \sum_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})} \frac{g_k}{\omega_k} \boldsymbol{\epsilon}(\boldsymbol{k}) \left(\hat{a}_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{r}} + \hat{a}^\dagger_{\boldsymbol{k}, \boldsymbol{\epsilon}(\boldsymbol{k})} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{r}} \right). \tag{2}$$

The electric field strength of a photon is defined as $g_k = \sqrt{\frac{\omega_k}{2\epsilon_0 V}}$ with vacuum permittivity ϵ_0 and quantization volume V[1] Next, this Hamiltonian is transformed into the

[1] When quantizing the electromagnetic field, this is usually done by considering a finite cube, but final results should not depend on this volume.

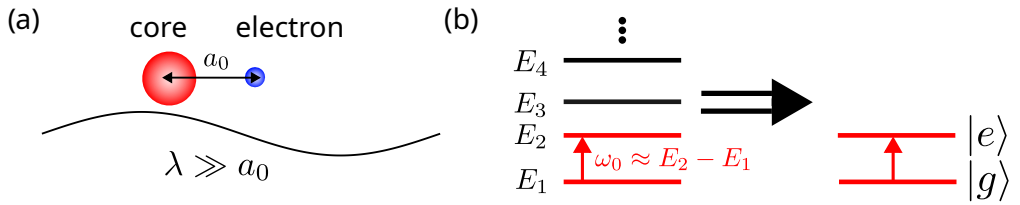


Figure 2: Here we show the approximations that will be made one after another to arrive at the dipolar coupling Hamiltonian. In (a), the dipolar approximation is performed with an illustration showing a small variation of a plane wave over the distance between the electron and the nucleus. A plane wave corresponding to wavelength λ is almost constant on the scale of the Bohr radius a_0 since $\lambda \gg a_0$. In (b), it is shown how only two levels from many electronic levels may be relevant for the system physics.

length gauge where the dipolar coupling emerges via the unitary transformation[2]

$$U = \exp\left(-\mathrm{i}e \sum_{j=1}^{N} \int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j} \mathrm{d}\boldsymbol{r} \cdot \hat{\boldsymbol{A}}(\boldsymbol{r})\right), \tag{3}$$

which removes the gauge field from the momentum terms. This term commutes with all potential terms, but it transforms the operators

$$\begin{aligned}
U\boldsymbol{P}_j U^\dagger &= \boldsymbol{P}_j + e\boldsymbol{A}(\boldsymbol{R}_j) \\
U\boldsymbol{p}_j U^\dagger &= \boldsymbol{p}_j - e\boldsymbol{A}(\boldsymbol{r}_j) \\
U\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} U^\dagger &= \hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} + \mathrm{i} \sum_{j=1}^{N} \frac{e g_k}{\omega_k} \int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j} \mathrm{d}\boldsymbol{r} \cdot \boldsymbol{\epsilon}(\boldsymbol{k}) \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{r}} \\
U\hat{\boldsymbol{A}}(\boldsymbol{r}) U^\dagger &= \hat{\boldsymbol{A}}(\boldsymbol{r}) + \sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} \frac{g_k}{\omega_k} \boldsymbol{\epsilon}(\boldsymbol{k}) \left(\mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} \sum_{j=1}^{N} \frac{e g_k}{\omega_k} \int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j} \mathrm{d}\boldsymbol{r} \cdot \boldsymbol{\epsilon}(\boldsymbol{k}) \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{r}} + \mathrm{h.c.} \right).
\end{aligned} \tag{4}$$

We omit the corrections to $\boldsymbol{A}(\boldsymbol{r})$ here since these terms would also vanish in the electric dipole approximation, which we perform later. In a more careful treatment, it becomes apparent that these terms are related to higher orders in the multipole expansion and are

[2] This is a version of the Power-Zienau-Woolley transformation, which is usually performed on the level of the action [88].

related to the magnetic field [88, 89]. The Hamiltonian in the length gauge then becomes

$$\begin{aligned}\mathcal{H} = &\sum_{j=1}^{N}\left[\frac{\boldsymbol{P}_j^2}{2M}+\frac{\boldsymbol{p}_j^2}{2m}+V_{ec}(\boldsymbol{R}_j,\boldsymbol{r}_j)\right]+\mathcal{H}_{aa}\\ &+\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\omega_k\left(\hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}-\mathrm{i}\sum_{j=1}^{N}\frac{eg_k}{\omega_k}\int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j}\mathrm{d}\boldsymbol{r}\cdot\boldsymbol{\epsilon}(\boldsymbol{k})\mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{r}}\right)\\ &\qquad\times\left(\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}+\mathrm{i}\sum_{j=1}^{N}\frac{eg_k}{\omega_k}\int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j}\mathrm{d}\boldsymbol{r}\cdot\boldsymbol{\epsilon}(\boldsymbol{k})\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{r}}\right).\end{aligned} \tag{5}$$

The dipolar approximation is then an approximation of the operator-valued integral $\int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j}\mathrm{d}\boldsymbol{r}\cdot\boldsymbol{\epsilon}(\boldsymbol{k})\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{r}}\approx(\boldsymbol{R}_j-\boldsymbol{r}_j)\cdot\boldsymbol{\epsilon}(\boldsymbol{k})\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}_j}$, where the phase in the integral was approximated to be constant over the range $[\boldsymbol{r}_j,\boldsymbol{R}_j]$, i.e. assuming that the relevant wavelengths are much longer than the distance of the electron from the core, see Fig. 2(a) for a pictorial representation. We also introduce the relative displacement of the electron with respect to the core $\boldsymbol{x}_j=\boldsymbol{R}_j-\boldsymbol{r}_j$. Multiplying this expression out then yields

$$\begin{aligned}&\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\omega_k\left(\hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}-\mathrm{i}\sum_{j=1}^{N}\int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j}\mathrm{d}\boldsymbol{r}\cdot\boldsymbol{\epsilon}(\boldsymbol{k})g_k\mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{r}}\right)\left(\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}+\mathrm{i}\sum_{j=1}^{N}\int_{\boldsymbol{r}_j}^{\boldsymbol{R}_j}\mathrm{d}\boldsymbol{r}\cdot\boldsymbol{\epsilon}(\boldsymbol{k})g_k\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{r}}\right)\\ &\approx\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\omega_k\hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}-\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\sum_{j=1}^{N}\boldsymbol{E}(\boldsymbol{R}_j)(e\boldsymbol{x}_j)\\ &+\sum_{j=1}^{N}\sum_{j'=1}^{N}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\frac{e^2g_k^2}{\omega_k}(\boldsymbol{\epsilon}(\boldsymbol{k})\cdot\boldsymbol{x}_j)(\boldsymbol{\epsilon}(\boldsymbol{k})\cdot\boldsymbol{x}_{j'}),\end{aligned} \tag{6}$$

with the electric field operator

$$\hat{\boldsymbol{E}}(\boldsymbol{R})=\mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}g_k\boldsymbol{\epsilon}(\boldsymbol{k})\left(\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}}-\hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}}\right). \tag{7}$$

The last term in Eq. (6) proportional to the position square is a dipolar coupling term that exactly cancels the interaction $\mathcal{H}_{aa}$ in the dipolar approximation in free space (aside from contact terms for $j=j'$). This is another advantage of this gauge; the dipolar interactions are completely contained in this definition of the photon, and static terms are not required. In this gauge, the contact term is referred to as a polarization self-energy term, and in the length gauge [22] one obtains an A^2-term instead, which is relevant

for phase transitions in the ultrastrong coupling regimes leading to no-go theorems [90]. This term, however, is only relevant when the particle wave functions start to overlap, which would require a full field theoretical treatment [87].

The dipolar coupling Hamiltonian thus becomes

$$\mathcal{H} = \sum_{j=1}^{N} \left[\frac{\boldsymbol{P}_j^2}{2M} + \frac{\boldsymbol{p}_j^2}{2m} + V_{ec}(\boldsymbol{R}_j, \boldsymbol{r}_j) \right] + \mathcal{H}_{aa} + \sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} \omega_k \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} \hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} - \sum_{j=1}^{N} \hat{\boldsymbol{E}}(\boldsymbol{R}_j)(e\boldsymbol{x}_j). \tag{8}$$

As a further approximation, the core coordinate $\boldsymbol{R}_j$ is usually treated as the center of mass coordinate instead of the nuclear core coordinate, and $\boldsymbol{P}_j$ as the center of mass momentum, justified by the fact that the mass of the nucleus is much larger than that of the electronic cloud. This is referred to as the Born-Oppenheimer approximation in the literature since it decouples the center of mass and electronic degrees of freedom.

What follows in the last step is the two-level approximation sketched in Fig. 2(b), where an emitter has multiple available levels, but only two of these levels contribute to the physics of the system when driven at a given frequency. The two-level approximation does not hold for every atom, in particular, if there is a degeneracy in the level structure of the atom. Suppose such a degeneracy does not exist or has been manually lifted. In that case, the level structure of the atoms is usually well separated so that there is only a single electronic transition which is of interest for a particular experiment. In this context, a projection onto the relevant ground state $|g_j\rangle$ and excited state $|e_j\rangle$ can be performed to simplify Eq. (8). Under this two-level approximation, the dipolar coupling Hamiltonian takes the widely used form

$$\mathcal{H} = \sum_{j=1}^{N} \omega_0 \hat{\sigma}_j^\dagger \hat{\sigma}_j + \sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} \omega_k \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} \hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} + \sum_{j=1}^{N} \hat{\boldsymbol{E}}(\boldsymbol{R}_j) \left(\boldsymbol{d}\hat{\sigma}_j + \boldsymbol{d}^* \hat{\sigma}_j^\dagger \right), \tag{9}$$

where $\boldsymbol{d} = \langle e_j | -e\boldsymbol{x}_j | g_j \rangle$ is the transition matrix element of the position operator, which we assume to be identical for each atom. Since the matter part is described by an ensemble of N two-level systems it can be characterized by the matrices $\hat{\sigma}_j = |g_j\rangle \langle e_j|$ where $|e_j\rangle$ is the excited state for an atom at position $\boldsymbol{R}_j$ and $|g_j\rangle$ the ground state which is split from the excited state by frequency ω_0. No static electric dipole moment exists for atoms. Indeed, the matrix elements of the position operator read

$$\langle \alpha | -e\hat{\boldsymbol{x}} | \beta \rangle = -e \int \mathrm{d}\boldsymbol{x} \psi_\alpha^*(\boldsymbol{x}) \hat{\boldsymbol{x}} \psi_\beta(\boldsymbol{x}), \tag{10}$$

and since the wave functions ψ_α and ψ_β have fixed parity for atomic systems, $\langle\alpha|\hat{\boldsymbol{x}}|\alpha\rangle = 0$, i.e. atoms have no static electric dipole moment and only the offdiagonal matrix elements of $\boldsymbol{d}$ are nonzero. There are higher orders in the multipolar expansion that have been neglected, such as the magnetic dipole, the electric quadrupole, and so on.

We also introduce the notation for the positive and negative frequency components of the electric field

$$\hat{\boldsymbol{E}}^{(+)}(\boldsymbol{R}) = \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} g_k \boldsymbol{\epsilon}(\boldsymbol{k}) \hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} \tag{11}$$

and $\hat{\boldsymbol{E}}^{(-)}(\boldsymbol{R}) = \left(\hat{\boldsymbol{E}}^{(+)}(\boldsymbol{R})\right)^\dagger$ so that $\hat{\boldsymbol{E}}(\boldsymbol{R}_j) = \hat{\boldsymbol{E}}^{(+)}(\boldsymbol{R}) + \hat{\boldsymbol{E}}^{(-)}(\boldsymbol{R})$.

1.2 Coupling of an ensemble of two-level systems to light

A commonality in the research presented is the coupling between light and matter. This situation can often be simplified by deriving an effective model for the matter part of the system. By tracing out the photonic degrees of freedom, a master equation description for the matter subsystem can be derived. The starting point for the following discussion is the dipolar coupling Hamiltonian Eq. (9) between many two-level systems and light, which has been derived in the previous section.

Additionally, the systems we consider will usually be driven by a laser modeled by the Hamiltonian

$$\mathcal{H}^{\mathrm{TLS}}_{\mathrm{drive}} = \eta \sum_j \left(\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_\ell \boldsymbol{R}_j} \mathrm{e}^{\mathrm{i}\omega_\ell t} \hat{\sigma}_j + \mathrm{e}^{\mathrm{i}\boldsymbol{k}_\ell \boldsymbol{R}_j} \mathrm{e}^{-\mathrm{i}\omega_\ell t} \hat{\sigma}_j^\dagger\right), \tag{12}$$

where $\boldsymbol{k}_\ell$ is the laser wave vector and η the Rabi frequency, which can be written in terms of the classical field amplitude $\eta = \boldsymbol{E}^{(+)}_{\mathrm{in}} \cdot \boldsymbol{d}^*$.

This term can be formally derived by applying a coherent displacement on the electric field mode $\boldsymbol{k}_\ell$, i.e. applying the unitary $U = \mathrm{e}^{\frac{\eta}{g_k d}\left(\hat{a}_{\boldsymbol{k}_\ell,\boldsymbol{\epsilon}(\boldsymbol{k}_l)} + \hat{a}^\dagger_{\boldsymbol{k}_l,\boldsymbol{\epsilon}(\boldsymbol{k}_\ell)}\right)}$ to Eq. (9). Note that this also leads to additional terms, but these correspond to the classical energy of the laser field and can be omitted after the elimination of the photonic vacuum.

As illustrated in Fig. 3(a), the system is now split into two parts, the photonic and matter systems, and the photonic system is now considered as a vacuum bath to the matter system. To obtain a Lindblad master equation for the system with a photonic bath, one performs second-order perturbation theory in the system-bath coupling and assumes Markovian dynamics for the system. We do not repeat this derivation, which can be found in several textbooks [91, 92]. Physically, this induces unitary dipole-dipole

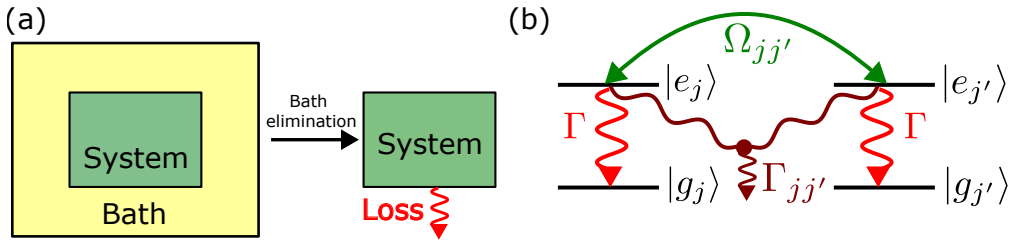


Figure 3: Here we illustrate in (a) the elimination of a bath from a coupled system-bath composite. The elimination of the bath in a Born-Markov approximation leads to effective system dynamics, including new unitary interactions in the system as well as loss described by dissipators. The loss necessitates a master equation description of the effective system dynamics. In (b), we show the effective description for two two-level systems after the photonic bath has been eliminated. The elimination of the bath leads to effective coherent electric dipole-dipole interactions $\Omega_{jj'}$ as well as spontaneous emission Γ and correlated spontaneous emission $\Gamma_{jj'}$ due to the coupling to the common bath.

interactions sketched in Fig. 3(b)

$$\begin{aligned} \mathcal{H} &= \mathcal{H}_0^{\mathrm{TLS}} + \mathcal{H}_{\text{d-d}}^{\mathrm{TLS}} \\ \mathcal{H}_0^{\mathrm{TLS}} &= \omega_0 \sum_{j=1}^{N} \hat{\sigma}_j^\dagger \hat{\sigma}_j \\ \mathcal{H}_{\text{d-d}}^{\mathrm{TLS}} &= \sum_{jj'}^{N} \Omega_{jj'} \hat{\sigma}_j^\dagger \hat{\sigma}_{j'}, \end{aligned} \tag{13}$$

where virtual photons are exchanged between emitters, as well as spontaneous emission in the form of

$$\mathcal{L}[\rho] = \sum_{jj'}^{N} \Gamma_{jj'} \left(\hat{\sigma}_j \rho \hat{\sigma}_{j'}^\dagger - \frac{1}{2} \left\{ \hat{\sigma}_j^\dagger \hat{\sigma}_{j'}, \rho \right\}_+ \right). \tag{14}$$

which corresponds to the correlated (or uncorrelated for the $j = j'$ terms) loss of excitation into the reservoir. Putting these two contributions together then yields system dynamics described by the quantum optical Lindblad master equation[3]

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\mathrm{i}[\mathcal{H}, \rho] + \mathcal{L}[\rho]. \tag{15}$$

[3]The loss of excitation into the reservoir is a stochastic process, so that a Master equation formalism becomes necessary.

Here, the dyadic Green's function[4] is used to write the induced interactions

$$\boldsymbol{G}(\boldsymbol{R}) = \frac{\mathrm{e}^{\mathrm{i}k_0 R}}{4\pi k_0^2}\left[\left(\frac{k_0^2}{R} + \frac{\mathrm{i}k_0}{R^2} - \frac{1}{R^3}\right)\mathbb{1} + \left(-\frac{k_0^2}{R} - \frac{3\mathrm{i}k_0}{R^2} + \frac{3}{R^3}\right)\frac{\boldsymbol{R}\otimes\boldsymbol{R}}{R^2}\right], \tag{16}$$

where $k_0 = \omega_0/c$ is the wave number corresponding to the transition frequency between the excited and ground state. The dyadic or outer product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined as $(\boldsymbol{a}\otimes\boldsymbol{b})_{jj'} = a_j b_{j'}$.

It will later become clear that it is also convenient to write the Green's function as

$$\boldsymbol{G}(\boldsymbol{R}) = \left(\mathbb{1} + \frac{1}{k_0^2}\nabla\otimes\nabla\right)\frac{\mathrm{e}^{\mathrm{i}k_0 R}}{4\pi R}. \tag{17}$$

This Green's function can be derived by eliminating the photonic degrees of freedom in the aforementioned perturbation theory [93, 94], but it also emerges in the classical multipolar expansion as a field that is emitted by a classical rotating dipole [95]. Indeed, the field emitted due to a classical dipole at position $\boldsymbol{R}_j$ with dipole moment $\boldsymbol{d}_j$ observed at $\boldsymbol{R}$ reads

$$\boldsymbol{E}^{(+)}(\boldsymbol{R}) = \epsilon_0^{-1}k_0^2\boldsymbol{G}(\boldsymbol{R}-\boldsymbol{R}_j)\boldsymbol{d}_j \tag{18}$$

and the interaction energy between two neighbouring dipoles at positions $\boldsymbol{R}_j$ and $\boldsymbol{R}_{j'}$ with dipole moments $\boldsymbol{d}_j$ and $\boldsymbol{d}_{j'}$ is

$$\Omega_{jj'} = -\epsilon_0^{-1}k_0^2\,\mathrm{Re}\left(\boldsymbol{d}_j^*\boldsymbol{G}(\boldsymbol{R}_j-\boldsymbol{R}_{j'})\boldsymbol{d}_{j'}\right). \tag{19}$$

In addition to the interaction energy between two transition dipoles, there is a term for their correlated decay

$$\Gamma_{jj'}/2 = \epsilon_0^{-1}k_0^2\,\mathrm{Im}\left(\boldsymbol{d}_j^*\boldsymbol{G}(\boldsymbol{R}_j-\boldsymbol{R}_{j'})\boldsymbol{d}_{j'}\right). \tag{20}$$

From here on, we will always assume that the dipole moments of all atoms are the same, $\boldsymbol{d}_j = \boldsymbol{d}_{j'}$. The limit of this expression at $\boldsymbol{R} = 0$ for identical dipole moments is half the spontaneous decay rate of the single atom $\Gamma = \frac{k_0^3 d^2}{3\pi\epsilon_0}$.

Such an ensemble of emitters already exhibits rich physics [13, 96, 97] as described in the introduction, and we elucidate this by studying only two atoms. Consider two atoms with their dipoles aligned in the z-direction. Then the Green's function $G_{ee}(\boldsymbol{R}) = \Omega_{12} - \mathrm{i}\Gamma_{12}/2$

[4]This Green's function contains both a real and imaginary part for the unitary and dissipative part of the master equation. It is a dyadic, i.e. a second rank tensor to account for the polarization degrees of freedom of light.

between the two dipoles at distance $\boldsymbol{R}$ becomes

$$G_{ee}(\boldsymbol{R}) = -\frac{3\Gamma}{4}\mathrm{e}^{-\mathrm{i}k_0 R}\left(\frac{1-\cos^2(\theta)}{k_0 R} - \frac{\mathrm{i}(1-3\cos^2(\theta))}{k_0^2 R^2} - \frac{1-3\cos^2(\theta)}{k_0^3 R^3}\right), \tag{21}$$

where θ is the angle of the dipole separation axis with respect to the dipole axis. This geometry is illustrated in Fig. 4(c).

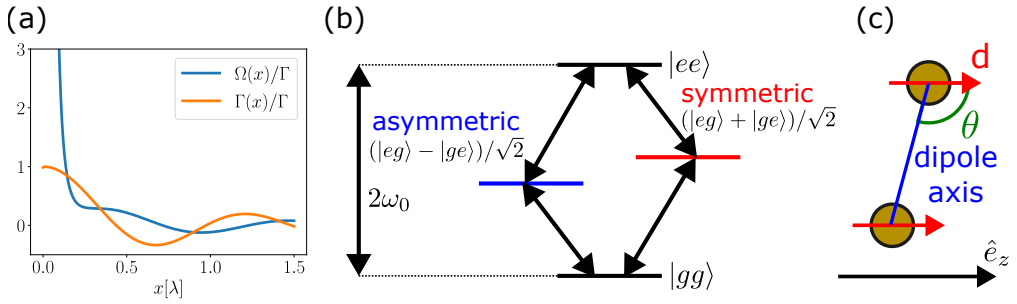


Figure 4: In (a), the functional dependence of the dipolar interaction $\Omega(\boldsymbol{R})$ and $\Gamma(\boldsymbol{R})$ in units of the natural linewidth Γ in the x-direction in units of λ is shown. The orientation of the two dipoles was chosen in the z-direction so that the symmetric state $|s\rangle$ has larger energy than the antisymmetric $|a\rangle$ state for small x. In (b) is a level diagram of the four eigenstates in Eq. (24) of the two coupled two-level systems in the regime of small distance $k_0 x \ll 1$ in the configuration in (a). The excited state $|ee\rangle$ is separated by $2\omega_0$ from the ground state $|gg\rangle$, and the symmetric and antisymmetric states are split by $\Omega(d)$, where d is the distance between the two emitters. See Ref. [2] for the seminal essay about super and subradiance and Ref. [1] for the original paper by Dicke. In (c), two dipoles with dipole moments $\boldsymbol{d}$ in red pointing in the z-direction and a dipole axis in blue are illustrated. The angle between them is indicated in green as θ.

If one is interested in the behaviour close to $|\boldsymbol{R}| = 0$, the exponential in front can be Taylor-expanded in $k_0 R$

$$\begin{aligned} G_{ee}(\boldsymbol{R}) &\approx -\frac{3\Gamma}{4}(1 - \mathrm{i}k_0 R - k_0^2 R^2)\left(\frac{1-\cos^2(\theta)}{k_0 R} - \frac{\mathrm{i}(1-3\cos^2(\theta))}{k_0^2 R^2} - \frac{1-3\cos^2(\theta)}{k_0^3 R^3}\right) \\ &= -\frac{\mathrm{i}\Gamma}{2} + \frac{3\Gamma}{4}\left[\frac{1-3\cos^2(\theta)}{k_0^3 R^3} - \frac{1-\cos^2(\theta)}{k_0 R}\right]. \end{aligned} \tag{22}$$

This implies that the dissipative component remains finite even for $|\boldsymbol{R}| = 0$ and becomes exactly Γ as shown in Fig. 4. On the other hand, the near-field unitary interaction diverges, but the exact value depends on the angle between the dipoles. The term $\Omega_{12} = \frac{3\Gamma}{4}\frac{1-3\cos^2(\theta)}{k_0^3 R^3}$ usually dominates, as long as the dipoles are not arranged in such a

way that this term cancels exactly at $\theta = \pm\arccos\left(1/\sqrt{3}\right)$. If we assume that this is not the case, the non-Hermitian Hamiltonian of these coupled two-level systems becomes in the basis $|ee\rangle, |ge\rangle, |eg\rangle, |gg\rangle$

$$H_2^{\mathrm{TLS}} = \begin{pmatrix} 2(\omega_0 - \mathrm{i}\Gamma/2) & 0 & 0 & 0 \\ 0 & \omega_0 - \mathrm{i}\Gamma/2 & \Omega_{12} - \mathrm{i}\Gamma/2 & 0 \\ 0 & \Omega_{12} - \mathrm{i}\Gamma/2 & \omega_0 - \mathrm{i}\Gamma/2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{23}$$

which is diagonalized by the eigenvectors and eigenvalues

$$\begin{aligned} &|ee\rangle && \lambda_{ee} = 2(\omega_0 - \mathrm{i}\Gamma/2) \\ &|s\rangle = 1/\sqrt{2}(|ge\rangle + |eg\rangle) && \lambda_s = -\mathrm{i}\Gamma + \omega_0 + \Omega_{12} \\ &|a\rangle = 1/\sqrt{2}(|ge\rangle - |eg\rangle) && \lambda_a = \omega_0 - \Omega_{12} \\ &|gg\rangle && \lambda_{gg} = 0. \end{aligned} \tag{24}$$

The eigenvalues λ_{ee} and λ_{gg} are not modified, indicating that the doubly excited state is still spaced $2\omega_0$ from the ground state, in contrast to the symmetric state eigenvalue λ_s as well as the antisymmetric state eigenvalue λ_a. The symmetric state is always completely superradiant with decay rate $\Gamma_s = \Gamma$, whereas the antisymmetric state does not decay at all with $\Gamma_a = 0$. Energetically, it depends on the orientation of the dipoles, which state has higher energy. If the dipolar axis and the separation axis are perpendicular, i.e. $\theta = \pi/2$, then the superradiant state is higher in frequency by $\frac{3\Gamma}{2k_0^3R^3}$. If the dipolar axis and the separation axis are aligned, then the superradiant state is lower in frequency by $\frac{3\Gamma}{k_0^3R^3}$. An example of a physical consequence is the different optical properties of H and J molecular dye aggregates, whose dipolar orientations lead to a higher or lower-lying symmetric state, respectively. This system is studied in more detail in Sect. 1.5 and Ref. [P3].

1.3 Generalization to atoms with magnetic substructure

If the multipolar expansion is continued to higher orders, the next order beyond the electric dipole contribution are the electric quadrupole and the magnetic dipole component. In principle, these terms can also contribute to atomic and molecular two-level systems, but the type of interactions that are relevant depend on the particular selection rules due to the symmetries of the transition. In this section, the focus is on magnetic dipole

contributions, which are relevant for Ref. [P1] and Ref. [P2].

For a two-level system, the most general expression for a system with magnetic dipole coupling to light is

$$\mathcal{H}_{\text{el+mag}} = \sum_j (\hat{\boldsymbol{\mu}}_{t,j} + \hat{\boldsymbol{\mu}}_{s,j}) \cdot \hat{\boldsymbol{B}}(\boldsymbol{R}_j), \tag{25}$$

where $\hat{\boldsymbol{\mu}}_{t,j} = \boldsymbol{\mu}_j \hat{\sigma}_j + \text{h.c.}$ are the offdiagonal and $\hat{\boldsymbol{\mu}}_{s,j} = \boldsymbol{\mu}_{j,e}\hat{\sigma}_j^\dagger\hat{\sigma}_j + \boldsymbol{\mu}_{j,g}\hat{\sigma}_j\hat{\sigma}_j^\dagger$ the diagonal matrix elements of the magnetic dipole operator $\hat{\boldsymbol{\mu}}$ using for these matrix elements $\boldsymbol{\mu}_j = \langle e_j|\hat{\boldsymbol{\mu}}_j|g_j\rangle$, $\boldsymbol{\mu}_{j,e} = \langle e_j|\hat{\boldsymbol{\mu}}_j|e_j\rangle$ and $\boldsymbol{\mu}_{j,g} = \langle g_j|\hat{\boldsymbol{\mu}}_j|g_j\rangle$. The magnetic dipole operator takes the form $\hat{\boldsymbol{\mu}} = -\frac{e}{2m}(\hat{\boldsymbol{L}} + g_s\hat{\boldsymbol{S}})$ where $\hat{\boldsymbol{L}}$ is the orbital angular momentum operator, $\hat{\boldsymbol{S}}$ is the spin operator and $g_s \approx 2$ is the gyromagnetic ratio for the electron so that $\boldsymbol{L} + g_s\boldsymbol{S}$ is the total angular momentum.

The quantized magnetic field operator is defined as

$$\hat{\boldsymbol{B}}(\boldsymbol{R}) = \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})} \frac{g_k}{\omega_k}(\boldsymbol{k}\times\boldsymbol{\epsilon}(\boldsymbol{k}))\left(\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}(\boldsymbol{k})}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}}\right). \tag{26}$$

Just as for the electric dipole interaction, the elimination of the photonic environment leads to effective terms in the Hamiltonian due to the magnetic interaction. First, we state the terms that emerge due to the static dipole moment. Since the excited and ground states have different static magnetic dipole moments, the frequency between the excited and ground state now depends on the induced magnetic field of the surrounding atoms, corresponding to a dynamical Zeeman effect, see Fig. 5(b). Formally, this interaction reads

$$\begin{aligned}\mathcal{H}^s_{\text{mag}} = &\frac{1}{8}\sum_{j\neq j'}\Omega^{e,e}_{jj'}(1+\hat{\sigma}^z_j)(1+\hat{\sigma}^z_{j'})\\ &+\frac{1}{8}\sum_{j\neq j'}\Omega^{g,g}_{jj'}(1-\hat{\sigma}^z_j)(1-\hat{\sigma}^z_{j'})\\ &+\frac{1}{4}\sum_{j\neq j'}\Omega^{g,e}_{jj'}(1-\hat{\sigma}^z_j)(1+\hat{\sigma}^z_{j'}),\end{aligned} \tag{27}$$

where $\hat{\sigma}^z_j = \hat{\sigma}^\dagger_j\hat{\sigma}_j - \hat{\sigma}_j\hat{\sigma}^\dagger_j$ is the inversion of the atom j.[5]

The interaction after the elimination of the photonic vacuum between two static magnetic dipoles is the same as the dipolar exchange energy in classical electrostatics

$$\Omega^{\alpha,\beta}_{jj'} = \frac{\mu_0}{4\pi R^3_{jj'}}\left[3\frac{(\boldsymbol{\mu}_{j,\alpha}\cdot\boldsymbol{R}_{jj'})(\boldsymbol{\mu}_{j',\beta}\cdot\boldsymbol{R}_{jj'})}{R^2_{jj'}} - \boldsymbol{\mu}_{j,\alpha}\cdot\boldsymbol{\mu}_{j',\beta}\right], \tag{28}$$

[5]The inversion terms here are simply the excited and ground state densities.

where μ_0 is the vacuum permeability.

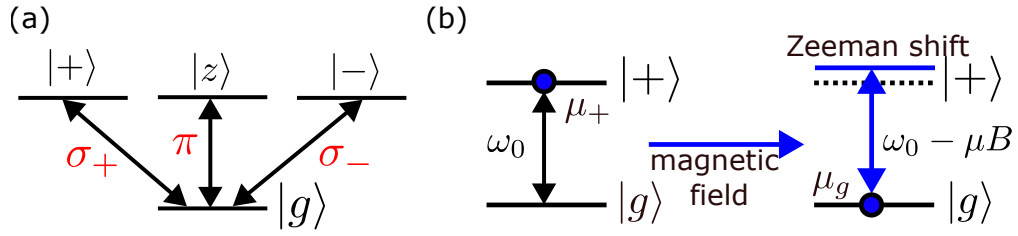


Figure 5: In (a), a four-level system with three excited states $|+\rangle, |z\rangle, |-\rangle$ of a $J = 1$ magnetic substructure is shown. On the degenerate transitions, the polarization of light that couples to these transitions is displayed. (b) illustrates how a static magnetic dipole interaction emerges between two two-level systems with $|+\rangle$ excited state levels due to the induced Zeeman shift of one scatterer on the other. The difference in dipole moment between the ground and excited state leads to a magnetic field that depends on the state of the atom and hence a state-dependent Zeeman shift for adjacent atoms.

The result for the interaction between two static dipoles is different from the Green's function for a magnetic/electric transition dipole moment since the photonic reservoir is evaluated at the frequency of the dipole in the Born-Markov approximation. The transition dipole moment rotates at $\omega = \omega_0$ whereas the static dipole moment, as the name suggests, is constant in time ($\omega = 0$).

The results when these terms are incorporated into a coupled dipole model and the low-excitation approximation are the main content in Ref. [P1]. In this context, Ref. [9] was a starting point for our investigations into this system. Furthermore, we observe modified vacuum Rabi splitting, which was reported experimentally in Ref. [98]. For dilute clouds, there exists an established scattering theory, see Ref. [99].

We also introduce the formalism for more complex electronic structures, which we discuss in the next section for regular arrangements of such emitters. First, consider an atom with different excited states with magnetic quantum numbers $m_J = -1, 0, 1$ for $J = 1$, whose interaction with the incident light depends on its polarization, see Fig. 5(a). The canonical choice of a quantization axis along the z-axis has been made at this point.

The terms in the Hamiltonian then take the form

$$\mathcal{H}_0^{\mathrm{4LS}} = \omega_0 \sum_{j,\alpha} \hat{\sigma}_{j,\alpha}^\dagger \hat{\sigma}_{j,\alpha} \tag{29}$$

$$\mathcal{H}_{\mathrm{drive}}^{\mathrm{4LS}} = \sum_{j,\alpha} \left(\eta_\alpha \mathrm{e}^{-\mathrm{i}\boldsymbol{k}_\ell \boldsymbol{R}_j} \hat{\sigma}_{j,\alpha}^\dagger + \eta_\alpha^* \mathrm{e}^{\mathrm{i}\boldsymbol{k}_\ell \boldsymbol{R}_j} \hat{\sigma}_{j,\alpha} \right) \tag{30}$$

$$\mathcal{H}_{\mathrm{d\text{-}d}}^{\mathrm{4LS}} = \sum_{j,j',\alpha,\alpha'} \Omega_{jj'}^{\alpha,\alpha'} \hat{\sigma}_{j,\alpha}^\dagger \hat{\sigma}_{j',\alpha'}. \tag{31}$$

The theoretical description now requires three transition operators between the ground state to each of the three excited states. As such, the operators $\hat{\sigma}_{j,\alpha} = |g_j\rangle \langle \alpha_j|$ correspond to the electronic transition operators between the ground state and the excited states α. A canonical naming scheme for these excited states is $\alpha = \{-, z, +\}$ which are the states with quantum numbers $m_j = \{-1, 0, +1\}$ respectively and the naming is due to the orientation of the corresponding dipole moments $\boldsymbol{d}_+ \parallel \hat{e}_+$, $\boldsymbol{d}_- \parallel \hat{e}_-$ and $\boldsymbol{d}_z \parallel \hat{e}_z$ where $\boldsymbol{d}_\alpha = \langle \alpha | -e\hat{\boldsymbol{x}} | g \rangle$ is a generalization to the two-level system electric dipole moment. Here the definition of the unit vectors $\hat{e}_+ = (1, \mathrm{i}, 0)^T/\sqrt{2}$ and $\hat{e}_- = (1, -\mathrm{i}, 0)^T/\sqrt{2}$ has been used. The splitting between the ground state and the three excited state levels is ω_0 since they are degenerate in the absence of an external magnetic field. A static magnetic field $\boldsymbol{B}$ would lead to m_j-dependent frequencies due to the Zeeman effect

$$H_{\mathrm{Zeeman}} = \sum_{j,\alpha} \omega_\alpha \hat{\sigma}_{j,\alpha}^\dagger \hat{\sigma}_{j,\alpha}, \tag{32}$$

where $\omega_\alpha = \omega_0 - (\boldsymbol{\mu}_\alpha - \boldsymbol{\mu}_g) \cdot \boldsymbol{B}$ for a static magnetic dipole moment $\boldsymbol{\mu}_\alpha$ for the excited state $|\alpha\rangle$ and external magnetic field $\boldsymbol{B}$[6]. Similar treatments are possible for larger quantum numbers J with $m_J = -J, \ldots, J$ leading to $2J + 1$ excited states. Since this magnetic substructure allows for control over the interaction of the emitter with polarization, we leverage these four-level systems in an array structure to implement optical elements in Sect. 1.4 and Ref. [P4]. Furthermore, the drive and dipole-dipole interaction also obtains a structure to reflect the coupling of different polarizations to these electronic transitions. The Rabi frequency now depends on the polarization of the incoming drive $\eta_\alpha = \boldsymbol{E}_{\mathrm{in}}^{(+)} \cdot \boldsymbol{d}_\alpha^*$ and so does the unitary part of the dipolar interaction

$$\Omega_{jj'}^{\alpha,\alpha'} = -k_0^2 \epsilon_0^{-1} \operatorname{Re} \left(\boldsymbol{d}_\alpha^* \boldsymbol{G}(\boldsymbol{R}_j - \boldsymbol{R}_{j'}) \boldsymbol{d}_{\alpha'} \right). \tag{33}$$

[6] If μB_α is large enough, the three excited states are so far split in frequency that an effective two-level model is re-obtained with static magnetic dipole moments as in Eq. (27).

The master equation must also be supplemented by the corresponding incoherent dissipators with

$$\Gamma_{jj'}^{\alpha,\alpha'}/2 = k_0^2\epsilon_0^{-1}\,\mathrm{Im}\left(\boldsymbol{d}_j^*\boldsymbol{G}(\boldsymbol{R}_j-\boldsymbol{R}_{j'})\boldsymbol{d}_{j'}\right), \tag{34}$$

leading to an equation of motion for the density matrix

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\mathrm{i}\big[\mathcal{H}_0^{\mathrm{4LS}}+\mathcal{H}_{\mathrm{drive}}^{\mathrm{4LS}}+\mathcal{H}_{\mathrm{d\text{-}d}}^{\mathrm{4LS}},\rho\big] + \sum_{j,j',\alpha,\alpha'}\Gamma_{jj'}^{\alpha,\alpha'}\left(\hat{\sigma}_{j,\alpha}\rho\hat{\sigma}_{j',\alpha'}^{\dagger} - \frac{1}{2}\Big\{\hat{\sigma}_{j',\alpha'}^{\dagger}\hat{\sigma}_{j,\alpha},\rho\Big\}_+\right). \tag{35}$$

What is left to discuss is the transition magnetic dipole moment in the Hamiltonian Eq. (25). For atoms, transitions either have a dominant magnetic or a dominant electric dipole moment, but not both since they require different selection rules (for the electric transition dipole moment to be nonzero, the parity of the wave functions must be different. For the magnetic transition dipole moment the parity must be the same). As such, it is rarely necessary to include both the magnetic and the electric dipole moment for a single atomic transition. This is not the case for molecules since molecular wavefunctions do not necessarily have a fixed parity, see Fig. 6. It is often the case in molecular systems that a transition has both an appreciable magnetic and electric dipole moment, which has significant physical consequences, such as chiral dichroism, which we will discuss in the following.

For a scatterer with an electric transition dipole moment, the Green's function for the quantum mechanical interaction between two dipoles is described by the classical Green's function from multipolar theory, which gives the electric field emitted from an electric point dipole. The same correspondence exists for magnetic dipolar transitions. To describe the entire system two Green's functions are required. The Green's function is already stated in Eq. (17) and an additional Green's function

$$\boldsymbol{F}(\boldsymbol{R}) = \frac{\mathrm{i}}{4\pi\omega_0}\boldsymbol{\nabla}\times\left(\mathbb{1}\frac{\mathrm{e}^{\mathrm{i}k_0R}}{R}\right) = \frac{(k_0R+\mathrm{i})\mathrm{e}^{\mathrm{i}k_0R}}{4\pi\omega_0R^3}\begin{pmatrix}0 & -z & y\\ z & 0 & -x\\ -y & x & 0\end{pmatrix}. \tag{36}$$

which is related to cross terms in the dipolar coupling, i.e. the electric field generated by a magnetic dipole and the magnetic field generated by an electric dipole. We now state for completeness all interaction matrix elements and all propagators for the electric and magnetic fields from electric dipoles $\boldsymbol{d}_1,\boldsymbol{d}_2$ and magnetic dipoles $\boldsymbol{\mu}_1,\boldsymbol{\mu}_2$ for an interaction between scatterers at position $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ and fields at position $\boldsymbol{R}$ due to a scatterer at position $\boldsymbol{R}_1$

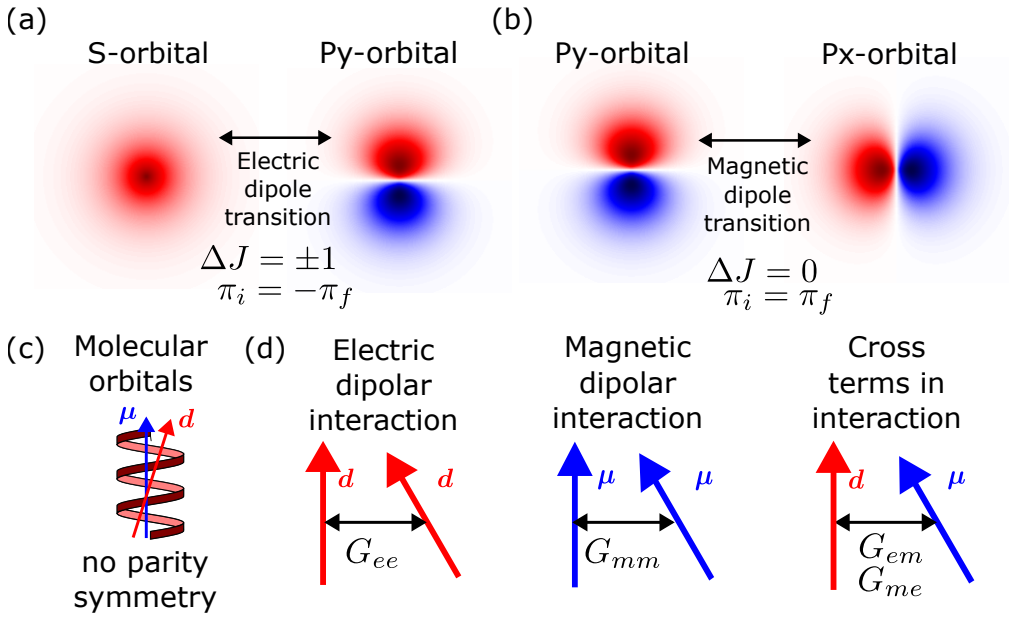


Figure 6: A comparison is shown between the symmetries required for an electric and a magnetic dipole transition and two examples where either can occur. For an electric dipole transition in (a), the difference in magnetic quantum number must be $\Delta J = \pm 1$, and the parity of the ground and excited state must be different $\pi_i = -\pi_f$. An example of such a transition is the transition between the S and P_y-orbital. For a magnetic dipole transition in (b), the magnetic quantum number must be constant $\Delta J = 0$ and the parity of ground and excited state must be the same $\pi_i = \pi_f$, such as for the P_x and P_y state. Contrasting this is a sketch of a helical molecule in (c) whose wavefunctions do not have parity symmetry so that it can have both a magnetic and electric transition dipole moment. In (d), all resulting interaction terms are sketched. The interaction between two electric transition dipoles is denoted by G_{ee}, the interaction between two magnetic transition dipole moments by G_{mm}, and the cross terms between the two by G_{me} and G_{em} respectively.

$$
\begin{aligned}
G_{ee}(\boldsymbol{R}_1 - \boldsymbol{R}_2) &= -\epsilon_0^{-1} k_0^2 \boldsymbol{d}_1^* \boldsymbol{G}(\boldsymbol{R}_1 - \boldsymbol{R}_2) \boldsymbol{d}_2 \\
G_{me}(\boldsymbol{R}_1 - \boldsymbol{R}_2) &= -\mu_0 k_0^2 \boldsymbol{\mu}_1^* \boldsymbol{F}(\boldsymbol{R}_1 - \boldsymbol{R}_2) \boldsymbol{d}_2 \\
G_{em}(\boldsymbol{R}_1 - \boldsymbol{R}_2) &= \epsilon_0^{-1} k_0^2 \boldsymbol{d}_1^* \boldsymbol{F}(\boldsymbol{R}_1 - \boldsymbol{R}_2) \boldsymbol{\mu}_2 \\
G_{mm}(\boldsymbol{R}_1 - \boldsymbol{R}_2) &= -\mu_0 k_0^2 \boldsymbol{\mu}_1^* \boldsymbol{G}(\boldsymbol{R}_1 - \boldsymbol{R}_2) \boldsymbol{\mu}_2 \\
\boldsymbol{E}^{(+)}_{e,\mathrm{scat}}(\boldsymbol{R}) &= \epsilon_0^{-1} k_0^2 \boldsymbol{G}(\boldsymbol{R} - \boldsymbol{R}_1) \boldsymbol{d}_1 \\
\boldsymbol{E}^{(+)}_{m,\mathrm{scat}}(\boldsymbol{R}) &= -\epsilon_0^{-1} k_0^2 \boldsymbol{F}(\boldsymbol{R} - \boldsymbol{R}_1) \boldsymbol{\mu}_1 \\
\boldsymbol{B}^{(+)}_{e,\mathrm{scat}}(\boldsymbol{R}) &= \mu_0 k_0^2 \boldsymbol{F}(\boldsymbol{R} - \boldsymbol{R}_1) \boldsymbol{d}_1 \\
\boldsymbol{B}^{(+)}_{m,\mathrm{scat}}(\boldsymbol{R}) &= \mu_0 k_0^2 \boldsymbol{G}(\boldsymbol{R} - \boldsymbol{R}_1) \boldsymbol{\mu}_1 .
\end{aligned}
\tag{37}
$$

Consider a single two-level scatterer with magnetic and electric transition dipole moment placed at the coordinate origin, driven by a classical field $\boldsymbol{E}_{\text{in}}^{(+)}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_\ell \boldsymbol{R}}$ as in Eq. (12). To remove the explicit time dependence of the drive, the Hamiltonian has to be transformed by

$$U(t) = \mathrm{e}^{-\mathrm{i}\omega_\ell t\hat{\sigma}^\dagger\hat{\sigma}}, \tag{38}$$

which removes the time-dependent phases from the drive, but also incurs a term $-\omega_\ell\hat{\sigma}^\dagger\hat{\sigma}$[7] and replaces the frequency difference ω_0 by the detuning $\Delta = \omega_0 - \omega_\ell$. After this procedure, an equation of motion is obtained

$$\dot{\beta} = (-\mathrm{i}\Delta + \Gamma/2)\beta - \mathrm{i}\boldsymbol{d}^*\cdot\boldsymbol{E}_{\text{in}}^{(+)} - \mathrm{i}\boldsymbol{\mu}^*\cdot\boldsymbol{B}_{\text{in}}^{(+)} \tag{39}$$

in the low-excitation limit $\langle\hat{\sigma}^z\rangle = -1$ with $\beta = \langle\hat{\sigma}\rangle$ being the expectation value of the transition operator. Equivalently, it is also the coherence of the density matrix of the two-level system. In the steady state $\dot{\beta} = 0$, the scattered field then becomes

$$\begin{aligned}\boldsymbol{E}_{\text{scat}}^{(+)}(\boldsymbol{r}) &= k_0^2\epsilon_0^{-1}\left[\boldsymbol{G}(\boldsymbol{R}-\boldsymbol{R}_1)\boldsymbol{d} + \boldsymbol{F}(\boldsymbol{R}-\boldsymbol{R}_1)\boldsymbol{\mu}\right]\beta \\ &= \frac{\mathrm{i}k_0^2\epsilon_0^{-1}}{\mathrm{i}\Delta - \Gamma/2}\left[\boldsymbol{G}(\boldsymbol{R})\cdot\boldsymbol{d} + \boldsymbol{F}(\boldsymbol{R})\cdot\boldsymbol{\mu}\right]\left[\boldsymbol{d}^*\cdot\boldsymbol{E}_{\text{in}}^{(+)} + \boldsymbol{\mu}^*\cdot\boldsymbol{B}_{\text{in}}^{(+)}\right].\end{aligned} \tag{40}$$

To see the effect of circular dichroism, consider an incoming plane wave with polarization $\boldsymbol{\epsilon} = \hat{e}_\pm$. Also, take the scattered plane-wave component $\boldsymbol{k}_s$ with the same polarization $\boldsymbol{\epsilon}_s = \hat{e}_\pm$ as the incoming field. We denote this quantity as $E_{\text{scat},\pm}^{(+)}(\boldsymbol{k}_s)$ where the $\pm$ subscript indicates the polarization. The magnetic field for an electric field with circular polarization is $H_{\text{scat},\pm}^{(+)}(\boldsymbol{k}) = \mp\frac{\mathrm{i}}{c}E_{\text{scat},\pm}^{(+)}(\boldsymbol{k})$. Then the scattered field component becomes

$$E_{\text{scat},\pm}^{(+)}(\boldsymbol{k}_s) = \frac{\mathrm{i}k_0^2\epsilon_0^{-1}G(\boldsymbol{k}_s)E_{\text{in}}}{\mathrm{i}\Delta - \Gamma/2}\left[\hat{e}_\pm\cdot\left(\boldsymbol{d}\mp\frac{\mathrm{i}}{c}\boldsymbol{\mu}\right)\left(\boldsymbol{d}^*\pm\frac{\mathrm{i}}{c}\boldsymbol{\mu}^*\right)\cdot\hat{e}_\pm\right]. \tag{41}$$

The quantity $G(\boldsymbol{k})$ accounts for the propagation of the plane wave component of the electromagnetic field and does not depend on whether the polarization is left or right-handed. Rotational averaging[8] for different emitter orientations of this gives

$$\left\langle E_{\text{scat},\pm}^{(+)}(\boldsymbol{k}_s)\right\rangle = \frac{\mathrm{i}k_0^2\epsilon_0^{-1}G(\boldsymbol{k}_s)E_{\text{in}}}{3(\mathrm{i}\Delta - \Gamma/2)}\left|\boldsymbol{d}\mp\frac{\mathrm{i}}{c}\boldsymbol{\mu}\right|^2. \tag{42}$$

The circular dichroism can be determined as the difference between the left-handed

[7] For a time-dependent unitary transformation, one can derive that the Hamiltonian of the system is modified as $H' = U(t)HU^\dagger(t) - \mathrm{i}\dot{U}^\dagger U$.

[8] The rotational average of tensors can be derived in general. For details on this, see Ref. [89].

field component and the right-handed field component due to the same-polarization incident field

$$\left\langle E^{(+)}_{\mathrm{scat},+}(\boldsymbol{k}_s)\right\rangle - \left\langle E^{(+)}_{\mathrm{scat},-}(\boldsymbol{k}_s)\right\rangle = \frac{4k_0^2 G(\boldsymbol{k}_s)E_{\mathrm{in}}}{3c(\mathrm{i}\Delta - \Gamma/2)}\,\mathrm{Im}\left(\boldsymbol{d}^*\cdot\boldsymbol{\mu}\right), \tag{43}$$

which depends on the so-called rotary strength $\mathrm{Im}\left(\boldsymbol{d}^*\cdot\boldsymbol{\mu}\right)$ of the transition. The non-vanishing of the scalar product between the electric and magnetic dipole moments indicates a helical structure of the currents associated with the dipole moments.

1.4 Coupling of atomic arrays to light

We now move to the scattering of light from regular structures, in particular from two-dimensional arrays of atoms. The interest in such structures mainly originates in the numerical discovery in Ref. [32], that an array of atoms can perfectly reflect incoming light, acting as a highly frequency-dependent mirror. Then analytical theory followed, which showed how this could be understood for infinite arrays in Ref. [31], and the experiment followed soon after [28]. Due to the regular arrangement of the atoms, this provides an interface between condensed matter physics and quantum optics, which is supported by the discovery of topological effects in such arrays [38, 39]. A recent review of the developments in this field is Ref [30]. We also show how the band structures of such arrays emerge and can be useful for the characterization of the light scattering from them. These basics are particularly relevant for Ref. [P4].

The starting point for this discussion is the Hamiltonian $\mathcal{H}_0^{\mathrm{TLS}} + \mathcal{H}_{\mathrm{d\text{-}d}}^{\mathrm{TLS}} + \mathcal{H}_{\mathrm{drive}}^{\mathrm{TLS}}$ from Eq. (13) with the associated dissipators for correlated spontaneous emission $L[\rho]$ from Eq. (14) arranged as in Fig. 7(a). There are now two limiting cases, both corresponding to a low-excitation sector in which this problem can be linearized and then solved by means of calculating the photonic band structure for the underlying lattice. On one hand, the problem becomes linear via a projection of the array Hilbert space onto the single excitation subspace. The resulting dynamics can be described by a non-Hermitian Hamiltonian, which can be diagonalized via a Fourier transform. Another way to approximate this linearly is to go into the linear optics regime, which is equivalent to performing a Holstein-Primakoff transformation [100] on the two-level systems and omitting all non-quadratic terms. The photonic band structure is the essential quantity since it is the result of the diagonalization of the problem in the linear regime and thus contains the frequencies and decay rates of all array eigenmodes. For the definitions related to lattices and reciprocal lattices, which we use in the following discussion, see

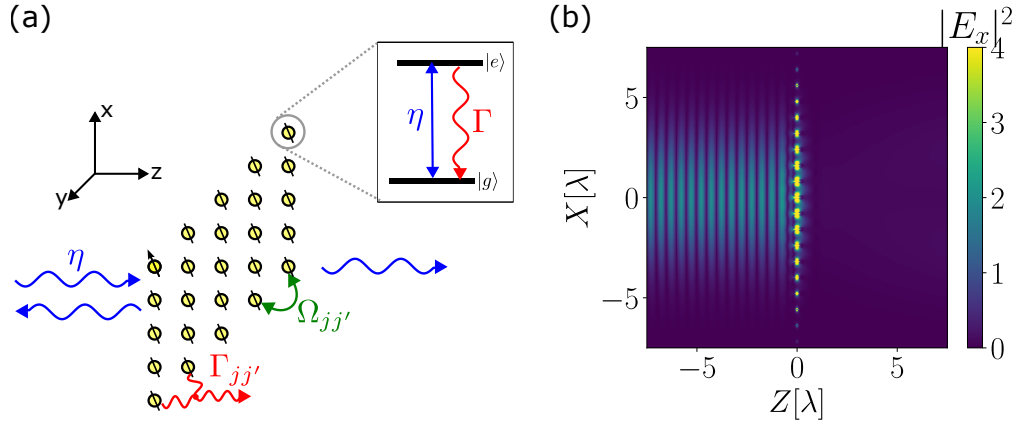


Figure 7: In (a), a sketch of two-level atoms with a dipole moment $\boldsymbol{d}$ and spontaneous decay rate Γ arranged in an array with an incident laser with wave vector $\boldsymbol{k}_\ell$ and Rabi frequency η which we assume to be a plane wave is shown. The normal of the array is chosen in $\hat{e}_z$-direction, and the array is infinitely extended in the xy-plane. Plotted in (b) is a simple simulation where a laser whose polarization is aligned with the dipole moments of the atoms $\hat{d} = \hat{e}_x$ is perfectly reflected from the array. For the simulation, a Gaussian beam polarized in the x-direction instead of a plane wave for a more realistic scenario and to reduce edge effects from the finite array was chosen. We consider a square lattice with spacing $a = 0.8\lambda$ and $N = 15^2$ emitters.

Fig. 8.

We now show that these two linearization procedures are equivalent. In the single excitation subspace[9] the system dynamics are governed by a non-Hermitian Hamiltonian

$$H_{\text{array}}^{(\text{single})} = \omega_0 \sum_{j=1}^{N} |j\rangle\langle j| + \sum_{jj'}^{N} \left[\Omega_{jj'} - \mathrm{i}\Gamma_{jj'}/2\right] |j'\rangle\langle j|\,, \tag{44}$$

where $|j\rangle = |g_1, \ldots, e_j, \ldots, g_N\rangle$ is the state where the jth atom is in the excited state. On the other hand, consider the time evolution of the optical coherences under the low-excitation approximation

$$\dot{\beta}_j = (-\mathrm{i}\omega_0 - \Gamma/2)\,\beta_j - \sum_{j'}^{N} \left[\mathrm{i}\Omega_{jj'} + \Gamma_{jj'}/2\right] \beta_{j'}. \tag{45}$$

The matrix elements of the non-Hermitian Hamiltonian and the dynamical matrix defined by $\dot{\boldsymbol{\beta}} = -\mathrm{i}\mathcal{H}\boldsymbol{\beta}$ coincide exactly. As such, the description of the single excitation

[9]The dynamics between the single excitation subspace are completely defined by a non-Hermitian Hamiltonian. In the full master equation, there would also be quantum jumps to the ground state. These are omitted in the non-Hermitian formulation.

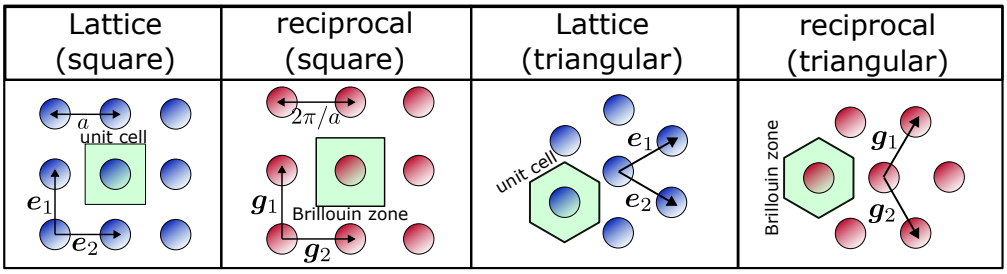


Figure 8: Illustrated here are the concepts of the Bravais lattice and the associated reciprocal lattice. The 2D lattice $\Lambda = \{n\boldsymbol{e}_1 + m\boldsymbol{e}_2 \text{ for } n, m \in \mathbb{Z}\}$ is the set of all points for linearly independent primitive vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. The choice of a primitive vector determines which lattice is formed. Such lattices are called Bravais lattices. The reciprocal lattice Λ^* is the Bravais lattice spanned by reciprocal lattice vectors $\boldsymbol{g}_1$ and $\boldsymbol{g}_2$ defined such that $\boldsymbol{e}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$. In the case of the square lattice, $\boldsymbol{e}_1 = (a, 0)$ and $\boldsymbol{e}_2 = (0, a)$ the reciprocal lattice is also a square lattice $\boldsymbol{g}_1 = 2\pi/a(1, 0)$ and $\boldsymbol{g}_2 = 2\pi/a(0, 1)$. It is a distinct feature of 2D lattices that the reciprocal lattice is of the same type as the original lattice. The momentum states mentioned in the text are the elements of the Brillouin zone $\mathcal{B}$, which is the set of momentum vectors that are closest to a particular point in reciprocal space, usually the origin (See Ref. [101] for more details). The Brillouin zone for the reciprocal lattices is indicated in green.

subsector and the linear approximation for the atomic scatterers are very similar. We proceed with the classical picture since it is more consistent with the treatments in our publications and the other parts of this thesis, though a single excitation subspace treatment is more common in the literature.

For a two-dimensional array of atoms, this problem is diagonalized by a Fourier transform $\widetilde{\beta}_{\boldsymbol{k}} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}_j \in \Lambda} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}_j} \beta_j$ where Λ is the set of all points in the Bravais lattice which constitutes the array and $\boldsymbol{k} \in \mathcal{B}$ are elements of the Brillouin zone. We focus on the square lattice which is a Bravais lattice, see Fig. 8. Consider now an array driven by a laser with frequency ω_ℓ and wave vector $\boldsymbol{k}_\ell$ in the rotating frame. Then the equation of motion becomes

$$\partial_t \widetilde{\beta}_{\boldsymbol{k}} = (-\mathrm{i}\Delta - \Gamma/2)\, \widetilde{\beta}_{\boldsymbol{k}} - \left(\mathrm{i}\widetilde{\Omega}_{\boldsymbol{k}} + \widetilde{\Gamma}_{\boldsymbol{k}}/2\right) \widetilde{\beta}_{\boldsymbol{k}} - \mathrm{i}\eta\sqrt{N}\delta_{\boldsymbol{k},\boldsymbol{k}_\ell}. \tag{46}$$

where we used the lattice Fourier transform of the dissipative and unitary part of the dipolar interaction, equivalent to the atomic coherences. The off-diagonal dipolar interactions now become collective frequency and linewidth modifications of the spin waves on the array. Consider the lattice Fourier transform of a quantity $A(\boldsymbol{R})$ defined on

the array[10]

$$\widetilde{A}(\boldsymbol{k}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}\in\Lambda} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}} A(\boldsymbol{R}). \tag{47}$$

It is well known that such lattice summations can be turned into sums over the reciprocal lattice via the Poisson summation formula, which physically corresponds to an expansion in scattering orders. Formally, the Poisson formula states that for a lattice Λ and its reciprocal lattice Λ^*, and defining $\det\Lambda$ as the volume of the unit cell of Λ,[11]

$$\widetilde{A}(\boldsymbol{k}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}\in\Lambda} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}} A(\boldsymbol{R}) = \frac{(2\pi)^2}{\det\Lambda} \sum_{\boldsymbol{g}\in\Lambda^*} \bar{A}(\boldsymbol{k}+\boldsymbol{g}), \tag{48}$$

where $\bar{A}(\boldsymbol{k})$ is the continuous 2D Fourier transform of $A(\boldsymbol{R})$ in the lattice plane[12]. Finding the values of $\Omega_{\boldsymbol{k}}$ and $\Gamma_{\boldsymbol{k}}$ as well as the structure of the scattered light $\boldsymbol{E}(\boldsymbol{R})$ requires the two-dimensional Fourier transform of the free space dipolar Green's function which is also known as the Weyl expansion of the plane wave

$$\frac{\mathrm{e}^{\mathrm{i}k_0 R}}{R} = \frac{\mathrm{i}}{2\pi} \int_{\mathbb{R}^2} \mathrm{d}\boldsymbol{q} \frac{1}{q_z} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_\parallel} \mathrm{e}^{\mathrm{i}q_z|Z|}, \tag{49}$$

where $\hat{e}_z$ direction is the array normal. The vector $\boldsymbol{q}$ is the two-dimensional projection of the wave vector on the array plane and can be understood as a momentum state on the array. Since the Green's function can be written as a derivative of a spherical wave, see Eq. (17), its two-dimensional Fourier transform can now be derived to be

$$\bar{\boldsymbol{G}}(\boldsymbol{q}; Z) = \frac{\mathrm{i}}{8\pi^2} \left(\mathbb{1} - \frac{\boldsymbol{v}(\boldsymbol{q}, Z) \otimes \boldsymbol{v}(\boldsymbol{q}, Z)}{k_0^2} \right) \frac{\mathrm{e}^{\mathrm{i}q_z|Z|}}{q_z}, \tag{50}$$

where $\boldsymbol{v}(\boldsymbol{q}, Z) = (q_x, q_y, \mathrm{sgn}(Z)\sqrt{k_0^2 - q^2})$ takes into account the absolute value in the normal direction in the Weyl expansion as well as the constraint that $\|\boldsymbol{v}(\boldsymbol{q}, Z)\| = k_0$ momentum is conserved so that one can parameterize $q_z = \sqrt{k_0^2 - q^2}$. It is now clear why it is useful to write the Green's functions as derivatives of a spherical wave. Then the lattice Fourier transform of the Green's function becomes

[10] To keep prefactors simple, we will formally consider N emitters in the array.
[11] The volume of the unit cell can be found as the cross product of the primitive vectors $\det\Lambda = \boldsymbol{p}_1 \times \boldsymbol{p}_2$.
[12] If the dimensionality of the structure is different, it will be the d-dimensional Fourier transform instead where d is the dimension of the structure.

$$\widetilde{\boldsymbol{G}}(\boldsymbol{q};Z)=\frac{1}{2\det\Lambda}\sum_{\boldsymbol{g}\in\Lambda^*}\left(\mathbb{1}-\frac{\boldsymbol{v}(\boldsymbol{q}+\boldsymbol{g},Z)\otimes\boldsymbol{v}(\boldsymbol{q}+\boldsymbol{g},Z)}{k_0^2}\right)\frac{\mathrm{e}^{\mathrm{i}\sqrt{k_0^2-(\boldsymbol{q}+\boldsymbol{g})^2}|Z|}}{\sqrt{k_0^2-(\boldsymbol{q}+\boldsymbol{g})^2}}. \tag{51}$$

The key observation for an analytical simplification is to consider the sign of the term in the square root $k_0^2-(\boldsymbol{q}+\boldsymbol{g})^2$. For the sake of simplicity assume that the array is square so that $\boldsymbol{g}=\frac{2\pi}{a}\left[n\hat{e}_x+m\hat{e}_y\right]$ with $n,m\in\mathbb{Z}$, $k_0=\frac{2\pi}{\lambda}$ and a parameterization $\boldsymbol{q}=\frac{2\pi}{\lambda}(\alpha,\beta)$ with $\alpha^2+\beta^2<1$. Then the condition $k_0^2-(\boldsymbol{q}+\boldsymbol{g})^2>0$ is equivalent to $\left(\alpha-n\frac{\lambda}{a}\right)^2+\left(\beta-m\frac{\lambda}{a}\right)^2<1$. This is always true when $n=m=0$. With the condition that $a<\lambda/2$, this is only true when $n=m=0$. For the completely symmetric mode $\boldsymbol{q}=0$ and with the conditions that $\alpha=\beta=0$ and $a<\lambda$, this is only true when $n=m=0$. This has consequences for the dipolar interaction and the electric field in the far-field $|\boldsymbol{R}|\gg\lambda$, making an analytical treatment for subwavelength atomic arrays possible[13].

In the case of the electric field in the far-field, one makes use of the fact that the positivity argument above means that only the $\boldsymbol{g}=0$ term contributes in $\widetilde{\boldsymbol{G}}(\boldsymbol{q};Z\gg\lambda)$ for $a<\lambda/2$ since the other terms become evanescent. Similarly, the spontaneous decay rate of the collective modes of the array is given by $\operatorname{Im}\widetilde{\boldsymbol{G}}(\boldsymbol{q};Z=0)$ to which only the $\boldsymbol{g}=0$ term contributes since the higher order terms are purely imaginary. For the collective frequency, all terms contribute, and it has to be evaluated numerically (although a few orders in this reciprocal lattice summation are required for quantitative results). We now present the analytical results that follow from these arguments.

The far-field propagator under the subwavelength condition becomes

$$\widetilde{\boldsymbol{G}}^{\mathrm{far}}(\boldsymbol{q};Z)\approx\frac{\mathrm{i}}{2\det\Lambda}\frac{\mathrm{e}^{\mathrm{i}q_z|Z|}}{q_z}\left(\mathbb{1}-\frac{\boldsymbol{v}(\boldsymbol{q},Z)\otimes\boldsymbol{v}(\boldsymbol{q},Z)}{k_0^2}\right), \tag{52}$$

confirming that the mode radiated in the far field is a plane wave. On resonance, this plane wave then perfectly interferes with the incoming field, leading to perfect reflection, as simulated in Fig. 7(b). The collective decay rate for array mode $\beta_{\boldsymbol{q}}$ becomes

$$\widetilde{\boldsymbol{\Gamma}}(\boldsymbol{q})\approx\frac{3\Gamma_0}{4\pi}\left(\frac{\lambda^2}{\det\Lambda}\right)\frac{k_0}{q_z}\left(1-\left|\hat{v}(\boldsymbol{q})\cdot\hat{d}\right|^2\right). \tag{53}$$

Notably, the term $\left|\hat{v}(\boldsymbol{q})\cdot\hat{d}\right|^2$ encodes the effects of polarization in the decay of the

[13]The previous in-depth discussion elucidates that the condition for being subwavelength only suffices for normal incidence and $a<\lambda/2$ is required for arbitrary incidence angles.

modes on the array and ensures that modes with dipole moment perpendicular to the radiated plane wave in the far-field do not radiate at all. For instance, the symmetric $\boldsymbol{q} = 0$ mode with dipoles oriented in $\hat{e}_z$-direction does not radiate for an infinite array.

These calculations were performed to illustrate the underlying principles when dealing with periodic subwavelength arrays. They can easily be extended to magnetic transition dipole elements by considering the off-diagonal Green's function in Eq. (36), to more complex level structures and more atoms per unit cell by introducing additional indices and then solving a local matrix problem. Effects that break translational invariance are more difficult to treat and may only be accessible in perturbation theory.

A particularly relevant reference is Ref. [29], which discusses electric dipole, magnetic dipole, electric quadrupole, and magnetic quadrupole interactions on an array of atoms and calculates the appropriate Fourier transforms for their analytical treatment. Furthermore, the theory of subwavelength atomic arrays in the linear regime is deeply connected to the more established field of metasurfaces, where many possible technological applications have already been explored [102–104].

1.5 Interaction of light with molecules

In the following, we discuss a simple model for a molecule and how it couples to light. This is particularly relevant for Ref. [P3], but also motivates the toy model used in Ref. [P5] to investigate the cooling of molecules in a cavity. This is also a starting point for future research and serves as a bridge to the outlook of this thesis. The groundwork for much of this was laid in Ref. [105], but Holstein-type Hamiltonians are well studied in the literature in general [106–108].

Consider, as a starting point to describe the physics governing a molecule, the Hamiltonian

$$H = \hat{T}_e + \hat{T}_n + \hat{U}(\hat{\boldsymbol{r}}, \hat{\boldsymbol{Q}}), \tag{54}$$

where $\hat{T}_e$ is the electronic kinetic energy and $\hat{T}_n$ is the nuclear kinetic energy. $\hat{U}(\hat{\boldsymbol{r}}, \hat{\boldsymbol{Q}})$ is the electronic and nuclear potential energy for a vector of all nuclear displacements $\hat{\boldsymbol{Q}}$ and a vector for the electronic displacements $\hat{\boldsymbol{r}}$. If the nuclear kinetic energy is ignored, the electronic Hamiltonian is obtained as $H_e = \hat{T}_e + \hat{U}(\hat{\boldsymbol{r}}, \boldsymbol{Q})$ where $\boldsymbol{Q}$ now only plays the role of a parameter. We can define the orthonormal wavefunctions that diagonalize this electronic Hamiltonian via the eigenvalue equation

$$H_e \Phi_n(\boldsymbol{r}, \boldsymbol{Q}) = \hat{V}_n(\boldsymbol{Q}) \Phi_n(\boldsymbol{r}, \boldsymbol{Q}). \tag{55}$$

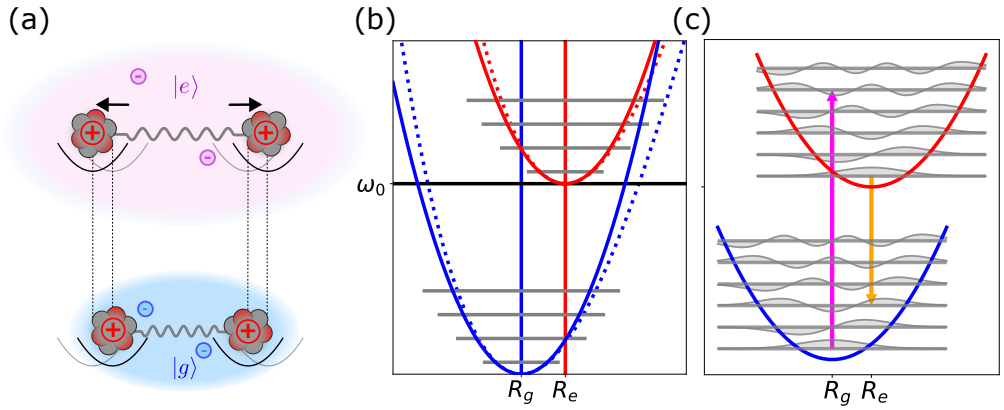


Figure 9: In (a) is a sketch of the equilibrium nuclear configurations of a vibrational mode in the ground and excited state. There is a particular emphasis on the fact that these equilibrium configurations can, in principle, be different since the underlying electronic configurations differ. In (b), we illustrate two potential energy surfaces of a molecule as Morse potentials in dotted lines. We also overlay two harmonic oscillators, which are displaced by the ground and excited state equilibrium displacement, and the corresponding equidistant energy levels. The equilibrium nuclear displacements are indicated as vertical lines. In (c), the Franck-Condon physics of these two potential energy surfaces in the harmonic approximation is illustrated. The overlap between the displaced wavefunctions of the harmonic oscillator gives the contribution for the orange emission and magenta absorption process, respectively.

The electronic energies $\hat{V}_n(\boldsymbol{Q})$ (also referred to as the potential energy surfaces) and eigenfunctions $\Phi_n(\boldsymbol{r}, \boldsymbol{Q})$ now depend parametrically on the nuclear coordinates $\boldsymbol{Q}$. The complete Schroedinger equation is solved by some wavefunction $\Psi(\boldsymbol{r}, \boldsymbol{Q})$

$$H\Psi(\boldsymbol{r}, \boldsymbol{Q}) = E\Psi(\boldsymbol{r}, \boldsymbol{Q}). \tag{56}$$

It is now useful to expand this wavefunction in the basis $\Phi_n(\boldsymbol{r}, \boldsymbol{Q})$, which is orthonormal in the electronic subspace for each choice of $\boldsymbol{Q}$

$$\Psi(\boldsymbol{r}, \boldsymbol{Q}) = \sum_{n=1}^{N_e} \chi_n(\boldsymbol{Q})\Phi_n(\boldsymbol{r}, \boldsymbol{Q}), \tag{57}$$

where $\chi_n(\boldsymbol{Q})$ is the nuclear wavefunction and $\Phi_n(\boldsymbol{r}, \boldsymbol{Q})$ is the electronic wavefunction. Here N_e denotes the number of orbitals. We will now perform the Born-Oppenheimer or "crude" adiabatic approximation[14] where we assume that the electronic and motional

[14] It is called crude since it implies that the electronic orbital remains the same over a given potential surface.

degrees of freedom decouple $\Phi_n(\boldsymbol{r}, \boldsymbol{Q}) \approx \Phi_n(\boldsymbol{r})$. This is usually motivated by the argument that the nuclear mass is much larger than the electronic mass, so that the electronic motion is slaved to the nuclear motion.

Since the electronic and nuclear problems decouple completely, the Schroedinger equation for the nuclear wavefunction becomes

$$\left[\hat{T}_N + \hat{V}_n(\hat{\boldsymbol{Q}}) - E\right] \chi_n(\boldsymbol{Q}) = 0. \tag{58}$$

We now project this problem onto two relevant orbitals, in a similar fashion as the two-level approximation for the atomic case, so that the Hamiltonian projected onto the excited and ground state manifold becomes

$$\mathcal{H} = \hat{T}_N \mathbb{1} + \begin{pmatrix} \hat{V}_e(\hat{\boldsymbol{Q}}) & 0 \\ 0 & \hat{V}_g(\hat{\boldsymbol{Q}}) \end{pmatrix}, \tag{59}$$

with an excited state potential surface $\hat{V}_e(\hat{\boldsymbol{Q}})$ and a ground state potential surface $\hat{V}_g(\hat{\boldsymbol{Q}})$. Examples of such potential surfaces in one dimension are illustrated in Fig. 9(b). In many cases, it is now a good approximation to perform a Taylor expansion of these potentials around the equilibrium positions, which are minima of the potentials, up to quadratic order to obtain a harmonic oscillator model. Thus,

$$\begin{aligned} \hat{V}_e(\boldsymbol{Q}) &\approx V_e(\boldsymbol{Q}_e) + \frac{1}{2}\sum_{p=1}^{\mathcal{N}} (\hat{Q}_p - Q_{e,p})^2 \frac{\partial^2 V_e(\boldsymbol{Q}_e)}{\partial Q_p^2} \\ \hat{V}_g(\boldsymbol{Q}) &\approx V_g(\boldsymbol{Q}_g) + \frac{1}{2}\sum_{p=1}^{\mathcal{N}} (\hat{Q}_p - Q_{g,p})^2 \frac{\partial^2 V_g(\boldsymbol{Q}_g)}{\partial Q_p^2}, \end{aligned} \tag{60}$$

where the nuclear coordinates are chosen such that the mixed partial derivatives of the second order vanish. $\mathcal{N}$ denotes the number of nuclear coordinates[15] enumerated by p. Here, $Q_{g/e,p}$ are the ground state and excited state equilibrium positions of the p-th nuclear coordinate. The vector $\boldsymbol{Q}_{g/e}$ contains the equilibrium coordinates of all modes. This approximation to the Hamiltonian is illustrated in Fig. 9(b). Note that the ground state potential surface and the excited state potential surface do in principle not have the same equilibrium configuration $\boldsymbol{Q}_{e,p} \neq \boldsymbol{Q}_{g,p}$ which is illustrated in Fig. 9(a) and Fig. 9(b). Since the electronic configuration is different in the ground and the excited state, the minimum energy configuration of the nuclei also changes as a consequence. The coordinate origin is chosen at the position of the ground state equilibrium coordinate $\boldsymbol{Q}_{g,p} = 0$. The

[15] Given N atoms, the number of nuclear normal modes is $\mathcal{N} = 3N - 6$ for a non-planar molecule.

potential energy spacing is associated with the two-level splitting $\omega_0 = V_e(\boldsymbol{Q}_e) - V_g(0)$. Furthermore, the curvatures in the excited and ground states are assumed to be the same $\frac{\partial^2 V_e(\boldsymbol{Q})}{\partial Q_p^2} = \frac{\partial^2 V_g(\boldsymbol{Q})}{\partial Q_p^2}$. Each vibrational mode corresponds to an effective mass μ_p so that the curvature can be written as $\frac{\partial^2 V_g(\boldsymbol{Q})}{\partial Q_p^2} = \mu_p \nu_p^2$ with vibrational frequency ν_p. Then, the nuclear coordinates can be written in terms of bosonic creation and annihilation operators $\hat{Q}_p = Q_{\mathrm{zpm},p}(\hat{b}_p^\dagger + \hat{b}_p)$ and $\hat{P}_p = \mathrm{i}P_{\mathrm{zpm},p}(\hat{b}_p^\dagger - \hat{b}_p)$ with zero point motion $Q_{\mathrm{zpm},p} = 1/\sqrt{2\mu_p\nu_p}$ and zero point momentum $P_{\mathrm{zpm},p} = \sqrt{\mu_p\nu_p/2}$. Inserting all of these definitions and using Pauli matrices, the Hamiltonian takes the form

$$\mathcal{H} = \left(\omega_0 + \sum_{p=1}^{\mathcal{N}} s_p\right)\hat{\sigma}^\dagger\hat{\sigma} + \sum_{p=1}^{\mathcal{N}} \nu_p \hat{b}_p^\dagger \hat{b}_p - \sqrt{s_p}\nu_p\left(\hat{b}_p^\dagger + \hat{b}_p\right)\hat{\sigma}^\dagger\hat{\sigma}, \tag{61}$$

where the dimensionless Huang-Rhys factors are defined as

$$\sqrt{s_p} = \mu_p \nu_p Q_{e,p} Q_{\mathrm{zpm},p}\,. \tag{62}$$

This formulation of the Hamiltonian in terms of vibrational modes instead of continuous nuclear vibrational coordinates for a harmonic approximation to the potential energy surfaces is shown in Fig. 9(c), where each of the vibrations is now expressed in quanta of energy ν_p.

Depending on the environment, additional terms need to be included in this description. We consider three different scenarios. The molecules can be in a solid state matrix, in solution, or in a gaseous phase. The first two are fundamentally different from the gas phase. In a solid, for instance, DBT (dibenzoterrylene) in Anthracene, the molecule is also coupled to the lattice vibrations. The linear coupling of the phonons to the molecule then leads to modifications of the spectroscopic response, whereas higher-order terms lead to thermalization of the vibrational modes of the molecule and dephasing due to quadratic phonon couplings. This can be combined in the dissipators

$$\mathcal{L}[\rho] = \gamma_\mathrm{r} L\left[\hat{\sigma},\rho\right] + \gamma_\mathrm{d} L\left[\hat{\sigma}^\dagger\hat{\sigma},\rho\right] + \sum_p \Gamma_p L\left[\hat{b}_p - \sqrt{s_p}\hat{\sigma}^\dagger\hat{\sigma},\rho\right], \tag{63}$$

where $L[\hat{A},\rho] = \hat{A}\rho\hat{A}^\dagger - \frac{1}{2}\left\{\hat{A}^\dagger\hat{A},\rho\right\}_+$. The particular form of this dissipator might seem odd at first since in a phenomenological description, the dissipator $\hat{b}_p$ is more natural than $\hat{b}_p - \sqrt{s_p}\hat{\sigma}^\dagger\hat{\sigma}$ for the vibrational relaxation. A proper elimination of a thermal reservoir coupled to the vibrational degrees of freedom, however, reveals that one only obtains proper thermal relaxation using the "global" dissipator $\hat{b}_p - \sqrt{s_p}\hat{\sigma}^\dagger\hat{\sigma}$, which is

obtained using the global secular approximation outlined in Ref. [109]. This calculation is unpublished.

In Ref. [P3], we discuss the physics of dye aggregates in solution, which was inspired in particular by the seminal article in Ref. [50]. Thus, the vibrational relaxation of the modes in the aggregate is appreciable. The key addition to the terms described until now is the dipolar interaction between molecules in the aggregate, since they are only separated by a few nanometers. We employ the free space Green's function since the effect of the solution is usually only to change the refractive index, thus renormalizing the molecular distance.

The dipolar interaction Green's function from Eq. (19) also describes the unitary part of the dipolar interaction between molecules

$$H_{\text{d-d}}^{\text{TLS}} = \sum_{jj'} \Omega_{jj'} \hat{\sigma}_j^\dagger \hat{\sigma}_{j'} \,. \tag{64}$$

To treat a chain of emitters this dipolar interaction is first diagonalized via the momentum states $\hat{\sigma}_k = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \mathrm{e}^{-\mathrm{i}\pi \frac{kj}{N+1}} \hat{\sigma}_j$ on the chain and the transfer from the symmetric state $\hat{\sigma}_{\boldsymbol{k}=0}$ to the non-symmetric manifold $\left\{\hat{\sigma}_k, k \neq 0\right\}$ is calculated to first order in the Huang Rhys factors (which is technically a second-order perturbation theory in the coupling terms $\nu_j \sqrt{s_j}$). This procedure was performed in Ref. [P3] and analytical formulas in the nearest neighbor coupling regime were found. The results then physically represent rate equations for Kasha's rule in an H-aggregate where the symmetric state has a larger frequency than the non-symmetric states.

1.6 Doppler cooling of molecules

In Ref. [P5], we investigated the problem of Doppler cooling of an ensemble of molecules. From the scenarios described above, we consider now that the molecules are in a gaseous phase so that vibrational relaxation occurs extremely slowly compared to the optical decay rate $\Gamma_p \ll \Gamma$. This illustrates the fundamental difficulty with the laser cooling of molecules. Most cooling techniques rely on the fact that the emitter in question relaxes back into the electronic ground state after excitation, so that the next cooling cycle can begin. However, molecules generally contain rovibrational manifolds with long-lived states where excitations can get stuck. These states then need to be repumped to the electronic ground state to continue cooling. This has been done for smaller molecules with few vibrational modes for which this is feasible and for which the population of these modes is already unlikely to begin with [75, 76, 80, 110].

Due to recent experiments [86] where a cavity was used to Purcell enhance the zero-vibration line, we investigated this setup for cooling. Additionally, we investigated effects due to the collective cooperativity of an ensemble of molecules in a cavity. Since the description of molecules using the model derived in the previous section is still too complex for a description of many emitters, we modeled the molecules as three-level Λ-type systems with a ground state $|g\rangle$, an optical excited state $|e\rangle$ and a reservoir state $|i\rangle$ to model irreversible loss into the rovibrational manifold as illustrated in Fig. 10(a).

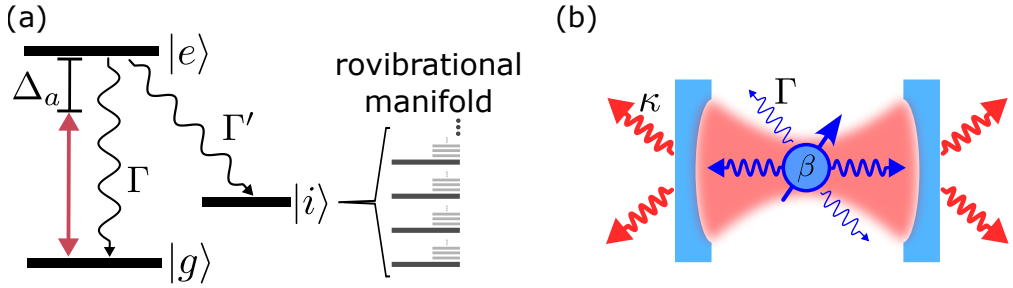


Figure 10: In (a), the toy three-level Λ system that is used to model molecular systems is shown, where excitations are irreversibly lost to the reservoir state $|i\rangle$ at rate Γ'. In (b), a cavity-atom system is sketched where the atom can either directly emit into free space or via the cavity, leading to a Purcell enhancement of spontaneous emission.

The Hamiltonian, which describes an ensemble of molecules coupled to a cavity including the motional degrees of freedom, is

$$\mathcal{H}_{\text{cooling}} = \sum_j \left[\frac{\hat{p}_j^2}{2m} + \Delta_a \hat{\sigma}_j^\dagger \hat{\sigma}_j \right] + \Delta_c \hat{a}^\dagger \hat{a} \; + \sum_j g(\hat{x}_j) \left[\hat{\sigma}_j^\dagger \hat{a} + \hat{\sigma}_j \hat{a}^\dagger \right] + \eta(\hat{a} + \hat{a}^\dagger), \quad (65)$$

where $g(x) = g\cos(k_c x)$ for a planar cavity with a standing wave as a mode function and light-matter coupling constant $g = \sqrt{\frac{\omega_c}{2\epsilon_0 V}} d$ for a cavity volume V and molecular transition dipole moment magnitude $d = \|\boldsymbol{d}\|$. The cavity is driven directly with amplitude η. In addition to an atomic detuning $\Delta_a = \omega_0 - \omega_\ell$, we must also account for the cavity detuning $\Delta_c = \omega_c - \omega_\ell$ where ω_c is the resonance frequency of the cavity and k_c is the corresponding cavity wave number. The problem is reduced to one dimension along the cavity axis with a particle momentum $\hat{p}_j$ and particle position $\hat{x}_j$ along this axis. The intermediate reservoir level is only included via dissipative terms

$$\mathcal{L}[\rho] = \Gamma \sum_j L[\mathrm{e}^{-\mathrm{i}k_0 \hat{x}_j} \hat{\sigma}_j, \rho] + \Gamma' \sum_j L[|i_j\rangle \langle e_j|, \rho] + \kappa L[\hat{a}, \rho], \quad (66)$$

where Γ is the spontaneous decay rate of the $e - g$ transition and Γ' the irreversible loss

rate into the reservoir for each scatterer as illustrated in Fig. 10. The cavity loss is also included via the decay rate κ, which describes the tunneling of photons through the cavity mirrors.

The recoil from spontaneous emission is included by choosing $\mathrm{e}^{-\mathrm{i}k_0\hat{x}_j}\hat{\sigma}_j$ as the dissipator instead of $\hat{\sigma}_j$. On the mean-field level, however, this is not relevant, and we proceeded with neglecting these terms since the final velocity for these three-level systems is much larger than the velocity associated with the Doppler temperature.

1.7 Conclusions

We used this introduction to give some overarching context on the work performed for this cumulative thesis and made some connections between the works which may have seemed disjoint before. Furthermore, we have given the most relevant references for the particular publications in the chapters dedicated to them.

To conclude, we summarize some of the results of our publications and connect the scientific introduction to the papers. First, we investigated an ensemble of Dysprosium scatterers in a cold cloud and the effect of the static magnetic dipole moments of the atoms in Ref. [P1]. We found modifications of superradiance and sub-radiance, and in particular a modification of the vacuum Rabi splitting. Sect. 1.2 introduces the concepts needed to understand most of this paper, and in Sect. 1.3 we discuss more complex scatterers. Then, we investigated atomic subwavelength arrays in the two publications Ref. [P4] and Ref. [P2] for technological applications, i.e. linear optical elements, and chiral sensing. The analytical treatment of infinite two-dimensional arrays of atoms is outlined in Sect. 1.4. Lastly, in Ref. [P3] and Ref. [P5] we discuss the Kasha rule for H-aggregates of molecules and Purcell-enhanced Doppler cooling of molecules. The Holstein model for molecules used in Ref. [P3] is introduced and discussed in Sect. 1.5 and the toy model used for Ref. [P5] is introduced in Sect. 1.6. In both Ref. [P3] and Ref. [P5] we find analytical theories which describe the physics well in the regime of interest. To make clear in which context and under which approximations the theory presented is valid, we started with a derivation of the dipolar coupling Hamiltonian in Sect. 1.1, laying the foundation for everything else.

1.8 Outlook

We outline four different projects that extend the topics discussed in the thesis. In Ref. [P1], we considered a cold atomic cloud of localized scatterers. When the temperature

is low enough, the quantum statistics of the atoms need to be taken into account since their wavefunction have non-negligible overlap. The fundamentals for treating systems where this is the case were laid out in Ref. [87], and as a consequence, many effects were discovered in quantum gases that were described by theory long before. Such effects include Pauli blocking in a Fermi gas, superradiant emission from a Bose-Einstein condensate, and many more. With the advent of optical tweezers and unprecedented control over single atoms in them, we investigate the question of cooling a few particles in the quantum degenerate regime, i.e. what is the role of Pauli blocking, superradiance, and subradiance in the degenerate regime for sideband cooling in a tweezer trap. We reviewed the relevant formalism for this problem in Ref. [P1]. To give a flavor of this, the excited and ground state atoms are now represented by field theoretical operators

$$\begin{aligned} \hat{g}_{\boldsymbol{n}} &= \int \mathrm{d}\boldsymbol{R}\, [\phi_{\boldsymbol{n}}(\boldsymbol{R})]^* \, \hat{\Psi}_g(\boldsymbol{R}) \\ \hat{e}_{\boldsymbol{n}} &= \int \mathrm{d}\boldsymbol{R}\, [\phi_{\boldsymbol{n}}(\boldsymbol{R})]^* \, \hat{\Psi}_e(\boldsymbol{R}), \end{aligned} \tag{67}$$

with two atomic species e and g for ground and excited state atoms, which have fermionic/bosonic commutation relations. From the commutation relations, many results can already be derived from a Fermi's golden rule treatment for the spontaneous emission from an ensemble of fermions or bosons. Similarly, one can extend the usual sideband cooling treatment to account for the particle statistics, taking special care of density correlations that develop due to the dipolar interactions and drive. Note that there is literature in this direction already, see Ref. [25, 111].

In both Ref. [P2] and Ref. [P4], atomic arrays were described in the low-excitation regime. Another application of atomic arrays that goes beyond the low-excitation regime is in the implementation of the DLCZ (Duan-Lukin-Cirac-Zoller) scheme [112], which can be used to entangle two atomic arrays in metastable states so that they can act as quantum repeater endpoints. The usual DLCZ scheme is performed for an ensemble of Λ-type systems just as for the molecular cooling toy model in Ref. [P5] and Sect. 1.6. The entanglement is then achieved via a spontaneous Raman transfer from $|g\rangle$ to the intermediate state $|s\rangle$ when pumping the $g - e$ transition. The output of two ensembles is then entangled via a beam splitter. The fundamental question is then if the explicit inclusion of dipolar interactions in such a structured configuration of atoms will lead to an improvement in the directionality of the output as well as the overall operational parameters of the DLCZ gate.

Instead of extending the work on atomic arrays in Ref. [P2] and Ref. [P4] into the

direction of quantum information, making a clearer connection with the community of metasurfaces would be useful. As mentioned previously, there is a correspondence between the Green's functions from classical multipolar theory and the elimination of the photonic vacuum in a Master equation formalism when applied to magnetic and electric dipoles. It thus stands to reason that the coupled dipole formalism in the linear regime should also apply to dielectric nanostructures. This is the subject of Ref. [113], and we aim to understand a system of dielectric chiral scatterers (helices) by modeling its electric and magnetic dipole moment. In the experiment, large chiral dichroism is observed. We aim to investigate this effect for spatially and temporally coherent light as well as light with finite coherence length and time.

In all treatments of the molecular Hamiltonian discussed in this thesis and in Ref. [P3] and Ref. [P5], the Born-Oppenheimer approximation was made so that the Hamiltonian is diagonal in the electronic degrees of freedom as in Eq. (59). Going beyond the Born-Oppenheimer approximation leads to nonadiabatic coupling matrix elements. The physical consequences of such a term include nonradiative, incoherent transitions between two potential surfaces. If these two surfaces are the excited and ground state, this leads to a reduction in quantum efficiency[16] of the molecule since some absorbed light is not radiated. The Hamiltonian for a molecule with these off-diagonal couplings reads[17]

$$\mathcal{H} = \mathcal{H}_0 + \left[C + \sum_{p=1}^{\mathcal{N}} C_p^{(1)}(\hat{b}_p^\dagger + \hat{b}_p)\right](\hat{\sigma}^\dagger + \hat{\sigma}), \tag{68}$$

where the diabatic coupling here is Taylor expanded up to linear order with coupling constants C for the constant and $C_p^{(1)}$ for the linear coupling to mode p. Then the nonadiabatic decay rate can be found by tracing out the vibrational degrees of freedom in a Born-Markov approximation. This is discussed in Ref. [114]. This publication was not included in the cumulative part of the thesis, since it has not been peer-reviewed yet. This is an extension of the well-known theory from Ref. [115].

More generally, emitter metasurfaces provide promising platforms for quantum simulation and quantum information purposes. Molecular systems, on the other hand, provide quantum technological platforms that are yet to be fully understood. As such, there are many directions beyond these projects that we have listed where continued research would be fruitful.

[16] The quantum efficiency is the fraction of the absorbed light that is radiated and the total absorbed light.

[17] The off-diagonal term can, in general, only approximately be represented as a potential term. This process is called diabatization in the literature.

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2 Publications

In the following, I will outline my contributions and include copies of the publications. The articles included are the published versions including supplemental materials.

2.1 P1: Cooperative effects in dense cold atomic gases including magnetic dipole interactions

N. S. Baßler, I. Varma, M. Proske, P. Windpassinger, K. P. Schmidt, and C. Genes, "Cooperative effects in dense cold atomic gases including magnetic dipole interactions", Phys. Rev. Res. **6**, 023147 (2024)

In this publication, my contributions were the following:

- All calculations and numerical simulations.
- Production of Figures 1-4.
- Writing Appendix B-E.
- Initial draft of main text sections II and III.
- Contributions to rewrites and finalization of main text sections II and III (30%).
- Literature research and contributions to the introduction (40%).

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Cooperative effects in dense cold atomic gases including magnetic dipole interactions

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We theoretically investigate cooperative effects in cold atomic gases exhibiting both electric and magnetic dipole-dipole interactions, such as occurs, for example, in clouds of dysprosium atoms. After introducing a general framework capturing both the quantum degenerate and nondenegerate cases, we focus on the emergence of tailorable spin models in the quantum nondegenerate regime. In the low-excitation limit, we provide analytical and numerical results detailing the effect of magnetic interactions on the directionality of scattered light and characterize sub and superradiant effects.

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I. INTRODUCTION

Scattering of light off atomic gases necessarily involves aspects of quantum cooperativity, arising from the common, hybrid reaction of closely positioned and mutually coupled quantum emitters to the external stimulation [1]. The optical response can be very complex, as it strongly depends on the gas density and temperature, the type of atoms comprising the gas, as well as on the strength of the driving field. For low density and high temperature, an independent scattering regime can be obtained where the gas response can be deduced from the single atom response [2,3]. For higher density, weak excitation, and still high temperature, cooperative aspects such as super [4,5] and subradiance [6] (spontaneous emission rates larger or smaller than that of an isolated particle) emerge. Many features of such ensembles have already been investigated theoretically and experimentally [7]. A dense ensemble exhibits a cooperative lineshift [8–10] that cannot be predicted from standard electrodynamics of a dielectric [11]. Superradiant emission is directional [12] and robust to many perturbations of the system [13]. The superradiant level structure is, however, completely quantum mechanical [14] and not simply due to synchronization of dipoles as proven by single-photon superradiance experiments [15,16]. There exist promising applications of superradiant ensembles such as the generation of pulse trains using the burst [17]. There are also investigations into the consequences of additional effects such as more complex level structures and atomic motion in different regimes [18,19]. The results for some dense cloud experiments are, however, still not in complete agreement with theory [20]. High driving powers lead then to nonlinear optical effects [21], as the atomic transitions can saturate and atoms become fundamentally nonlinear elements.

In all cases discussed so far, a simple quantum optics approach suffices, based on the open quantum system formalism, where most of the relevant features and behaviors are obtained in a two-level system approach where atoms are treated as pseudospins 1/2 (with transitions between ground and excited electronic orbitals) responding to an external stimulation.

In the low-temperature limit, the motional wavepackets of the atoms comprising the gas can overlap, leading to a change in the theoretical framework, which must necessarily include a many-body formulation to the problem [22–25]. This is illustrated in Fig. 1 where the nonoverlapping motional wavepackets case corresponding to a thermal, classical gas is shown in Fig. 1(b) while the quantum degenerate limit involving overlapping wave functions is depicted in Fig. 1(c). The quantum degenerate gas has been shown theoretically [26] and experimentally [8–10] to lead to fermionic subradiance via Pauli blocking [27–29] and to superradiance in Bose-Einstein condensates [30,31].

We introduce here analytical tools to describe both the quantum nondegenerate and quantum degenerate cases, however, with a focus on the first of these. To this end we extend the formulation of the cooperative response to include magnetic interactions on top of the standard electric dipole-dipole exchanges, as, for example, strongly present in experiments with dysprosium atoms [32,33]. In the arbitrarily strong drive limit, for classical gases, we show the emergence of a spin Hamiltonian with tunable parameters. In the weak driving limit, a simpler coupled dipoles approach suffices. This can be seen also as a reduction of the dynamics onto the bottom of the Bloch sphere involving only single excitation collective states. The results show slight modifications in the superradiant response.

The paper is organized as follows. In Sec. II we introduce the light-matter interaction model at the Hamiltonian level for both degenerate (Sec. II B) and nondegenerate gases

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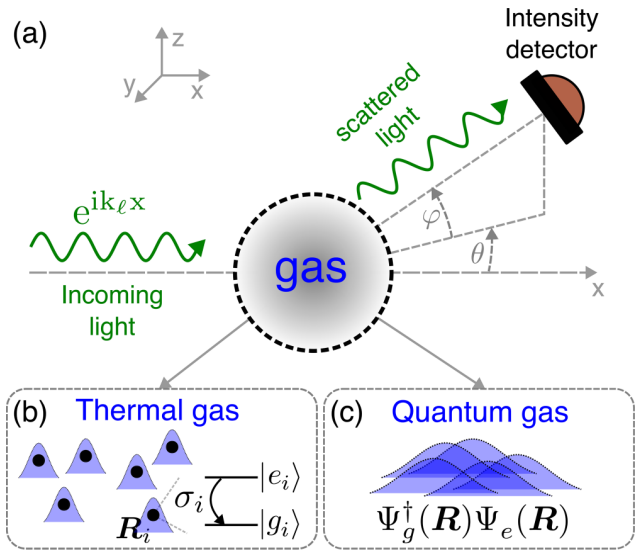


FIG. 1. (a) Schematics of the system. (b) In the first case, the de Broglie wavelength of each atom is much smaller than the average interparticle separation, allowing one to treat the system as a classical thermal gas far from quantum degeneracy. (c) In the opposite limit, quantum degeneracy is achieved by lowering the temperature and consequently producing particles with overlapping de Broglie wavepackets. To compute the optical response, a quantum many-body approach to the problem is necessary.

(Sec. II A) including both electric and magnetic dipole-dipole interactions. In Sec. III we reduce our treatment to the nondegenerate case, where we first analyze the strong driving limit and show the emergence of a many-particle spin Hamiltonian with interactions of tunable strength. We then exemplify the magnetic dipole interactions effect on the light scattered in the weak driving regime and find small deviations from a purely electrically interacting gas.

II. MODEL

We consider $\mathcal{N}$ atoms (positioned at $\boldsymbol{R}_i$ with index i running from 1 to $\mathcal{N}$) trapped in an external potential (see Table I for a list of notation used in this article). Optical addressing of each atom (both by classical fields and the quantum electromagnetic vacuum modes) is achieved by coupling to its single valence electron, which we denote by its position $\boldsymbol{r}_i$ measured with respect to the position of the nucleus. The atomic cloud is illuminated by a laser with Rabi frequency Ω, wave vector k_ℓ, and frequency $\omega_\ell = ck_\ell$ propagating in the x direction as depicted in Fig. 1(a). Two distinct regimes emerge: (i) a *quantum nondegenerate limit* where atomic motion is described by a thermal distribution of velocities and (ii) a *quantum degenerate limit* where atomic motion is quantized and atoms become indistinguishable, thus the quantum degenerate case. The distinction between the two cases is illustrated in Figs. 1(b) and 1(c) where the overlap of individual atomic de Broglie wavepackets indicates the criterion for the transition between the two limits. In both cases, we will be interested in the spectroscopic quantities obtained from a detector positioned in the far-field regime with angles θ and ϕ with respect to the incoming laser.

TABLE I. Table of definitions.

Quantity	Definition
$\Gamma(\Gamma_\mu)$	Electric (magnetic) radiative loss rate
f_{ij}^d	Dissipative electric dipole interaction
g_{ij}^d	Coherent electric dipole interaction
$\Omega_{i,j}^{\alpha,\beta}$	Static magnetic dipolar interaction
$\eta_{n,m}$	Franck-Condon factors
$\boldsymbol{k}_\ell$	Laser wave vector
k_ℓ	Laser wave vector magnitude
ω_ℓ	Laser frequency
ω_0	Transition frequency
Δ	Laser detuning
g_k	Photon coupling strength
$\epsilon_{\boldsymbol{k}}$	Photon polarization vector
$\hat{\boldsymbol{d}}_i$	Electric transition dipole operator
$\hat{\boldsymbol{\mu}}_i$	Magnetic transition dipole operator
$\hat{\boldsymbol{\mu}}_{s,i}$	Magnetic static dipole operator
$\boldsymbol{d}_i$	Electric transition dipole moment
$\boldsymbol{\mu}_i$	Magnetic transition dipole moment
$\boldsymbol{\mu}_{i,e}$	Static magnetic dipole moment of excited state
$\boldsymbol{\mu}_{i,g}$	Static magnetic dipole moment of ground state
σ_i	Transition operator
σ_i^z	Inversion operator
$\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}$	Photon annihilation operator
$\hat{\boldsymbol{r}}_i$	Relative electron position of transition
$\hat{\boldsymbol{L}}_i$	Angular momentum operator
$\hat{\boldsymbol{E}}(\boldsymbol{R})$, $(\hat{\boldsymbol{B}}(\boldsymbol{R}))$	Electric (magnetic) field operator at position $\boldsymbol{R}$
$\Psi_g(\boldsymbol{R})$	Annihilation operator for ground state atom at $\boldsymbol{R}$
$\Psi_e(\boldsymbol{R})$	Annihilation operator for excited state atom at $\boldsymbol{R}$
$\boldsymbol{v}_{\text{drive}}$	Drive vector for small drive
$\mathcal{M}$	Interaction matrix for small drive
$\delta\omega_j$	Frequency shift due to static magnetic field
μ_e	Excited state magnetic moment
μ_g	Ground state magnetic moment
Ω_μ	Magnetic interaction frequency
$\boldsymbol{r}_s$	Detector position
$\boldsymbol{k}_s$	Detector wave vector
$\mathcal{N}_c$	Number of runs over configurations
$\mathcal{N}$	Number of total atoms
τ	Optical depth
θ	Detection angle
$\hat{\Sigma}(\boldsymbol{R})$	Transition operator acting at position $\boldsymbol{R}$
$\hat{e}_{\boldsymbol{m}}$	Annihilation operator for excited state $\boldsymbol{m}$
$\hat{g}_{\boldsymbol{m}}$	Annihilation operator for ground state $\boldsymbol{m}$
$\Sigma_{\boldsymbol{n},\boldsymbol{m}}$	Transition operator between $\boldsymbol{n}$ and $\boldsymbol{m}$
$\mathcal{N}_g$	Number of atoms in the ground state
$\mathcal{N}_e$	Number of atoms in the excited state
$I(\boldsymbol{k}_s)$	Intensity at detection vector $\boldsymbol{k}_s$
$\omega_{\boldsymbol{n}}$	Frequency of state $\boldsymbol{n}$
$\boldsymbol{R}_i$	Center of mass position of atom i

For simplicity of presentation, we proceed with a set of approximations which are usually performed in treating light-matter interactions. We perform the dipole approximation which assumes that the size of the electronic orbital is negligible with respect to any relevant optical transition wavelength λ. In addition, we assume nonoverlapping electronic orbitals between neighboring atoms. The electric and

magnetic fields are quantized in a fictitious box of volume $\mathcal{V}$ and expressed in terms of photon creation and annihilation bosonic operators $\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}$ with frequencies $\omega_k = c|\boldsymbol{k}|$ and commutators $[\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}_k}, \hat{a}^\dagger_{\boldsymbol{k}',\boldsymbol{\epsilon}'_k}] = \delta_{\boldsymbol{k},\boldsymbol{k}'}\delta_{\boldsymbol{\epsilon}_k,\boldsymbol{\epsilon}'_k}$ (where $\boldsymbol{\epsilon}_k$ is the polarization for the mode $\boldsymbol{k}$). The free-space quantization yields the electric and magnetic field operators

$$\hat{\boldsymbol{E}} = \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}} g_k \omega_k \boldsymbol{\epsilon}_{\boldsymbol{k}} (\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}}), \tag{1}$$

$$\hat{\boldsymbol{B}} = \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}} (\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}}) g_k (\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}}), \tag{2}$$

with $g_k = 1/\sqrt{2\omega_k \mathcal{V}\epsilon_0}$ being the photon coupling strength. From here on the treatment is distinct for the two assumed limits of either a gas with a classical distribution of velocities or a quantum degenerate gas of indistinguishable atoms.

A. Quantum nondegenerate gas approach

We follow the standard quantum optics approach where the internal electronic dynamics of each atom is treated in terms of Pauli matrices. To this end, we restrict the dynamics of the electron to a ground state $|g\rangle_i$ and a single excited state $|e\rangle_i$, separated by frequency ω. Notice that, in practice, this assumption is a very good approximation for optically pumped atoms. An example based on dysprosium atoms is shown in Appendix A where the ground state is represented by a magnetic sublevel with $J = 8$ and $m_J = -8$ and the excited state with $J' = 9$ and $m'_J = -9$.

The Pauli matrices represent transitions in the electronic degrees of freedom and are defined as $\sigma_i = |g\rangle_i\langle e|_i$. The coupling to electromagnetic waves occurs via either the transition electric $\hat{\boldsymbol{d}}_i = \boldsymbol{d}_i\sigma_i + \boldsymbol{d}^*_i\sigma^\dagger_i$ and magnetic $\hat{\boldsymbol{\mu}}_i = \boldsymbol{\mu}_i\sigma_i + \boldsymbol{\mu}^*_i\sigma^\dagger_i$ dipole operators or via the static magnetic dipole operator $\hat{\boldsymbol{\mu}}_{s,i} = \boldsymbol{\mu}_{i,e}\sigma^\dagger_i\sigma_i + \boldsymbol{\mu}_{i,g}\sigma_i\sigma^\dagger_i$. The static components of the magnetic dipole $\boldsymbol{\mu}_{i,e}$ and $\boldsymbol{\mu}_{i,g}$ are computed within their respective electronic states, as $\boldsymbol{\mu}_{i,e} = -g\mu_B\langle e_i|\hat{\boldsymbol{L}}_i|e_i\rangle$ and $\boldsymbol{\mu}_{i,g} = -g\mu_B\langle g_i|\hat{\boldsymbol{L}}_i|g_i\rangle$. The operator $\hat{\boldsymbol{L}}_i$ is the angular momentum of atom i, μ_B is the Bohr magneton, and g is the Landé factor. The transition dipole matrix elements are computed between orbitals $\boldsymbol{d}_i = -e\langle e_i|\hat{\boldsymbol{r}}_i|g_i\rangle$ and $\boldsymbol{\mu}_i = -g\mu_B\langle g_i|\hat{\boldsymbol{L}}_i|e_i\rangle$. Due to the fixed parity of hydrogen-like orbitals, the static electric dipole moment must vanish, which is not the case for the static magnetic dipole moment. The resulting Hamiltonian

$$\mathcal{H} = \mathcal{H}_{\mathrm{at}} + \mathcal{H}_{\mathrm{em}} + \mathcal{H}_{\mathrm{el+mag}} + \mathcal{H}_{\mathrm{drive}}, \tag{3}$$

is a sum over the free Hamiltonians of the atoms $\mathcal{H}_{\mathrm{at}}$, the electromagnetic vacuum Hamiltonian $\mathcal{H}_{\mathrm{em}}$, the electric and magnetic dipole coupling to the radiation field $\mathcal{H}_{\mathrm{el+mag}}$, and the semiclassical drive $\mathcal{H}_{\mathrm{drive}}$.

The first two terms are explicitly written as

$$\mathcal{H}_{\mathrm{at}} + \mathcal{H}_{\mathrm{em}} = \omega\sum_i \sigma^\dagger_i\sigma_i + \sum_{\boldsymbol{k},\boldsymbol{\epsilon}} \omega_k \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}_k}\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}_k}, \tag{4}$$

where we set the zero of the energy at the ground electronic state level and ignored the zero-point energy of the vacuum modes. The coupling of the atomic system to the electric and magnetic quantum fields is

$$\mathcal{H}_{\mathrm{el+mag}} = \sum_i \hat{\boldsymbol{d}}_i\cdot\hat{\boldsymbol{E}}(\boldsymbol{R}_i) + (\hat{\boldsymbol{\mu}}_{t,i} + \hat{\boldsymbol{\mu}}_{s,i})\cdot\hat{\boldsymbol{B}}(\boldsymbol{R}_i). \tag{5}$$

The semiclassical drive is characterized by the Rabi frequency Ω and is written as

$$\mathcal{H}_{\mathrm{drive}} = \Omega\sum_i (e^{-\mathrm{i}\boldsymbol{k}_\ell\boldsymbol{R}_i}\sigma_i + e^{\mathrm{i}\boldsymbol{k}_\ell\boldsymbol{R}_i}\sigma^\dagger_i), \tag{6}$$

where $\boldsymbol{k}_\ell = k_\ell\hat{e}_x$ and $\hat{e}_x$ is the unit vector in the x-direction. The Hamiltonian above is the starting point for the models analyzed in Sec. III.

B. Quantum degenerate gas approach

In the opposite limit of a quantum degenerate gas, it is more convenient to introduce a two-species model where field operators $\Psi^\dagger_{e,g}(\boldsymbol{R})$ create atoms at some position $\boldsymbol{R}$ with their electron either in the ground or in the excited state. The dipole moments can be written in terms of field operators as the action of the Pauli matrices is now expressed by combinations of the field operators. For example, the operator $\hat{\Sigma}(\boldsymbol{R}) = \Psi^\dagger_g(\boldsymbol{R})\Psi_e(\boldsymbol{R})$ is the field-theoretical equivalent of the matrix operator σ_i where the particle location, which was previously denoted by an index, is now denoted by a continuous variable due to the replacement of localized scatterers by fields. The meaning is that an atom in the excited state is destroyed while another atom in the ground state is created in exactly the same position. The commutation relations are the standard ones $[\Psi_\alpha(\boldsymbol{R}), \Psi^\dagger_\beta(\boldsymbol{R}')]_\zeta = \delta(\boldsymbol{R}-\boldsymbol{R}')\delta_{\alpha,\beta}$, where $\alpha,\beta\in\{e,g\}$ are species indices while ζ specifies commutation or anticommutation relations depending on the bosonic or fermionic nature of the gas. The single-particle motional Hamiltonian is written as

$$\mathcal{H}_0 = -\nabla^2/2M + V_{\mathrm{ext}}(\boldsymbol{R}), \tag{7}$$

where $V_{\mathrm{ext}}(\boldsymbol{R})$ is an externally applied potential (assumed quadratic in the following) that allows for the use of a simpler notation using the trap basis states. Notice that, with a state-independent choice for $V_{\mathrm{ext}}(\boldsymbol{R})$, the motional wave functions are the same for the excited-type and ground-state-type atoms.

In the second quantization, the total system Hamiltonian is obtained as an integration of the Hamiltonian density. As a next step, the field operators can be expanded in a conveniently chosen basis. In particular, we will consider the trap basis defined by the eigenvectors $\mathcal{H}_0\phi_{\boldsymbol{n}} = \omega_{\boldsymbol{n}}\phi_{\boldsymbol{n}}$. Particle creation and annihilation operators can then be defined as

$$\begin{aligned}\hat{g}_{\boldsymbol{n}} &= \int d\boldsymbol{R}[\phi_{\boldsymbol{n}}(\boldsymbol{R})]^*\Psi_g(\boldsymbol{R}),\\ \hat{e}_{\boldsymbol{n}} &= \int d\boldsymbol{R}[\phi_{\boldsymbol{n}}(\boldsymbol{R})]^*\Psi_e(\boldsymbol{R}).\end{aligned} \tag{8}$$

The operators can be interpreted in the following way: when $\hat{g}^\dagger_{\boldsymbol{n}}$ is applied to the vacuum, it creates an atom in the trap state $\boldsymbol{n}$ and in the electronic ground state. Similarly, $\hat{e}^\dagger_{\boldsymbol{n}}$ creates an atom in the trap state $\boldsymbol{n}$ and in the electronic excited state. The fermionic or bosonic nature of the atoms is consistently taken into account by the commutation relations of these operators. The transition from free particles to trapped ones is performed by the tuning of the single-particle trapping potential, which,

in turn, affects the shape of the trap basis eigenvectors. With this, the free Hamiltonian of the atoms can be written as

$$\mathcal{H}_{\rm at} = \sum_{\boldsymbol{n}} \omega_{\boldsymbol{n}} \hat{g}^\dagger_{\boldsymbol{n}} \hat{g}_{\boldsymbol{n}} + \sum_{\boldsymbol{n}} (\omega_0 + \omega_{\boldsymbol{n}}) \hat{e}^\dagger_{\boldsymbol{n}} \hat{e}_{\boldsymbol{n}}. \tag{9}$$

The electromagnetic modes Hamiltonian $\mathcal{H}_{\rm em}$ is the same as before. The coupling of the atomic system to the electric and magnetic quantum fields is described by the following Hamiltonian:

$$\begin{aligned}\mathcal{H}_{\rm el+mag} = &\sum_{\boldsymbol{n},\boldsymbol{m}} (\hat{\Sigma}_{\boldsymbol{n},\boldsymbol{m}} + \hat{\Sigma}^\dagger_{\boldsymbol{n},\boldsymbol{m}}) \langle \boldsymbol{n}|\boldsymbol{d}\cdot\boldsymbol{E}(\boldsymbol{R})|\boldsymbol{m}\rangle \\ &+ \sum_{\boldsymbol{n},\boldsymbol{m}} (\hat{\Sigma}_{\boldsymbol{n},\boldsymbol{m}} + \hat{\Sigma}^\dagger_{\boldsymbol{n},\boldsymbol{m}}) \langle \boldsymbol{n}|\boldsymbol{\mu}\cdot\boldsymbol{B}(\boldsymbol{R})|\boldsymbol{m}\rangle \\ &+ \sum_{\boldsymbol{n},\boldsymbol{m}} (\hat{g}^\dagger_{\boldsymbol{n}} \hat{g}_{\boldsymbol{m}} \boldsymbol{\mu}_g + \hat{e}^\dagger_{\boldsymbol{n}} \hat{e}_{\boldsymbol{m}} \boldsymbol{\mu}_e) \cdot \langle \boldsymbol{n}|\boldsymbol{B}(\boldsymbol{R})|\boldsymbol{m}\rangle, \end{aligned} \tag{10}$$

where $\hat{\Sigma}_{\boldsymbol{n},\boldsymbol{m}} = \hat{g}^\dagger_{\boldsymbol{n}} \hat{e}_{\boldsymbol{m}}$ are ladder operators destroying an excited atom in trap state $\boldsymbol{m}$ and creating a ground-state atom in trap state $\boldsymbol{n}$. The semiclassical drive is now written as

$$\mathcal{H}_{\rm drive} = \Omega \sum_{\boldsymbol{n},\boldsymbol{m}} \hat{\Sigma}_{\boldsymbol{n},\boldsymbol{m}} \eta_{\boldsymbol{n},\boldsymbol{m}} + \hat{\Sigma}^\dagger_{\boldsymbol{n},\boldsymbol{m}} \eta_{\boldsymbol{n},\boldsymbol{m}}, \tag{11}$$

where we define the Franck-Condon factors $\eta_{\boldsymbol{n},\boldsymbol{m}} = \langle \boldsymbol{n}|e^{-{\rm i}\boldsymbol{k}_\ell \boldsymbol{R}}|\boldsymbol{m}\rangle = \int d\boldsymbol{R} [\phi_{\boldsymbol{n}}(\boldsymbol{R})]^* e^{-{\rm i}\boldsymbol{k}_\ell \boldsymbol{R}} \phi_{\boldsymbol{m}}(\boldsymbol{R})$. For tight trapping conditions, where the localization of the atoms is on a level much smaller than the wavelength $2\pi/k_\ell$, the exponential can be approximated with unity and the Franck-Condon factors are equal to $\delta_{\boldsymbol{nm}}$. In addition, to take into account the first-order correction, a Lamb-Dicke limit approximation can be made and only the matrix elements of the linear term ${\rm i}\boldsymbol{k}_\ell \boldsymbol{R}$ would need to be considered.

III. QUANTUM NONDEGENERATE GAS LIMIT

Let us now focus on the semiclassical case, which assumes classical dynamics for the atomic motion, while the electronic dynamics is described in a quantum fashion in terms of Pauli matrices. As a function of the drive intensity, the emergent physics can be quite distinct. In a first case, where a high-intensity drive is assumed, high-excitation levels can be reached, with many atoms being excited at the same time. This is the standard regime for Dicke superradiance, i.e., the quick burst of spontaneous emission from an initial fully excited ensemble of closely spaced atoms. The other limit we consider is the weak-excitation limit, where there are hardly any excitations present in the system, rendering it possible to perform a transformation to a fully classical coupled dipole model. Despite its simplicity, the weak-excitation limit gives insights into the emergence of cooperative effects and their role in modifying the directional scattering of light. In particular, we emphasize the role of magnetic dipole-dipole interactions and their competition with the widely studied electric counterpart.

A. Effective spin Hamiltonian

From the Hamiltonian listed in Eq. (3), one can derive a master equation for the evolution of the $\mathcal{N}$ electronic systems under the approximation of a frozen gas. Such an approximation can hold for low-enough temperatures, where motion evolution is slow compared to the time taken by any radiative processes. The derivation is based on the elimination of the photonic degrees of freedom [22–24,34,35], as outlined in detail in Appendixes B, C, D, and E. The effective Hamiltonian describing dynamics in the reduced subspace of dimension $2^{\mathcal{N}}$ of the electronic degrees of freedom reads

$$\begin{aligned}\mathcal{H}_{\rm eff} = &\Delta \sum_i \sigma_i^\dagger \sigma_i + \sum_{i\neq j} \big(g^{(d)}_{ji} + g^{(\mu)}_{ji}\big) \sigma_i^\dagger \sigma_j \\ &+ \Omega \sum_i (e^{-{\rm i}\boldsymbol{R}_i \boldsymbol{k}_\ell} \sigma_i + e^{{\rm i}\boldsymbol{R}_i \boldsymbol{k}_\ell} \sigma_i^\dagger) \\ &+ \frac{1}{8} \sum_{i\neq j} \Omega^{e,e}_{i,j} \big(1+\sigma_i^z\big)\big(1+\sigma_j^z\big) \\ &+ \frac{1}{8} \sum_{i\neq j} \Omega^{g,g}_{i,j} \big(1-\sigma_i^z\big)\big(1-\sigma_j^z\big) \\ &+ \frac{1}{4} \sum_{i\neq j} \Omega^{g,e}_{i,j} \big(1-\sigma_i^z\big)\big(1+\sigma_j^z\big). \end{aligned} \tag{12}$$

This describes the unitary part of the interaction corresponding to a full-fledged spin-1/2 XXZ model. Here the detuning is defined as $\Delta = \omega - \omega_\ell$. The second term gives the usual electric and magnetic dipole-dipole interactions allowing for the hopping of excitations within the whole ensemble, describing the XY part. The last three terms represent the contribution from the static magnetic interactions and lead to frequency shifts conditioned on the occupancy of the pair of atoms involved in the interaction, which is typically referred to as an effective Ising interaction. The coherent photon exchange via electric dipole-dipole interactions $g^{(d)}_{ji}$ has been widely studied [19]. The magnetic transition dipole-dipole couplings $g^{(\mu)}_{ji}$ generally can be ignored as they are of very small magnitude compared to the electric ones. However, other important couplings occur

$$\Omega^{\alpha,\beta}_{i,j} = \frac{\mu_0}{4\pi R^3_{ij}} \left[3 \frac{(\boldsymbol{\mu}_{i,\alpha}\cdot\boldsymbol{R}_{ij})(\boldsymbol{\mu}_{j,\beta}\cdot\boldsymbol{R}_{ij})}{R^2_{ij}} - \boldsymbol{\mu}_{i,\alpha}\cdot\boldsymbol{\mu}_{j,\beta} \right], \tag{13}$$

with $\alpha, \beta \in \{e, g\}$, which describe magnetic dipole interactions between atoms owing to the static magnetic dipoles, illustrated in Fig. 2 on the right side. This is an interaction between the electronic ground and excited state densities as opposed to a spin-flip interaction for transition dipoles.

With the elimination of the electromagnetic vacuum, the system is characterized by open system dynamics where the collective dissipation is included in the master equation

$$\frac{d\rho}{dt} = -{\rm i}[\mathcal{H}_{\rm eff}, \rho] + \mathcal{L}_\mu[\rho] + \mathcal{L}_d[\rho]. \tag{14}$$

We assume standard Lindblad form for the loss terms which we write as follows:

$$\mathcal{L}_\alpha[\rho] = \sum_{i,j=1}^{\mathcal{N}} f^{(\alpha)}_{ij} (2\sigma_i \rho \sigma_j^\dagger - \{\sigma_j^\dagger \sigma_i, \rho\}). \tag{15}$$

The independent radiative loss rates are $\Gamma = k_0^3 d^2/(3\pi\hbar\epsilon_0)$ and $\Gamma_\mu = k_0^3 \mu^2/3\pi\hbar\epsilon_0$ stemming from electric and magnetic

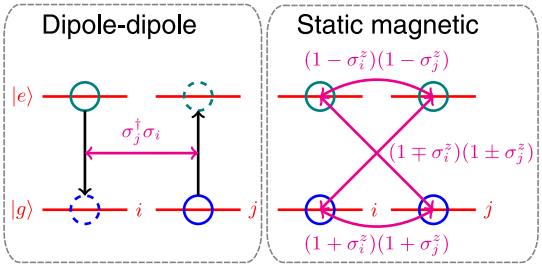


FIG. 2. Illustration of interactions present in the effective Hamiltonian Eq. (12). On the left is the excitation hopping due to the electronic dipole interaction between sites i and j. Initial occupations are indicated in filled open circles and the final occupations in dotted open circles. As indicated by the filled circle, the initially excited state on site i is changed to the ground state by σ_i. Similarly, the ground state on site j, indicated by a filled circle is excited due to $\sigma_j^\dagger$. On the right side is the density-density interaction due to the static magnetic dipole interaction with the same color coding. Here, either the ground or excited state can be occupied on i and j, leading to different interactions indicated by arrows between them and annotated with the appropriate interaction terms.

contributions. Moreover, collective dissipation at rates $f_{ij}^{(d)}$ (electric) and $f_{ij}^{(\mu)}$ (magnetic) are also present, with the exact expressions listed in Appendix E. The magnitude of Γ depends on the particular transition we consider, but it is around $\Gamma \approx 1$ MHz so that $f_{ij}^{(d)} \approx 1$ MHz particles at distance λ. We usually also choose Δ to be of the order of Γ, while $f_{ij}^{(\mu)}$ is negligible.

B. Weak-excitation limit: Coupled dipoles model

The Hamiltonian in Eq. (12) is of general validity in the quantum nondegenerate regime. However, by considering the weak-excitation limit, the evolution of the system is restricted to a very small subspace where the analytical description of the dynamics can be greatly simplified. In addition, we set the magnetic transition dipole moment to zero so that $f_{ij}^{(\alpha)} = g_{ij}^{(\alpha)} = 0$ since it is negligible. To do this, we linearize the time evolution of the system by replacing all population operators with -1 and factorizing all two operator correlations. We gather all expectation values of atomic coherences in a single vector $\boldsymbol{v} = (\langle\sigma_1\rangle, \ldots, \langle\sigma_{\mathcal{N}}\rangle)^\top$ and write an effective first-order differential equation

$$\dot{\boldsymbol{v}}(t) = -i\mathcal{M}\boldsymbol{v}(t) + \Omega\boldsymbol{v}_{\text{drive}}, \tag{16}$$

which is derived in Appendices E and F where the drive vector incorporates all the drive phases $\boldsymbol{v}_{\text{drive}} = (e^{i\boldsymbol{k}_\ell\boldsymbol{R}_1}, \ldots, e^{i\boldsymbol{k}_\ell\boldsymbol{R}_{\mathcal{N}}})^\top$. The weak driving approximation $\Omega \ll |\boldsymbol{f}^{(d)}|$ assumes that the Rabi frequency is much weaker than the dissipative part of the dipole-dipole interaction. Notice also that the equation above can be solved both in the steady state to derive spectroscopic features of the ensemble as well as in the time domain (by imposing a time dependence on Ω). The matrix $\mathcal{M}$ incorporates both coherent and dissipative cooperative behavior

$$\mathcal{M}_{jj'} = [\Delta + \delta\omega_j]\delta_{jj'} - \left[g_{jj'}^{(d)} + i f_{jj'}^{(d)}\right]. \tag{17}$$

We use the low-excitation condition in order to approximate $\langle\sigma_j^z\sigma_{j'}\rangle \approx -\langle\sigma_{j'}\rangle$. This is always true if $j = j'$, but only approximately valid under the condition that very few excitations are present in the system such that, on average, each site has a much lower than unit population in the excited state. This leads to the definition of a local frequency shift

$$\begin{aligned}\delta\omega_j &= \sum_{j'\neq j}\left(\Omega_{jj'}^{e,g} - \Omega_{jj'}^{g,g}\right) \\ &= \frac{(\mu_e-\mu_g)\mu_g\mu_0}{4\pi}\sum_{j'\neq j}\frac{3z_{jj'}^2/R_{jj'}^2 - 1}{R_{jj'}^3},\end{aligned} \tag{18}$$

where $\delta\omega_j$ denotes the total shift acquired owing to magnetic interactions. Since the particle positions are drawn from random configurations due to the thermal motion, this amounts to an effect of frequency disorder. We assume that all magnetic dipole moments in the ground state are $\mu_g \parallel \hat{e}_z$ and all magnetic moments in the excited state are $\mu_e \parallel \hat{e}_z$. We also take the electric dipole moments to be the same and belonging to a transition that emits circularly polarized light $\boldsymbol{d}_i \parallel (1, \pm i, 0)^T$ (the sign does not matter due to the symmetry of the cloud) and a laser with the same polarization. This leads to the definition of the magnetic interaction rate

$$\Omega_\mu = \frac{\mu_0(\mu_e-\mu_g)\mu_g}{4\pi}. \tag{19}$$

For $\mu_g = 10$ an estimate of the effect of the magnetic interactions leads to $\Omega_\mu = 0.\,815$ MHz $\times$ nm^3. Additionally, we neglect contributions from the magnetic transition dipole moments' couplings as they are much smaller than the electrically mediated couplings and we neglect the magnetic field induced by the electric dipole moments. We then consider the coherent part of the far-field intensity radiated by a system defined by these dipoles in the xy plane

$$I(\boldsymbol{r}_s) \propto \left|\sum_i e^{-i\boldsymbol{k}_s\boldsymbol{R}_i}v_i\right|^2, \tag{20}$$

where $\boldsymbol{r}_s$ indicates the detection angle, $\boldsymbol{k}_s \parallel \boldsymbol{r}_s$ and $|\boldsymbol{k}_s| = k_0$ is the wave vector for the propagation in the detection direction.

In the steady state, assuming time-independent driving, the linear system obtained from setting the left-hand side of Eq. (16) to zero has to be solved. In practice, this equation needs to be solved for many different configurations. Since we assume the cloud to be thermal, we assume this particle distribution with respect to the center of the cloud to be Gaussian. When the atomic motion timescale is much slower than $1/\Gamma$, i.e., the Doppler shift ω_d is much smaller than Γ, averaging over these different configurations corresponds to a thermal average. The velocities do not contribute since the Doppler shift is negligible and a detector will average over the different configurations the cloud assumes.

In Fig. 3(a), we can see that static magnetic dipolar interaction reduces the forward enhancement due to constructive interference of the emitted light in the laser direction. The inhomogeneous broadening reduces this constructive interference so that the forward enhancement is reduced as well. Alternatively, this can be reinterpreted when we consider that the frequency disorder generates matrix elements between different phase configurations in the atomic ensemble so that

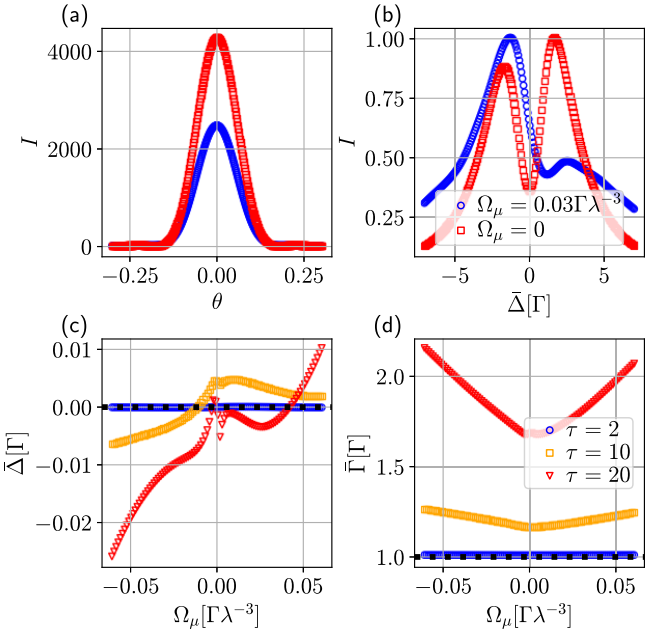


FIG. 3. Numerical results based on solution of Eq. (16) in the steady state. We consider multiple configurations of a cloud where the particles are distributed according to a Gaussian probability distribution with optical depth $\tau = 3\mathcal{N}/(2k_0^2R^2)$, simulating averaging over atomic motion. The number of configurations is $\mathcal{N}_c = 1000$ and $\mathcal{N} = 1000$. The optical depth is $\tau = 15$. Ω_μ quantifies the strength of the magnetic interaction. In (a) we show how the forward enhancement is modified by the magnetic interaction by plotting the intensity variation versus the in-plane angle θ on the single-particle resonance $\Delta = 0$. There is a reduction in the forward enhancement for $\Omega_\mu = 0.03\Gamma\lambda^3$ compared to no magnetic interaction. In (b) we investigate the lineshape of the cloud by changing the laser detuning Δ and compare $\Omega_\mu = 0.03\Gamma\lambda^3$ with $\Omega_\mu = 0$ at an in-plane detection angle of $\theta = \pi/10$. The double peak structure in the nonmagnetic case becomes asymmetric when the magnetic interaction is turned on. Changing the sign flips the effect on the two peaks (not shown). In (c) and (d) we investigate the effect of the magnetic interaction on the cooperative properties of the sample. To this end we determine the collective lineshift $\bar{\Delta}$ and the collective linewidth $\bar{\Gamma}$ in the forward direction $\theta = 0$ by calculating the lineshapes and then extracting these parameters from a Lorentzian fit. The lineshift and linewidth enhancement are shown for different optical densities τ with the magnetic interaction strength on the x-axis.

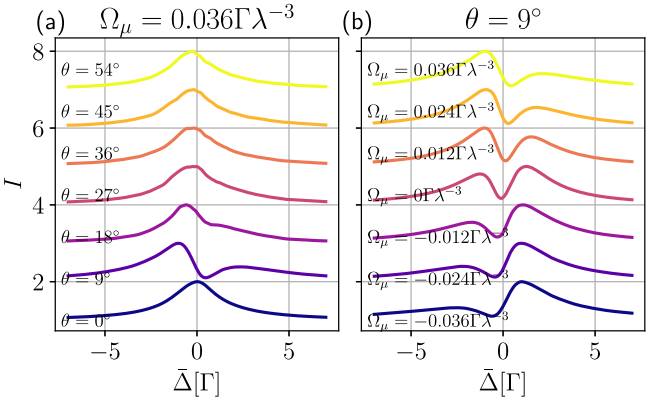


FIG. 4. Numerical results based on the solution of Eq. (16) in the steady state. We show for $N = 1000$ particles and averaging over $N_c = 1000$ runs the normalized lineshapes for different transversal angles θ in (a) and different magnetic interaction strengths Ω_μ in (b). The calculations are performed for a spherical Gaussian cloud with optical density $\tau = 25$.

more light is scattered into angles other than the laser-driven direction. Furthermore, we observe a double peak structure of the lineshape at a shallow detection angle of $\pi/10$ in Fig. 3(b), which was previously attributed to vacuum Rabi splitting in Ref. [36] and also observed in Ref. [18]. We discuss this in more detail when interpreting Fig. 4. In Figs. 3(c) and 3(d) we perform studies of the lineshift and linewidth modification due to the presence of the magnetic interaction for different optical densities τ. From the lineshift we can estimate the total symmetric contribution to the frequency shift of the ensemble, which causes a lineshift that depends on the sign of the magnetic dipole-dipole interaction. From the linewidth enhancement one can see the effect of the asymmetric contribution to the frequency disorder, i.e., the inhomogeneous broadening. From the results for $\tau = 20$ it is fairly clear that there are two different regimes. For large magnetic interaction Ω_μ the large frequency disorder simply leads to an inhomogeneous linewidth enhancement. For smaller $\Omega_\mu \approx 0.005\lambda^3\Gamma$ the linewidth is decreased. One explanation for this is that frequency disorder facilitates scattering into states that are naturally more subradiant than the originally driven state without disorder. Furthermore, we make connections between the quantities plotted here and the scaling of the features observed with the particle number N, the optical density τ, and the peak density n_0 of the cloud. First, consider Ref. [36] where it is shown that the optical density is the relevant parameter when investigating effects due to the electric dipole transition such as the observed double peak structure. Keeping the optical density τ constant, the peak density will then scale with the particle number $n_0 \propto \sqrt{N}\tau$ so that the density can be tuned via particle number at constant optical density. We then identify that the magnitude of the static magnetic dipolar interaction depends on the density of the cloud so that the magnetic interaction strength Ω_μ acts as a proxy for both density and particle number.

IV. CONCLUSION

We described the emission and scattering properties of a dense atomic cloud for distinguishable, classically moving particles. We made use of an open quantum system approach where we showed that tunable spin Hamiltonians can be designed. In addition, under the assumption of weak external drive, and in particular, for atoms where magnetic interactions are strong, we characterized their effects on the properties of scattered light. For quantum degenerate gases, we introduced the relevant Hamiltonian including electric and magnetic dipole interactions and provided a quick outlook of how one would proceed in analyzing their emission properties in an externally dictated trap basis. In the future, we will augment this approach to the full open system formulation to derive the interplay between quantum statistics and quantum

electromagnetic vacuum effects such as super and subradiance. We expect that our findings are of relevance for dense atomic dysprosium clouds for which a short description of a possible experimental implementation is presented in Appendix A. The setup is, in principle, fairly standard consisting of a cooling setup where Dysprosium atoms are Zeeman-slowed and then trapped and laser-cooled in a magnetooptical trap (MOT). This allows the tuning of the cloud density and particle number and hence the parameters varied in our simulations. If the sample is sufficiently cold, the quantum degenerate regime can also be reached in this setup so that both the quantum nondegenerate and degenerate regimes can be investigated.

ACKNOWLEDGMENTS

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APPENDIX A: POSSIBLE EXPERIMENTAL REALIZATION

To experimentally observe the effects discussed in the main body of the article, we developed and built an apparatus to cool and trap neutral dysprosium [37]. Dysprosium belongs to the group of lanthanide elements whose characteristic open f-shell electron configuration $[Xe]4f^{10}6s^2$ with spin $S = 2$, orbital angular momentum $L = 6$, and total angular momentum $J = 8$ gives rise to its high magnetic moment of 10 Bohr magnetons ($\mu_g \sim 10\mu_B$). Compared to alkali atoms whose magnetic moments are on the order of only $\mu_g \sim 1\mu_B$, the magnetic dipole-dipole interactions in ultracold gases of dysprosium is about 100 times stronger, making it an ideal candidate for investigating its contribution to cooperative effects. In addition, there is an almost equal abundance of stable bosonic and fermionic isotopes allowing for the creation of both types of quantum degenerate gases [38,39].

The general experimental scheme for producing laser cooled samples of lanthanide atoms is implemented as follows [40,41]: A strong optical transition in the blue spectrum range is used to precool the atoms in a Zeeman slower (ZS) before capturing them in a narrow line magnetooptical trap (MOT). Operating a MOT on transitions with natural linewidths on the order of 100 kHz is required to reach Doppler temperatures 5 μK, which allows one to directly transfer the atoms from the MOT into an optical dipole trap (ODT). In our setup, we employ a strong $J = 8 \rightarrow J' = 9$ transition at 421 nm, with a natural linewidth of $\Gamma_{421} = 2\pi \cdot 32$ MHz for precooling. A thermal beam of atomic dysprosium, transversally cooled on this broad transition, reaches the ZS with initial velocities of several hundred meters per second. The atoms are then longitudinally decelerated to a velocity of about 24 m/s in the spin-flip configuration ZS before entering the main chamber, where they are captured in a six beam three-dimensional (3D) MOT setup. The MOT transition, on the other hand, is a closed, narrow linewidth $J = 8 \rightarrow J' = 9$ transition at 626 nm with a natural linewidth of $\Gamma_{626} = 2\pi \times 136$ kHz which corresponds to a Doppler temperature of 3.2 μK.

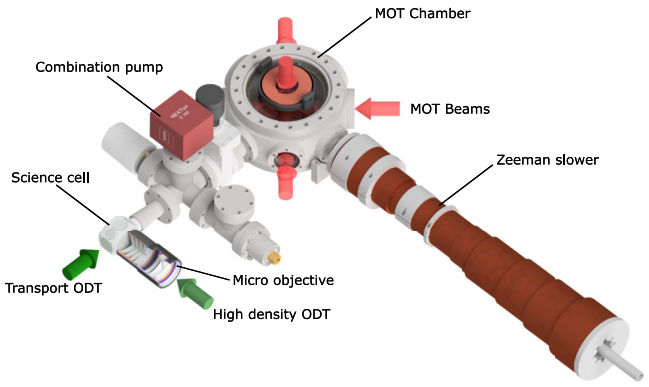


FIG. 5. CAD render of the vacuum system. Dysprosium atoms reach the ZS with a velocity of few hundred meters per second. They are longitudinally slowed in the ZS with resonant laser light at the 421 nm before being captured in a narrow line 3D MOT at 626 nm. An achromatic lens in conjunction with an air-bearing translation stage is used to focus down the optical transport beam at the position of the MOT and create a deep ODT. The focal spot of this beam is then moved from the MOT chamber to the center of the science cell by moving the translation stage, thereby transferring the atoms alongside. To retain a substantial number of atoms in the trap after this sequence, the process of transport and retrapping of atoms needs to be done on a timescale that is shorter than the lifetime of the ODT.

Our experimental setup consists of a high numerical aperture (NA) science cell connected to an ultrahigh vacuum (UHV) chamber (MOT) and a dedicated laser system capable of generating tunable and frequency stabilized laser radiation. Consequently, the laser frequency can be readily adapted to capture either bosons or fermions without any optical or mechanical adjustments. Figure 5 shows a CAD render of the vacuum system. To reduce the influence of undesired external magnetic fields, the vacuum system is assembled exclusively from nonmagnetic stainless steel, titanium, and ceramic glass. Additionally, a commercial, three-axis magnetic-field compensation system (Stefan Mayer Instruments GmbH: MR-3) is used to drive a 3D coil system and compensate low frequency magnetic-field disturbances.

The next step for creating dense samples of atomic dysprosium is to transfer atoms to the science cell attached to the MOT chamber. The cell offers a high optical access (with nine optical view ports), is designed in-house, and made out of the machinable glass-ceramic MACOR (Corning Inc.). A deep ODT at the MOT position will be created, using a diode pumped solid-state (DPSS) laser (Coherent: Mephisto MOPA 55 W) to transport the atoms. This will be achieved by using an achromatic lens to focus down the transport beam to a beam waist of $w_0 \sim 66$ μm and two mirrors mounted on an air-bearing translation stage (Aerotech: ABL1500-300), which enables reproducible and precise movement of the focal spot of the beam, thereby, transporting the atoms alongside [42]. The atoms are retrapped inside the science cell using a custom made multilens micro-objective, designed to offer diffraction-limited performance at the trapping wavelength of 1064 nm [43,44]. A five-lens configuration consisting solely of commercial singlets was chosen to achieve a high NA of 0.53 and a working distance of 22 mm. Simulation and

optimization of objective parameters like lens curvature, relative spacing, and thickness was done using a ray-tracing software (OPTICSTUDIO) to minimize optical aberrations and achieve diffraction-limited performance. The beam waist at the focus of the micro-objective was measured using a piezo-controlled knife edge and found to be $w_{0h} = 5.94 \pm 1.18$ μm and $w_{0v} = 6.99 \pm 0.67$ μm, in the horizontal and vertical directions, respectively. We estimate that an ODT created with this objective should provide a sufficiently tight confinement for a few hundred to a few thousand dysprosium atoms to reach densities on the order of $\sim 10^{13}$ atoms/cm^3. This condition would then allow us to explore dynamics in the high-density regime where the interparticle/interemitter spacing is lower than the wavelength of the exciting transition, giving rise to effects like Dicke sub and superradiance.

APPENDIX B: MAGNETIC INTERACTION TERMS IN LIGHT MATTER HAMILTONIAN FROM FIRST PRINCIPLES

We will be using the following definitions for the gauge field, electric, and magnetic field operators

$$\hat{\boldsymbol{A}} = \sum_{\boldsymbol{k},\boldsymbol{\epsilon}} g_k \boldsymbol{\epsilon}_{\boldsymbol{k}} (\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} + \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}}), \tag{B1}$$

$$\hat{\boldsymbol{E}} = \mathrm{i} \sum_{\boldsymbol{k},\boldsymbol{\epsilon}} g_k \omega_k \boldsymbol{\epsilon}_{\boldsymbol{k}} (\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}}), \tag{B2}$$

$$\hat{\boldsymbol{B}} = \mathrm{i} \sum_{\boldsymbol{k},\boldsymbol{\epsilon}} (\boldsymbol{k} \times \boldsymbol{\epsilon}_{\boldsymbol{k}}) g_k (\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}}), \tag{B3}$$

with $g_k = 1/\sqrt{2\omega_k \mathcal{V} \epsilon_0}$ photon coupling strength with photonic operators as defined in the main text.

If we omit terms where the square of the gauge field appears and the term where the atom core momentum is coupled to the gauge field (amounting to a Born-Oppenheimer and subsequently a rotating wave approximation), the simplified Hamiltonian is composed of three parts, written in the Coulomb gauge

$$\mathcal{H} \approx \mathcal{H}_0 + \mathcal{H}_{\mathrm{af}} + \mathcal{H}_{\mathrm{dipole}}, \tag{B4}$$

$$\mathcal{H}_0 = \sum_i^N \frac{\hat{\boldsymbol{P}}_i^2}{2M} + \frac{\hat{\boldsymbol{p}}_i^2}{2m} + V_{ec}(\boldsymbol{R}_i, \boldsymbol{r}_i), \tag{B5}$$

$$\mathcal{H}_{af} = \frac{1}{m} \sum_i^N \hat{\boldsymbol{A}}(\boldsymbol{r}_i) \cdot \hat{\boldsymbol{p}}_i, \tag{B6}$$

$$\mathcal{H}_{\mathrm{dipole}} = \frac{1}{2} \sum_{i \neq j} \sum \frac{\boldsymbol{d}_i \cdot \boldsymbol{d}_j - 3(\boldsymbol{R}_{ij} \boldsymbol{d}_i)(\boldsymbol{R}_{ij} \cdot \boldsymbol{d}_j)}{R_{ij}^3}. \tag{B7}$$

where we already performed a dipole approximation for the Coulomb interaction between different atoms leading to the explicit form of $\mathcal{H}_{\mathrm{dipole}}$. The Hamiltonian $\mathcal{H}_{af}$ is the canonical coupling between the electronic degree of freedom and the transverse modes of the light-field. We have not completed the dipolar approximation in the gauge field which still contains the position of the electron $\boldsymbol{r}_i$, which is a quantum operator in this description. The Hamiltonian $\mathcal{H}_0$ contains the electron kinetic energy, the center-of-mass kinetic energy, and the interaction between the electron and the core. The electron kinetic energy and the core-electron potential give rise to the level structure of the atom. We consider a Born-Oppenheimer picture where $\boldsymbol{R}_i$ represents the center of mass position.

Instead of making the usual dipolar approximation for the gauge field, we consider one order higher in the Taylor approximation of the electron position around the center-of-mass position $e^{\mathrm{i}\boldsymbol{k}\boldsymbol{r}_i} \approx e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_i}[1 + \mathrm{i}\boldsymbol{k}\boldsymbol{x}_i]$ with the distance vector $\boldsymbol{x}_i = \boldsymbol{r}_i - \boldsymbol{R}_i$. The atom field Hamiltonian then splits into two contributions

$$\mathcal{H}_{af} = \frac{1}{m} \sum_i^N \hat{\boldsymbol{p}}_i \cdot \hat{\boldsymbol{A}}(\boldsymbol{R}_i) + \mathrm{i} \sum_{\boldsymbol{k},\boldsymbol{\epsilon}} g_k (\hat{\boldsymbol{p}}_i \cdot \boldsymbol{\epsilon}_{\boldsymbol{k}})(\boldsymbol{k}\boldsymbol{x}_i)(\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_i} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}_i}). \tag{B8}$$

The first term is the simple dipole approximation which leads to the coupled dipole model after some additional approximations and a gauge transformation. The second term is of higher order and contains the magnetic and quadrupole electric moment as we will see. To see that the double dot product indeed yields the magnetic and quadrupole term, it needs to be rearranged

$$\begin{aligned}(\hat{\boldsymbol{p}}_i \cdot \boldsymbol{\epsilon}_{\boldsymbol{k}})(\boldsymbol{k} \cdot \boldsymbol{x}_i) &= (\hat{\boldsymbol{p}}_i \cdot \boldsymbol{k})(\boldsymbol{x}_i \cdot \boldsymbol{\epsilon}_{\boldsymbol{k}}) + (\boldsymbol{x}_i \times \hat{\boldsymbol{p}}_i)(\boldsymbol{k} \times \boldsymbol{\epsilon}_{\boldsymbol{k}}) \\ &= \tfrac{1}{2}((\hat{\boldsymbol{p}}_i \cdot \boldsymbol{\epsilon}_{\boldsymbol{k}})(\boldsymbol{k} \cdot \boldsymbol{x}_i) + (\hat{\boldsymbol{p}}_i \cdot \boldsymbol{k})(\boldsymbol{x}_i \cdot \boldsymbol{\epsilon}_{\boldsymbol{k}})) \\ &\quad + \tfrac{1}{2}\hat{\boldsymbol{L}}(\boldsymbol{k} \times \epsilon_{\boldsymbol{k}}).\end{aligned} \tag{B9}$$

The first term is the electronic quadrupole term whereas the second term contains the angular momentum implying that it is indeed the magnetic coupling. Summarizing the calculations above, the terms that must be added to the Hamiltonian if the dipole approximation is performed to higher order are

$$\begin{aligned}\mathcal{H}_m &= \mathrm{i} \sum_i^N \frac{\hat{\boldsymbol{L}}_i}{2m} \cdot \sum_{\boldsymbol{k},\boldsymbol{\epsilon}} g_k (\boldsymbol{k} \times \boldsymbol{\epsilon}_{\boldsymbol{k}})(\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_i} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}_i}) \\ &= \sum_i^N \hat{\boldsymbol{\mu}}_i \cdot \hat{\boldsymbol{B}}(\boldsymbol{R}_i), \\ \mathcal{H}_q &= \sum_{i,\boldsymbol{k},\boldsymbol{\epsilon}} \frac{\mathrm{i}|\boldsymbol{k}|}{2m} (p_{\boldsymbol{k}} r_{\boldsymbol{\epsilon}} + p_{\boldsymbol{\epsilon}} r_{\boldsymbol{k}}) g_k (\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_i} - \hat{a}^\dagger_{\boldsymbol{k},\boldsymbol{\epsilon}} e^{-\mathrm{i}\boldsymbol{k}\boldsymbol{R}_i}).\end{aligned} \tag{B10}$$

Here we avoid complications with spin degrees of freedom by making the association $\hat{\boldsymbol{\mu}}_i = -\frac{\hat{\boldsymbol{L}}_i}{2m}$ and noting that spin degrees of freedom need to be explicitly inserted into the Hamiltonian which ultimately change the definition of $\hat{\boldsymbol{\mu}}_i$. Since we go into a two-level picture this is, however, already taken care of and we need not consider these complications in detail.

APPENDIX C: REPRESENTATION OF THE HAMILTONIAN IN THE TWO-LEVEL APPROXIMATION

We perform a two-level approximation in which we assume there exist two relevant levels per atom, the ground state $|g_i\rangle$ and the excited state $|e_i\rangle$ separated by a frequency ω_0. Taking the projection of the momentum and magnetic moment operator onto this subspace, we can explicitly write the Hamiltonian using the matrix elements within this subspace. One relevant

point here is that the momentum operator has no diagonal components whereas the magnetic moment operator has no such symmetry restriction.

Omitting the quadrupole term, the atom-field Hamiltonian then becomes

$$\mathcal{H}_{\rm af} = \mathrm{i}\sum_i^N (\boldsymbol{d}_i\sigma_i - \boldsymbol{d}_i^*\sigma_i^\dagger)\cdot\hat{\boldsymbol{A}}(\boldsymbol{R}_i) + (\boldsymbol{\mu}_i\sigma_i + \boldsymbol{\mu}_i^*\sigma_i^\dagger + \boldsymbol{\mu}_{i,e}\sigma^\dagger\sigma + \boldsymbol{\mu}_{i,g}\sigma\sigma^\dagger)\cdot\hat{\boldsymbol{B}}(\boldsymbol{R}_i), \tag{C1}$$

with $\boldsymbol{d}_i = \langle g_i|\hat{x}_i|e_i\rangle$ and $\boldsymbol{\mu}_{i,e} = \langle g_i|\hat{\boldsymbol{\mu}}_i|e_i\rangle$ for the transition dipoles and $\boldsymbol{\mu}_{i,e} = \langle e_i|\hat{\boldsymbol{\mu}}_i|e_i\rangle$, $\boldsymbol{\mu}_{i,g} = \langle g_i|\hat{\boldsymbol{\mu}}_i|g_i\rangle$ for the static magnetic dipole moment. The sigma matrices are transition operators of the type $\sigma_i = |g_i\rangle\langle e_i|$.

APPENDIX D: UNITARY TRANSFORMATION INTO DIPOLAR GAUGE

The next step in deriving the light-matter Hamiltonian is the gauge transformation

$$U = \exp\left(-\mathrm{i}\sum_i^N \int_{\boldsymbol{r}_i}^{\boldsymbol{R}_i} d\boldsymbol{r}\cdot\hat{\boldsymbol{A}}(\boldsymbol{r})\right). \tag{D1}$$

We note here that the gauge-field operator commutes with the magnetic-field operator $[\hat{\boldsymbol{A}}(\boldsymbol{r}), \hat{\boldsymbol{B}}(\boldsymbol{r}')] = 0$ so that H_m is not affected by this transformation. The effect on the dipole electronic transition and the static dipole $\mathcal{H}_{\rm dipole}$ is as usual so that, after also performing a rotating wave approximation, the Hamiltonian is

$$\mathcal{H} = \omega\sum_i \sigma_i^\dagger\sigma_i + \sum_{\boldsymbol{k},\boldsymbol{\epsilon}} \omega_k \hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}^\dagger \hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} + \sum_i \hat{\boldsymbol{d}}_i\cdot\hat{\boldsymbol{E}}(\boldsymbol{R}_i) + (\hat{\boldsymbol{\mu}}_{t,i} + \hat{\boldsymbol{\mu}}_{s,i})\cdot\hat{\boldsymbol{B}}(\boldsymbol{R}_i), \tag{D2}$$

with definitions $\hat{\boldsymbol{d}}_j = \boldsymbol{d}\sigma_j + \boldsymbol{d}^*\sigma_j^\dagger$, $\hat{\boldsymbol{\mu}}_{t,i} = \boldsymbol{\mu}_i\sigma_i + \boldsymbol{\mu}_i^*\sigma_i^\dagger$ and $\hat{\boldsymbol{\mu}}_{s,i} = \boldsymbol{\mu}_{i,e}\sigma^\dagger\sigma + \boldsymbol{\mu}_{i,g}\sigma\sigma^\dagger$.

APPENDIX E: ELIMINATION OF PHOTONIC DEGREES OF FREEDOM FROM HAMILTONIAN

The last step in determining the effective system dynamics is to perform an adiabatic elimination of the light modes for this Hamiltonian Eq. (D2). The first step is to determine the equation of motion for the photonic operators

$$\frac{d\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}}{dt} = -\mathrm{i}\omega_k\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}} + \sum_i g_k\omega_k(\hat{\boldsymbol{d}}_i\cdot\boldsymbol{\epsilon}_{\boldsymbol{k}})e^{-\mathrm{i}\boldsymbol{k}\hat{\boldsymbol{R}}_i} + \sum_i g_k([\hat{\boldsymbol{\mu}}_{t,i} + \hat{\boldsymbol{\mu}}_{s,i}]\cdot[\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}}])e^{-\mathrm{i}\boldsymbol{k}\hat{\boldsymbol{R}}_i}, \tag{E1}$$

which can then be integrated to yield

$$\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}(t) = \hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}(0)e^{-\mathrm{i}\omega_k t} + \int_0^t ds e^{-\mathrm{i}\omega_k(t-s)} \times \sum_i g_k\omega_k(\hat{\boldsymbol{d}}_i(s)\cdot\boldsymbol{\epsilon}_{\boldsymbol{k}})e^{-\mathrm{i}\boldsymbol{k}\hat{\boldsymbol{R}}_i} + \int_0^t ds e^{-\mathrm{i}\omega_k(t-s)} \times \sum_i g_k([\hat{\boldsymbol{\mu}}_{t,i}(s) + \hat{\boldsymbol{\mu}}_{s,i}(s)]\cdot[\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}}])e^{-\mathrm{i}\boldsymbol{k}\hat{\boldsymbol{R}}_i}. \tag{E2}$$

A simplification for the calculation that follows is to consider negative and positive frequency components of the magnetic and electric field, in particular,

$$\hat{\boldsymbol{B}}^\pm = \pm\mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}} (\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}})g_k\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}e^{\pm\mathrm{i}\boldsymbol{k}\boldsymbol{R}},$$
$$\hat{\boldsymbol{E}}^\pm = \pm\mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon}} g_k\omega_k\boldsymbol{\epsilon}_{\boldsymbol{k}}\hat{a}_{\boldsymbol{k},\boldsymbol{\epsilon}}e^{\pm\mathrm{i}\boldsymbol{k}\boldsymbol{R}}, \tag{E3}$$

so that we only have to consider the equation of motion for one of the photonic operators. To shorten the notation we also introduce the total magnetic dipole operator $\hat{\boldsymbol{\mu}}_i(t) = \hat{\boldsymbol{\mu}}_{t,i}(t) + \hat{\boldsymbol{\mu}}_{s,i}(t)$ leading to expressions for the positive field components of the electric and magnetic fields

$$\hat{\boldsymbol{E}}^+(\boldsymbol{R}_i) = \hat{\boldsymbol{E}}_0^+(\boldsymbol{R}_i) + \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon},j}(g_k\omega_k)^2 e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_{ij}}\boldsymbol{\epsilon}_{\boldsymbol{k}}\int_0^t ds e^{-\mathrm{i}\omega_k(t-s)}(\hat{\boldsymbol{d}}_j(s)\cdot\boldsymbol{\epsilon}_{\boldsymbol{k}}) + \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon},j} g_k^2\omega_k e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_{ij}}\boldsymbol{\epsilon}_{\boldsymbol{k}}\int_0^t ds e^{-\mathrm{i}\omega_k(t-s)}(\hat{\boldsymbol{\mu}}_i(s)\cdot[\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}}]),$$
$$\hat{\boldsymbol{B}}^+(\boldsymbol{R}_i) = \hat{\boldsymbol{B}}_0^+(\boldsymbol{R}_i) + \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon},j}(\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}})g_k^2\omega_k e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_{ij}}\int_0^t ds e^{-\mathrm{i}\omega_k(t-s)}(\hat{\boldsymbol{d}}_j(s)\cdot\boldsymbol{\epsilon}_{\boldsymbol{k}}) + \mathrm{i}\sum_{\boldsymbol{k},\boldsymbol{\epsilon},j}(\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}})g_k^2 e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_{ij}}\int_0^t ds e^{-\mathrm{i}\omega_k(t-s)}(\hat{\boldsymbol{\mu}}_i(s)\cdot[\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}}]), \tag{E4}$$

where the zero subscript indicates that the photonic operators in the field expression were taken at time zero. Thus, both fields have similar terms. First, the vacuum time evolution, then the interaction with the proper dipole moment, and then a cross term between the magnetic and electric components of the interaction. There will be cross terms corresponding to the magnetic field produced by an electric dipole and the magnetic field produced by a magnetic dipole. Since we will neglect the magnetic transition dipole at the end of this treatment, we omit these cross terms and only keep the magnetic transition dipole terms to elucidate the difference to the magnetic static dipole term.

After the cross-terms are omitted, the total magnetic and electric field operators can be obtained from the positive frequency components by adding the hermitian conjugate so that after performing the usual summation over the polarization vectors

$$\hat{\boldsymbol{E}}(\boldsymbol{R}_i) = \hat{\boldsymbol{E}}_0(\boldsymbol{R}_i) + \mathrm{i}\sum_{\boldsymbol{k},j}(g_k\omega_k)^2 e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_{ij}}\int_0^t ds e^{-\mathrm{i}\omega_k(t-s)}\cdot\left[\hat{\boldsymbol{d}}_j(s) - \frac{\boldsymbol{k}(\hat{\boldsymbol{d}}_j(s)\cdot\boldsymbol{k})}{k^2}\right] + \text{H.c.}, \tag{E5}$$

$$\hat{\boldsymbol{B}}(\boldsymbol{R}_i) = \hat{\boldsymbol{B}}_0(\boldsymbol{R}_i) + \mathrm{i}\sum_{\boldsymbol{k},j}(g_k\omega_k)^2 e^{\mathrm{i}\boldsymbol{k}\boldsymbol{R}_{ij}}\int_0^t ds e^{-\mathrm{i}\omega_k(t-s)}\cdot\left[\hat{\boldsymbol{\mu}}_i(s) - \frac{\boldsymbol{k}(\hat{\boldsymbol{\mu}}_i(s)\cdot\boldsymbol{k})}{k^2}\right] + \text{H.c.}, \tag{E6}$$

so that the magnetic and electric fields have the same structure. Now, the continuum limit is taken for the electromagnetic field modes $\frac{1}{V}\sum_{\boldsymbol{k}} \to \frac{1}{(2\pi)^3}\int d\boldsymbol{k}$ and the angular integral is performed in spherical coordinates so that, defining the dipolar Green's function

$$\begin{aligned}\mathbf{G}_k(\mathbf{R}) &= \left(\mathbb{1} + \frac{1}{k^2}\nabla\otimes\nabla\right)\frac{e^{\mathrm{i}kR}}{4\pi R}\\ &= \frac{e^{\mathrm{i}kR}}{4\pi k^2}\left[\left(\frac{k^2}{R} + \frac{\mathrm{i}k}{R^2} - \frac{1}{R^3}\right)\mathbb{1} + \left(-\frac{k^2}{R} - \frac{3\mathrm{i}k}{R^2} + \frac{3}{R^3}\right)\frac{\mathbf{R}\otimes\mathbf{R}}{R^2}\right],\end{aligned} \tag{E7}$$

with the usual shorthand $\boldsymbol{G}(\boldsymbol{R}) = \boldsymbol{G}_{k_0}(\boldsymbol{R})$ and the decomposition $\boldsymbol{G}_k(\boldsymbol{R}) = \boldsymbol{\Omega}_k(\boldsymbol{R}) - \mathrm{i}\boldsymbol{\Gamma}_k(\boldsymbol{R})$. The expressions for the electric and magnetic field become

$$\hat{\boldsymbol{E}}(\boldsymbol{R}_i) = \hat{\boldsymbol{E}}_0(\boldsymbol{R}_i) + \frac{\mathrm{i}}{\pi\epsilon_0}\sum_j\int_0^\infty dk\int_0^t ds k^2\boldsymbol{\Gamma}_{\boldsymbol{k}}(\boldsymbol{R}_{ij})\cdot\hat{\boldsymbol{d}}_j(s)e^{-\mathrm{i}\omega_k(t-s)} + \text{H.c.}, \tag{E8}$$

$$\hat{\boldsymbol{B}}(\boldsymbol{R}_i) = \hat{\boldsymbol{B}}_0(\boldsymbol{R}_i) + \frac{\mathrm{i}}{\pi\epsilon_0}\sum_j\int_0^\infty dk\int_0^t ds k^2\boldsymbol{\Gamma}_{\boldsymbol{k}}(\boldsymbol{R}_{ij})\cdot\hat{\boldsymbol{\mu}}_j(s)e^{-\mathrm{i}\omega_k(t-s)} + \text{H.c.}. \tag{E9}$$

The key is now that this elimination cannot be performed exactly and one must make a perturbative ansatz by plugging the time evolution of the dipole operators without light-matter interactions into these integrals. The free-space time evolution of the dipole operators is explicitly

$$\begin{aligned}\hat{\boldsymbol{d}}_j(s) &= \boldsymbol{d}\sigma_j(s) + \boldsymbol{d}^*\sigma_j^\dagger = \boldsymbol{d}\sigma_j(t)e^{-\mathrm{i}\omega(s-t)} + \boldsymbol{d}^*\sigma_j(t)^\dagger e^{\mathrm{i}\omega(s-t)},\\ \hat{\boldsymbol{\mu}}_{t,j}(s) &= \boldsymbol{\mu}_j\sigma_j(s) + \boldsymbol{\mu}_j^*\sigma_j^\dagger(s) = \boldsymbol{\mu}_j\sigma_j(t)e^{-\mathrm{i}\omega(s-t)} + \boldsymbol{\mu}_j^*\sigma_j^\dagger(t)e^{\mathrm{i}\omega(s-t)},\\ \hat{\boldsymbol{\mu}}_{s,j}(s) &= \boldsymbol{\mu}_{j,e}\sigma_j^\dagger(s)\sigma_j(s) + \boldsymbol{\mu}_{j,g}\sigma_j(s)\sigma_j^\dagger(s) = \hat{\boldsymbol{\mu}}_{s,j}(t).\end{aligned} \tag{E10}$$

This implies that the fundamental difference between the static and transition dipole moments is given by the frequency they rotate at. All these time evolutions will lead to integrations of the type

$$\zeta(\omega) = \mathrm{i}\int_0^\infty e^{\mathrm{i}\omega t}dt = \mathcal{P}\frac{1}{\omega} + \mathrm{i}\pi\delta(\omega), \tag{E11}$$

so that splitting the magnetic field into a transition part $\hat{\boldsymbol{B}} = \hat{\boldsymbol{B}}_t + \hat{\boldsymbol{B}}_s$ while keeping the vacuum contribution in the transition part

$$\begin{aligned}\hat{\boldsymbol{E}}(\boldsymbol{R}_i) &= \hat{\boldsymbol{E}}_0(\boldsymbol{R}_i) + \frac{1}{\pi\epsilon_0}\sum_j\int_0^\infty dk k^2\boldsymbol{\Gamma}_k(\boldsymbol{R}_{ij})\cdot\boldsymbol{d}(\sigma_j\zeta(\omega_k+\omega) + \sigma_j^\dagger\zeta(\omega_k-\omega)) + \text{H.c.},\\ \hat{\boldsymbol{B}}_t(\boldsymbol{R}_i) &= \hat{\boldsymbol{B}}_0(\boldsymbol{R}_i) + \frac{1}{\pi\epsilon_0}\sum_j\int_0^\infty dk k^2\boldsymbol{\Gamma}_k(\boldsymbol{R}_{ij})\cdot\boldsymbol{\mu}_{t,j}(\sigma_j\zeta(\omega_k+\omega) + \sigma_j^\dagger\zeta(\omega_k-\omega)) + \text{H.c.},\\ \hat{\boldsymbol{B}}_s(\boldsymbol{R}_i) &= \frac{1}{\pi\epsilon_0}\sum_j\int_0^\infty dk k^2\boldsymbol{\Gamma}_k(\boldsymbol{R}_{ij})\cdot\boldsymbol{\zeta}(\omega_k)(\boldsymbol{\mu}_{j,e}\sigma_j^\dagger\sigma_j + \boldsymbol{\mu}_{j,g}\sigma_j\sigma_j^\dagger) + \text{H.c.}.\end{aligned} \tag{E12}$$

The previous statement about the rotational frequencies now becomes clear as the transition fields contain zeta functions of the form $\zeta(\omega_k \pm \omega)$ where the Green's function around $\boldsymbol{\Gamma}_{k_0}(\boldsymbol{R})$ matters. The static contributions, however, contain a zeta function of the form $\zeta(\omega_k)$ where $\boldsymbol{\Gamma}_{k=0}(\boldsymbol{R})$ matters. Since the zeta function essentially only evaluates the Kramers-Kronig relations for the complex function $\boldsymbol{G}(\boldsymbol{R})$, i.e., $(k'^2)\Omega_{k'}(\boldsymbol{R}) = \int dk\frac{k^2\Gamma_k(\boldsymbol{R})}{\omega'-\omega}$. Ultimately, this leads to the explicit representation of the fields, using

the fact that $\lim_{k\to 0} k^2 G_k(\boldsymbol{R}) = \frac{1}{4\pi R_{ij}^3}(3\frac{\boldsymbol{R}_{ij}\boldsymbol{R}_{ij}}{R_{ij}R_{ij}} - 1)$ so that only the real part remains

$$\hat{\boldsymbol{E}}(\boldsymbol{R}_i) = \hat{\boldsymbol{E}}_0(\boldsymbol{R}_i) + \frac{k_0^2}{\epsilon_0}\sum_j [(\boldsymbol{\Gamma}(\boldsymbol{R}_{ij}) - \mathrm{i}\boldsymbol{\Omega}(\boldsymbol{R}_{ij}))\sigma_j + (\boldsymbol{\Gamma}(\boldsymbol{R}_{ij}) + \mathrm{i}\boldsymbol{\Omega}(\boldsymbol{R}_{ij}))\sigma_j^\dagger]\cdot\boldsymbol{d},$$

$$\hat{\boldsymbol{B}}(\boldsymbol{R}_i) = \hat{\boldsymbol{B}}_0(\boldsymbol{R}_i) + \frac{k_0^2}{\epsilon_0}\sum_j [(\boldsymbol{\Gamma}(\boldsymbol{R}_{ij}) - \mathrm{i}\boldsymbol{\Omega}(\boldsymbol{R}_{ij}))\sigma_j + (\boldsymbol{\Gamma}(\boldsymbol{R}_{ij}) + \mathrm{i}\boldsymbol{\Omega}(\boldsymbol{R}_{ij}))\sigma_j^\dagger]\cdot\boldsymbol{\mu}_t,$$

$$\hat{\boldsymbol{B}}_s(\boldsymbol{R}_i) = \frac{\mu_0}{4\pi}\sum_j \frac{1}{R_{ij}^3}\left(3\frac{\boldsymbol{R}_{ij}\boldsymbol{R}_{ij}}{R_{ij}R_{ij}} - 1\right)\cdot\hat{\boldsymbol{\mu}}_{s,j}. \tag{E13}$$

The next step is to plug these expressions into the equations of motion, to use the rotating wave approximation and the definitions $\Gamma = k_0^3 d^2/(3\pi\hbar\epsilon_0)$ and $\Gamma_\mu = k_0^3\mu^2/3\pi\hbar\epsilon_0$ to define the Green's function for the matter problem. Indeed, the definitions

$$f_{ij}^{(\mu)} + \mathrm{i}g_{ij}^{(\mu)} = \frac{3\Gamma_v}{4}\left[\left(\frac{k^2}{R_{ij}} + \frac{\mathrm{i}k}{R_{ij}^2} - \frac{1}{R_{ij}^3}\right)\hat{\mu}_i\cdot\hat{\mu}_j + \left(-\frac{k^2}{R_{ij}} - \frac{3\mathrm{i}k}{R_{ij}^2} + \frac{3}{R_{ij}^3}\right)\frac{(\hat{\mu}_j\cdot\mathbf{R}_{ij})(\hat{\mu}_i\cdot\mathbf{R}_{ij})}{R_{ij}^2}\right]\frac{e^{\mathrm{i}kR_{ij}}}{k^2},$$

$$f_{ij}^{(g)} + \mathrm{i}g_{ij}^{(g)} = \frac{3\Gamma}{4}\left[\left(\frac{k^2}{R_{ij}} + \frac{\mathrm{i}k}{R_{ij}^2} - \frac{1}{R_{ij}^3}\right)\hat{d}_i\cdot\hat{d}_j + \left(-\frac{k^2}{R_{ij}} - \frac{3\mathrm{i}k}{R_{ij}^2} + \frac{3}{R_{ij}^3}\right)\frac{(\hat{d}_j\cdot\mathbf{R}_{ij})(\hat{d}_i\cdot\mathbf{R}_{ij})}{R_{ij}^2}\right]\frac{e^{\mathrm{i}kR_{ij}}}{k^2}, \tag{E14}$$

and Eq. (13) lead precisely to Eqs. (12), (14), and (15) after using the rotating wave approximation.

After adding a classical drive just as in the main text, the equations of motion for the transition operators become

$$\frac{d\sigma_k}{dt} = -\mathrm{i}[\Delta + \delta\hat{\omega}_k]\sigma_k + \mathrm{i}\sigma_k^z\sum_{i\neq k}\left(g_{ki}^{(d)} + g_{ki}^{(\mu)}\right)\sigma_i + \sigma_k^z\sum_i\left(f_{ki}^{(d)} + f_{ki}^{(\mu)}\right)\sigma_i - \mathrm{i}\Omega e^{\mathrm{i}\boldsymbol{k}_0\boldsymbol{R}_i}\sigma_k^z + \sigma_{k,\mathrm{in}}, \tag{E15}$$

where $\sigma_{k,\mathrm{in}}$ is the canonical input noise due to spontaneous emission of an atom

$$\sigma_{k,\mathrm{in}} = \mathrm{i}\sigma_k^z\boldsymbol{d}\cdot\boldsymbol{E}_0^+(\boldsymbol{r}_k), \tag{E16}$$

with expectation value zero and a frequency operator

$$\delta\hat{\omega}_k = \frac{1}{2}\sum_{i\neq k}\left[\left(\Omega_{ik}^{e,e} - \Omega_{ik}^{g,g}\right) + \left(\Omega_{ik}^{e,e} + \Omega_{ik}^{g,g} - 2\Omega_{ik}^{e,g}\right)\sigma_i^z\right]. \tag{E17}$$

This is a configuration and density-dependent Zeeman shift of the kth atom's frequency.

APPENDIX F: WEAK-EXCITATION LIMIT

The weak-excitation limit, i.e., $\langle\sigma_j^z\sigma_{j'}\rangle \approx -\langle\sigma_{j'}\rangle$, the equations of motion for the expectation values of the transition operators become

$$\frac{d\langle\sigma_k\rangle}{dt} = -\mathrm{i}[\Delta + \delta\omega_k]\langle\sigma_k\rangle - \mathrm{i}\sum_{i\neq k}\left(g_{ki}^{(d)} + g_{ki}^{(\mu)}\right)\langle\sigma_i\rangle - \sum_i\left(f_{ki}^{(d)} + f_{ki}^{(\mu)}\right)\langle\sigma_i\rangle + \mathrm{i}\Omega e^{\mathrm{i}\boldsymbol{k}_0\boldsymbol{R}_i}, \tag{F1}$$

which is written in matrix notation in Eq. (16).

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2.2 P2: Metasurface-Based Hybrid Optical Cavities for Chiral Sensing

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In this publication, my contributions were the following:

- First calculation of all results except S9 (calculations were discussed with and checked by Michael Reitz).
- Extension of S9 to slowly varying fields.
- Production of Fig S1-S3.
- Production of Fig 1 (b), (c), (e) and Fid 2 (b), (c).
- First draft of chapters S2, S3, S4, S5, S6, S7 and S8.
- Writing an initial draft of the main text except for introduction (40%).
- Contributions to rewrites of main text and supplement (20%).

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Metasurface-Based Hybrid Optical Cavities for Chiral Sensing

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Quantum metasurfaces, i.e., two-dimensional subwavelength arrays of quantum emitters, can be employed as mirrors towards the design of hybrid cavities, where the optical response is given by the interplay of a cavity-confined field and the surface modes supported by the arrays. We show that stacked layers of quantum metasurfaces with orthogonal dipole orientation can serve as helicity-preserving cavities. These structures exhibit ultranarrow resonances and can enhance the intensity of the incoming field by orders of magnitude, while simultaneously preserving the handedness of the field circulating inside the resonator, as opposed to conventional cavities. The rapid phase shift in the cavity transmission around the resonance can be exploited for the sensitive detection of chiral scatterers passing through the cavity. We discuss possible applications of these resonators as sensors for the discrimination of chiral molecules. Our approach describes a new way of chiral sensing via the measurement of particle-induced phase shifts.

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Conventional isotropic (e.g., metallic) mirrors reverse the handedness (or helicity) of circularly polarized light by turning right-circularly polarized (RCP) light into left-circularly polarized (LCP) light and vice versa [1–3]. This makes it impossible to realize helicity-preserving (HP) cavities or even chiral cavities (i.e., cavities only supporting light modes of a certain handedness), with conventional mirrors [4]. There is, however, a great current scientific and technological interest in the design of HP or chiral mirrors, metasurfaces, and resonators [5–11], in particular, for the enhancement of so-called dichroic effects. Dichroism refers to the (typically weak) differential absorption of circularly polarized light by chiral scatterers such as molecular enantiomers [12]. Enhancing dichroic effects with optical resonators can result in better sensitivities for the discrimination of molecular enantiomers [13–17], a desired task for biochemical applications. Other techniques for chiral sensing include measurements of photoelectron circular dichroism (PECD) [18], three- or multiwave mixing [19,20], or chirality-induced spin selectivity (CISS) [21]. In the strong light-matter coupling regime, chiral cavities have recently been proposed to create novel light-dressed states of matter by breaking the time-reversal symmetry in materials, leading to the emerging field of chiral polaritonics [22,23]. In this Letter, we show that HP cavities can be implemented with quantum metasurfaces employed as mirrors. These structures have emerged as platforms for achieving strong and highly directional light-matter interactions and can most prominently be realized with cold atoms trapped in optical lattices [24]. They can exhibit close to perfect reflection of incoming light [24–29] and have numerous other applications, e.g., as platforms for topological quantum optics [30–32], nonlinear quantum optics [33–38], or quantum information processing [39–44]. The main ingredient of our approach is to manipulate the polarization of the incoming light field via the orientation of the effective two-level systems that make up the metasurfaces, which can, for instance, be tuned via an external magnetic field for certain atomic transitions. We, however, strongly emphasize that our treatment is in principle valid for any array of point dipoles and could be analogously implemented with arrays of classical dipoles such as plasmonic lattices, in which case no external magnetic field control is needed as the polarization can be controlled by the geometry of the individual elements [45].

More generally, this work falls within the scope of *hybrid cavities*, i.e., the design of optical resonators going beyond the simple textbook picture of a single electromagnetic mode confined between two nonreactive mirrors. Instead, strongly dispersive optical elements such as photonic crystals or plasmonic metasurfaces are used as reflectors [46–50] with the aim to surpass the performance of standard cavities, implying a highly non-Markovian behavior of the cavity as characterized by non-Lorentzian, typically Fano-type line shapes [51].

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HP mirror.—Let us present an implementation procedure for a HP mirror using a stacked system of quantum metasurfaces. We start by introducing the formalism for a single metasurface [52]. To this end, we consider a 2D quasi-infinite quantum emitter array comprised of $\mathcal{N}$ emitters at transition frequency ω_0 situated in the xy plane at positions $\mathbf{r}_j$. For simplicity, one may imagine a square lattice; however, most of the results derived in the following are equally valid for other Bravais lattices. For the HP mirror, in the following we require that the transition dipoles of the emitters in the array can be identically oriented along a certain direction. One possible implementation of this lies in employing the magnetic substructure of a $J=0 \to J=1$ transition [24,37]. In the Cartesian polarization basis, the transition dipole operator for each emitter can then be written as $\mathbf{d}=\sum_\nu \mathbf{d}_\nu \sigma_\nu + \text{H.c.}$ with $\mathbf{d}_\nu = \langle g|\mathbf{d}|\nu\rangle$, and $\sigma_\nu = |g\rangle\langle\nu|$ is the corresponding lowering operator for each electronic transition ($\nu = x, y, z$). An external magnetic field along the ν direction can be used to "select" the ν-transition dipole (by making all other transitions orthogonal to the magnetic field off-resonant [52]). Other implementations could rely on artificial, nanoprinted elements (see previous work [53]), photonic crystal mirrors [49], or solid-state platforms [54]. We consider a laser drive entering from the left in the form of a plane wave with positive-frequency amplitude $\mathbf{E}_{\text{in}}^{(+)} = (E_{\text{in},x}, E_{\text{in},y}, 0)^\top$ and laser frequency $\omega_l = 2\pi c/\lambda = c k_l$, where λ and k_l are the laser wavelength and wave number, respectively.

In a frame rotating at the laser frequency, the Hamiltonian describing the dynamics of the emitter array is given by the sum of the free evolution and the dipole-dipole interaction ($\hbar = 1$)

$$\mathcal{H}_0 + \mathcal{H}_{\text{d-d}} = -\Delta \sum_{j,\nu} \sigma_{j,\nu}^\dagger \sigma_{j,\nu} + \sum_{j,j',\nu,\nu'} \Omega_{jj'}^{\nu\nu'} \sigma_{j,\nu}^\dagger \sigma_{j',\nu'}, \quad (1)$$

with the laser detuning $\Delta = \omega_l - \omega_0$ and $\sigma_{j,\nu}$ the lowering operator for the ν transition within a particular emitter j. Assuming normally incident illumination, the laser drive adds as $\mathcal{H}_l = \sum_{j,\nu}(\eta_\nu \sigma_{j,\nu}^\dagger + \text{H.c.})$ with Rabi frequencies $\eta_\nu = d_\nu E_{\text{in},\nu}^{(+)}$ and $\eta_z = 0$. In addition to the coherent processes, the collective loss of excitations due to spontaneous emission is described by the Lindblad term

$$\mathcal{L}[\rho] = \sum_{j,j',\nu,\nu'} \Gamma_{jj'}^{\nu\nu'} \left[\sigma_{j,\nu} \rho \sigma_{j',\nu'}^\dagger - \frac{1}{2} \{ \sigma_{j,\nu}^\dagger \sigma_{j',\nu'}, \rho \} \right], \quad (2)$$

where the last term denotes an anticommutator and the diagonal elements describe the independent spontaneous emission of the emitters $\Gamma_{jj}^{\nu\nu'} = \Gamma_0 \delta_{\nu\nu'}$ with $\Gamma_0 = \omega_0^3 d^2/(3\pi\epsilon_0 c^3)$ (we assume the dipole moments to be identical in the following $d_\nu \equiv d$). The rates $\Omega_{jj'}^{\nu\nu'}$, $\Gamma_{jj'}^{\nu\nu'}$ describe coherent or incoherent scattering of photons between emitters j and j' and between transitions ν and ν' and can be derived as real and imaginary parts of the photonic Green's tensor (see Supplemental Material [55]) as

$$\Omega_{jj'}^{\nu\nu'} - \text{i}\frac{\Gamma_{jj'}^{\nu\nu'}}{2} = -\mu_0 \omega_0^2 \mathbf{d}_\nu^* \cdot \mathbf{G}(\mathbf{r}_{jj'}) \cdot \mathbf{d}_{\nu'}, \quad (3)$$

expressed in terms of the vacuum permeability μ_0 and depending on the interparticle separation $\mathbf{r}_{jj'} = \mathbf{r}_{j'} - \mathbf{r}_j$. The Green's tensor is defined such that the real part of the self-interaction at $j = j'$ vanishes. From the steady-state solution of the quantum master equation $\dot{\rho} = \text{i}[\rho, \mathcal{H}_0 + \mathcal{H}_{\text{d-d}} + \mathcal{H}_l] + \mathcal{L}[\rho]$, the dipole amplitudes and thereby the transmitted and reflected fields can be computed (see Supplemental Material [55]). The transmission amplitude of the metasurface for a single Cartesian component $E_{\text{in},\nu}$ is then simply given by [25,26]

$$t_m = 1 + \frac{\text{i}\tilde{\Gamma}(0)/2}{\tilde{\Omega}(0) - \Delta - \text{i}\tilde{\Gamma}(0)/2}, \quad (4)$$

where $\tilde{\Omega}(0) = \sum_j \Omega_{0j}^{\nu\nu}$, $\tilde{\Gamma}(0) = \sum_j \Gamma_{0j}^{\nu\nu}$ describe the dipole-induced collective frequency shift and decay rate arising from the ν transition dipoles. The complex transmission and reflection amplitudes are connected as $t_m = 1 + r_m$ while $|t_m|^2 + |r_m|^2 = 1$. Most notably, if the laser frequency matches the collective metasurface resonance $\omega_l = \omega_\nu + \tilde{\Omega}(0)$, perfect reflection of incoming light is obtained as $|r_m|^2 = 1$. In the following, for the sake of clarity, we proceed with the simplified two-level description. Finally, to obtain a HP mirror, we consider now two copies of metasurfaces separated by a distance ℓ_m, one with dipoles pointing in x direction and one with dipoles pointing in the y direction with a path length difference of $k_l \ell_m = \phi_m = 2\pi n_m + \pi/2$ ($n_m \in \mathbb{N}_0$) between the two polarizations. The combination of these two mirrors is a helicity-preserving mirror. The path length difference leads to a total phase difference of π between the x- and y-polarization components in reflection, thereby reversing the mirror operation which does not conserve helicity for an ordinary mirror. A full transfer matrix calculation showing this can be found in the Supplemental Material [55].

HP cavity.—A HP optical cavity can now be simply implemented by two HP metasurface mirrors separated by a distance ℓ [see Fig. 1(a)]. The two layers making up a HP mirror consist of dipoles with perpendicular dipole orientations, leading to vanishing interactions between the x and y subcavities, which also holds in the near field. For simplicity, we thus continue the discussion for a single cavity while keeping in mind that the actual setup consists of two noninteracting copies. A full discussion for both polarization components can be found in the Supplemental Material [55]. Solving the coupled-dipole equations and neglecting the contributions from all evanescent terms, a

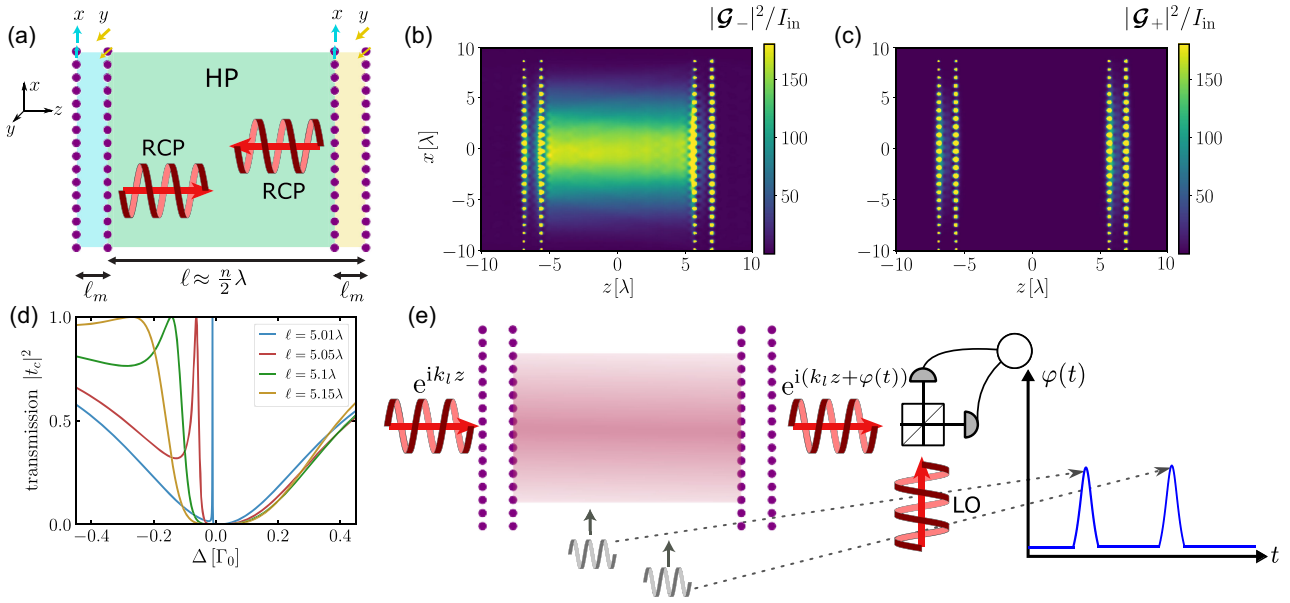


FIG. 1. Helicity-preserving metasurface cavity. (a) A HP metasurface cavity can be constructed with composite mirrors of orthogonal dipole orientation (e.g., along x and y), with an appropriate relative phase separation of $\phi_m = k_l \ell_m = \pi/2 + 2\pi n_m$. (b),(c) Absolute value squared of RS vectors $\mathcal{G}_\pm(\mathbf{R})$ for a cavity length $\ell = 12.505\lambda$, $\ell_m = 5\lambda/4$, for square lattices with lattice spacing $a = 0.8\lambda$ illuminated by RCP light with input polarization vector $\mathbf{E}_{\rm in}^{(+)} = (1, \mathrm{i}, 0)^\top/\sqrt{2}$ close to the cavity resonance. The plots are normalized to the input intensity $I_{\rm in} = |\mathbf{E}_{\rm in}^{(+)}|^2$. The metasurfaces are defined with a finite curvature radius, leading to Gaussian confinement of the mode (beam waist $w_0 = 8\lambda$). (d) Cavity transmission $|t_c|^2$ as a function of the laser detuning Δ for different cavity lengths ℓ. (e) Chiral sensing with HP metasurface cavities: Ideal chiral scatterers passing through a cavity with the same handedness yield a phase shift in the cavity transmission output which can be measured by homodyne detection [local oscillator (LO)] and leads to a phase detection sequence as schematically illustrated in the diagram.

simple expression for the total transmitted field can be obtained as $E^{(+)}(z > \ell) = t_c E_{\rm in}^{(+)} e^{ik_l z}$ with the cavity transmission coefficient (assuming $k_0 \approx k_l$, for derivation see Supplemental Material [55])

$$t_c = \frac{(\Delta - \tilde{\Omega}(0))^2}{\left(\Delta - \tilde{\Omega}(0) + \mathrm{i}\frac{\tilde{\Gamma}(0)}{2}\right)^2 + \frac{\tilde{\Gamma}(0)^2}{4} e^{2ik_l \ell}}. \tag{5}$$

We remark that instead of solving the coupled-dipole equations for the two arrays, the same result can be obtained from classical transfer matrix theory [38,47,60], where the transfer matrix of a single metasurface can be expressed in terms of the mirror polarizability $\zeta_m = -\mathrm{i} r_m/t_m = \tilde{\Gamma}(0)/[2(\tilde{\Omega}(0) - \Delta)]$ as

$$\mathbf{T}_m = \begin{pmatrix} 1 + \mathrm{i}\zeta_m & \mathrm{i}\zeta_m \\ -\mathrm{i}\zeta_m & 1 - \mathrm{i}\zeta_m \end{pmatrix}. \tag{6}$$

The total transfer matrix is then simply obtained as $\mathbf{T} = \mathbf{T}_m \mathbf{T}_f \mathbf{T}_m$ with the free space propagation matrix $\mathbf{T}_f = \mathrm{diag}(e^{ik_l \ell}, e^{-ik_l \ell})$. The condition that the transmission ought to equal unity at the cavity resonance $|t_c|^2 = 1$, yields the following expression for the cavity resonance:

$$\Delta - \tilde{\Omega}(0) = -\frac{\tilde{\Gamma}(0)}{2} \tan(k_l \ell). \tag{7}$$

To demonstrate that the resulting cavity consisting of two HP mirrors indeed preserves the helicity, we compute the Riemann-Silberstein (RS) vectors [61]

$$\mathcal{G}_\pm(\mathbf{R}) = \frac{1}{\sqrt{2}}(\mathbf{E}(\mathbf{R}) \pm \mathrm{i}\mathcal{Z}\mathbf{H}(\mathbf{R})), \tag{8}$$

which describe the combined electromagnetic field of chiral polarization and $\mathcal{Z} = (\epsilon_0 c)^{-1}$ is the vacuum impedance. The absolute value of these quantities is plotted in Figs. 1(b) and 1(c), for RCP light entering the cavity, confirming that the cavity preserves the helicity while also showing a strong field enhancement for this component [62]. The cavity itself is, however, not chiral as *any* elliptical input polarization is supported. The magnetic field $\mathbf{H}(\mathbf{R})$ is determined via Maxwell's equations from the excitations of the electric dipoles on the metasurface as detailed in the Supplemental Material [55].

The transmission profile around the cavity resonance is illustrated in Fig. 1(d) for different cavity lengths for a lattice spacing of $a = 0.8\lambda$, where the dipole-induced collective frequency shift is close to zero, i.e., $\tilde{\Omega}(0) \approx 0$. If the cavity length ℓ exactly matches $n\lambda/2$ ($n \in \mathbb{N}$), the

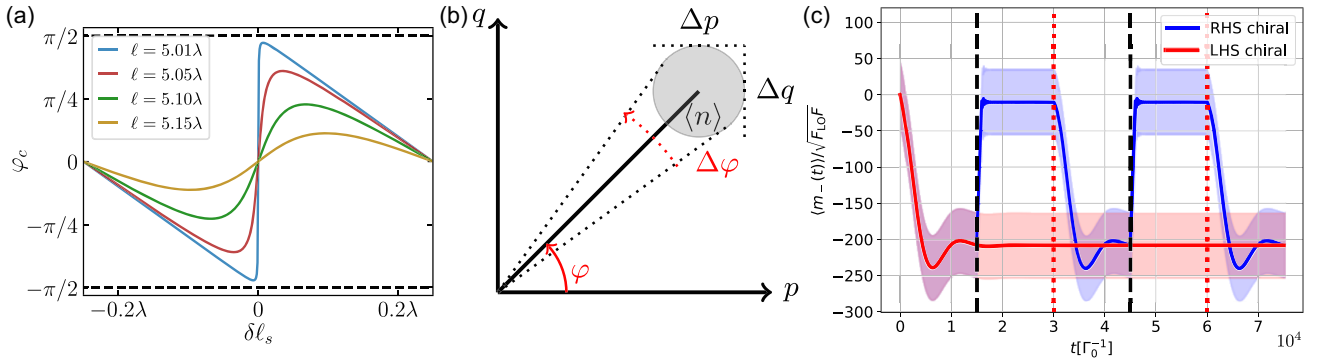


FIG. 2. Chiral sensing.—(a) Cavity phase $\varphi_c = \arg(t_c)$ versus small perturbations of the cavity length $\delta\ell_s$ for different cavity lengths ℓ (square lattice, $a = 0.8\lambda$). For each ℓ, the resonance condition described by Eq. (7) is fulfilled. (b) Illustration of phase uncertainty in phase space for a coherent state. (c) Homodyne detection signal $\langle m_-(t)\rangle$ for (ideal) right- and left-handed chiral scatterers passing through a RCP cavity with corresponding shot noise shown as shaded area. The dashed black line indicates the entry of a particle into the cavity and the dotted red line indicates its exit. In addition, a rotational average was performed as detailed in the Supplemental Material [55]. We have used $\Delta_s = 10\Gamma_0$, $\gamma_s = \Gamma_0$, with an average of one photon within the time interval Γ_0^{-1}, i.e., $F = \Gamma_0$, an integration time of $T\Gamma_0 = 2000$ and a detuning of $\Delta = \Gamma_0/100$.

cavity resonance coincides with the resonance of an individual layer and no transmission is obtained as all the light is reflected. If ℓ becomes slightly larger, a narrow transmission window opens up as the mirrors and the cavity now possess different resonance frequencies. Further increasing the cavity length leads to a strongly asymmetric Fano-type profile with a larger linewidth and the cavity resonance drifting towards infinity for $\ell \to (n+\frac{1}{2})\lambda/2$. Once the next multiple of $\lambda/2$ is approached, the cavity linewidth becomes narrow again and the cavity resonance shifts towards $\omega_\nu + \tilde{\Omega}(0)$ as $\tan(k_l\ell) \to 0$. The distance between the zero and the maximum of the transmission can be used as a measure for the cavity linewidth $\kappa = \tilde{\Gamma}(0)|\tan(k_l\ell)|$. We present a coupled-modes theory for the input-output description of cavities made from quantum metasurface mirrors in the Supplemental Material [55].

Chiral sensing.—We now consider the scenario depicted in Fig. 1(e) where chiral scatterers with radiative linewidth γ_s are sent through the cavity. We assume the resonance of the scatterers ω_s to be far-detuned from the cavity resonance $|\Delta_s| \gg |\Delta_c|, \gamma_s$ with $\Delta_{s/c} = \omega_l - \omega_{s/c}$. In this case, the effect of a scatterer with the same helicity as the cavity is to increase the path length of light passing through the cavity and thereby shift the cavity length by a small amount $\delta\ell_s = -\arctan(\gamma_s/2\Delta_s)/k_l$ (for derivation see Supplemental Material [55]), such that the total effective cavity length is now given by $\ell + \delta\ell_s$. Because of the quick phase switch around the cavity resonance for cavity lengths close to $n\lambda/2$, a small perturbation of the cavity length can lead to a considerable phase shift of the cavity transmission [see Fig. 2(a)].

One can proceed to compute the relative phase change in the cavity transmission between lengths ℓ and $\ell + \delta\ell_s$ on the cavity resonance (assuming $\ell \approx n\lambda/2$)

$$\varphi = \arg\frac{t_c(\ell+\delta\ell_s)}{t_c(\ell)}\bigg|_{\mathrm{res},\ell\approx n\frac{\lambda}{2}} \approx \arctan\left(\cot(2k_l\delta\ell_s)\right), \tag{9}$$

which reaches a value of $\pi[\theta(k_l\delta\ell_s) - 1/2]$ as $\delta\ell_s \to 0$, implying a phase jump from $-\pi/2$ to $\pi/2$ around the cavity resonance for lengths close to $n\lambda/2$ [$\theta(x)$ is the Heaviside function]. If the cavity length departs from $n\lambda/2$, the cavity linewidth increases and the phase shift gets diminished, as illustrated in Figs. 1(d) and 2(a).

An experimental setup to measure this phase is homodyne detection as illustrated in Fig. 1(e), where the phase between a local oscillator (for instance, obtained from beam splitting the input field) is compared to the phase of the output field. Suppose we consider a signal beam with which we drive the cavity $\alpha(t) = \sqrt{F}\exp(-i\omega_l t + i\theta)$ and the local oscillator field $\alpha_{\mathrm{LO}}(t) = \sqrt{F_{\mathrm{LO}}}\exp(-i\omega_l t + i\theta_{\mathrm{LO}})$ with intensities (number of photons per unit of time) F and F_{LO}. Then considering homodyne detection for a Fabry-Pérot cavity leads to an uncertainty in the measured phase for an integration time T of the measurement and a quantum efficiency η_Q, which is encoded in the intensity difference m_- with variance $(\Delta m_-)^2_{\mathrm{res}} = \eta_Q T F_{\mathrm{LO}}$ and expectation value (assuming the phase variation to happen on a timescale much slower than the optical frequency)

$$\langle m_-(t,T)\rangle = 2\eta_Q\sqrt{FF_{\mathrm{LO}}}|t_c|\int_t^{t+T} dt' \sin\left(\varphi(t')\right). \tag{10}$$

Here, we have approximated that $F_{\mathrm{LO}} \gg F$ and have taken $\theta + \theta_{\mathrm{LO}} = 2\pi n$, which can be obtained by phase matching the local oscillator and the signal beam. Since the cavity is a linear element, the resulting phase uncertainty [see Fig. 2(b)] is independent of any cavity properties and only depends on the properties of the state of the incoming beam, which is assumed to be classical for this calculation.

As an alternative to describing the passage of particles through the cavity with transfer matrix theory, the dipole theory can be extended to include the presence of an additional chiral scatterer which can be represented by coupled electric and magnetic dipoles (see Supplemental Material [55]). These equations of motion are simulated in Fig. 2(c) for right- and left-handed scatterers (RHS and LHS) entering a RCP cavity, showing a clear distinction in the resulting signal. Here, the classical shot noise of a coherent input field is used to estimate the phase error for a homodyne detection.

Let us finally briefly discuss the applicability of the chiral sensing scheme to the discrimination of molecular enantiomers. Chiral molecules are, in general, not perfect chiral scatterers. This is manifested in the fact that their circular dichroism $\mathrm{CD} = (A_+ - A_-)/(A_+ + A_-)$, i.e., the difference in absorbance between RCP and LCP light, is not unity but some small finite value. This implies that the assumption of coupling to only a single polarization component is unrealistic. Instead, both enantiomers will lead to small differential shifts in optical path length. Aside from the absolute magnitude of this change in path length, however, the presented strategy retains generality and the applicability is a question of detailed system parameters.

Conclusions and outlook.—We have proposed a novel approach for chiral sensing based on the interferometric detection of phase shifts in HP metasurface cavities. In a first step, we have shown that HP mirrors and cavities can be created from stacked quantum metasurfaces with orthogonal dipole orientation. We then proposed to use these narrow-linewidth HP cavity modes for the optical sensing of chiral scatterers by discussing how the phase of the output field is modified by an off-resonant scatterer passing through the cavity. We also discussed briefly the applicability to the discrimination of molecular enantiomers. Our approach constitutes a fully optical approach to chiral sensing in the linear regime as compared to other approaches relying on nonlinear techniques [20]. Furthermore, it does not require strong coupling or resonance between the scatterers and the cavity mode in contrast to other recent treatments of cavity-based enantiomer discrimination [17]. We provide a discussion of possible errors leading to, e.g., mixing of helicities in the Supplemental Material [55]. We have previously discussed derogating effects such as motion, vacancies, and nonlinearities for single metasurfaces in Ref. [52].

We also remark that, in addition to the use proposed in this Letter, layered metasurfaces can enable a host of other applications. For instance, stacking many of these layers leads to Bragg-mirror physics which could be used to tailor the frequency windows of optical elements based on quantum metasurfaces. Even more general polarization structures of the cavity mode, such as Faraday cavities [22], might also be implementable.

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Supplemental material for "Metasurface-based hybrid optical cavities for chiral sensing"

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S1. DYADIC GREEN'S FUNCTION

The free-space photonic Green's tensor is given by (evaluated at the resonance of the emitters $\omega_0 = ck_0$)

$$\mathbf{G}(\mathbf{R}) = \left(\mathbb{1} + \frac{1}{k_0^2}\nabla\otimes\nabla\right)\frac{\mathrm{e}^{\mathrm{i}k_0 R}}{4\pi R} - \frac{\mathbb{1}}{3k_0^2}\delta(\mathbf{R}), \tag{S.1}$$

where $\otimes$ denotes the dyadic product, $R = |\mathbf{R}|$ and the last term removes the divergence on the self-interaction terms at $\mathbf{R} = 0$. This can be expressed more explicitly as

$$\mathbf{G}(\mathbf{R}) = \frac{\mathrm{e}^{\mathrm{i}k_0 R}}{4\pi k_0^2}\left[\left(\frac{k_0^2}{R} + \frac{\mathrm{i}k_0}{R^2} - \frac{1}{R^3}\right)\mathbb{1} + \left(-\frac{k_0^2}{R} - \frac{3\mathrm{i}k_0}{R^2} + \frac{3}{R^3}\right)\frac{\mathbf{R}\otimes\mathbf{R}}{R^2}\right] - \frac{\mathbb{1}}{3k_0^2}\delta(\mathbf{R}), \tag{S.2}$$

where $\mathbb{1}$ is the 3×3 identity matrix and $\mathbf{R} = (x, y, z)^\top$. For the scattering problem considered here, a Fourier decomposition of the Green's tensor with respect to the in-plane wave vector components $\mathbf{q} = (q_x, q_y)^\top$ is useful. Decomposing also $\mathbf{R} = (\mathbf{r}_\parallel, z)^\top$ into lattice plane and out-of-plane coordinate, one can make use of the Weyl expansion [S1]

$$\frac{\mathrm{e}^{\mathrm{i}k_0 R}}{R} = \frac{\mathrm{i}}{2\pi}\int \mathrm{d}\mathbf{q}\,\frac{1}{q_z}\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}_\parallel}\mathrm{e}^{\mathrm{i}q_z|z|}, \tag{S.3}$$

with $q_z = \sqrt{k_0^2 - q^2}$. The Green's tensor (including the divergent self-interaction) can now be written as

$$\mathbf{G}(\mathbf{R}) = \frac{\mathrm{i}}{8\pi^2 k_0^2}\int \mathrm{d}\mathbf{q}\,\frac{k_0^2\mathbb{1} - \bar{\mathbf{q}}\otimes\bar{\mathbf{q}}}{q_z}\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}_\parallel}\mathrm{e}^{\mathrm{i}q_z|z|}, \tag{S.4}$$

where $\bar{\mathbf{q}} = (q_x, q_y, \mathrm{sgn}(z)\sqrt{k_0^2 - q^2})^\top$.

From Maxwell's equations for monochromatic fields, one can find the magnetic Green's tensor (including the self-interaction) as

$$\mathbf{G}_m(\mathbf{R}) = \frac{\mathrm{i}}{\omega_0}\boldsymbol{\nabla}\times\mathbf{G}(\mathbf{R}) = \frac{\mathrm{i}}{4\pi\omega_0}\boldsymbol{\nabla}\times\left(\mathbb{1}\frac{\mathrm{e}^{\mathrm{i}k_0 R}}{R}\right) = \frac{(k_0 R + \mathrm{i})\mathrm{e}^{\mathrm{i}k_0 R}}{4\pi\omega_0 R^3}\begin{pmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0\end{pmatrix}. \tag{S.5}$$

which can be used to determine the magnetic field emitted from an electric dipole. Here, the curl of a matrix $\mathbf{A}$ is defined (using Einstein sum convention) as $(\boldsymbol{\nabla}\times\mathbf{A})_{il}=\varepsilon_{ijk}\partial_j A_{kl}$ with ε_{ijk} the Levi-Civita symbol.

S2. FOURIER SPACE TREATMENT OF BILAYER SYSTEM: INTRA- AND INTERLAYER INTERACTIONS

As mentioned in the main text, the HP cavity setup can be decomposed into two cavities consisting of two metasurfaces each. We will thus restrict our discussion here to bilayers. The bilayer system forming the cavity can be treated analytically by a Fourier transform of the dipole operators on both left (L) and right (R) sublattice. We define the Fourier transformation for the Pauli operators (considering a single polarization degree of freedom) as

$$\sigma_{\mathbf{q}}^{L/R}=\sum_{j\in L/R}\sigma_j \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{r}_j},\qquad \sigma_j^{L/R}=\frac{1}{\mathcal{N}}\sum_{\mathbf{q}}\sigma_{\mathbf{q}}^{L/R}\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}_j^{L/R}}. \tag{S.6}$$

The effective non-Hermitian Hamiltonian describing the free propagation of a single excitation on the two independent arrays may then be expressed in Fourier space as

$$\widetilde{\mathcal{H}}_{\mathrm{eff}}=\sum_{\mathbf{q}}\left(-\Delta+\widetilde{\Omega}(\mathbf{q})-\mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{q})}{2}\right)\left(\sigma_{\mathbf{q}}^{R,\dagger}\sigma_{\mathbf{q}}^{R}+\sigma_{\mathbf{q}}^{L,\dagger}\sigma_{\mathbf{q}}^{L}\right), \tag{S.7}$$

where $\widetilde{\Omega}(\mathbf{q})$, $\widetilde{\Gamma}(\mathbf{q})$ denote the in-plane frequency shifts and decay rates, respectively (definition see below in Eq. (S.10)). Importantly however, in addition to Eq. (S.7), one is left with a term describing the interlayer interaction between the arrays which is discussed in more detail below. The laser drive under general oblique incidence with respect to the array plane is included as

$$\mathcal{H}_l=\sum_{j\in L}\left(\eta^L\mathrm{e}^{\mathrm{i}\mathbf{k}_\parallel\cdot\mathbf{r}_j}\sigma_j^\dagger+\mathrm{h.c.}\right)+\sum_{j\in R}\left(\eta^R\mathrm{e}^{\mathrm{i}\mathbf{k}_\parallel\cdot\mathbf{r}_j}\sigma_j^\dagger+\mathrm{h.c.}\right), \tag{S.8}$$

where $\mathbf{k}_\parallel$ is the wave vector of the laser parallel to the array plane, $k_z=\sqrt{k_l^2-k_\parallel^2}$, and the Rabi drive for the two arrays differs by a phase factor due to the free propagation of the incoming field along the z-direction $\eta^R=\eta^L\mathrm{e}^{\mathrm{i}k_z\ell}$. The linearized equations of motion for the expectation values of the dipole amplitudes $\beta_{\mathbf{q}}^{L/R}=\left\langle\sigma_{\mathbf{q}}^{L/R}\right\rangle$ can be expressed in the weak excitation limit $\left\langle\sigma_j^z\sigma_{j'}\right\rangle\approx-\left\langle\sigma_{j'}\right\rangle$ in Fourier space as

$$\frac{\mathrm{d}\beta_{\mathbf{q}}^L}{\mathrm{d}t}=\mathrm{i}(\Delta-\mathcal{M}_{\mathbf{q}})\beta_{\mathbf{q}}^L-\mathrm{i}\mathcal{M}_{\mathbf{q}}^{LR}\beta_{\mathbf{q}}^R-\mathrm{i}\mathcal{N}\delta_{\mathbf{q},\mathbf{k}_\parallel}\eta^L, \tag{S.9a}$$

$$\frac{\mathrm{d}\beta_{\mathbf{q}}^R}{\mathrm{d}t}=\mathrm{i}(\Delta-\mathcal{M}_{\mathbf{q}})\beta_{\mathbf{q}}^R-\mathrm{i}\mathcal{M}_{\mathbf{q}}^{LR}\beta_{\mathbf{q}}^L-\mathrm{i}\mathcal{N}\delta_{\mathbf{q},\mathbf{k}_\parallel}\eta^R, \tag{S.9b}$$

from which we can see that an incident laser only probes the surface modes corresponding to $\mathbf{q}=\mathbf{k}_\parallel$. The term $\mathcal{M}_{\mathbf{q}}=\widetilde{\Omega}(\mathbf{q})-\mathrm{i}\widetilde{\Gamma}(\mathbf{q})/2$ describing the collective frequency shifts and decay rates of an individual array and is given by a sum over all in-plane interactions (for an arbitrary index 0 on one of the two arrays)

$$\mathcal{M}_{\mathbf{q}}=\sum_{j\in\text{in-plane}}\left(\Omega_{0j}-\mathrm{i}\frac{\Gamma_{0j}}{2}\right)\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{r}_j}=\widetilde{\Omega}(\mathbf{q})-\mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{q})}{2}=-\frac{3}{2}\Gamma_0\lambda_0\widetilde{G}(\mathbf{q};0), \tag{S.10}$$

which can be expressed in terms of the Fourier transform of the Green's tensor (single polarization component) with respect to its first two arguments $\widetilde{G}(\mathbf{q};z)=\sum_j\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{r}_j}G(\mathbf{R})$. The polarizability of a single metasurface can be expressed as $\alpha_{\mathbf{q}}^{L/R}=-\beta_{\mathbf{q}}^{L/R}d/(\mathcal{N}E_{\mathrm{in}}^{(+)})=d^2(\mathcal{M}_{\mathbf{q}}-\Delta)^{-1}$. The term $\mathcal{M}_{\mathbf{q}}^{LR}$, which describes the interaction between the two arrays, involves a summation over all coherent and dissipative out-of-plane interactions with the emitters of the opposing lattice

$$\mathcal{M}_{\mathbf{q}}^{LR}=\sum_{j\in\text{out-of-plane}}\left(\Omega_{0j}-\mathrm{i}\frac{\Gamma_{0j}}{2}\right)\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{r}_j}=-\frac{3}{2}\Gamma_0\lambda_0\widetilde{G}(\mathbf{q};\ell), \tag{S.11}$$

which simply corresponds to evaluating the 2D lattice transform of the Green's tensor at $z = \ell$. In steady state, the equations of motion for the coherences (S.9) become

$$\left[\Delta - \mathcal{M}_{\mathbf{q}} - \frac{\left(\mathcal{M}_{\mathbf{q}}^{LR}\right)^2}{\Delta - \mathcal{M}_{\mathbf{q}}}\right] \beta_{\mathbf{q}}^{L} = \mathcal{N}\delta_{\mathbf{q},\mathbf{k}_\parallel}\left(\eta^{L} + \frac{\mathcal{M}_{\mathbf{q}}^{LR}\eta_{\mathbf{q}}^{R}}{\Delta - \mathcal{M}_{\mathbf{q}}}\right), \tag{S.12a}$$

$$\left[\Delta - \mathcal{M}_{\mathbf{q}} - \frac{\left(\mathcal{M}_{\mathbf{q}}^{LR}\right)^2}{\Delta - \mathcal{M}_{\mathbf{q}}}\right] \beta_{\mathbf{q}}^{R} = \mathcal{N}\delta_{\mathbf{q},\mathbf{k}_\parallel}\left(\eta^{R} + \frac{\mathcal{M}_{\mathbf{q}}^{LR}\eta_{\mathbf{q}}^{L}}{\Delta - \mathcal{M}_{\mathbf{q}}}\right). \tag{S.12b}$$

By making use of Poisson's summation formula, the Fourier transform of the Green's tensor can be turned into a sum over all vectors $\mathbf{g}$ of the reciprocal lattice Λ^* and can be approximated in the far field $z \gg \lambda$ as (denoting the unit cell area of the real space lattice by $\mathcal{A}$)

$$\widetilde{G}(\mathbf{q}; z) = \frac{\mathrm{i}}{2\mathcal{A}} \sum_{\mathbf{g}\in\Lambda^*} \frac{\mathrm{e}^{\mathrm{i}\sqrt{k_0^2 - (\mathbf{q}+\mathbf{g})^2}|z|}}{\sqrt{k_0^2 - (\mathbf{q}+\mathbf{g})^2}} \approx \frac{\mathrm{i}}{2\mathcal{A}} \frac{\mathrm{e}^{\mathrm{i}q_z|z|}}{q_z}, \tag{S.13}$$

where we have only kept the $\mathbf{g} = 0$ contribution and took all other contributions to be evanescent as it is the case for subwavelength lattices. The interaction between the arrays is then simply described by the plane wave term

$$\mathcal{M}_{\mathbf{q}}^{LR} = -\mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{q})}{2}\mathrm{e}^{\mathrm{i}q_z\ell}, \tag{S.14}$$

where the amplitude of the interaction is governed by the effective in-plane decay rate $\widetilde{\Gamma}(\mathbf{q}) = 3\Gamma_0/(4\pi)(\lambda_0^2/\mathcal{A})(k_0/q_z)$. This leads to the following expressions for the dipole amplitudes on left and right layer:

$$\beta_{\mathbf{q}}^{L} = \mathcal{N}\delta_{\mathbf{q},\mathbf{k}_\parallel}\eta^{L}\left[\frac{\Delta - \mathcal{M}_{\mathbf{q}} - \mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{q})}{2}\mathrm{e}^{2\mathrm{i}q_z\ell}}{(\Delta - \mathcal{M}_{\mathbf{q}})^2 + \frac{\widetilde{\Gamma}(\mathbf{q})^2}{4}\mathrm{e}^{2\mathrm{i}q_z\ell}}\right], \qquad \beta_{\mathbf{q}}^{R} = \mathcal{N}\delta_{\mathbf{q},\mathbf{k}_\parallel}\eta^{L}\left[\frac{\left(\Delta - \mathcal{M}_{\mathbf{q}} - \mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{q})}{2}\right)\mathrm{e}^{\mathrm{i}q_z\ell}}{(\Delta - \mathcal{M}_{\mathbf{q}})^2 + \frac{\widetilde{\Gamma}(\mathbf{q})^2}{4}\mathrm{e}^{2\mathrm{i}q_z\ell}}\right]. \tag{S.15}$$

S3. TRANSMISSION, INTRACAVITY FIELD, AND RESONANCE CONDITION

The total electric field can be computed as the sum of the incident field and the dipole-scattered field $E^{(+)}(\mathbf{R}) = E_{\mathrm{in}}^{(+)}\mathrm{e}^{\mathrm{i}kz} + E_{\mathrm{dip}}^{(+)}(\mathbf{R})$. The dipole field is given by the sum over all individually emitted fields and can be expressed as a sum over all wave vectors as (only considering the field along the z-direction)

$$E_{\mathrm{dip}}^{(+)}(z) = -\frac{3\pi\Gamma_0}{k_0 d}\sum_j G(z\hat{e}_z - \mathbf{r}_j)\beta_j = -\frac{3\pi\Gamma_0}{k_0 d\mathcal{N}}\sum_{\mathbf{q}}\left[\widetilde{G}^{L}(\mathbf{q}; z)\beta_{\mathbf{q}}^{L} + \widetilde{G}^{R}(\mathbf{q}; z)\beta_{\mathbf{q}}^{R}\right], \tag{S.16}$$

where $\widetilde{G}^{L}$, $\widetilde{G}^{R}$ denote the Fourier transforms of the Green's tensor with respect to the left and right sublattice. This leads to the following expressions for the transmitted and intracavity field (neglecting all contributions from evanescent terms)

$$E^{(+)}(z > \ell) = \left[E_{\mathrm{in}}^{(+)}\mathrm{e}^{\mathrm{i}k_z z} - \mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{k}_\parallel)}{2d}\left(\beta_{\mathbf{k}_\parallel}^{L} + \mathrm{e}^{-\mathrm{i}q_z\ell}\beta_{\mathbf{k}_\parallel}^{R}\right)\mathrm{e}^{\mathrm{i}q_z z}\right] \approx t_c E_{\mathrm{in}}^{(+)}\mathrm{e}^{\mathrm{i}k_z z}, \tag{S.17a}$$

$$E^{(+)}(0 < z < \ell) = \left[E_{\mathrm{in}}^{(+)}\mathrm{e}^{\mathrm{i}k_z z} - \mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{k}_\parallel)}{2d}\left(\beta_{\mathbf{k}_\parallel}^{L}\mathrm{e}^{\mathrm{i}q_z z} + \mathrm{e}^{\mathrm{i}q_z(\ell - z)}\beta_{\mathbf{k}_\parallel}^{R}\right)\right], \tag{S.17b}$$

where the cavity transmission coefficient reads

$$t_c = \frac{\left[\Delta - \mathcal{M}_{\mathbf{k}_\parallel} - \mathrm{i}\frac{\widetilde{\Gamma}(\mathbf{k}_\parallel)}{2}\right]^2}{(\Delta - \mathcal{M}_{\mathbf{k}_\parallel})^2 + \frac{\widetilde{\Gamma}(\mathbf{k}_\parallel)^2}{4}\mathrm{e}^{2\mathrm{i}k_z\ell}}, \tag{S.18}$$

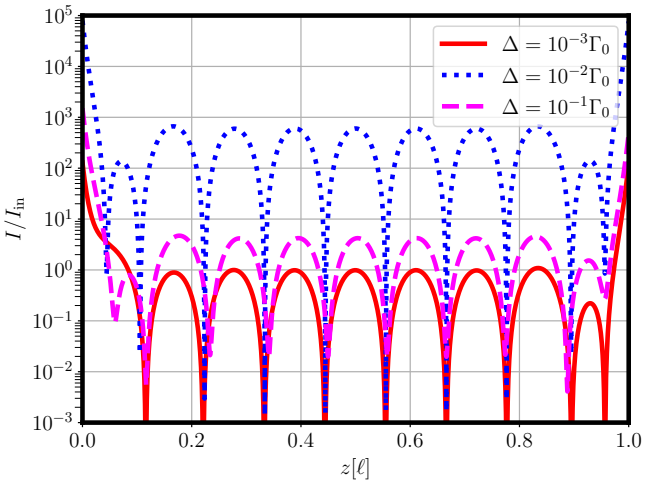


FIG. S1. Plots of intracavity field intensity for a bilayer system where the resonance condition for Δ is always satisfied. The z-coordinate is normalized to the cavity length so that the different cases can be compared. The plots are generated by considering the infinite square lattice with $a = 0.8\lambda$ in Fourier space. Eq. (S.13) was evaluated until the reciprocal lattice sum converged, so that evanescent contributions to the interlayer interaction and cavity field are also included.

and we have assumed $q_z \approx k_z$. The condition that the cavity transmission should equal unity at the cavity resonance $|t_c|^2 = 1$ yields a quadratic equation for the collective resonance $\Delta - \widetilde{\Omega}(\mathbf{k}_\parallel)$ with only a single solution given by

$$\Delta - \widetilde{\Omega}(\mathbf{k}_\parallel) = -\frac{\widetilde{\Gamma}(\mathbf{k}_\parallel)}{2}\frac{\sin(2k_z\ell)}{1+\cos(2k_z\ell)} = -\frac{\widetilde{\Gamma}(\mathbf{k}_\parallel)}{2}\tan(k_z\ell), \tag{S.19}$$

implying that the resonance frequency of the cavity is given by $\omega_c = \omega_0 + \widetilde{\Omega}(\mathbf{k}_\parallel) - \widetilde{\Gamma}(\mathbf{k}_\parallel)\tan(k_z\ell)/2$. The results presented in the manuscript correspond to $\mathbf{k}_\parallel = 0$, i.e., perpendicular illumination of the cavity. In Fig. S1 we plot the intensity of the intracavity field at different resonances including the contribution of the evanescent terms, leading to a field distribution which is maximal at the mirror surfaces and exhibiting a sine-like behavior in between.

One can account for additional loss channels (dephasing, non-radiative decay,...) by adding an additional decay term to the intralayer interaction term $\mathcal{M}'_{\mathbf{q}} = \mathcal{M}_{\mathbf{q}} - i\Gamma_{\rm nr}/2$ (also see section S5 below for further discussions of derogating effects such as helicity errors). The results can furthermore be easily generalized to several polarization components (e.g., in the Cartesian polarization basis) by considering the vector of dipole amplitudes $\boldsymbol{\beta}_{\mathbf{q}} = (\beta_{\mathbf{q},x}, \beta_{\mathbf{q},y}, \beta_{\mathbf{q},z})^\top$ with $\beta_{\mathbf{q},\nu} = \langle\sigma_{\mathbf{q},\nu}\rangle$ and by taking into account the full Green's tensor Eq. (S.4).

S4. COMPARISON OF FOURIER SPACE SOLUTION WITH FINITE-SIZE REAL SPACE SIMULATION

The analytical results derived above are valid for infinite lattices. In Figs. S2(a)-(c), we compare the result of Eq. (S.17a) with the transmission profile obtained by a real space simulation of the coupled-dipole equations for a finite number of emitters on the layers, for different cavity lengths. For a finite lattice, also other modes than $\mathbf{q} = 0$ are excited by the incoming laser. The scattering contributions from these modes lead to a transmission larger than unity and vanish for larger array sizes, converging towards the analytical solution. While the results presented below correspond to plane wave illumination, for a Gaussian beam with beam width smaller than the array size, the infinite lattice case is approached more quickly as boundary effects are avoided.

S5. TRANSFER MATRIX FOR HP MIRROR

In this section, we show that the proposed setup indeed preserves the helicity by making use of the transfer matrix method. We start by constructing the transfer matrix for a single HP mirror consisting of two metasurface layers with dipoles oriented along x and y. From this, the cavity properties can then be determined in a straightforward manner.

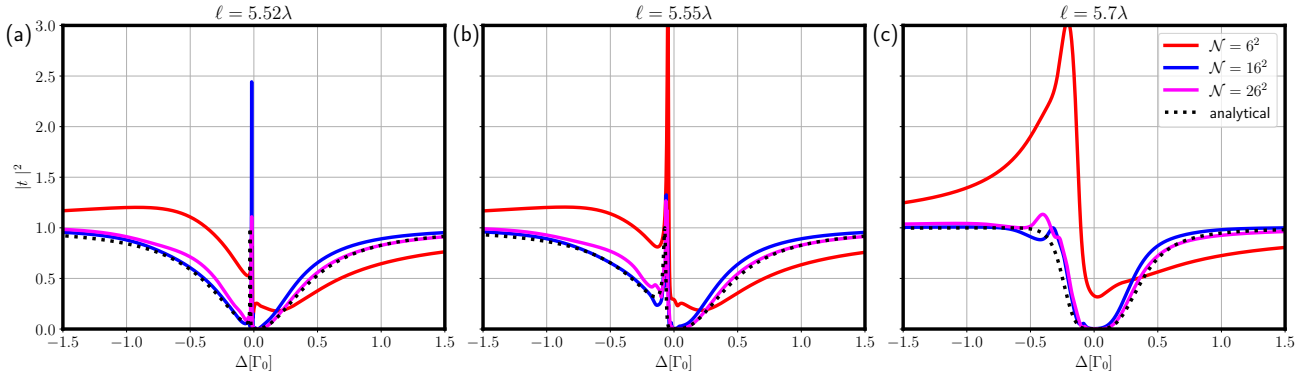


FIG. S2. Comparison between analytical expression of transfer matrix/Fourier space transmission lineshapes (Eq. (S.18), dotted black curve) with real space simulation (solid curves) for square lattices with $a = 0.8\lambda$ for different cavity lengths (a) $\ell = 5.52\lambda$, (b) $\ell = 5.55\lambda$, and (c) $\ell = 5.7\lambda$. The number of emitters per metasurface is given by $\mathcal{N}$. For the numerical simulations, a small array curvature with radius $R \approx 3.5 \cdot 10^8 \lambda$ was implemented since the finite size of the array would otherwise lead to instabilities due to transverse waveguiding modes [S2]. Simulations for other curvature radii show similar results.

Consider the setup of mirrors with corresponding transfer matrices of

$$\mathbf{T}_m = \begin{pmatrix} 1 + \mathrm{i}\zeta_m & \mathrm{i}\zeta_m \\ -\mathrm{i}\zeta_m & 1 - \mathrm{i}\zeta_m \end{pmatrix}, \tag{S.20}$$

where ζ_m is the polarizability of the metasurface. The first mirror only affects the x-polarization and the second only the y-polarization component. Between them the light propagates freely according to

$$\mathbf{T}_f = \begin{pmatrix} \mathrm{e}^{\mathrm{i}k\ell_m} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}k\ell_m} \end{pmatrix}, \tag{S.21}$$

for both polarization components. In the basis $(E_x^{\leftarrow}, E_x^{\rightarrow}, E_y^{\leftarrow}, E_y^{\rightarrow})^\top$, where the subscript denotes the polarization and the superscript the propagation direction of the electric field, the total transfer matrix can be expressed in block notation as

$$\mathbf{T}_{\mathrm{HP}} = \begin{pmatrix} \mathbf{T}_m & 0 \\ 0 & \mathbb{1} \end{pmatrix} \begin{pmatrix} \mathbf{T}_f & 0 \\ 0 & \mathbf{T}_f \end{pmatrix} \begin{pmatrix} \mathbb{1} & 0 \\ 0 & \mathbf{T}_m \end{pmatrix}. \tag{S.22}$$

To see now that this matrix represents a HP mirror for $\ell_m = \lambda/4$, we must perform a basis transformation which rotates $(E_x^{\leftarrow}, E_x^{\rightarrow}, E_y^{\leftarrow}, E_y^{\rightarrow})^\top$ into $(E_+^{\leftarrow}, E_+^{\rightarrow}, E_-^{\leftarrow}, E_-^{\rightarrow})^\top$. This is given by

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\mathrm{i} & 0 & \mathrm{i} & 0 \\ 0 & \mathrm{i} & 0 & -\mathrm{i} \end{pmatrix}. \tag{S.23}$$

Applying this basis transformation to the transfer matrix yields

$$\widetilde{\mathbf{T}}_{\mathrm{HP}} = \mathbf{U}^\dagger \mathbf{T}_{\mathrm{HP}} \mathbf{U} = \begin{pmatrix} 1 + \mathrm{i}\zeta_m & \mathrm{i}\zeta_m & 0 & 0 \\ -\mathrm{i}\zeta_m & 1 - \mathrm{i}\zeta_m & 0 & 0 \\ 0 & 0 & 1 + \mathrm{i}\zeta_m & \mathrm{i}\zeta_m \\ 0 & 0 & -\mathrm{i}\zeta_m & 1 - \mathrm{i}\zeta_m \end{pmatrix} \begin{pmatrix} \mathrm{i} & 0 & 0 & 0 \\ 0 & -\mathrm{i} & 0 & 0 \\ 0 & 0 & \mathrm{i} & 0 \\ 0 & 0 & 0 & -\mathrm{i} \end{pmatrix}, \tag{S.24}$$

which is indeed block-diagonal in the circular basis with identical subblocks for the two polarization components. The diagonal matrix on the right-hand side accounts for the free propagation. In total, it can thus be formulated that the resulting optical element describes a "*helicity-preserving mirror plus free propagation of* $\lambda/4$".

The transfer matrix formalism is also very well-suited to study the effect of different types of losses/errors. One can generally distinguish between (i) 'global' errors that reduce the reflectivity of the individual metasurfaces and therefore

also the finesse of the cavity but preserve the helicity and (ii) helicity errors that introduce a mixing of helicities. One can easily account for global errors (e.g., due to dephasing, non-radiative losses,...) by introducing a renormalized mirror polarizability $\zeta'_m = \widetilde{\Gamma}(0)/[2(\widetilde{\Omega}(0) - \Delta - \mathrm{i}\Gamma_{\mathrm{nr}}/2)]$ containing an additional non-radiative loss channel Γ_{nr}.

Helicity errors can e.g., be caused by unbalanced mirror reflectivities. Considering different polarizabilites for x and y (ζ_m^x, ζ_m^y) with average $\bar{\zeta}_m = (\zeta_m^x + \zeta_m^y)/2$ and relative difference $\delta\zeta = \zeta_m^x - \zeta_m^y$, one obtains

$$\widetilde{\mathbf{T}}'_{\mathrm{HP}} = \begin{pmatrix} 1+\mathrm{i}\bar{\zeta}_m & \mathrm{i}\bar{\zeta}_m & -\delta\zeta_m/2 & \delta\zeta_m/2 \\ -\mathrm{i}\bar{\zeta}_m & 1-\mathrm{i}\bar{\zeta}_m & \delta\zeta_m/2 & -\delta\zeta_m/2 \\ -\delta\zeta_m/2 & \delta\zeta_m/2 & 1+\mathrm{i}\bar{\zeta}_m & \mathrm{i}\bar{\zeta}_m \\ \delta\zeta_m/2 & -\delta\zeta_m/2 & -\mathrm{i}\bar{\zeta}_m & 1-\mathrm{i}\bar{\zeta}_m \end{pmatrix} \begin{pmatrix} \mathrm{i} & 0 & 0 & 0 \\ 0 & -\mathrm{i} & 0 & 0 \\ 0 & 0 & \mathrm{i} & 0 \\ 0 & 0 & 0 & -\mathrm{i} \end{pmatrix}, \tag{S.25}$$

thereby leading to a helicity error given by the difference in polarizability. Another cause of helicity errors can be fluctuations δ_m of the inter-mirror path length separation $k_l\ell_m + \delta_m$ away from $\pi/2$, leading to

$$\widetilde{\mathbf{T}}'_{\mathrm{HP}} = \begin{pmatrix} 1+\mathrm{i}\zeta_m & \mathrm{i}e^{\mathrm{i}\delta_m}\zeta_m\cos(\delta_m) & 0 & e^{\mathrm{i}\delta_m}\zeta_m\sin(\delta_m) \\ -\mathrm{i}e^{-\mathrm{i}\delta_m}\zeta_m\cos(\delta_m) & 1-\mathrm{i}\zeta_m & e^{-\mathrm{i}\delta_m}\zeta_m\sin(\delta_m) & 0 \\ 0 & e^{\mathrm{i}\delta_m}\zeta_m\sin(\delta_m) & 1+\mathrm{i}\zeta_m & \mathrm{i}e^{\mathrm{i}\delta_m}\zeta_m\cos(\delta_m) \\ e^{-\mathrm{i}\delta_m}\zeta_m\sin(\delta_m) & 0 & -\mathrm{i}e^{-\mathrm{i}\delta_m}\zeta_m\cos(\delta_m) & 1-\mathrm{i}\zeta_m \end{pmatrix} \begin{pmatrix} \mathrm{i}e^{\mathrm{i}\delta_m} & 0 & 0 & 0 \\ 0 & -\mathrm{i}e^{-\mathrm{i}\delta_m} & 0 & 0 \\ 0 & 0 & \mathrm{i}e^{\mathrm{i}\delta_m} & 0 \\ 0 & 0 & 0 & -\mathrm{i}e^{-\mathrm{i}\delta_m} \end{pmatrix}, \tag{S.26}$$

where the matrix on the right-hand side again accounts for the free propagation. The left-hand matrix can be Taylor expanded around $\delta_m = 0$ yielding in first order

$$\widetilde{\mathbf{T}}'_{\mathrm{HP}} = \widetilde{\mathbf{T}}_{\mathrm{HP}} + \begin{pmatrix} 0 & -1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \delta_m\zeta_m, \tag{S.27}$$

which is a linear helicity-error in reflection as would be expected from this treatment. The difference between this error and the error due to $\delta\zeta_m$ is in the fact that the phase error only leads to errors in reflection whereas the $\delta\zeta_m$ term leads to errors in transmission as well.

Global polarizability errors essentially mean, that the HP mirror has decreased reflectivity, leading to an increase in the cavity linewidth as well as a an attenuation in the phase sensitivity as illustrated in Fig. (2a) of the main text. First-order helicity errors can be filtered with a polarizer behind the cavity. Second-order or higher terms in the helicity error lead to a renormalized interaction for both enantiomers since the cavity mode is now a superposition of both LHS and RHS light. Hence, the shifts in optical path lengths caused by both enantiomers becomes more similar with increasing helicity error.

S6. CAVITY PHASE SHIFT DUE TO PRESENCE OF CHIRAL SCATTERER

For further discussions, we focus on one of the equivalent subblocks in Eq. (S.24). A chiral molecule can be interpreted as a scatterer that induces different path lengths for the two polarization components $\delta l_s^{\pm}$.

The transfer matrix for a single scatterer expresses as

$$\mathbf{T}_s = \begin{pmatrix} 1+\mathrm{i}\zeta_s & \mathrm{i}\zeta_s \\ -\mathrm{i}\zeta_s & 1-\mathrm{i}\zeta_s \end{pmatrix}, \tag{S.28}$$

where the polarizability of a single scatterer with linewidth γ_s and detuning from the laser frequency $\Delta_s = \omega_l - \omega_s$ is given by $\zeta_s = -(\gamma_s/2)/(\mathrm{i}\gamma_s/2 + \Delta_s)$. In the limit $|\Delta_s| \gg \gamma_s$, this can be understood as an effective path length added to the free propagation inside the cavity due to the approximation

$$\mathbf{T}_s \approx \begin{pmatrix} \mathrm{e}^{\mathrm{i}k_l\delta\ell_s} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}k_l\delta\ell_s} \end{pmatrix}, \tag{S.29}$$

with the effective length change of the cavity caused by the off-resonant scatterer

$$\delta\ell_s = -\frac{\arctan(\gamma_s/2\Delta_s)}{k_l}. \tag{S.30}$$

The cavity transmission coefficient is obtained from the last entry of the total transfer matrix $t_c = 1/T_{22}$ and reads

$$t_c(\ell+\delta\ell_s) = \frac{\left(\Delta-\widetilde{\Omega}(0)\right)^2}{\left(\Delta-\widetilde{\Omega}(0)+\mathrm{i}\frac{\widetilde{\Gamma}(0)}{2}\right)^2+\frac{\widetilde{\Gamma}(0)^2}{4}\mathrm{e}^{2\mathrm{i}k_l(\ell+\delta\ell_s)}}, \tag{S.31}$$

and the relative phase shift due to the presence of the scatterer in the cavity can be expressed as (on the cavity resonance $\Delta-\widetilde{\Omega}(0) = -\frac{\widetilde{\Gamma}(0)}{2}\tan(k_l\ell)$)

$$\begin{aligned}\varphi = \arg\left[\frac{t_c(\ell+\delta\ell_s)}{t_c(\ell)}\right]_{\mathrm{res}} &= -\operatorname{arccot}\Big(\cot(2k_l\delta\ell_s)-\csc(2k_l\delta\ell_s)\sec(2k_l\ell)^2\Big) \\ &\approx \arctan(\cot(2k_l\delta\ell_s)) - \frac{k_l^2\bar{\ell}^2}{2}\cot\big(k_l\bar{\ell}\big),\end{aligned} \tag{S.32}$$

where the last approximation holds for $\ell \approx n\lambda/2$ and $\bar{\ell} = \ell - \lfloor 2\ell \rceil/2$ is the relative distance from this half-integer factor of the wavelength.

Enantiomers can now be differentiated by illuminating the cavity with circularly polarized light. Within the polarization subblocks, the two enantiomers will lead to different effective path lengths $\delta\ell_s^{\pm}$ and hence to a different signal in the homodyne measurement.

S7. COUPLED-MODES THEORY

$c_\lambda(\omega,t)$	bath operator of mode λ	
b_λ	system oscillator mode λ	
ω_λ	frequency of oscillator mode λ	
$b_{\mathrm{in},\lambda}$	input operator of port λ	$b_{\mathrm{in},\lambda}(t) = \lim_{t_1\to\infty}\frac{1}{2\pi}\int \mathrm{d}\omega\, \mathrm{e}^{-\mathrm{i}\omega(t-t_1)}c_\lambda(\omega,t_1)$
$b_{\mathrm{out},\lambda}$	output operator of port λ	$b_{\mathrm{out},\lambda}(t) = \lim_{t_0\to-\infty}\frac{1}{2\pi}\int \mathrm{d}\omega\, \mathrm{e}^{-\mathrm{i}\omega(t-t_0)}c_\lambda(\omega,t_0)$
$W_{\lambda,\lambda'}$	coupling between oscillator mode λ and input operator λ'	
t_c^{TM}	transmission function from transfer matrix theory	
t_c^{CM}	transmission function from coupled-modes theory	
κ	cavity mode linewidth	
ω_c	cavity mode frequency	

TABLE SI. Definitions for section S7.

A great simplification of the scattering problem considered here can stem from decomposing the hybrid cavity system into only a few quantized, lossy modes corresponding to a cavity mode as well as the surface modes of the mirrors.

To this end, we assume a phenomenological coupled-modes model (as introduced in Ref. [S3]) where the cavity mode and surface modes of the metasurface are modeled by harmonic oscillators coupled to common vacuum modes. We follow the notation in Ref. [S4] which assumes equations of motion $\dot{b}_\lambda = -\mathrm{i}\omega_\lambda b_\lambda - \mathrm{i}\sum_{\lambda'} W_{\lambda,\lambda'} b_{\mathrm{in},\lambda'}$ for the oscillator modes b_λ and input noise operators $b_{\mathrm{in},\lambda}$ (see table of definitions SI above). Then, the input-output relations may be formulated in compact matrix-vector form in frequency domain as

$$\mathbf{b}_{\mathrm{out}}(\omega) = \mathbf{S}(\omega)\mathbf{b}_{\mathrm{in}}(\omega), \tag{S.33}$$

where the scattering matrix assumes the following form in the Markovian limit

$$\mathbf{S}(\omega) = \mathbb{1} - 2\pi\mathrm{i}\mathbf{W}^\dagger\mathbf{D}^{-1}(\omega)\mathbf{W}, \tag{S.34}$$

with

$$[\mathbf{D}(\omega)]_{\lambda\lambda'} = (\omega-\omega_\lambda)\delta_{\lambda\lambda'} + \mathrm{i}\pi[\mathbf{W}\mathbf{W}^\dagger]_{\lambda\lambda'}. \tag{S.35}$$

The transmission can then be found as matrix elements of the scattering matrix $t_c^{\mathrm{CM}} = S_{12}$.

Now, we make a physical assumption for the form of this coupling matrix, see for instance [S3]

$$\mathbf{W} = \begin{pmatrix} \sqrt{\frac{\kappa}{4\pi}} & \sqrt{\frac{\kappa}{4\pi}} \\ \sqrt{\frac{\widetilde{\Gamma}(0)}{2\pi}} & 0 \\ 0 & \sqrt{\frac{\widetilde{\Gamma}(0)}{2\pi}} \end{pmatrix}. \tag{S.36}$$

By this choice we identify b_1 as the cavity mode and b_2 and b_3 as the surface modes of the metasurface. Thus, we also choose $\omega_2 = \omega_3 = \omega_0 + \widetilde{\Omega}(0)$. This satisfies the condition that the cavity transmission ought to be zero at the metasurface resonance $t_c^{\mathrm{CM}}(\omega_0 + \widetilde{\Omega}(0)) = 0$. The other condition that needs to be satisfied is that the cavity transmission becomes unity at the cavity resonance $|t_c^{\mathrm{CM}}(\omega_c)|^2 = 1$. It turns out that any choice $\omega_1 = \omega_c + \delta$ and $\kappa = \left(\frac{\widetilde{\Gamma}(0)}{\omega_c} + 4\frac{\omega_c}{\widetilde{\Gamma}(0)}\right)\delta$ will satisfy this condition. The natural choice $\delta = 0$ will however not lead to a physical solution. Checking against the transfer matrix theory, we find that the choice $\delta = \omega_c$ gives the correct linewidth.

The comparison between the transfer matrix results and the result for the coupled-modes theory is shown in Fig. S3 for different cavity lengths ℓ. We can see that for $\ell \approx n\lambda/2$, the coupled-modes theory perfectly fits the transmission around the resonance. For larger deviations from this condition, the transmission functions show differences. This can be interpreted as the contributions of the transmission that occur via the free space modes instead of via the cavity-confined mode for a bad cavity (the transmission of a metasurface becomes unity for large detuning).

S8. CHIRAL RESPONSE OF FAR-DETUNED MOLECULAR SCATTERER IN HP METASURFACE CAVITY

Consider now that we place a chiral molecular scatterer with electric moment $\mathbf{d}_s$ and magnetic moment $\boldsymbol{\mu}_s$ inside the cavity. We now assume that both these moments correspond to the same transition, implying cross-terms in the polarizability for the electric and magnetic components, i.e., an electric field will also excite the magnetic dipole moment.

In order to continue, we first need to consider the magnetic far field due to an array of electric dipoles due to the cross term in the Green's function given in Eq. (S.5) (the Green's function for the electric field generated by a magnetic dipole is the same but multiplied by minus one). The procedure to calculate the far field in the subwavelength regime is essentially the same as for the purely electric Green's function. To see this, consider that both Green's functions can be written as a differential operator acting on the scalar Green's function. As such,

$$\mathcal{M}_{\mathbf{q}}^{m} \approx -\mathrm{i}\frac{k_0^3}{3\pi\sqrt{\epsilon_0\mu_0}}\frac{\boldsymbol{\mu}_s^* \cdot (\hat{e}_z \times \mathbf{d})}{2}\mathrm{sgn}(z)\mathrm{e}^{\mathrm{i}q_z|z|} \tag{S.37}$$

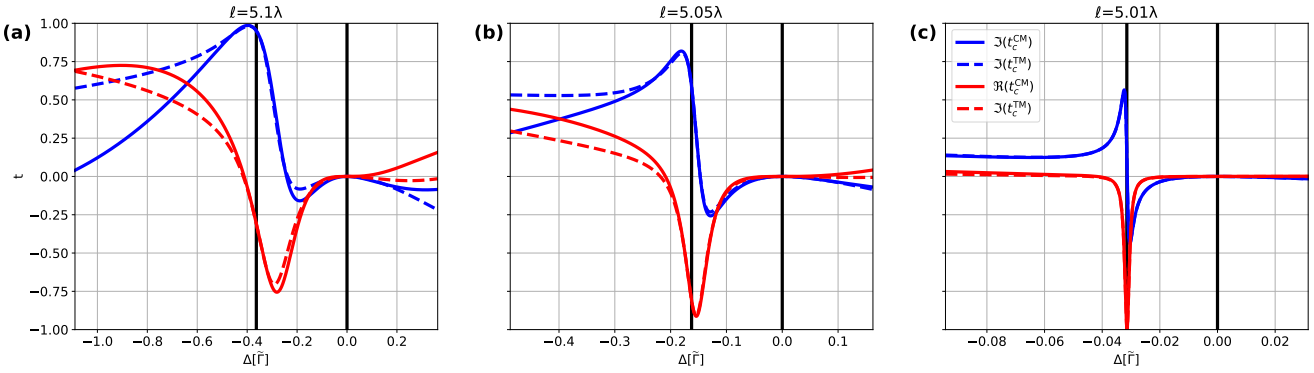


FIG. S3. Comparison between the cavity transmission function t_c^{TM} obtained from transfer matrix theory/coupled-dipole formalism (dashed curves) and the cavity transmission function t_c^{CM} of the coupled-modes theory (solid curves). Here we chose $\widetilde{\Omega}(0) = 0$ as the origin. The real part of the transmission functions is shown in red and the imaginary part in blue. From left to right, we show the comparison for decreasing cavity lengths (a) $\ell = 5.1\lambda$, (b) $\ell = 5.05\lambda$, and (c) $\ell = 5.01\lambda$, which corresponds to increasing reflectivity of the metasurface at the resonance. The transmission is shown within the frequency window $[\omega_c - \kappa, \omega_c + \kappa]$. The vertical line at $\Delta = 0$ indicates the metasurface resonance and the other vertical line indicates the cavity resonance.

ω_s	transition frequency of molecule	
$\mathbf{d}_s$	electric dipole moment of molecule	
$\boldsymbol{\mu}_s$	magnetic dipole moment of molecule	
ℓ_s	position of molecule along the z-axis	
β_s	coherence of molecular transition	$\beta_s = \langle \lvert e_s\rangle \langle g_s\rvert\rangle$
$\mathbf{d}_i$	dipole moment of ith metasurface in stack	
β_i	coherence of ith metasurface in stack	
ℓ_i	position of ith metasurface in stack	
Δ_s	detuning of molecule	$\Delta_s = \omega_l - \omega_s$
γ_s	total molecule linewidth	$\gamma_s = \gamma_s^m + \gamma_s^e$
γ_s^m	molecule magnetic linewidth	$\gamma_s^m = \frac{k_0^3}{3\pi\mu_0}\mu_s^2$
γ_s^e	molecule electric linewidth	$\gamma_s^e = \frac{k_0^3}{3\pi\epsilon_0}d_s^2$
$\hat{e}_\pm$	circular basis vectors	$\hat{e}_\pm = (\hat{e}_x \pm \mathrm{i}\hat{e}_y)/\sqrt{2}$
ξ	polarization of incoming light (RCP/LCP)	$\xi = \pm$
$\langle\cdot\rangle_{\mathrm{rot}}$	rotational average	see Ref. [S5]
ℓ_m	HP mirror metasurface spacing	
β_L^x	coherence of left metasurface with dipole moment in x-direction	
β_L^y	coherence of left metasurface with dipole moment in y-direction	
β_R^x	coherence of right metasurface with dipole moment in x-direction	
β_R^y	coherence of right metasurface with dipole moment in y-direction	
β_L^ξ	circular superposition of left metasurfaces	$\beta_L^\xi = \beta_L^x + \xi\beta_L^y$
β_R^ξ	circular superposition of right metasurfaces	$\beta_R^\xi = \beta_R^x + \xi\beta_R^y$
$\hat{v}$	a hat indicates that the vector is normalized (unit vector)	$\hat{v} = \mathbf{v}/\lVert\mathbf{v}\rVert$
$\mathcal{M}_{\mathbf{q}}^m$	interaction between $\mathbf{q}$-mode of a metasurface and magnetic moment of molecule in far field	
$\mathbf{E}_{\mathrm{in}}$	incoming electric field	
$\mathbf{B}_{\mathrm{in}}$	incoming magnetic field	
d	magnitude of dipole moments of HP mirror metasurfaces	$d = \lVert\mathbf{d}_i\rVert$
$\widetilde{\Gamma}$	linewidth of metasurfaces	$\widetilde{\Gamma} = \widetilde{\Gamma}(0) = \frac{k_0^3}{3\pi\epsilon_0}d^2$

TABLE SII. Definitions for section S8.

gives the interaction due to light propagating from the metasurface to the molecule located at position z. Here, $\mathbf{d}$ indicates the dipole moment of the metasurface (also see table of definitions SII). Now, the interaction between N metasurface layers with dipole moments $\mathbf{d}_i$ and a single molecule can be written as (denoting for simplicity $\widetilde{\Gamma}(0) \equiv \widetilde{\Gamma}$ and omitting the coupling of the molecule to other surface modes of the array with $\mathbf{q} \neq 0$)

$$\frac{\mathrm{d}\beta_i}{\mathrm{d}t} = (\mathrm{i}\Delta - \widetilde{\Gamma}/2)\beta_i - \frac{k_0^3}{3\pi\epsilon_0}\sum_{j\neq i}\mathbf{d}_i^*\cdot\mathbf{d}_j\mathrm{e}^{\mathrm{i}k_z|\ell_i-\ell_j|}\beta_j - \mathrm{i}\mathbf{d}_i^*\cdot\mathbf{E}_{\mathrm{in}}\mathrm{e}^{\mathrm{i}k_l\ell_i}$$
$$- \frac{k_0^3}{3\pi\sqrt{\epsilon_0\mu_0}}\mathrm{sgn}(\ell_i-\ell_s)\mathbf{d}_i^*\cdot(\hat{e}_z\times\boldsymbol{\mu}_s)\mathrm{e}^{\mathrm{i}k_z|\ell_i-\ell_s|}\beta_s - \frac{k_0^3}{3\pi\epsilon_0}\mathbf{d}_i^*\cdot\mathbf{d}_s\mathrm{e}^{\mathrm{i}k_z|\ell_i-\ell_s|}\beta_s, \tag{S.38a}$$

$$\frac{\mathrm{d}\beta_s}{\mathrm{d}t} = (\mathrm{i}\Delta_s - \gamma_s/2)\beta_s - \frac{k_0^3}{3\pi\epsilon_0}\sum_i\mathbf{d}_s^*\cdot\mathbf{d}_i\mathrm{e}^{\mathrm{i}k_z|\ell_i-\ell_j|}\beta_i - \mathrm{i}\mathbf{d}_s^*\cdot\mathbf{E}_{\mathrm{in}}\mathrm{e}^{\mathrm{i}k_l\ell_s} - \mathrm{i}\boldsymbol{\mu}_i^*\cdot\mathbf{B}_{\mathrm{in}}\mathrm{e}^{\mathrm{i}k_l\ell_s}$$
$$- \frac{k_0^3}{3\pi\sqrt{\epsilon_0\mu_0}}\sum_i\mathrm{sgn}(\ell_i-\ell_s)\boldsymbol{\mu}_s^*\cdot(\hat{e}_z\times\mathbf{d}_i)\mathrm{e}^{\mathrm{i}k_z|\ell_i-\ell_s|}\beta_i, \tag{S.38b}$$

where β_i are the coherences of the metasurfaces and β_s is the coherence of the molecular transition. ℓ_i and ℓ_s denote the positions of the metasurfaces and the molecular scatterer along the z-axis. This neglects the coupling between the molecule and all other surface modes with different momenta. This can be taken into account by renormalizing the decay rate $\gamma_s = \frac{k_0^3}{3\pi\epsilon_0}d_s^2 + \frac{k_0^3}{3\pi\mu_0}\mu_s^2$ which will however only affect the total transmission which is of no particular interest to us.

For the proposed setup of four metasurface layers with dipole moments $(d\hat{e}_x, d\hat{e}_y, d\hat{e}_x, d\hat{e}_y)$ at positions $(0, \ell_m, \ell, \ell+\ell_m)$ with $\ell_m = \lambda/4$, it is thus clear that the superpositions $\beta_L^x + \xi\beta_L^y$ and $\beta_R^x + \xi\beta_R^y$ couple to a specific polarization component. For left or right-handed illumination, we thus consider the equations of motion only in this subspace

$$\frac{\mathrm{d}\beta_L^\xi}{\mathrm{d}t} = (\mathrm{i}\Delta - \widetilde{\Gamma}/2)\beta_L^\xi - \frac{\widetilde{\Gamma}}{2}\mathrm{e}^{\mathrm{i}k_z\ell}\beta_R^\xi - \frac{\sqrt{\widetilde{\Gamma}}}{2}\mathrm{e}^{\mathrm{i}k_z\ell_s}\hat{e}_{-\xi}\cdot\left[\sqrt{\gamma_s^e}\hat{d}_s + \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s\right]\beta_s - \mathrm{i}dE_{\mathrm{in}}, \tag{S.39a}$$

$$\frac{\mathrm{d}\beta_R^\xi}{\mathrm{d}t} = (\mathrm{i}\Delta - \widetilde{\Gamma}/2)\beta_R^\xi - \frac{\widetilde{\Gamma}}{2}\mathrm{e}^{\mathrm{i}k_z\ell}\beta_L^\xi - \frac{\sqrt{\widetilde{\Gamma}}}{2}\mathrm{e}^{\mathrm{i}k_z(\ell-\ell_s)}\hat{e}_{\xi}\cdot\left[\sqrt{\gamma_s^e}\hat{d}_s + \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s\right]\beta_s - \mathrm{i}dE_{\mathrm{in}}\mathrm{e}^{\mathrm{i}k_z\ell}, \tag{S.39b}$$

$$\begin{aligned}\frac{\mathrm{d}\beta_s}{\mathrm{d}t} &= [\mathrm{i}\Delta_s - (\gamma_s^m + \gamma_s^e)/2]\beta_s - \frac{\sqrt{\widetilde{\Gamma}}}{2}\mathrm{e}^{\mathrm{i}k_z\ell_s}\left[\sqrt{\gamma_s^e}\hat{d}_s^* - \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s^*\right]\cdot\hat{e}_{-\xi}\beta_L^\xi \\ &\quad - \frac{\sqrt{\widetilde{\Gamma}}}{2}\mathrm{e}^{\mathrm{i}k_z(\ell-\ell_s)}\left[\sqrt{\gamma_s^e}\hat{d}_s^* - \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s^*\right]\cdot\hat{e}_{\xi}\beta_R^\xi.\end{aligned} \tag{S.39c}$$

Here, $\xi = \pm$ indicates the incoming polarization and we have split the decay rate of the molecular transition into magnetic and electric part $\gamma_s = \gamma_s^m + \gamma_s^e$. The vectors $\hat{e}_\pm$ are the circular basis vectors. Assuming steady state for β_s in the dispersive regime $|\Delta_s| \gg \gamma_s$, one obtains two terms for the left and right side. One term that renormalizes the coupling between the metasurface layers and another term which renormalizes the metasurface linewidth and frequency. The second term is of the form

$$-\mathrm{i}\frac{\widetilde{\Gamma}}{4\Delta_s}\mathrm{e}^{2\mathrm{i}k_z\ell_s}\hat{e}_{-\xi}\cdot\left[\sqrt{\gamma_s^e}\hat{d}_s + \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s\right]\left[\sqrt{\gamma_s^e}\hat{d}_s^* - \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s^*\right]\cdot\hat{e}_{-\xi} \tag{S.40}$$

for the left and

$$-\mathrm{i}\frac{\widetilde{\Gamma}}{4\Delta_s}\mathrm{e}^{2\mathrm{i}k_z(\ell-\ell_s)}\hat{e}_{\xi}\cdot\left[\sqrt{\gamma_s^e}\hat{d}_s + \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s\right]\left[\sqrt{\gamma_s^e}\hat{d}_s^* - \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s^*\right]\cdot\hat{e}_{\xi} \tag{S.41}$$

for the right.

It is well known that a rotational average of the terms in brackets yields the same result for both terms

$$\left\langle\hat{e}_{\pm\xi}\cdot\left[\sqrt{\gamma_s^e}\hat{d}_s + \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s\right]\left[\sqrt{\gamma_s^e}\hat{d}_s^* - \mathrm{i}\xi\sqrt{\gamma_s^m}\hat{\mu}_s^*\right]\cdot\hat{e}_{\pm\xi}\right\rangle_{\mathrm{rot}} = \left[\gamma_s + 2\xi\sqrt{\gamma_s^m\gamma_s^e}\,\mathrm{Im}(\hat{d}_s^*\cdot\hat{\mu}_s)\right], \tag{S.42}$$

where $\mathrm{i}\sqrt{\gamma_s^m\gamma_s^e}\,\mathrm{Im}(\hat{d}_s^*\cdot\hat{\mu}_s)$ is usually identified as the rotary strength of the molecular transition which is nonzero for chiral molecules and zero for achiral molecules. The limit of a perfect chiral scatterer for $\hat{d}_s \in \mathbb{R}^3$ and $\hat{\mu}_s \in \mathrm{i}\mathbb{R}^3$ is when the two are aligned and have the same magnitude so that $\sqrt{\gamma_s^m\gamma_s^e}\,\mathrm{Im}(\hat{d}_s^*\cdot\hat{\mu}_s) = \pm\gamma_s/2$. This implies that the scatterer only couples to one of the polarization components.

We produce Fig. 2(c) of the main text by simulating the time evolution of Eqs. (S.39) and then calculating the observable defined below in Eqs. (S.48) and (S.49) from the the cavity output field including the additional electric and magnetic dipole of the chiral scatterer, similar to Eq. (S.17a). The electric field behind the resonator is obtained as

$$E_{\mathrm{out}}^{(+)}(z > l) = E_{\mathrm{in}}^{(+)}\mathrm{e}^{\mathrm{i}k_z l} - \frac{\mathrm{i}}{2}\left[\frac{(2\pi)^3}{3\pi\epsilon_0}\sum_{i=1}^{N}\mathbf{d}_i\beta_i\mathrm{e}^{-\mathrm{i}k_z\ell_i} + \left(\frac{(2\pi)^3}{3\pi\epsilon_0}\mathbf{d}_s - \frac{(2\pi)^3}{3\pi\hbar\mu_0}\hat{e}_z\times\boldsymbol{\mu}_s\right)\beta_s\mathrm{e}^{-\mathrm{i}k_z\ell_s}\right]\mathrm{e}^{\mathrm{i}k_z z}. \tag{S.43}$$

Then we apply the error computed in Eq. (S.57) and add it as shaded area.

S9. DERIVATION OF PHASE UNCERTAINTY FOR HOMODYNE DETECTION

We consider balanced homodyne detection and we follow closely section 6.11 of [S6]. First, we define the intensity difference operator

$$M_-(t,T) = \mathrm{i}\int_t^{t+T}\mathrm{d}t'\left[t_c^* a^\dagger(t')a_{\mathrm{LO}}(t') - t_c a_{\mathrm{LO}}^\dagger(t')a(t')\right], \tag{S.44}$$

$M_-(t,T)$	intensity difference operator	
m_-	detected intensity difference	
F	intensity of signal beam (#photons/time)	unit [Hz]
$F_{\rm LO}$	intensity of LO beam (#photons/time)	unit [Hz]
θ	phase of signal beam	
$\theta_{\rm LO}$	phase of LO	
T	detector integration time	
$t_{\rm RT}$	cavity round-trip time	$2\ell/c$
η_Q	quantum efficiency of detectors	$0 \leq \eta_Q \leq 1$
t_c	cavity transmission amplitude	
φ_c	phase of cavity transmission	

TABLE SIII. Definitions for section S9.

where T is the detector integration time (T and t are not to be confused with the cavity transmission function t_c), $a(t)$ is given by

$$a(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{d}\omega \, a(\omega) \mathrm{e}^{-\mathrm{i}\omega t}. \tag{S.45}$$

and $a_{\rm LO}(t)$ is the annihilation operator of the local oscillator field, still defined, mutatis mutandis, by Eq. (S.45). The signal and the LO are prepared in the *monochromatic* coherent states $|\alpha\rangle$ and $|\alpha_{\rm LO}\rangle$, respectively, such that

$$a(t)\,|\alpha\rangle = \alpha(t)\,|\alpha\rangle\,, \qquad \text{and} \qquad a_{\rm LO}(t)\,|\alpha_{\rm LO}\rangle = \alpha_{\rm LO}(t)\,|\alpha_{\rm LO}\rangle\,, \tag{S.46}$$

where

$$\alpha(t) = \sqrt{F}\exp(-\mathrm{i}\omega_l t + \mathrm{i}\theta), \qquad \text{and} \qquad \alpha_{\rm LO}(t) = \sqrt{F_{\rm LO}}\exp(-\mathrm{i}\omega_l t + \mathrm{i}\theta_{\rm LO}). \tag{S.47}$$

By definition F and $F_{\rm LO}$ are the number of photons per unit of time of the signal and the LO beams, respectively. In a typical experimental setup $F_{\rm LO} \gg F$.

The detected intensity difference m_- is calculated as

$$m_- = \eta_Q \left\langle M_-(t,T) \right\rangle, \tag{S.48}$$

where η_Q is the quantum efficiency of the detectors and here and hereafter $\langle O \rangle = \langle \alpha, \alpha_{\rm LO}|O|\alpha, \alpha_{\rm LO}\rangle$. A straightforward calculation gives

$$\left\langle M_-(t,T) \right\rangle = 2\sqrt{F F_{\rm LO}}|t_c| \int_t^{t+T} \mathrm{d}t' \sin\left(\theta - \theta_{\rm LO} + \varphi_c(t')\right), \tag{S.49}$$

where we have defined $t_c = |t_c|\exp(\mathrm{i}\varphi_c(t))$. Since we consider the case of a particle passing through the cavity, we keep the phase time-dependent but assume that the transmission is not affected significantly by the passage.

The difference photocount variance $(\Delta m_-)^2$ is given by

$$(\Delta m_-)^2 = \eta_Q^2 \left\{ \left\langle \left[M_-(t,T)\right]^2 \right\rangle - \left\langle M_-(t,T) \right\rangle^2 \right\} + \eta_Q(1-\eta_Q) \int_t^{t+T} \mathrm{d}t' \left\langle a_{\rm LO}^\dagger(t') a_{\rm LO}(t') - |t_c|^2 a^\dagger(t') a(t') \right\rangle, \tag{S.50}$$

where

$$\left\langle \left[M_-(t,T)\right]^2 \right\rangle = |t_c|^2 \left[T\left(F + F_{\rm LO}\right) + 4F F_{\rm LO} \left(\int_t^{t+T} \mathrm{d}t' \sin\left(\theta - \theta_{\rm LO} + \varphi_c(t')\right) \right)^2 \right]. \tag{S.51}$$

Gathering all the pieces together, we eventually find

$$(\Delta m_-)^2 = \eta_Q T \Big[\eta_Q |t_c|^2 \left(F + F_{\rm LO}\right) + (1-\eta_Q) \left(|t_c|^2 F + F_{\rm LO} \right) \Big]. \tag{S.52}$$

Small changes in the transmission function of the Fabry-Pérot cavity due to the passage of the particle produce small changes in the mean photocounts at the two detectors in the homodyne setup. The limiting resolution is determined by the intrinsic uncertainty in the homodyne detection, quantified by the difference photocount variance $(\Delta m_-)^2$. The uncertainty $\Delta\varphi_c$ in the transmission function caused by the homodyne uncertainty is obtained from

$$\Delta\varphi_c = \frac{\sqrt{(\Delta m_-)^2}}{\left|\dfrac{\mathrm{d}m_-}{\mathrm{d}\varphi_c}\right|}. \tag{S.53}$$

From Eqs. (S.48) and (S.49) we have

$$\left|\frac{\mathrm{d}m_-}{\mathrm{d}\varphi_c}\right| = 2\eta_Q T\sqrt{FF_{\mathrm{LO}}}|t_c|\left|\int_t^{t+T}\mathrm{d}t'\cos\left(\theta-\theta_{\mathrm{LO}}+\varphi_c(t')\right)\right|. \tag{S.54}$$

For a signal and a local oscillator matched with one of the cavity resonances $\omega_n = 2\pi(n-1/2)/t_{\mathrm{RT}}$ (round-trip time $t_{\mathrm{RT}} = 2\ell/c$), we have $|t_c| = 1$ and $\theta - \theta_{\mathrm{LO}} + \varphi_c(t) = \delta\varphi(t)$, so that

$$(\Delta m_-)^2_{\mathrm{res}} = \eta_Q T\left(F+F_{\mathrm{LO}}\right), \qquad \text{and} \qquad \left|\frac{\mathrm{d}m_-}{\mathrm{d}\varphi_c}\right|_{\mathrm{res}} = 2\eta_Q T\sqrt{FF_{\mathrm{LO}}}\left|\int_t^{t+T}\mathrm{d}t'\cos\left(\delta\varphi(t')\right)\right|. \tag{S.55}$$

Substituting Eq. (S.55) into (S.53) we obtain

$$\Delta\varphi_c = \frac{\sqrt{F+F_{\mathrm{LO}}}}{2\sqrt{\eta_Q T}\sqrt{FF_{\mathrm{LO}}}\left|\int_t^{t+T}\mathrm{d}t'\cos\left(\delta\varphi(t')\right)\right|} \approx \frac{1}{2(\eta_Q TF)^{1/2}\left|\int_t^{t+T}\mathrm{d}t'\cos\left(\delta\varphi(t')\right)\right|}, \tag{S.56}$$

where the last approximate equality holds when $F_{\mathrm{LO}} \gg F$. Similarly, in this regime

$$\frac{\sqrt{(\Delta m_-)^2_{\mathrm{res}}}}{\sqrt{F_{\mathrm{LO}}}} = \sqrt{\eta_Q T}. \tag{S.57}$$

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2.3 P3: Scaling Law for Kasha's Rule in Photoexcited Molecular Aggregates

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In this publication, my contributions were the following:

- Contributions to rewrites and finalization of the draft (15%).
- Simulation of aggregation of molecular aggregates (not included in draft).
- Discussion of calculations presented in main text and appendix.
- Discussion of results in Fig 3 and Fig 4.

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Scaling Law for Kasha's Rule in Photoexcited Molecular Aggregates

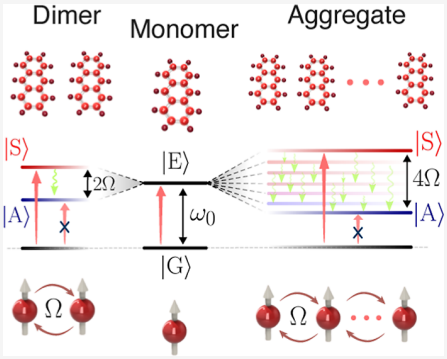
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ABSTRACT: We study the photophysics of molecular aggregates from a quantum optics perspective, with emphasis on deriving scaling laws for the fast nonradiative relaxation of collective electronic excitations, referred to as Kasha's rule. Aggregates exhibit an energetically broad manifold of collective states with delocalized electronic excitations originating from near-field dipole−dipole exchanges between neighboring monomers. Photoexcitation at optical wavelengths, much larger than the monomer−monomer average separation, addresses almost exclusively symmetric collective states, which for an arrangement known as H-aggregate show an upward hypsochromic shift. The extremely fast subsequent nonradiative relaxation via intramolecular vibrational modes populates lower energy, subradiant states, resulting in effective inhibition of fluorescence. Our analytical treatment allows for the derivation of an approximate scaling law of this relaxation process, linear in the number of available low-energy vibrational modes and directly proportional to the dipole−dipole interaction strength between neighboring monomers.



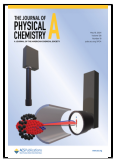
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INTRODUCTION

Molecular aggregates[1−3] are self-ordered arrangements of monomers showing strong collective optical transition dipole strengths. Owing to the dense packing of monomers, with monomer−monomer separations at the order of a nanometer and total aggregate lengths of tens or hundreds of nanometers, thus much below an optical wavelength and despite inhomogeneous broadening and separation disorder, they exhibit delocalized excitons.[4] This allows strong coupling to external light modes, resulting in collectively modified fluorescence rates. Following the discovery of J- and H-aggregates in the 1930s by Scheibe et al.[5] and Jelley,[6] their standard understanding is based on the original approach introduced by Kasha in the 1960s.[7] Currently, J-aggregates are widely employed in light-matter coupling experiments aiming at the modification of material properties via the manipulation of the electromagnetic vacuum mode density around electronic resonances.[8] Numerous approaches to understand their aggregation behavior and the subsequent response to external illumination have been taken, as thoroughly reviewed in ref 1.

There has been recently a new surge in interest in theoretically understanding the behavior of organic molecular systems under the action of confined vacuum fields, i.e., within the context of cavity quantum electrodynamics.[9−12] Among many approaches taken, a purely quantum optics formalism has been introduced, which is based on a simple model for electron−vibron interactions contained within the formalism of the Holstein Hamiltonian and which can incorporate loss of photons or vibrations via an open system approach following either a master equation or quantum Langevin equation approaches.[13,14] Within such a quantum optics approach, rate equations for the dynamics of a molecular aggregate upon photoexcitation with light at visible wavelengths λ can be derived. As pointed out in ref 1, the dynamics of the exciton in the aggregate can be understood in terms of bright (or symmetric) and dark (or asymmetric) states. In the simplest example, a molecular dimer in the side-by-side H-aggregate configuration (see Figure 1) exhibits a symmetric superposition $|S\rangle = (|eg\rangle + |ge\rangle)/\sqrt{2}$ sitting at energy $\omega_0 + \Omega$. Here $|eg\rangle$ denotes that the first monomer is in the excited electronic state while the second one is in the ground state and Ω is the nearest neighbor dipole−dipole interaction scaling with the monomer separation as d^{-3}. As d is typically much smaller than λ, the asymmetric state, at energy $\omega_0 - \Omega$, is decoupled from external light and instead is reached indirectly from the symmetric state $|S\rangle$ via redistribution of energy into molecular vibrations, consistent with the standard understanding of a Kasha process. This can be easily generalized to a mesoscopic-sized aggregate, where an additional large number of dark states fill the 4 Ω frequency split between the photoexcited symmetric state and the lowest energy state. The Kasha incoherent loss of energy is mediated by a multitude of vibrational relaxation pathways. For a molecule with $\mathcal{N}$ atoms, a large number of the order $3\mathcal{N} - 6$ vibrational modes are available; some of them, up to some index n_{max}, find themselves

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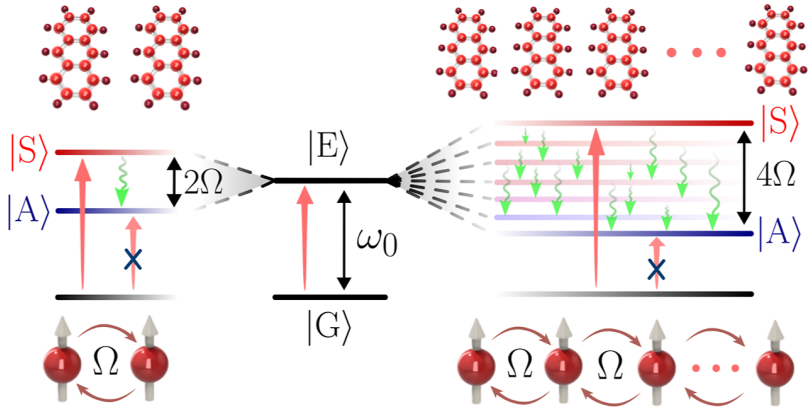


Figure 1. Illustration of the Kasha process in an H-aggregate for a dimer (left) and larger aggregate (right). The population migrates from the optically addressed state $|S\rangle$ (superradiant, therefore color-coded in red) toward the lower and subradiant collective states $|A\rangle$ (color-coded toward blue) via a multitude of vibrational relaxation channels (indicated as green arrows). The energy splitting in the large aggregate is four times the nearest neighbor dipole−dipole coupling strength Ω and strongly depends on the aggregate separation at the level of nanometers. A resulting effective relaxation rate from the high energy optically addressed state to the bottom of the energy manifold results, on the femtosecond time scale.

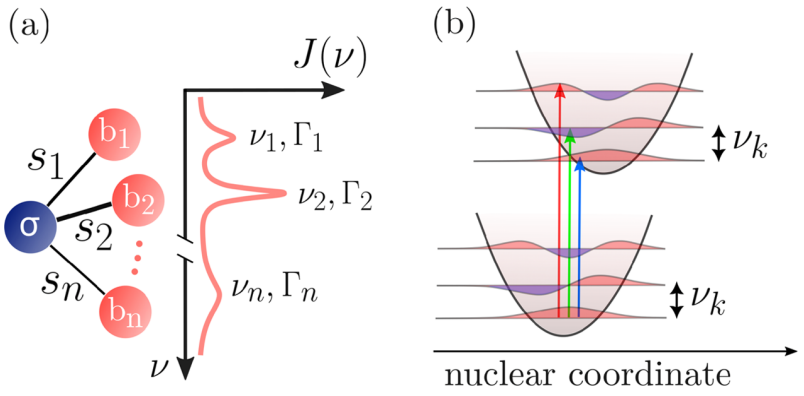


Figure 2. (a) Diagrammatic description of electron−vibron interactions within each monomer. A single optically addressable electronic transition operator σ is coupled to n vibrational mode operators b_m with frequencies ν_m, line widths Γ_m, and Huang−Rhys factors s_m, where m runs from 1 to n. The vibrational spectral density $J(\nu)$ has a number of vibrational frequencies ν_m and associated relaxation rates Γ_m which dictate their line widths. (b) Ground and excited electronic potential energy landscapes along a single nuclear coordinate under the harmonic assumption. The vertical lines illustrate the Franck−Condon principle where substantial transition rates only are possible when a match between the symmetry of the nuclear wavepacket in the ground and excited electronic states.

within the interval 4 Ω, thus providing resonant transfer between the set of collective electronic states.

We show that an approach based on rate equations allows for an approximate expression of the rate of a Kasha process as

$$\kappa_S \approx \frac{4s\Omega}{3}\frac{(n_{\max}+1)(2n_{\max}+1)}{n_{\max}}. \quad (1)$$

The simplicity of the result lies in the small number of parameters involved: the average estimated Huang−Rhys factor $s = \sum_{j=1}^{n_{\max}} s_j/n_{\max}$, the monomer separation d (appearing via Ω), and the estimated number of intramolecular vibrations $n_{\max}$ of relevant frequencies. As an example, we consider the aqueous dye molecule Cresyl Violet (CV) which aggregates in an H-configuration, featuring more than 30 vibrational modes in the range ∼5−50 THz with a zero-phonon transition wavelength λ_0 = 590 nm and a monomer fluorescence line width γ_0 = 454 MHz.[15] The nearest neighbor dipole−dipole coupling is estimated at around Ω = 23 THz and the vibrational spectrum is extracted from ref 16. We estimate around $n_{\max}$ = 12 vibrational modes to have a non-negligible Huang−Rhys factor larger than 0.01 and to fall into the range of the electronic energy band [0, 4 Ω]; this leads to an average estimated Huang−Rhys factor s = 0.09. The estimated time scale for the Kasha process is then around $\kappa_S^{-1} \approx$ 13 fs. Similar estimates for Rhodamine 800[17] and Bacteriochlorophyll a (BChla)[18] give values for the Kasha process time scale of order 30 and 34 fs, respectively.

METHODS

The minimal model we consider involves a one-dimensional molecular chain of identical monomers with a typical intermonomer distance d in the nm range. As photoexcitation with wavelength $2\pi/k_0$ (with k_0 the wavevector) is in the μm range, the condition $k_0 d \ll 1$ is fulfilled, meaning that the chain is uniformly illuminated. Each monomer is assumed to undergo a single electronic transition, which in turn is coupled to n vibrational modes (see Figure 2). Each vibrational mode has a frequency ν_m and relaxation rate Γ_m, where m runs from 1 to n. The monomer can undergo spontaneous emission at a rate γ_0, owing to the coupling to the electromagnetic environment. For each monomer j, the electronic transition is at frequency splitting ω_0 ($\hbar$ = 1) and is described by the

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collapse operator $\sigma_j = |g\rangle_j\langle e|_j$. The vibrational degrees of freedom are described by bosonic operators b_{jm} satisfying the commutation relations $[b_{jm}, b^\dagger_{j',m'}] = \delta_{jj'}\delta_{mm'}$. The vibronic couplings are illustrated in Figure 2a as links between the electronic and vibration operators with magnitudes characterized by the Huang–Rhys factors s_m. The electronic and vibrational degrees of freedom are subject to loss quantified by spontaneous emission rate γ_0 and vibrational relaxation rates Γ_m, respectively. The Hamiltonian for all $\mathcal{N}$ monomers is obtained as a sum over each particle's Hamiltonian

$$h^{(j)} = \bar{\omega}_0\sigma_j^\dagger\sigma_j + \sum_{m=1}^{n} \nu_m(b_{jm}^\dagger b_{jm} - \sqrt{s_m}\sigma_j^\dagger\sigma_j(b_{jm}^\dagger + b_{jm})). \tag{2}$$

Notice that the bare frequency is Stokes shifted $\bar{\omega}_0 = \omega_0 + \sum_m s_m\nu_m$. This shift will later be eliminated after a polaron transformation (see Supporting Information for more details). The model and its validity and relevance are extensively discussed in ref 13.

Dipole–dipole exchanges at rates $\Omega_{jj'}$ have a strong imprint at nm distances, owing to their scaling with the inverse cube of the particle separation $|\vec{r}_j - \vec{r}_{j'}|^{-3}$ in the near-field region.[19] This can be listed in the Hamiltonian as $\mathcal{H}_{d-d} = \sum_{j\neq j'}\Omega_{jj'}\sigma_j^\dagger\sigma_{j'}$ and describes an excitation transfer between pairs of monomers via a virtual photon exchange. The coherent exchange is mediated by the dipole–dipole frequency shifts $\Omega_{jj'}$, which in units of the optical emission rate γ_0 are given by $\Omega_{jj'}/\gamma_0 = -3\pi/k_0\vec{\mu}^*\cdot\mathrm{Re}\boldsymbol{G}(\vec{r}_j, \vec{r}_{j'}, \omega_0)\cdot\vec{\mu}$, namely, proportional to the real part of the Green's tensor in free space (shown in the Supporting Information). In the following, we will consider the particular case of side-by-side arrangement, where all transition dipoles $\vec{\mu}$ are parallel to each other and perpendicular to the chain direction.

The model can be better tackled on a collective basis[14] resulting from the diagonalization of the first excitation subspace including dipole–dipole interactions. This can be done either in the discrete space where states are indexed from 1 to $\mathcal{N}$ (see Figure 3a) or alternatively, with the quasi-momentum index $q = 2\pi k/(\mathcal{N}d)$ obtained by a rescaling of the index of the mode $k = \pm1, ..., \pm(\mathcal{N}-1)/2$ (see Figure 3b). This is based on the observation that, for aggregates with $\mathcal{N}$ of the order of a hundred monomers, the mesoscopic limit can be assumed such that the system can be considered translationally invariant and periodic boundary conditions can be invoked. A single symmetric mode can be distinguished with a state vector obtained by the application of the symmetric operator $\mathcal{S}^\dagger = \sum_j\sigma_j^\dagger/\sqrt{\mathcal{N}}$ to the collective ground state $|G\rangle$. To a good approximation, the system can be considered to be in the Dicke limit where a single superradiant emission rate roughly estimated by $\gamma_S = \mathcal{N}\gamma_0$ characterizes this 'bright' state. In addition, the other orthogonal $\mathcal{N}-1$ asymmetric states are obtained via the application of asymmetric operators $\mathcal{A}_q^\dagger = \sum_{j=1}^{\mathcal{N}} e^{iqjd}\sigma_j^\dagger/\sqrt{\mathcal{N}}$ to $|G\rangle$. These states are nonradiative at such deep subwavelength molecular separations, and we dub them therefore as 'dark'.

The collective excitations are eigenstates of the dipole–dipole interaction Hamiltonian $\mathcal{H}_{d-d}\mathcal{A}_q^\dagger|g\rangle^{\otimes\mathcal{N}} = \Omega_q\mathcal{A}_q^\dagger|g\rangle^{\otimes\mathcal{N}}$ where by definition we fix the symmetric shift $\Omega_S \equiv \Omega_{q=0}$.

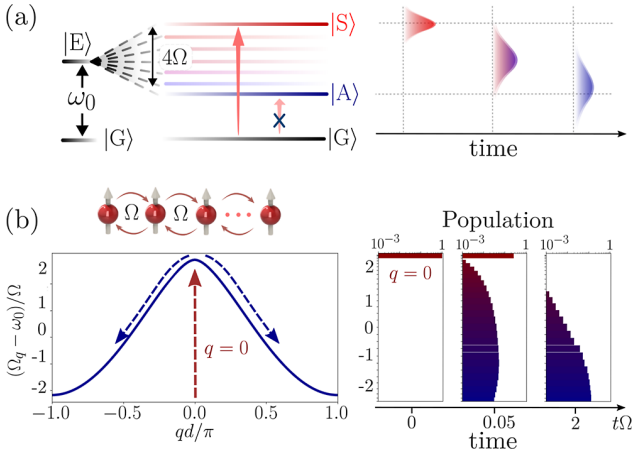


Figure 3. Dynamics of the collective state populations upon photoexcitation of the symmetric state in H-aggregates. (a) Illustration of the population migration in time from the optically excited state (superradiant and therefore color-coded in red) toward the lower and subradiant collective states (color-coded toward blue) via a multitude of vibrational relaxation channels. (b) Exact numerical time dynamics for a chain of $\mathcal{N} = 100$ monomers. In such a mesoscopic limit, the analysis can be performed on a band diagram where the symmetric state, photoexcitable, sits at $q = 0$. The numerical simulation shows the time dynamics of populations on a time scale normalized to Ω^{-1}.

With periodic boundary conditions imposed, in the mesoscopic limit, one can derive the collective shifts as $\Omega_q = 2\sum_j\Omega_{1j}\cos(q(j-1)d)$.[14,20] Further simplifications occur by considering the nearest neighbor approximation and the collective eigenenergies become $\Omega_q = 2\Omega\cos(qd)$, where the nearest-neighbor coupling is simply denoted by $\Omega \equiv\Omega_{12}$. Closely following the procedure introduced in ref 14, one can analyze the coupling between states of different symmetries via electron–vibron couplings by an additional transformation to a collective basis for the vibrational degrees of freedom as well. This is done by introducing collective vibrational modes $Q_q^{(m)} = \sum_{j=1}^{\mathcal{N}}(b_{jm} + b_{jm}^\dagger)e^{iqjd}/\sqrt{\mathcal{N}}$, with the momentum quadratures satisfying $[Q_q^{(m)}, P_q^{(m)}] = i$ and m labels the vibrational mode running from 1 to n.

RESULTS

Owing to the small dimension of a molecular aggregate compared to an optical wavelength, under photoexcitation, only the symmetric collective mode is directly accessible. This immediately leads to a rescaling of the Rabi driving by a factor of $\sqrt{\mathcal{N}}$: this can be seen equivalently as an increase of the oscillator strength by $\sqrt{\mathcal{N}}$, thus rendering aggregates of any kind as good candidates for strong light-matter coupling. In addition, the particularity of the subsequent aggregate electronic dynamics lies within the shape of the energy band. For example, J-aggregates present an energy band where the symmetric mode lies at the bottom of the band, thus leading to a bathochromic frequency shift (to the left of the bare monomer frequency) and subsequently shows not only an enhanced absorption cross section but also enhanced fluorescence at a superradiant rate. In contrast, H-aggregates have a symmetric state located at the top of the energy band, corresponding to a hypsochromic shift. Most importantly, quick dynamics follows the optical excitation involving the

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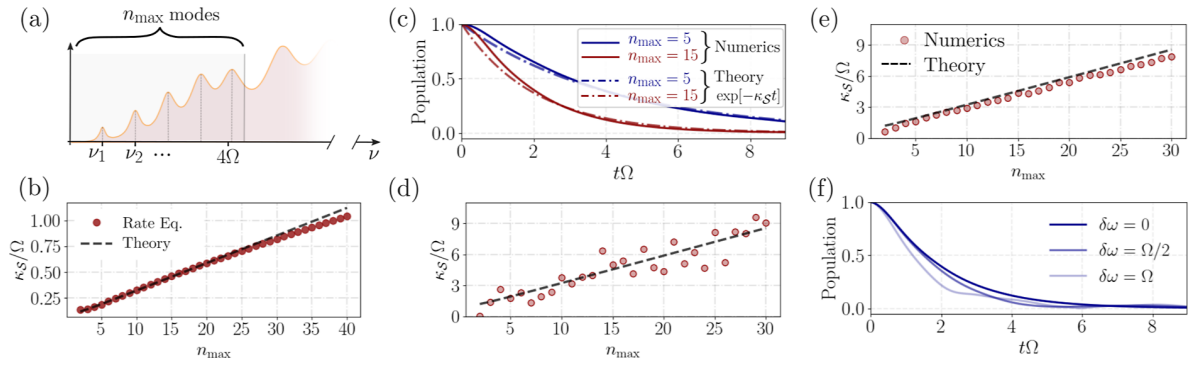


Figure 4. (a) Only a limited number of vibrational modes, from 1 to n_{max} can efficiently mediate resonant transfer between the symmetric state and the dark manifold. These modes are within the 4Ω bandwidth. (b) Linear scaling of the transfer rate κ_S as a function of the number of vibrational modes, with the same parameters as in (c) and $\mathcal{N} = 20$ molecules at $k_0 d = 0.0126$ separation. The scaling law from eq 5 provides a very good fit to the rate equations in eq 4. (c) Time dynamics of 20 molecules initialized in the symmetric state. Numerical results in the single excitation manifold show excellent agreement with an exponential decay given by $e^{-\kappa_S t}$, and governed by the analytical formula in eq 5 (Same parameters as in (d),(e)). (d) Numerical results for κ_S plotted against increasing n_{max} for a randomly drawn set of vibrational frequencies $\nu_1,\cdots\nu_{n_{max}}$ in the window 0−4Ω and Huang−Rhys factors $s_1,\cdots s_{n_{max}}$ in the interval [0,0.2]. The fit is performed with the analytical scaling in eq 5 for evenly spaced vibrational frequencies and Huang−Rhys factor at the level of the distribution average $s = 0.1$. (e) Further comparison of the analytical scaling from eq 5 with an average of over 200 random realizations shows almost perfect agreement. (f) Decay of the initial symmetric state population under the influence of random static frequency disorder with fluctuation $\delta\omega$ around ω_0. Further parameters in all plots: $\mathcal{N} = 20$ molecules, $k_0 d = 0.0126$, $\Gamma_m = \nu_m/10$, $\Omega \approx 3.759 \times 10^5 \gamma_0$.

relaxation of the collective state toward low-energy dark states. This dynamics is illustrated in Figure 3 in both the discrete and continuous cases. In Figure 3, a numerical simulation for $\mathcal{N} = 100$ at an intermolecular spacing $k_0 d = 0.0126$. We have considered a large n spanning the range of frequencies from 0 to 4Ω, with $\Gamma_m = \nu_m/10$ and identical $s_m = 0.01$. Such time dynamics can be analytically tackled at the level of rate equations. We largely follow the derivation in ref 14 which we generalize here to incorporate the crucial aspect that many vibrational modes have to be taken into account. Under the assumption that the vibrational relaxation rates are fast compared to the coherent couplings and radiative loss rates, a set of rate equations for the populations of the symmetric state $p_S = \langle \mathcal{S}^\dagger \mathcal{S} \rangle$ and all dark states $p_q = \langle \mathcal{A}_q^\dagger \mathcal{A}_q \rangle$ can be derived (see Supporting Information). With the definitions κ_S —total loss rate of the symmetric state, κ_q—loss rate for dark state q, $\kappa_{q\to S}$—incoherent repopulation rate from the dark to the bright state and $\kappa_{q'\to q}$—incoherent rate for redistribution of energy within the dark state manifold, one can write

$$\dot{p}_S = -(\gamma_S + \kappa_S)p_S + \sum_{q\neq 0} \kappa_{q\to S} p_q, \tag{3a}$$

$$\dot{p}_q = -\kappa_q p_q + \sum_{q'\neq q} \kappa_{q'\to q} p_{q'}. \tag{3b}$$

The rate equations show that the symmetric state energy spills into the whole dark state manifold via rate κ_S and, in addition, higher energy dark states spill into the lower energy ones via κ_q. The quasi-unidirectionality of the process is ensured by the fact that, in this perturbative treatment, the coherent coupling between states is followed by quick vibrational relaxation, making the reverse process, governed by rates $\kappa_{q\to S}$ and $\kappa_{q'\to q}$ from lower energy state to higher ones, very unlikely (as shown in ref 14 for the two monomer case). Mathematically, the condition is $\sqrt{s_m}\nu_m/\sqrt{\mathcal{N}} \ll \Gamma$. Analytically, one can get an expression for the transfer rate between the symmetric mode and any dark mode q mediated by vibrational mode m as

$$\kappa_{S\to q}^{(m)} = \frac{2 s_m \nu_m^2 (\Gamma_m + \gamma_S)/\mathcal{N}}{(\Gamma_m + \gamma_S)^2 + 4(\Omega_S - \Omega_q - \nu_m)^2}. \tag{4}$$

and construct the total rate to all states spanned by the index q by summing $\kappa_S = \sum_{m=1}^{n_{max}} \sum_q \kappa_{S\to q}^{(m)}$ over all vibrations up to an index n_{max} within the frequency interval covered by 4Ω where the electronic collective states are positioned in energy. Moreover, we will consider the standard underdamped harmonic oscillator model for the molecular vibrations, i.e., the dissipation rate is much smaller than the resonance frequency for any mode $\Gamma_m \ll \nu_m$. Equivalently, one can state that the quality factor of any vibrational mode is much larger than unity $\nu_m/\Gamma_m \gg 1$.

In order to further proceed with analytical estimates, let us first make some comments regarding typical time scales. Given that vibrational relaxation is on the order of tens to hundreds of GHz while spontaneous emission is in the range of tens of MHz, a very quick nonradiative path from the symmetric to low energy asymmetric states can be achieved on ps time scales. For monomer separations in the nanometer range, expected near-field shifts Ω in the range of 1 THz to tens of THz are expected. This means that only a few, low-energy, molecular vibrations can fit in the window of 4Ω (see Figure 4a) and aid the relaxation process. This allows us to derive an approximate scaling law for κ_S as a function of a given number of vibrational modes n_{max} that can efficiently mediate the relaxation of the symmetric state into the dark state manifold. We proceed by first considering a given vibrational mode m and asking for the condition that this mode can transfer excitation from the symmetric mode to any of the dark states. We notice that with the condition that $\Gamma_m \ll \nu_m$ (it is also implied that $\gamma_S \ll \nu_m$ even for large $\mathcal{N}$) the resonance condition requires that the states to which resonant transfer can take place are only in the vicinity of the mode q fulfilling

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$\Omega_q = \Omega_S - \nu_m$. Of course, in this case, one can immediately observe that any modes with $\nu_m > 4\Omega$ cannot take part in this transfer. Assuming a constant density of all $\mathcal{N}$ collective states spread within the interval 4Ω, we can then estimate that a number of approximately $\epsilon_m = \Gamma_m \mathcal{N}/(4\Omega)$ states fall close to the resonance $\Omega_q = \Omega_S - \nu_m$, i.e., within the line width Γ_m. Summing all of these contributions gives the total rate for all transitions mediated by mode m to states close to q. Summing over all possible relaxation paths that participate in the transfer giving thus an estimate for the total rate

$$\kappa_S = \sum_q \kappa_{S\to q}^{(m)} \approx \sum_{m=1}^{n_{\max}} \frac{\Gamma_m}{2\Omega} \frac{s_m \nu_m^2}{(\Gamma_m + \gamma_S)}. \tag{5}$$

In order to estimate this quantity, knowledge of the particular nature of the monomer's frequencies, Huang−Rhys factors, and vibrational relaxation rates is necessary. However, while we will later numerically investigate random distributions of frequencies and Huang−Rhys factors, we aim first at deriving a simple scaling law. To this end, we will proceed by making some simplifying assumptions, among which the first is that the vibrational spectrum is equally spaced in the interval from 0 up to 4Ω. We denote the frequency of mode m by $\nu_m = m4\Omega/n_{\max}$. Let us also consider that all Huang−Rhys factors are equal to s (later we compare with a randomized distribution with an average s). Moreover, we neglect the contribution of γ_S as it is much smaller than Γ_m (this should be typically very well fulfilled as γ_0/Γ_m is expected to be around 10^{-5}). Summing over all vibrational modes within the interval of 4Ω gives us an approximated scaling law as the expression already introduced in eq 1. The result shows the independence of the total number of monomers $\mathcal{N}$ and a quasi-linear dependence on the total number of available low-frequency vibrational modes which can resonantly participate in Kasha's relaxation process from the high energy symmetric state to the bottom of the dark state manifold. Notice that the predicted time scale is dictated by the nearest neighbor dipole−dipole coupling strength Ω which in turn depends on the inverse cube of the monomer−monomer separation. In the first step, we can estimate that the analytical scaling is in very good agreement with the results of the rate equations, as shown in Figure 4b.

The validity of the analytically obtained result can be easily checked against the numerical simulations. In the first step, we compare the numerical results with the analytical scaling of eq 1, for an equidistant vibrational spectrum, in Figure 4c. A very good agreement is obtained showing that the time evolution of the symmetric state is well reproduced by an exponential decay following $e^{-\kappa_S t}$. In the next step we pick a set of randomly drawn vibrational frequencies $\nu_1,...\nu_{n_{\max}}$ in the window $0-4\Omega$ and Huang−Rhys factors $s_1,...s_{n_{\max}}$ in the interval $[0,0.2]$ (with an average $s = 0.1$). The results plotted in Figure 4a show that a good fit is obtained with the simplified result of eq 1 which assumes evenly spaced vibrational frequencies and a Huang−Rhys factor at the level of the distribution average $s = 0.1$. Furthermore, an average of the numerical results over 200 random realizations predicts excellent agreement to the linear fit predicted by eq 1.

As aggregates are immersed in solvents and usually under room temperature conditions, they are expected to present a large inhomogeneous broadening at the level of THz. Moreover, monomers are not placed in an exactly equidistant chain, resulting in further shifts in molecular energies. We consider a distribution of the $\mathcal{N}$ monomer frequencies around ω_0 such that the frequency of each monomer becomes $\omega_0 + \delta_j$ where δ_j is randomly drawn from a distribution of width $\delta\omega$. On a collective basis, the symmetric state couples to any asymmetric state and acquires a shift as well.[21] The coupling of $\mathcal{S}$ to a state q mediated by disorder is simply given by the Fourier transform of the distribution $\delta_q = \frac{1}{\sqrt{\mathcal{N}}} \sum_{j=1}^{\mathcal{N}} e^{iqjd} \delta_j$, while the shift of the collective state $\delta_S = 1/\sqrt{\mathcal{N}} \sum_{j=1}^{\mathcal{N}} \delta_j$ is the average of the distribution, thus close to zero. According to ref 21, disorder-induced couplings introduce an additional loss channel, thus slightly increasing the Kasha rate. This is indeed consistent with numerical simulations shown in Figure 4f where the dynamics of the symmetric state without disorder and with considerable disorder at $\delta\omega = \Omega$ and $\delta\omega = 2\Omega$ are compared. The result is that the analytically derived loss rate holds well even for considerable disorder levels.

Finally, let us provide some estimates of the expected time scale for such a Kasha process in the aqueous dye molecule CV, Rhodamine 800, and BChla. The results are listed in Table 1 with parameters estimated from refs 15−18. The upshot is that analytically estimated values for the Kasha time scale run in the tens of femtoseconds.

Table 1. Estimates of the Time Scale κ_S^{-1} (Femtoseconds) of the Kasha Process According to Eq 1[a]

dye molecule	d (nm)	$\lvert\vec{\mu}\rvert$ (D)	γ_0 (MHz)	$n_{\max}$	s	κ_S^{-1} (fs)
CV	∼2	1.8	455	12	0.09	13
rhodamine 800	∼2	1.8	525	16	0.067	30
BChl a	∼1	6.4	335	25	0.07	34

[a]The specific dyes considered are CV with $\Omega_0 \approx 23$ THz,[15,16] rhodamine 800 with $\Omega_0 \approx 8$ THz,[17] and BChla with $\Omega_0 \approx 6$ THz, measured at room temperature.[18]

DISCUSSION AND CONCLUSIONS

We investigated the time dynamics following the photoexcitation of molecular aggregates in the side-by-side configuration. These are a perfect showcase of cooperative phenomena as their photophysics is naturally characterized by coherent and incoherent effects brought on by the positioning of individual monomers in the near field of each other. Effects widely explored in quantum optics, such as Dicke superradiance and subradiance, naturally occur in the theoretical description of such compounds, albeit in the presence of more complex, additional interactions between electrons and a vast number of molecular vibrations. We have provided a theoretical approach to the dynamics of collective electronic states making the connection, at the analytical and numerical level, with the physical mechanism introduced long ago by Kasha, stating that photon emission (fluorescence or phosphorescence) occurs in appreciable yield only from the lowest excited state of a given multiplicity. Our analytical conclusions predict that the Kasha loss rate from the symmetric, high energy, optically addressable state to the lower energy states of an H-aggregate is roughly independent of the number of monomers but strongly dependent on the number of low-energy vibrational modes which can be excited

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and then dissipate the accumulated energy afterward. Further investigations will focus on aspects, such as the role of quantum coherence in such systems. An important direction is the application of the methods presented in this manuscript to photosynthetic systems under various conditions of illumination, ranging from spatially and time coherent laser light to spatially and time incoherent light sources.

■ ASSOCIATED CONTENT

Supporting Information

The following files are available free of charge. The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpca.4c00342.

Analytical derivation of the results in the main text and description of the quantum optical formalism (PDF)

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Notes

The authors declare no competing financial interest.

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Supporting Information for Scaling Law for Kasha's Rule in Photoexcited Molecular Aggregates

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S1. Vibronic coupling

Let us justify the form of the Holstein Hamiltonian in Eq. 2 by following a first-principle derivation for a single nuclear coordinate R of effective mass μ. We assume that, along the nuclear coordinate, the equilibria for ground (coordinate R_g , state vector $|g\rangle$) and excited (coordinate R_e and state vector $|e\rangle$) electronic orbitals are different. Assuming equilibrium positions R_g and R_e for the potential surfaces of electronic ground and excited states, one can

write the total molecular Hamiltonian describing both electronic and vibrational dynamics as

$$\mathcal{H}_{\text{mol}} = \left[\omega_0 + \frac{\hat{P}^2}{2\mu} + \frac{1}{2}\mu\nu^2\left(\hat{R} - R_e\right)^2\right]\sigma^\dagger\sigma + \left[\frac{\hat{P}^2}{2\mu} + \frac{1}{2}\mu\nu^2\left(\hat{R} - R_g\right)^2\right]\sigma\sigma^\dagger, \tag{S.1}$$

where μ is the reduced mass of the vibrational mode and $\sigma = |g\rangle\langle e|$. The kinetic and potential energies are written in terms of the position $\hat{Q}$ and momentum operator $\hat{P}$ describing the nuclear coordinate under consideration, with commutation $[\hat{Q}, \hat{P}] = i$. Introducing oscillations around the equilibria $\hat{Q} = \hat{R} - R_{\text{g}}$ and subsequently $\hat{R} - R_{\text{e}} = \hat{Q} + R_{\text{g}} - R_{\text{e}} =: \hat{Q} - R_{\text{ge}}$ we obtain

$$\mathcal{H}_{\text{mol}} = \frac{\hat{P}^2}{2\mu} + \frac{1}{2}\mu\nu^2\hat{Q}^2 + \omega_0\sigma^\dagger\sigma - \mu\nu^2\hat{Q}R_{\text{ge}}\sigma^\dagger\sigma + \frac{1}{2}\mu\nu^2R_{\text{ge}}^2\sigma^\dagger\sigma. \tag{S.2}$$

We can now rewrite the momentum and position operators in terms of bosonic operators $\hat{Q} = q_{\text{zpm}}(b^\dagger + b)$, $\hat{P} = ip_{\text{zpm}}(b^\dagger - b)$. The bosonic operators satisfy the usual commutation relation $[b, b^\dagger] = 1$ and the zero-point motion displacement and momentum are defined as $q_{\text{zpm}} = 1/\sqrt{2\mu\nu}$ and $p_{\text{zpm}} = \sqrt{\mu\nu/2}$. Reexpressing the terms above yields the Holstein Hamiltonian

$$\mathcal{H}_{\text{mol}} = (\omega_0 + s\nu)\sigma^\dagger\sigma + \nu b^\dagger b - \sqrt{s}\nu(b^\dagger + b)\sigma^\dagger\sigma. \tag{S.3}$$

The dimensionless vibronic coupling strength s is given by $\sqrt{s} = \mu\nu R_{\text{ge}}q_{\text{zpm}}$ (s is called the Huang-Rhys factor and is typically on the order of $\sim 0.01 - 1$).

S2. Radiative and vibrational loss

In a master equation formulation for the system density operator ρ written as $\partial_t\rho = i[\rho, \mathcal{H}] + \mathcal{L}[\rho]$ loss can be included via the Lindblad superoperator

$$\mathcal{L}_\gamma[\rho] = \gamma_{\mathcal{O}}/2\left[2\mathcal{O}\rho(t)\mathcal{O}^\dagger - \mathcal{O}^\dagger\mathcal{O}\rho(t) - \rho(t)\mathcal{O}^\dagger\mathcal{O}\right], \tag{S.4}$$

describing decay at generic rate $\gamma_{\mathcal{O}}$ through a single channel with a generic collapse operator $\mathcal{O}$. For vibrational loss, the collapse rate for each mode m is Γ_m and the corresponding collapse operator is $b_{jm} - \sqrt{s_m}\sigma_j^\dagger\sigma_j$. This form for the collapse operator is derived in analogy to the dissipative physics of optomechanical systems in the ultrastrong coupling regime.[1] The radiative loss is not in diagonal Lindblad form but achieves the following expression $\mathcal{L}_e[\rho] = \sum_{j,j'}\gamma_{jj'}/2\left[2\sigma_j\rho\sigma_{j'}^\dagger - \sigma_j^\dagger\sigma_{j'}\rho - \rho\sigma_j^\dagger\sigma_{j'}\right]$. This form can be diagonalized and it shows the emergence of $\mathcal{N}$ independent decay channels, each corresponding to some collective electronic superposition state.[2] At very small separation, deep into the subwavelength regime, the fully symmetric superposition decays at a superradiant rate roughly equal to $\mathcal{N}\gamma_0$ while all other states have vanishingly small decay rates (which we will assume in the following to be exactly zero). This is by no means a limitation of our treatment as one can easily generalize this to the case of non-zero decay rates of the dark manifold.[2,3]

S3. Hamiltonian in the collective basis

The Hamiltonian coupling the symmetric state to the dark state manifold is then given by

$$\mathcal{H}_{\text{int}}^{\mathcal{SA}} = -\sum_{m=1}^{n}\sum_{q\neq 0}\frac{\sqrt{s_m}\nu_m}{\sqrt{\mathcal{N}}}\left(Q_q^{(m)}\mathcal{S}^\dagger\mathcal{A}_q + \text{h.c.}\right), \tag{S.5}$$

via collective vibrations. This coupling is responsible for funneling population into the long lived dark state manifold after the initial driving of the fully symmetric state under uniform illumination. This mechanism is fundamental to understand the dynamics associated with Kasha's rule which we tackle in the main text. In addition, within the dark state manifold an all-to-all coupling Hamiltonian acts with the following form

$$\mathcal{H}_{\text{int}}^{\mathcal{AA}} = -\sum_{m=1}^{n}\sum_{q\neq q'}\frac{\sqrt{s_m}\nu_m}{\sqrt{\mathcal{N}}}\left(Q_{q-q'}^{(m)}\mathcal{A}_q^\dagger\mathcal{A}_{q'} + \text{h.c.}\right), \tag{S.6}$$

and the sum implies that $q, q' \neq 0$. This indicates that a redistribution of energy takes place within the whole manifold of dark states. After transforming the system Hamiltonian, the energies of the collective states are shifted by the contribution of the symmetric vibrational mode $-\sum_m \sqrt{s_m}\nu_m Q_0^{(m)}/\sqrt{\mathcal{N}}$. The energy shifts can be removed by the collective polaron transformation $U = \prod_q \prod_{m=1}^{n} e^{i\sqrt{s_m}/\sqrt{\mathcal{N}} P_0^{(m)} \mathcal{A}_q^\dagger \mathcal{A}_q}$ which leads to a renormalization of the collective state energies as $\bar{\omega}_q = \omega_0 + \sum_m s_m \nu_m/2 + \Omega_q$.

S4. Vacuum mediated dipole-dipole coupling rates

The vacuum mediated dipole-dipole interactions for an electronic transition at wavelength λ_0 (corresponding wave vector $k_0 = 2\pi/\lambda_0$) between identical pairs of emitters separated by r_{ij} is given in terms of the free-space electromagnetic Green's tensor $\mathbf{G}(\vec{r}_i - \vec{r}_j, \omega_0) \equiv \mathbf{G}(\vec{r}_{ij}, \omega_0)$, with $\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$, which reads

$$\mathbf{G}(\vec{r}, \omega_0) = \frac{e^{ik_0 r}}{4\pi k_0^2 r^3}\Big[(k_0^2 r^2 + ik_0 r - 1)I + (-k_0^2 r^2 - 3ik_0 r + 3)\frac{\vec{r} \otimes \vec{r}}{r^2}\Big], \tag{S.7}$$

where $r = |\vec{r}|$. The Green's function $\mathbf{G}_{\alpha\beta}$ is a tensor quantity, with $\{\alpha, \beta\} = \{x, y, z\}$ which is determined by the polarization direction of the dipoles. In order to obtain the dipole-dipole couplings for H-aggregates we chose linear polarization in z-direction for all molecules, namely we take $\vec{\mu}_z^* \cdot \mathbf{G} \cdot \vec{\mu}_z$.

S5. Deriving rate equations

Starting from the Holstein Hamiltonian in Eqs. (5)-(6) for $\mathcal{N}$ identical molecules with n vibrational modes each. The Heisenberg equations for the collective electronic modes are given by

$$\dot{\mathcal{S}} = -i\Big(\Omega_{\mathcal{S}} - \frac{\gamma_{\mathcal{S}}}{2}\Big)\mathcal{S} + \frac{i\sqrt{s}_m\nu_m}{\sqrt{\mathcal{N}}}\sum_{m=1}^{n}\sum_{q} Q_q^{(m)}\mathcal{A}_q + \text{noise}, \tag{S.8a}$$

$$\dot{\mathcal{A}}_q = -i\Omega_q\mathcal{A}_q + \sum_{m=1}^{n}\frac{i\sqrt{s_m}\nu_m}{\sqrt{\mathcal{N}}}\Big(Q_q^{(m)\dagger}\mathcal{S} + \sum_{q'\neq q} Q_{q-q'}^{(m)}\mathcal{A}_{q'}\Big) + \text{noise}. \tag{S.8b}$$

The collective noise terms will be neglected from now on as they do not contribute to the transfer process.

To calculate the transfer rate from the symmetric state to the antisymmetric states we assume some initial population in the symmetric state and no population in the antisymmetric states, additionally we assume that the symmetric state decays independently and formally integrate

$$\mathcal{S}(t) = S(0)e^{-(i\Omega_S+\gamma_S/2)t}, \tag{S.9a}$$

$$\mathcal{A}_q(t) = A_q(0)e^{-i\Omega_q t} + \sum_{m=1}^{n}\frac{i\sqrt{s_m}\nu_m}{\sqrt{\mathcal{N}}}\int_0^t dt' e^{-i\Omega_q(t-t')}\Big(Q_q^{(m)}(t')\mathcal{S}(t') + \sum_{q'\neq q} Q_{q-q'}^{(m)}(t')\mathcal{A}_{q'}(t')\Big), \tag{S.9b}$$

and for the expectation value of the populations we get

$$\langle\dot{\mathcal{S}^\dagger\mathcal{S}}\rangle = -\gamma_S\langle\mathcal{S}^\dagger\mathcal{S}\rangle - \sum_{m=1}^{n}\sum_{q}\frac{2\sqrt{s_m}\nu_m}{\sqrt{\mathcal{N}}}\,\mathrm{Im}\langle\mathcal{S}^\dagger\mathcal{A}_q Q_q^{(m)}\rangle, \tag{S.10a}$$

$$\langle\dot{\mathcal{A}_q^\dagger\mathcal{A}_q}\rangle = -\sum_{m=1}^{n}\frac{2\sqrt{s_m}\nu_m}{\sqrt{\mathcal{N}}}\,\mathrm{Im}\Big(\langle\mathcal{A}_q^\dagger\mathcal{S}Q_q^{(m)}\rangle + \sum_{q'\neq q}\langle\mathcal{A}_q^\dagger\mathcal{A}_{q'}Q_{q'-q}^{(m)}\rangle\Big). \tag{S.10b}$$

Therefore the terms $-2\sqrt{s}_m\nu_m/\sqrt{\mathcal{N}}\,\mathrm{Im}\langle\mathcal{A}_q^\dagger\mathcal{S}Q_q^{(m)}\rangle$ will be responsible for population transfer from the symmetric to the antisymmetric state with quasi-momentum q at a rate $\kappa_{\mathcal{S}\to q}^{(m)}$. We can calculate the rates explicitly up to order $\mathcal{O}(s_m\nu_m^2)$ and assuming that correlations between vibronic and electronic operators factorize.

$$-2\sqrt{s}_m\nu_m/\sqrt{\mathcal{N}}\langle\mathcal{A}_q^\dagger\mathcal{S}Q_q^{(m)}\rangle = -is_m\nu_m^2\int_0^t dt' e^{-\Omega_q(t-t')}\langle Q_q^{(m)}(t')Q_q^{(m)}(t)\rangle\langle\mathcal{S}^\dagger(0)\mathcal{S}(0)\rangle e^{-\epsilon_S t'}e^{-\epsilon_S^* t}$$
$$= -is_m\nu_m^2\langle\mathcal{S}^\dagger(0)\mathcal{S}(0)\rangle\frac{e^{-\gamma_S t}-e^{-((\Gamma_m+\gamma_S)/2+i(\Omega_S-\Omega_q-\nu_m))t}}{(\Gamma_m+\gamma_S)/2+i(\Omega_S-\Omega_q-\nu_m)}, \quad \text{(S.11)}$$

where we defined $\epsilon_S = -(\gamma_S/2 - i\Omega_S)$ and used the fact that different vibrational modes are uncorrelated at all times, i.e. $\langle Q_q^{(m')}(t')Q_q^{(m)}(t)\rangle = 0$ for $m' \neq m$. The correlations for $Q_q^{(m)}$ are evaluated assuming free evolution of the vibrations (to lowest order) and zero temperature for the vibrational modes:

$$\langle Q_q^{(m)}(t')Q_q^{(m)}(t)\rangle = \frac{1}{\mathcal{N}}\sum_{j=1}^{\mathcal{N}}\langle b_{jm}(t')b_{jm}^\dagger(t)\rangle = e^{-(\Gamma_m/2-i\nu_m)(t-t')}. \quad \text{(S.12)}$$

The transfer rate can be written as

$$\kappa_{\mathcal{S}\to q}^{(m)} = \frac{2s_m\nu_m^2(\Gamma_m+\gamma_\mathcal{S})/\mathcal{N}}{(\Gamma_m+\gamma_\mathcal{S})^2+4(\Omega_\mathcal{S}-\Omega_q-\nu_m)^2}, \quad \text{(S.13)}$$

given fast vibrational relaxation rates $\Gamma_m \gg \gamma_S$ compared to the electronic decay rates.

S6. Single excitation subspace

The numerical diagonalization and subsequent time dynamics are evaluated in the single-excitation sector for both the electronic and vibrational degrees of freedom. This allows to rewrite the effective Hamiltonian in non-hermitian form as ($\hbar = 1$)

$$\mathcal{H}_{\text{eff}} = \sum_{j=1}^{\mathcal{N}}\Big(h^{(j)} + \sum_{j'=1}^{\mathcal{N}}\Big(\Omega_{jj'} - i\frac{\gamma_{jj'}}{2}\Big)\sigma_j^\dagger\sigma_{j'} - \frac{i}{2}\sum_{m=1}^{n}\Gamma_m\mathcal{O}_{jm}^\dagger\mathcal{O}_{jm}\Big), \quad \text{(S.14)}$$

where $h^{(j)}$ is defined in Eq. 1 and $\mathcal{O}_{jm} = b_{jm} - \sqrt{s_m}\sigma_j^\dagger\sigma_j$. The dynamics of the electron-vibron density matrix ρ can be described by a von Neumann equation of the form

$$i\frac{\mathrm{d}}{\mathrm{dt}}\rho(t) = [\mathcal{H}_{\mathrm{eff}}\rho - \rho\mathcal{H}_{\mathrm{eff}}], \tag{S.15}$$

and the expectation value of observable $\mathcal{O}$ becomes $\bar{\mathcal{O}} = \mathrm{tr}(\rho\mathcal{O})$. However, instead of solving the von Neumann equation directly, one can use the quantum jump formalism to evaluate single stochastic quantum trajectories using the Monte Carlo wave function method (MCWF). For large numbers of trajectories, the statistical average then approximates the result of the Master equation. The huge advantage is that instead of describing the state of the quantum system by a density matrix of size $\mathcal{N}^4 \times n^2$ these trajectories work in terms of state vectors of size $\mathcal{N}^2 \times n$. This is somewhat counteracted by the stochastic nature of the formalism which makes it necessary to repeat the simulation until the wanted accuracy is reached. It turns out, however, that for many cases, especially for high dimensional quantum systems, the necessary number of repetitions is much smaller than the system size $\mathcal{N}^2 \times n$ and therefore using the MCWF method is advantageous.

The system size stems from the fact, that in the single excitation subspace for both electronic and vibronic modes a general state vector can be written as

$$|\Psi\rangle = \sum_{j=1}^{\mathcal{N}} \alpha_{jj'}^{(m)} |g, g, ...e_j, ...\rangle \otimes \sum_{m=1}^{n}\sum_{j'=1}^{\mathcal{N}} |0, 0, ...1_{j'}, ...\rangle^{(m)}, \tag{S.16}$$

with coefficients $\alpha_{jj'}^{(m)}$ and where the first part refers to the electronic excitation of molecule j and the second part to the excitation of the m-th vibrational mode of molecule j'. Thus, the single excitation assumption substantially reduces the Hilbert space dimension from $2^{\mathcal{N}} \times n_{\mathrm{cut}}^{\mathcal{N}} \times n$ to $\mathcal{N}^2 \times n$ (where n_{cut} is the cut-off of the Fock space dimension for the vibrational modes), allowing the simulation of mesoscopic numbers of molecules.

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2.4 P4: Linear optical elements based on cooperative subwavelength emitter arrays

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- All calculations and numerical simulations.
- Production of all plots.
- Writing initial draft of the manuscript except for introduction.
- Contributions to rewrites and finalization of the main text (20%).

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Linear optical elements based on cooperative subwavelength emitter arrays

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Abstract: We describe applications of two-dimensional subwavelength quantum emitter arrays as efficient optical elements in the linear regime. For normally incident light, the cooperative optical response, stemming from emitter-emitter dipole exchanges, allows the control of the array's transmission, its resonance frequency, and bandwidth. Operations on fully polarized incident light, such as generic linear and circular polarizers as well as phase retarders can be engineered and described in terms of Jones matrices. Our analytical approach and accompanying numerical simulations identify optimal regimes for such operations and reveal the importance of adjusting the array geometry and of the careful tuning of the external magnetic fields amplitude and direction.

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1. Introduction

Subwavelength arrays of emitters react cooperatively to coherent laser illumination, owing to the strong intrinsic near-field dipole-dipole interactions [1–4]. In essence, their linear optical response is similar to that of arrays of classical dipoles, such as is the case for plasmonic lattices [5–9], in the fact that light couples to delocalized surface excitations with tunable resonances. In addition, the linewidths of these collective resonances can also be tuned: the collective coupling of all emitters to the ubiquitous electromagnetic vacuum, in which they are inherently embedded, leads to super- or subradiant behavior, i.e., spontaneous emission rates smaller or larger than the independent emitter radiative rate [10,11]. Theoretical and experimental proposals are currently exploring the fascinating properties of emitter arrays with potential use as metasurfaces which can e.g. exhibit perfect reflectivity [12–16], for applications as photon storage devices and in quantum information processing [17–23], or as atom-thick membranes with extremely small mass, useful for optomechanical applications [24,25]. In the direction of topological quantum optics [26–29], it has been proposed to combine the subradiant protection of excitation against radiative losses with the topological protection of excitation against structural defects. The intrinsic nonlinearity of electronic transitions adds further possibilities into applications in nonlinear optics [30–33]. Possible experimental implementations of quantum emitter arrays include atoms trapped in optical lattices [14,34] as well as localized excitons in semiconductor structures [35,36] where each platform is subject to different kinds of defects and disorder.

Here, we theoretically characterize the functionality of two-dimensional subwavelength emitter arrays as linear optical elements, such as polarizers or waveplates. More specifically, we show that tuning of the array geometry combined with the application of an externally controllable magnetic field allows for the implementation of quasi general Jones matrices. For fully polarized incident light, such 2×2 matrices describe polarizers, i.e., filters for light polarization perpendicular to some chosen axis, or phase retarders exhibiting controllable delays $e^{i\phi_x}$ and $e^{i\phi_y}$ for the orthogonal x- and y-polarization components. The main tuning knob revealed by our theoretical analysis is the possibility of manipulating the band structure of the collective excitations, residing on the surface

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of the array, by the control of the magnitude and direction of an externally applied magnetic field. In particular, for the simplified situation of normal, uniform plane-wave illumination, only the zero-momentum mode surface excitation is driven, i.e., a symmetric collective mode of all emitter dipoles oscillating in phase. In such a case, the tuning up of the magnetic field allows to create a large gap between the optimal points of the reflection windows for orthogonal polarizations. For example, a large magnetic field in the x direction can help shift the reflectivity window of the y-polarized light thus allowing it to be completely transmitted while at the same time fully reflecting the x-polarization component. The situation is roughly depicted in Fig. 1 which illustrates the array geometry, the internal electronic structure of an individual quantum emitter, the field intensity for the two orthogonal polarizations and the frequency gap in transmission peaks between the two polarizations, introduced by a controllable externally applied magnetic field.

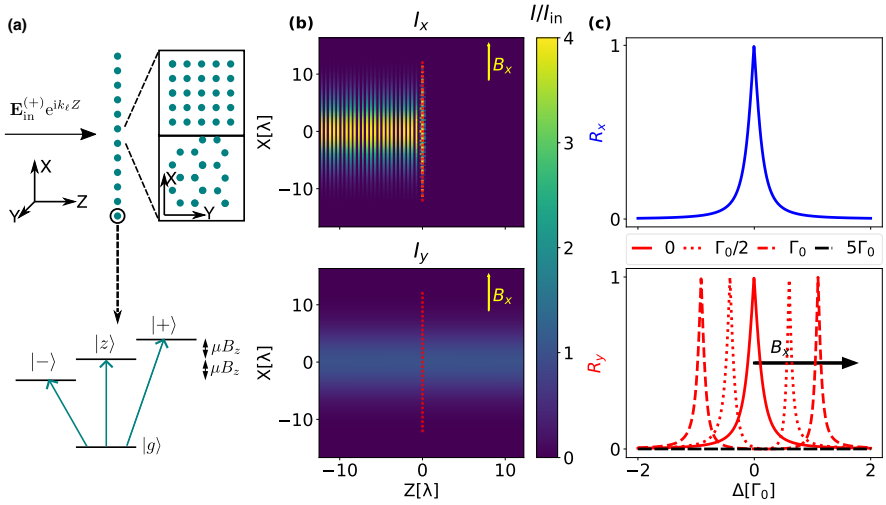


Fig. 1. (a) An incoming plane wave propagating along the z direction is impinging, at normal incidence, on a two-dimensional subwavelength quantum emitter array. The inset shows two possible arrangements of the emitters comprising the metasurface. The lower panel shows the internal electronic structure of each emitter, where the excited electronic states can be split in energy by an externally controlled magnetic field (here pointing in the z direction). (b) Field intensity distribution for the x polarization (upper plot) and y polarization (lower plot) for a metasurface acted upon with a magnetic field in the x direction. The array acts as a perfect polarizer, fully reflecting the x-polarized light while allowing full transmission for the orthogonal y-polarization component. (c) Scan of the reflectivity for the x-polarized light (upper plot) and y-polarized light (lower plot) as a function of the incoming field frequency (quantified by the detuning Δ with respect to the bare array resonance). An increasing magnetic field in the x direction allows for the progressive shift of the reflection peak for the y polarization far enough from the reflectivity window, thus allowing for the implementation of a perfect polarizer.

The analytical procedure involves the following steps: i) connecting the outgoing, far field to the cooperative response of the emitters, ii) computing the response of the array to the incident field in both real space and Fourier domains and iii) connecting the total outgoing, transmitted field to the incoming field via a transmission matrix such that $\mathbf{E}^{(+)}(Z{>}0) = \mathcal{T}\mathbf{E}_{\rm in}^{(+)}(Z{<}0)$. On the way, a reduction from a three-component treatment to two components is performed owing to the subwavelength nature of the array, which imposes that the z-polarized component of the outgoing field always vanishes. This allows the derivation of an analytical expression of a 2×2

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Jones matrix directly connecting the two components of the outgoing field to the ones of the incoming field [37,38]. These analytical expressions allows to identify the tuning knobs for the on-demand implementation of a subset of polarizers or waveplates and to manipulate the effective macroscopic polarizability tensor of the array.

The paper is organized as follows: The analytical formalism is introduced in Sec. 2. The far field emitted as a cooperative response to normal incident illumination is connected to the dipolar response of the array in the weak driving limit. The array response is computed with an open system approach, in the form of a master equation, to light-matter interactions. In Sec. 3, the reduction to a two-by-two description is detailed which allows then for the derivation of a general expression for the Jones matrices under the action of an external magnetic field. The implementation procedure for polarizers and waveplates is detailed in Secs. 3.2 and 3.3. Additional aspects relevant for experimental implementations such as the effect of thermal motion, defects and the departure from the weak excitation condition are discussed in Sec. 4.

2. Linear optical response of a two-dimensional array

We consider the situation depicted in Fig. 1(a) showing a fully polarized incident field with wave vector $k_\ell \hat{e}_z$, frequency $\omega_\ell = 2\pi c/\lambda$ (with wavelength $\lambda = 2\pi/k_\ell$) and electric field amplitudes $E_{\text{in},x}$ and $E_{\text{in},y}$ (such that the total polarization points in some direction θ in the xy plane). The field impinges, at normal incidence, upon a subwavelength quantum emitter array located in the xy plane at $z = 0$. Later, we will generalize the approach to oblique incidence. The array is comprised of $\mathcal{N}$ identical emitters, each with an internal structure described by a $J = 0$ to $J = 1$ transition, such that an externally applied magnetic field $\mathbf{B}$ can split the degeneracy of the three excited sublevels by Zeeman frequency shifting. This is depicted in the lower panel of Fig. 1(a). While the transitions used in current experiments exhibit a more complex magnetic substructure [14], this transition provides the minimal increase in complexity from the standard two-level system approximation. The results presented in the next section for the implementation of polarizers are completely generic as we show that in the limit of large magnetic fields, the dynamics can always be mapped onto an effective two-level description. For the implementation of waveplates, one has to be more careful since the magnetically-induced couplings between different transitions become important. Denoting the four states by $|g\rangle$ and $|\nu\rangle$ (with $\nu = 0, \pm$), the transition dipole operator for each emitter can be written as $\mathbf{d} = \sum_\nu \mathbf{d}_\nu \sigma_\nu$ + h.c. where $\mathbf{d}_\nu = \langle g|\mathbf{d}|\nu\rangle$ and $\sigma_\nu = |g\rangle\langle\nu|$ is the corresponding lowering operator for the transition. Alternatively, we can define combinations $\sigma_x = \sigma_- + \sigma_+$ and $\sigma_y = \mathrm{i}(\sigma_- - \sigma_+)$ and rewrite $\mathbf{d} = \mathbf{d}^{(+)} + \mathbf{d}^{(-)}$, in terms of positive and negative frequency components. The positive frequency component is then expressed in a Cartesian basis as $\mathbf{d}^{(+)} = d_x\sigma_x\hat{e}_x + d_y\sigma_y\hat{e}_y + d_z\sigma_z\hat{e}_z$ (with the negative component obtained as its Hermitian conjugate). We will denote by the index $\alpha = x, y, z$ any components in the Cartesian basis. The scattering from such atomic arrays has been discussed in many previous works [12,13,39] and we essentially follow the formalism introduced in Ref. [13].

2.1. Dynamics of a driven array

We assume that the frequency splitting (for the degenerate case, in the absence of an externally applied magnetic field) for all 3 possible optical transitions is ω_0 ($\hbar = 1$) which is detuned by $\Delta = \omega_\ell - \omega_0$ from the laser frequency. Denoting now each lowering operator in the Cartesian basis as $\sigma_{j,\alpha}$ (for the α transition within the particular emitter j) we can write the Hamiltonian of a driven emitter in the frame of the laser as a sum between the free evolution part $\mathcal{H}_0$ and the driving part $\mathcal{H}_\ell$

$$\mathcal{H}_0 + \mathcal{H}_\ell = -\sum_{j,\alpha} \Delta\sigma^\dagger_{j,\alpha}\sigma_{j,\alpha} + \sum_{j,\alpha}\left(\eta_\alpha\sigma^\dagger_{j,\alpha} + \eta^*_\alpha\sigma_{j,\alpha}\right). \tag{1}$$

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We have assumed constant, uniform illumination such that the Rabi frequencies $\eta_x = d_x E_{\text{in},x}$, $\eta_y = d_y E_{\text{in},y}$, and $\eta_z = 0$ are independent of the emitter position.

As the emitters are closely positioned with respect to each other, dipole-dipole shifts will occur, which can be encompassed in the following Hamiltonian

$$\mathcal{H}_{\text{d-d}} = \sum_{j,j',\alpha,\alpha'} \Omega_{jj'}^{\alpha,\alpha'} \sigma_{j,\alpha}^{\dagger} \sigma_{j',\alpha'}, \tag{2}$$

describing an exchange of excitation between two emitters indexed by j and j' and between transitions α and α'. This can be derived from the general free space photonic Green's tensor (see Appendix A for more details)

$$\mathbf{G}(\mathbf{R}) = \left(\mathbb{1} + \frac{1}{k_0^2}\nabla \otimes \nabla\right) \frac{\mathrm{e}^{\mathrm{i}k_0 R}}{4\pi R} - \frac{\mathbb{1}}{3k_0^2}\delta(\mathbf{R}), \tag{3}$$

where $\omega_0 = ck_0$, the symbol $\otimes$ denotes the dyadic product, $R = |\mathbf{R}|$ and the last term removes the divergence on the self interaction terms at $\mathbf{R} = 0$. The exchange rate $\Omega_{jj'}^{\alpha,\alpha'}$ is then given as [40–42]

$$\Omega_{jj'}^{\alpha,\alpha'} \equiv \Omega^{\alpha,\alpha'}(\mathbf{r}_{jj'}) = -\mu_0 \omega_0^2 \, \mathbf{d}_{\alpha}^{*} \cdot \text{Re}[\mathbf{G}(\mathbf{r}_{jj'})] \cdot \mathbf{d}_{\alpha'}, \tag{4}$$

which strongly depends on the inter-emitter separation $\mathbf{r}_{jj'} = \mathbf{r}_j - \mathbf{r}_{j'}$ and scales as $|\mathbf{r}_{jj'}|^{-3}$ in the near-field region.

In addition, an externally applied magnetic field $\mathbf{B}$ adds to the Hamiltonian by differentially shifting the excited levels with magnetic moment μ. For example, the z component B_z shifts the $|\pm\rangle$ levels by $\pm\Delta_B^{(z)} = \pm\mu B_z$. In the Cartesian basis, this can be reexpressed as

$$\mathcal{H}_{\text{B}}^{(z)} = \mathrm{i} \sum_j \Delta_B^{(z)} (\sigma_{j,y}^{\dagger}\sigma_{j,x} - \sigma_{j,x}^{\dagger}\sigma_{j,y}), \tag{5}$$

which can be seen as a mixing or excitation hopping between different levels within the same emitter. The expression for $\mathcal{H}_{\text{B}}^{(x,y)}$ is similar with the cyclic permutation of indexes (see Appendix B).

To the coherent processes in the Hamiltonian, one adds the collective decay imposed by the interaction with the electromagnetic vacuum

$$\mathcal{L}[\rho] = \sum_{j,j',\alpha,\alpha'} \Gamma_{jj'}^{\alpha,\alpha'} \left[\sigma_{j,\alpha}\rho\sigma_{j',\alpha'}^{\dagger} - \frac{1}{2}\left\{\sigma_{j,\alpha}^{\dagger}\sigma_{j',\alpha'}, \rho\right\}\right], \tag{6}$$

which describes a superoperator acting on the density operator of the system ρ. The decay terms are also obtained from the Green's tensor by [40–42]

$$\Gamma_{jj'}^{\alpha,\alpha'} \equiv \Gamma^{\alpha,\alpha'}(\mathbf{r}_{jj'}) = 2\mu_0 \omega_0^2 \, \mathbf{d}_{\alpha}^{*} \cdot \text{Im}[\mathbf{G}(\mathbf{r}_{jj'})] \cdot \mathbf{d}_{\alpha'}. \tag{7}$$

The independent decay of an individual emitter is given by $\Gamma_{jj}^{\alpha,\alpha'} = \Gamma_0 \delta_{\alpha,\alpha'}$ with the spontaneous emission rate $\Gamma_0 = \omega_0^3 d^2/(3\pi\epsilon_0 c^3)$ (we assume all transitions to have identical dipole moment $d_\alpha = d$ in the following). The dynamics of the system is then described by an open system master equation for ρ which reads

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \mathrm{i}[\rho, \mathcal{H}_0 + \mathcal{H}_\ell + \mathcal{H}_{\text{d-d}} + \mathcal{H}_{\text{B}}] + \mathcal{L}[\rho]. \tag{8}$$

We will make use of this master equation in Sec. 2.3 to compute the linear optical response as quantified by a macroscopic polarizability tensor for the whole two-dimensional array.

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2.2. Far radiated field

The field emitted by a two-dimensional metasurface comprised of $\mathcal{N}$ emitters is the sum of the individually emitted dipole fields. However, the high density, i.e., small emitter-emitter separations, resulting in collective phase shifts and collective dissipation, will give rise to a more complex, cooperative response. Let us denote the individual emitters by an index j and compute the classical electric field amplitude in the far field at some position on the z axis at distance $Z>0$. For simplicity, we restrict the discussion to uniform illumination at normal incidence, with a plane wave of positive frequency component $\mathbf{E}_{\text{in}}^{(+)}\mathrm{e}^{\mathrm{i}k_\ell z}$. The input field can be written as a three-component vector $(E_{\text{in},x}, E_{\text{in},y}, 0)^\top$. The far field vector is then related to the array response as

$$\mathbf{E}^{(+)}(Z) = \mathbf{E}_{\text{in}}^{(+)}\mathrm{e}^{\mathrm{i}k_\ell Z} - \frac{3\pi\Gamma_0}{k_0 d}\sum_{j=1}^{\mathcal{N}} \mathbf{G}^{\text{far}}(Z\hat{e}_z - \mathbf{r}_j)\boldsymbol{\beta}_j, \tag{9}$$

where each of the $\boldsymbol{\beta}_j = \langle \boldsymbol{\sigma}_j \rangle$ is a three-component vector, containing the Cartesian components of the expectation values of the lowering operators $\boldsymbol{\sigma}_j$. The sum above applies to all the emitters in the array situated at positions $\mathbf{r}_j$. The dipolar response is computed via the far-field component of the Green's tensor $\mathbf{G}^{\text{far}}(Z\hat{e}_z - \mathbf{r}_j)$, which is a simplified version of the already invoked Green's tensor under the assumption that $k_0 Z \gg 1$. Generally, for an arbitrary array, the dipole-emitted field could have a component in the direction of propagation. As it will be shown in Sec. 3, in the particular case of subwavelength arrays, the far field contains only xy components, even if the array dipoles have a z component.

2.3. Surface modes of the array

In the low-excitation limit where each electronic transition is only weakly excited $\langle \sigma_{j,\alpha}^z \rangle \approx -1$, the response of the emitter array can be computed from the master equation approach to open quantum systems, as detailed in the previous subsection. In particular, we consider the limit of a weak classical drive where all system correlation functions factorize and we consider only the quantum averages. One can alternatively consider the single-excitation subspace which leads to similar equations of motion described by an effective non-Hermitian Hamiltonian [27–29]. The coupling of the incident field to the electronic transitions is quantified by the vector of Rabi frequencies $\boldsymbol{\eta}$ with components $(\eta_x, \eta_y, 0)$. We will proceed in analyzing the dipolar response both in the position space and via a Fourier transform allowing the understanding in terms of quasi-momenta residing in the reciprocal space. In the position space, one can derive a set of equations of motion for the three-component coherence of each j-indexed emitter

$$\dot{\boldsymbol{\beta}}_j = -\mathrm{i}\sum_{j'} \mathbf{M}_{jj'}\boldsymbol{\beta}_{j'} - \mathrm{i}\boldsymbol{\eta}. \tag{10}$$

The matrix $\mathbf{M}_{jj'} = -\Delta\mathbb{1}\delta_{j,j'} + \boldsymbol{\Omega}_{jj'} - \mathrm{i}\boldsymbol{\Gamma}_{jj'}/2 + \mathbf{M}_{\text{B}}\delta_{j,j'}$ contains the laser detuning, the effect of the coherent and incoherent interactions induced by the electromagnetic vacuum and the couplings introduced by the magnetic field in the Cartesian basis (explicitly specified in Appendix D). Under the assumption of uniform illumination and that the array is quasi-infinite, all emitters are identically excited in steady-state $\boldsymbol{\beta}_j = \boldsymbol{\beta}$ with the vector of amplitudes

$$\boldsymbol{\beta} = -\boldsymbol{\mathcal{M}}^{-1}\boldsymbol{\eta}, \tag{11}$$

where the matrix $\boldsymbol{\mathcal{M}} = \sum_{j'} \mathbf{M}_{0j'}$ does not depend on the index j for an infinite array which is why we set it to zero. This expression will later allow us to simply compute the response of the array in the next section by replacing it into Eq. (9). Notice that $\boldsymbol{\mathcal{M}}$ is a 3×3 matrix and even if the incoming field has a vanishing z component, the dipole pattern imprinted onto the array can have a z component.

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The solution discussed in the previous paragraph corresponds to uniform, perpendicular illumination, where only the symmetric surface mode is activated. More generally, for non-normal incidence of the laser with a wave vector $\mathbf{k}_\parallel$ parallel to the array plane, the equations of motion can be expressed in Fourier space as

$$\dot{\tilde{\boldsymbol{\beta}}}_{\mathbf{q}} = -\mathrm{i}\boldsymbol{\mathcal{M}}(\mathbf{q})\tilde{\boldsymbol{\beta}}_{\mathbf{q}} - \mathrm{i}\mathcal{N}\boldsymbol{\eta}\delta_{\mathbf{q},\mathbf{k}_\parallel}, \tag{12}$$

where the evolution matrix is now given by $\boldsymbol{\mathcal{M}}(\mathbf{q}) = -\Delta\mathbb{1} + \tilde{\boldsymbol{\Omega}}(\mathbf{q}) - \mathrm{i}\tilde{\boldsymbol{\Gamma}}(\mathbf{q})/2 + \mathbf{M}_\mathrm{B}$ and contains the matrices describing the effective frequency shifts and decay rates as a function of the in-plane momentum $\tilde{\boldsymbol{\Omega}}(\mathbf{q}) - \mathrm{i}\tilde{\boldsymbol{\Gamma}}(\mathbf{q})/2$ which are modified by the presence of the magnetic field $\mathbf{M}_\mathrm{B}$. Diagonalization of the matrix $\boldsymbol{\mathcal{M}}(\mathbf{q})$ gives rise to the photonic band structure of the array (see paragraph below). Equation (12) indicates that only the mode corresponding to $\mathbf{q} = \mathbf{k}_\parallel$ is driven by the laser. The symmetric illumination scenario discussed in the previous paragraph then simply refers to $\mathbf{q} = 0$. The real and imaginary parts of the matrices describing the effective frequency shift and decay rate of a given $\mathbf{q}$ mode of the array are obtained by the Fourier transforms $\tilde{\boldsymbol{\Omega}}(\mathbf{q}) = \sum_{\mathbf{r}\in\Lambda} \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{r}}\boldsymbol{\Omega}(\mathbf{r})$ and $\tilde{\boldsymbol{\Gamma}}(\mathbf{q}) = \sum_{\mathbf{r}\in\Lambda} \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{r}}\boldsymbol{\Gamma}(\mathbf{r})$ involving a summation over all vectors of the lattice $\mathbf{r} \in \Lambda$. This can be compactly expressed in terms of the Green's tensor as

$$\tilde{\boldsymbol{\Omega}}(\mathbf{q}) - \mathrm{i}\frac{\tilde{\boldsymbol{\Gamma}}(\mathbf{q})}{2} = -\frac{3}{2}\Gamma_0\lambda_0 \sum_{\mathbf{r}\in\Lambda} \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{r}}\mathbf{G}(\mathbf{r}). \tag{13}$$

While analytical expressions for the dipole-dipole interactions are not available analytically, for subwavelength arrays excited at normal incidence, the relevant decay rate matrix at zero quasi-momentum can be approximated by (for derivation see Appendix C)

$$\tilde{\boldsymbol{\Gamma}}(0) = \frac{3\Gamma_0}{4\pi}\frac{\lambda_0^2}{\mathcal{A}}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{14}$$

where $\mathcal{A}$ is the area of the unit cell of the real-space lattice. This result implies that for the symmetric mode of subwavelength lattices, spontaneous emission of the dipoles along the z direction is completely suppressed while it assumes the simple expression $\tilde{\Gamma}(0) = 3(\Gamma_0/4\pi)(\lambda_0^2/\mathcal{A})$ for the other two directions.

The modes of the array define the response to external illumination via the polarizability tensor. This can be expressed in the Fourier domain simply as $\boldsymbol{\alpha}(\mathbf{q}) = d^2\boldsymbol{\mathcal{M}}^{-1}(\mathbf{q})$. For a single incident polarization this assumes a simple form derived in Refs. [12,13] while here, owing to its cross-coupling components, it can describe changes in one polarization owed to the other orthogonal one. The photonic band structure and polarizability are illustrated in Fig. 2 in the absence (left column) and with an applied magnetic field (right column). The photonic band structure is obtained from the diagonalization of the matrix $\boldsymbol{\mathcal{M}}(\mathbf{q})$, giving rise to three energy bands (for lattices containing a single emitter per unit cell). While in Figs. 2(a),(b) the color coding indicates the decay rate, in Figs. 2(c),(d) the color shows the relative $x/y/z$ content of a respective mode. In the absence of an external field, at the Γ point (corresponding to zero quasi-momenta both in the x and y directions) the x and y bands are completely degenerate, owing to the symmetry of the square lattice, which is also reflected in the fact that the cross term of the polarizability tensor α_{xy} is zero, implying that the bands do not hybridize. The z band does not have this symmetry constraint and its resonance is at a different frequency at the Γ point. Going away from the Γ point, the xy bands start to hybridize slightly leading to different band shapes. Different bands for x and y also emerge due to the breaking of the xy symmetry by picking a wave vector which does not obey this symmetry. Turning on a magnetic field in the x direction

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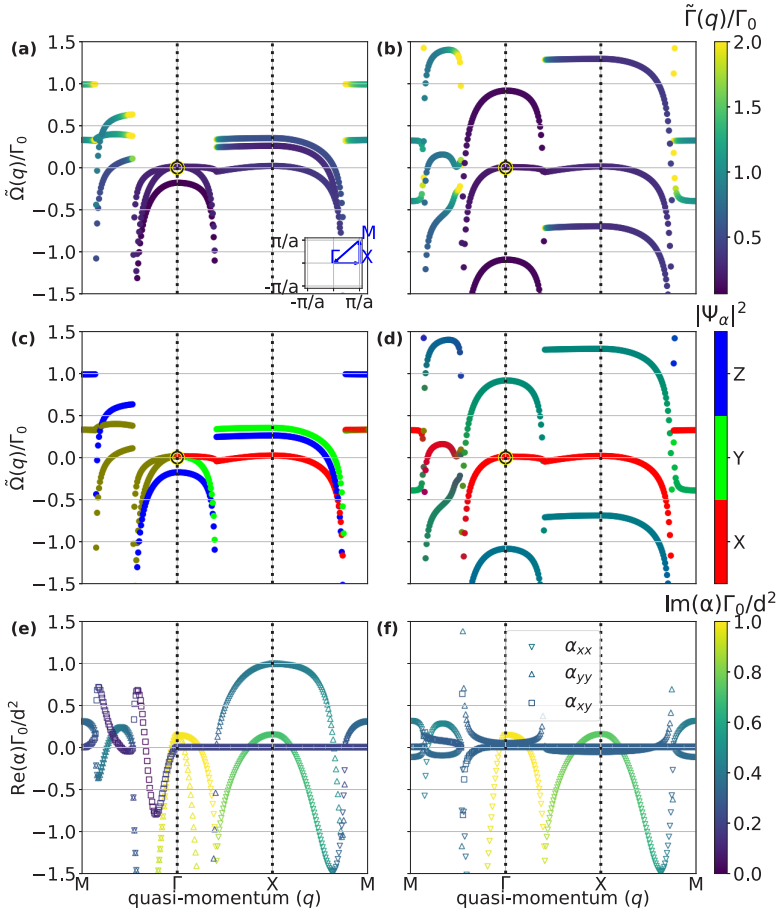


Fig. 2. (a)-(d) Photonic band structures (eigenvalues of $\boldsymbol{\mathcal{M}}(\mathbf{q})$ for $\Delta = 0$) and (e)-(f) polarizability of the emitter array in the absence (left column) and with an applied magnetic field in x direction (right column) for a square lattice with spacing $a = 0.8\lambda$. The inset in (a) shows the path taken between points in the 2D reciprocal space. (a) Real part of the photonic band structure with color coding indicating the corresponding imaginary part, i.e., the decay rate which determines how sub- or superradiant a mode is at the particular quasi-momentum. For $a = 0.8\lambda$, the decay rate at the Γ point is subradiant for instance. The pointer indicates the perfect reflectivity point situated at the Γ point for both incoming field polarizations. In (b), the magnetic field shifts the reflectivity point for the y component but not for the x component. This is more evident in (c) and (d) where the color coding indicates the overlap of the eigenstate with the Cartesian components such that the RGB color is defined as $(|\Psi_x|^2, |\Psi_y|^2, |\Psi_z|^2)$. The applied magnetic field hybridizes the yz bands and shifts the resonances of these bands away from the zero energy point. In (e) and (f) we illustrate the real part and, in the color scale, the imaginary part of certain components α_{xx}, α_{yy}, and α_{xy} of the polarizability tensor. The applied magnetic field for (b), (d), and (f) is $\mu B_x = \Gamma_0$.

has dramatic effects on all bands except the x band which is pure as seen in Figs. 2(b),(d). Indeed, the strong hybridization of the y and z band induced by the external magnetic field in x direction leads to a strong splitting of these bands around $\tilde{\Omega}(\mathbf{q}) = 0$ so that a laser with frequency ω_0 is only resonant with the x band. This can also be observed in the polarizability tensor plot on the

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bottom right where all matrix elements of the polarizability tensor are close to zero except α_{xx} for almost all $\mathbf{q}$. Indeed, the structure of $\alpha_{xx}(\mathbf{q})$ remains unchanged. This is also illustrated in Fig. 3(b) where we plot the entries of the polarizability tensor as a function of the magnetic field along the x direction μB_x, showing that the y component becomes close to zero for $\mu B_x > \tilde{\Gamma}(0)$. The gapping of the bands in the band-structure picture is therefore equivalent to an effective two-level system description in the polarizability tensor picture.

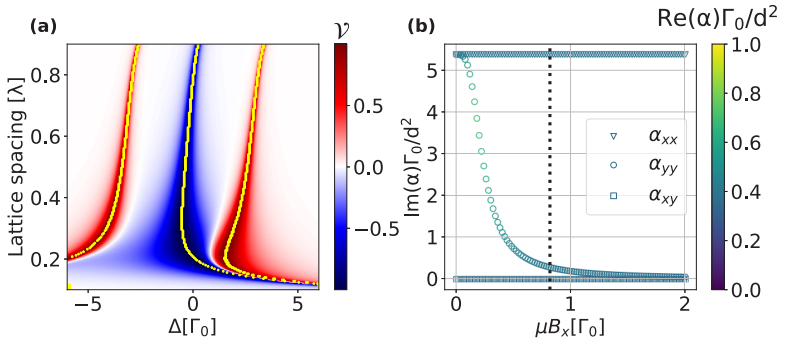


Fig. 3. (a) Visibility $\mathcal{V}$ of the x- and y-polarization components as a function of the laser detuning and lattice spacing (for a square lattice). An external magnetic field in the x direction with magnitude $\mu B_x = 3\Gamma_0$ is applied. The blue areas indicate a y-polarizer and the red regions an x-polarizer implementation. Optimal parameters (corresponding to the resonances of the bands) are highlighted by yellow lines. The condition for obtaining a perfect polarizer, for the $a = 0.8\lambda$ case, is when the laser is resonant to the collective surface excitation at the Γ point (see also Fig. 2(b)). (b) Components of the polarizability tensor (imaginary part on the y axis, real part in color) as a function of the applied magnetic field in x direction. With increasing field, α_{yy} is reduced until it is negligible for values much larger than the natural linewidth of the collective modes $\tilde{\Gamma}(0)$ indicated by the dotted black line.

3. Linear optical elements

The transmitted field amplitude through a subwavelength array is computed by plugging the result of Eq. (11) into Eq. (9). The dipole contribution then reads

$$\mathbf{E}^{(+)}_{\mathrm{dip}}(Z) = \frac{3\pi\Gamma_0}{k_0}\left[\sum_{j=1}^{N}\mathbf{G}^{\mathrm{far}}(Z\hat{e}_z - \mathbf{r}_j)\right]\boldsymbol{\mathcal{M}}^{-1}\mathbf{E}^{(+)}_{\mathrm{in}}, \tag{15}$$

thus simply requiring the estimate of the contribution within the square brackets (which we denote in the following by $\boldsymbol{\mathcal{I}}$). While generally, for arbitrarily chosen lattice geometries, the dipole field can contain many contributions, we will see in the next subsection that, subwavelength arrays impose the cancelations of all other modes except for the fundamental mode at $\mathbf{q} = 0$.

3.1. Derivation of the 2D transmission matrix

To estimate the sum in the brackets in Eq. (15) (see Appendix C for full details of the calculation), we make use of the wavevectors $\mathbf{g}$ in the reciprocal lattice Λ^* of the array. Any sum over individual sites in the lattice space Λ can then be evaluated by an equivalent sum performed in the reciprocal space Λ^*. Making use of the general form of the Green's tensor (no far field approximation is yet

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required) the sum becomes

$$\mathcal{I} = \frac{\mathrm{i}}{2k_0^2\mathcal{A}} \sum_{\mathbf{g}\in\Lambda^*} \frac{k_0^2\mathbb{1} - \mathbf{v}(\mathbf{g},Z)\otimes\mathbf{v}(\mathbf{g},Z)}{\sqrt{k_0^2-g^2}} \mathrm{e}^{\mathrm{i}\sqrt{k_0^2-g^2}|Z|} \tag{16}$$

where $\mathbf{v}(\mathbf{g},Z)$ is a three-dimensional vector with components of $\mathbf{g}$ in the x and y directions and component $\mathrm{sgn}(Z)\sqrt{k_0^2-g^2}$ in the z direction. The main observation comes here by a quick inspection of the exponent $\mathrm{e}^{\mathrm{i}\sqrt{k_0^2-g^2}|Z|}$. For any subwavelength lattice, only the value $g = 0$ can lead to a purely imaginary exponent thus describing a propagating plane wave; otherwise, only evanescent wave solutions are obtained as $g^2 > k_0^2$. Therefore, we can simply evaluate the contribution of $\mathbf{g} = (0,0)$ and obtain

$$\mathcal{I} = \frac{\mathrm{i}}{2k_0\mathcal{A}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathrm{e}^{\mathrm{i}k_0|Z|} \tag{17}$$

resulting in a total field (both in reflection, for $Z<0$, and in transmission, for $Z>0$) obtained as a sum between incident and scattered field

$$\mathbf{E}^{(+)}(Z) = \left[\mathbb{1}\mathrm{e}^{\mathrm{i}k_\ell Z} + \boldsymbol{\mathcal{S}}\mathrm{e}^{\mathrm{i}k_\ell|Z|}\right]\mathbf{E}_{\mathrm{in}}^{(+)}, \tag{18}$$

where we used $k_\ell \approx k_0$ which is also assumed in the rest of the text. The scattering matrix is defined as (see Appendix D for more details such as generalized expression for oblique incidence)

$$\boldsymbol{\mathcal{S}} = \mathrm{i}\frac{\tilde{\Gamma}(0)}{2}[\boldsymbol{\mathcal{M}}^{-1}]_{\mathrm{red}} = \mathrm{i}\frac{\tilde{\Gamma}(0)}{2d^2}\boldsymbol{\alpha}_{\mathrm{red}}, \tag{19}$$

being easily connected to the polarizability tensor. Here, both $[\boldsymbol{\mathcal{M}}^{-1}]_{\mathrm{red}}$ and $\boldsymbol{\alpha}_{\mathrm{red}}$ are the reduced 2×2 matrices obtained by a projection into the xy plane of the array. This stems from the action of the matrix $\boldsymbol{\mathcal{I}}$ which effectively reduces the problem to 2 dimensions. Therefore, we can now define the two-dimensional transmission matrix as

$$\boldsymbol{\mathcal{T}} = \mathbb{1} + \boldsymbol{\mathcal{S}} = \mathbb{1} + \mathrm{i}\frac{\tilde{\Gamma}(0)}{2d^2}\boldsymbol{\alpha}_{\mathrm{red}}. \tag{20}$$

Note that the subwavelength nature of the array guarantees that, for normal incidence, even if the dipolar response of the surface can have z-polarized components, the emitter far field will only contain the x- and y-polarization components. This allows an input-output formalism where the action of the array is described by a 2×2 Jones matrix.

3.2. Polarizers

The two simplest examples of Jones matrices [37]

$$\boldsymbol{\mathcal{T}}_x = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \boldsymbol{\mathcal{T}}_y = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \tag{21}$$

describe a polarizer in one direction (either x or y) while completely canceling the propagation of the other polarization component. In order to produce such an effect, we make use of a magnetic field oriented along a conveniently chosen direction. The effect of the magnetic field

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is then to shift the position of the maximum reflectivity points in such a way that the array is not simultaneously reflective/transmissive for both polarization directions. A useful, alternative picture is that of 'stripping' the multilevel system of additional levels, until a single two-level system with an appropriately chosen dipole moment orientation is obtained. More precisely, the magnetic field can be used to pull all adjacent undesired levels far from resonance with the incident laser frequency. To see this, we consider the transmission matrix assuming an applied magnetic field $\mathbf{B} = B_x \hat{e}_x$ and take the magnetic field towards large values compared to the decay rates such that (see Appendix D for full expression)

$$\lim_{B_x\to\infty} \mathcal{T} = \mathbb{1} + \frac{\mathrm{i}\tilde{\Gamma}(0)/2}{\tilde{\Omega}^{xx}(0) - \Delta - \mathrm{i}\tilde{\Gamma}(0)/2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \tag{22}$$

where we used that $\tilde{\Omega}^{xx}(0) = \tilde{\Omega}^{yy}(0)$ as it stands for a square lattice. For resonant illumination of the array $\omega_\ell = \omega_0 + \tilde{\Omega}^{xx}(0)$, this describes a y-polarizing action in transmission

$$\lim_{B_x\to\infty} \mathcal{T} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \mathcal{T}_y, \tag{23}$$

i.e., only the y component is transmitted while the x component is perfectly reflected (also compare Figs. 1(b),(c)). Close to perfect polarizers can be reached as long as the magnetically induced frequency shifts are much larger than $\tilde{\Gamma}(0)$. Analogously, an x polarizer described by the Jones matrix $\mathcal{T}_x$ can be implemented by a magnetic field $\mathbf{B} \parallel \hat{e}_y$. The action of a polarizer for different lattice constants at a magnetic field strength $\mu B_x = 3\Gamma_0$ is illustrated in Fig. 3(a), by plotting the visibility $\mathcal{V} = \left(I_x - I_y\right) / \left(I_x + I_y\right)$ of the intensity components. The particular value of μB_x is irrelevant as long as the splitting of the resulting bands is larger than the linewidth of all bands [cf. Figure 3(b)]. What can be seen in this plot are the frequencies at which a polarizer in x-direction and in y-direction performs optimally, indicated by yellow dots. What is also available is information on the linewidth of this polarizing action which is determined by the width of the Lorentzians which emerge when taking horizontal slices of this two-dimensional plot. It is well known that at 0.2λ and at 0.8λ there exist resonances at $\Delta = 0$ [12,13] which are indeed indicated by yellow dots in the blue-shaded area of the plot. Furthermore, aside from the resonance of the x band indicated by the blue region, one also has the resonances of the hybridized yz bands at positive and negative detunings. Since the z component does not participate for normal incidence, these bands also show the perfect reflection condition for resonant illumination, therefore implementing an x polarizer.

Finally, we stress that one is not only restricted to the x or y axes: a general polarizer for an arbitrary axis (e.g. for $\theta = \pm 45°$) can always be obtained by applying a magnetic field perpendicular to that respective axis. Further more, polarizers for circularly polarized light can be implemented by the application of a magnetic field in z direction. This can be easily understood in the circular basis: By tuning the laser to the distinct resonance of the $|+\rangle/|-\rangle$ band, a polarization filter for left(−)/right(+) circularly polarized light is obtained, respectively (the other component is again perfectly reflected). This is described by the following Jones matrices [37]

$$\mathcal{T}_- = \frac{1}{2}\begin{pmatrix} 1 & -\mathrm{i} \\ \mathrm{i} & 1 \end{pmatrix}, \quad \mathcal{T}_+ = \frac{1}{2}\begin{pmatrix} 1 & \mathrm{i} \\ -\mathrm{i} & 1 \end{pmatrix}. \tag{24}$$

While these expressions for the transmission matrices are obtained in the Cartesian polarization basis $\{\hat{e}_x, \hat{e}_y, \hat{e}_z\}$, one can equivalently describe the problem in the circular polarization basis $\{\hat{e}_+, \hat{e}_-, \hat{e}_z\}$ with $\hat{e}_\pm = (\hat{e}_x \pm \mathrm{i}\hat{e}_y)/\sqrt{2}$. In this basis, the matrices $\mathcal{T}_{+/-}$ are analogous to the matrices $\mathcal{T}_{x/y}$ in the Cartesian basis.

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All together, this enables the implementation of highly tunable polarizers for both linearly and circularly polarized light on subwavelength quantum emitter arrays.

3.3. Phase retarders

A generic waveplate (or phase retarder), without losses in either components, can be described by the Jones matrix

$$\mathcal{T}_\phi = \begin{pmatrix} e^{i\phi_x} & 0 \\ 0 & e^{i\phi_y} \end{pmatrix}, \tag{25}$$

which imprints a relative phase difference of $\Delta\phi = \phi_x - \phi_y$ onto the outgoing polarization components. Quarter- and half-waveplates would correspond to phase shifts of $\Delta\phi = \pi/2$ and π, respectively.

Let us now investigate the question of the possible implementation of general waveplates with subwavelength arrays. As analytical considerations are hindered by the cumbersome expressions for the scattering matrix of the array in the presence of external magnetic fields (see Appendix D for detailed expressions) we take recourse to numerical simulations and considerations. The phase retardation effect accompanied by the inevitable loss in transmitted intensity is illustrated in Fig. 4, for a square lattice. In Fig. 4(a), one can see that a reasonable phase rotation, which is indicated by red or blue color respectively is accompanied by a corresponding reduction in the outgoing intensity with respect to the ingoing intensity on the right indicated by blue. This fundamental problem can be traced back to the fact that the resonances of the array have a Lorentzian structure meaning that the real and imaginary part of the response are strongly dependent on each other. Taking the single-band limit of the expression in Eq. (18), i.e., considering only a single scattering band with effective decay rate $\tilde{\Gamma}(0)$, frequency shift $\tilde{\Omega}(0)$ and scattering amplitude $\mathcal{S}$, the phase shift of the transmitted light due to the interaction with the array can be written as

$$\Delta\phi = \arg(1 + \mathcal{S}) = \arctan\left(\frac{\tilde{\Gamma}(0)/2}{\tilde{\Omega}(0) - \Delta}\right). \tag{26}$$

The direction of the phase shift depends on whether the detuning of the laser with respect to the collective resonance is negative or positive. This implies that the contribution of a single band to the phase shift of a polarization component can at most account for $\pi/2$ close to the resonance of the Lorentzian. Around the resonance however, the array is also highly reflective meaning that a large phase shift is accompanied with significant intensity loss. If the laser is resonant with two or more bands, this argument does no longer hold exactly, but the general idea that a larger phase shift also implies more losses due to reflection holds. In this sense, high phase shift differences are always obtained only for very reflecting surfaces. For instance, one can show that for $\Delta\phi = \pi/4$ half of the ingoing intensity of the affected polarization component is always reflected. We can instead optimize the phase shift difference for an imperfect, lossy but balanced waveplate, where the intensity difference of the incoming polarizations is kept as constant as possible. This would imply that there is simply a prefactor in front of the scattering matrix corresponding to the absolute loss of light intensity due to reflection. An attempt at such an implementation is shown in Fig. 4(c) and Fig. 4(d), where we select the appropriate magnetic field configuration for a particular lattice and laser frequency in order to define a phase-tunable waveplate. We see analytically, that the x and y component of the magnetic field must be modified in a symmetric fashion in order to keep the intensity difference as constant as possible. Hence, we choose a point where ΔI_{out} is minimal for a magnetic field only pointing in x-direction and then parameterize the magnetic field symmetrically around this point $\mu\mathbf{B}(\varepsilon) = (-1.75\Gamma_0, 0, 0) + \varepsilon\Gamma_0\left(\hat{e}_x + \hat{e}_y\right)$ (with ε as parameter). As seen in Fig. 4(d) this leads to an extended region where the outgoing intensity is modified significantly, but the intensity difference ΔI_{out} is kept minimal. Thus there exist regions

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to the left and to the right of the resonance which provide a magnetic-field tunable waveplate which reduces the intensity in both polarization components in a balanced manner. In particular, we find that a quarter-waveplate is implemented by choosing $\mu\mathbf{B} = (-1.44, 0.31, 0)\Gamma_0$ with $I_{\text{out}} = 0.77 I_{\text{in}}$ and a half-waveplate is implemented at $\mu\mathbf{B} = (-1.13, 0.62, 0)\Gamma_0$ with $I_{\text{out}} = 0.14 I_{\text{in}}$. We remark that for more complex array set-ups it has been demonstrated that generic waveplates can be implemented without losses [23].

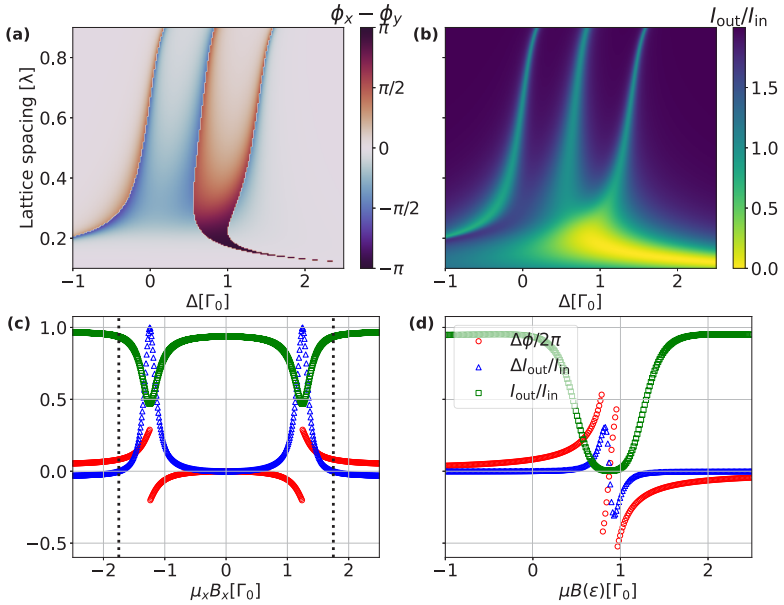


Fig. 4. (a) Phase difference $\Delta\phi = \phi_x - \phi_y$ between polarization components of the transmitted light and (b) the corresponding amount of total light transmitted by the emitter array as a function of the lattice spacing and detuning (for a square lattice). The incoming laser beam is equally polarized in x and y direction with no phase difference between them, i.e., $\Delta\phi = 0$ in the absence of the array. In (c) and (d) we show the phase difference $\Delta\phi$, the total intensity and the intensity difference $\Delta I_{\text{out}} = I_{x,\text{out}} - I_{y,\text{out}}$ for the x- and y-polarization components for different external magnetic fields for a square lattice with lattice spacing $a = 0.6\lambda$ and detuned by $\Delta = \Gamma_0$ from the bare emitter frequency. The incoming polarization is $(E_{\text{in},x}, E_{\text{in},y}) = (1, 1)$. In (c), the external magnetic field only has an x component for which we show the different quantities. Then in (d), we select the point on the left with $\mu B_x = -1.75\Gamma_0$ (as indicated by the dotted line in (c)), where the intensity difference is minimal and modify both B_x and B_y in a symmetric fashion on the right, i.e., the values on the x axis parameterize $\mu\mathbf{B}(\varepsilon)/\Gamma_0 = (\varepsilon - 1.75, \varepsilon, 0)$.

4. Discussions and extensions

A variety of additional factors can perturb the functionality of linear optical elements implemented on two-dimensional subwavelength quantum emitter arrays. We partially address a few of these aspects such as thermal effects and vacancies (which are a limiting factor for atoms trapped in optical lattices [12,14]) as well as nonlinear effects associated with high intensity driving. We discuss the influence of these effects onto the band shapes and the transmittivity/reflectivity of the array. We also discuss how non-Bravais lattices can be described which we illustrate for a honeycomb lattice.

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4.1. Effects of motion

There are different strategies to quantify the action of thermal disturbances in the equilibrium positions of the emitters in the array. In the limit in which the motion is fast compared to the light-matter interaction time scale given by Γ_0^{-1}, it can be accounted for by averaging over static disorder configurations around the equilibrium positions. Real space simulations of the effect of motion are shown in Fig. 5. The first observation is that no additional plane wave components aside from the ones propagating in z direction are generated. This is due to the fact that in the infinite-array case the averaging over configurations restores the translational symmetry such that there is no coupling between different momenta if one were to write down an effective description including motion. It is also clear that motion in-plane and out-of-plane have very different effects on the scattered light.

Including motion in z direction with Gaussian width Σ_z derogates the coherence of the backscattered light. The only effect of coupling to the z component at the Γ point in Fig. 5(f) is a small reduction of the frequency since the xy components now hybridize with the z component whose resonance is at a lower frequency at the Γ point. Including motion in the array plane with Gaussian width Σ_{xy} reduces the polarizer effect more than out-of-plane motion. From Fig. 5(e) we see that including motion in the array plane leads to a broadening of the array response indicated by a larger decay rate for larger Σ_{xy}. The broadening can be accounted for by considering that motion in the xy plane leads to coupling to other momentum modes at different frequencies. This broadening leads to a reduction in reflectivity. Note that, in order to calculate the modifications of the band diagram we define finite-lattice momentum states $|\mathbf{q}\rangle = \sum_{\mathbf{r}_i \in \Lambda} \mathrm{e}^{-\mathrm{i}\mathbf{r}_i\mathbf{q}}|i\rangle$. The band structure is then defined by calculating $E^c_{\mathbf{q}} = \langle \mathbf{q}|\mathbf{\Omega} - \mathrm{i}\mathbf{\Gamma}/2|\mathbf{q}\rangle$ for each configuration c with component matrices $\mathbf{\Omega}_{ij} = \mathbf{\Omega}(\mathbf{r}_{ij})$ and $\mathbf{\Gamma}_{ij} = \mathbf{\Gamma}(\mathbf{r}_{ij})$. The averaged band diagram is then taken to be $\bar{E}_{\mathbf{q}} = N_c/(\sum_c 1/E^c_{\mathbf{q}})$ which corresponds to an average over the polarizability of the N_c configurations. A more quantitative overview of the effect of in-plane and out-of-plane disorder is show in Fig. 5(g) where the reflectivity is shown for different disorder strengths. While the functional shape of both effects is similar, first parabolic and then becoming more constant, the reduction of the reflectivity with out-of-plane disorder is much faster than with in-plane disorder.

We also note that in this simplified description, the effect of quantum fluctuations and thermal fluctuations is completely equivalent since the wave packet of the harmonic oscillator ground state has a Gaussian shape. Similarly, thermal disorder corresponds to static Gaussian disorder.

4.2. Role of vacancies

Vacancies inherently break the translational invariance of the system. They are difficult to treat analytically and numerical studies show that their influence on the radiation pattern is significant. The effect of the breaking of translational symmetry can immediately be seen in Fig. 6 where we simulate the scattering of light on lattices with an emitter missing with probability p (from 0% to 10%) at each lattice site. In particular, for larger vacancy densities a lot of plane-wave components aside from the one propagating in z direction are generated. Since there is usually no averaging over different vacancy configurations, this implies that vacancies deprecate the polarizer effect in a completely different manner from motional disorder which does not break translational invariance. Starting from the translationally invariant picture, the vacancies induce coupling between the previously decoupled momentum modes. This leads to indirect driving of other modes coupled to the laser driven mode which then emit light into free space.

Quantitative estimations of effects that break the translational invariance are difficult since the effect depends completely on the physical configuration that is actually achieved in the system. While the plots in Fig. 6 are for vacancies, they also give qualitative statements about other effects which break the translational symmetry of the array. Since these kinds of effects introduce coupling between different momentum modes, the physics can be perturbatively understood in the original momentum space picture. The mode that is originally driven scatters into other modes

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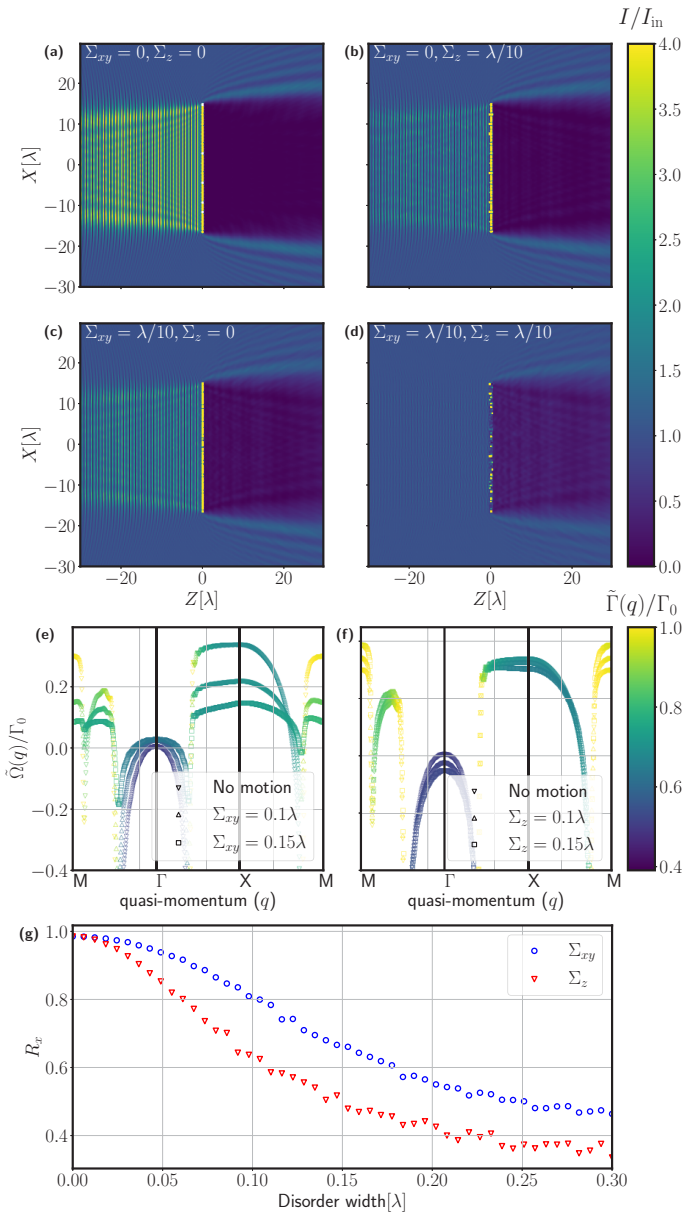


Fig. 5. Real space simulations of the electric field intensity of the x-polarization component of an incoming field with $(E_{\text{in},x}, E_{\text{in},y}) = (1, 1)$ for a square lattice with $a = 0.8\lambda$ and $\mathcal{N} = 40^2 = 1600$ emitters. The magnetic field is $\mu B_x = \Gamma_0$. The Gaussian widths of the static disorder are marked by the in-plane Σ_{xy} and out-of-plane Σ_z symbols. The numerical simulations are performed for $N_c = 100$ configurations and then averaged. (e), (f) Effective band diagram calculations including effects of motion. (e) Band diagrams for in-plane motion and (f) band diagram for out-of-plane motion for disorder widths of $\Sigma = 0, 0.1\lambda, 0.15\lambda$, respectively. (g) Plot of the average effect of in-plane and out-of-plane disorder on the reflectivity for different disorder strengths. Here we performed an average over $N_c = 1500$ configurations for each point for an array size $\mathcal{N} = 30^2$.

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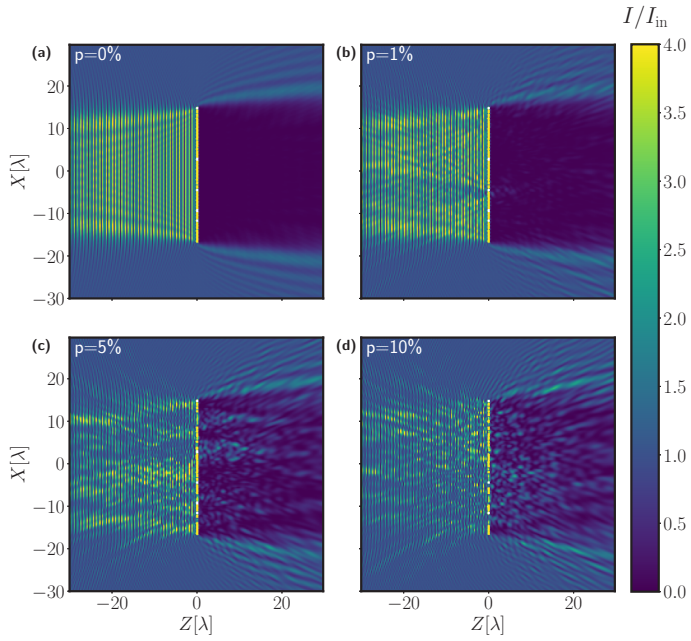


Fig. 6. Simulations of the field intensity accounting for single configurations where the lattice has a vacancy randomly located at any lattice site with a probability p. (a) 0%, (b) 1%, (c) 5%, and (d) 10%. The incoming polarization is $(E_{\mathrm{in},x}, E_{\mathrm{in},y}) = (1, 1)$ and all other parameters are kept as in Fig. 5.

in first order with the coupling strengths depending on the particular configuration. These modes then radiate into the particular plane wave outside modes that corresponds to the momentum vector of the mode.

4.3. *Beyond the linear regime*

The main focus of the paper has been devoted to the linear response of the array, i.e., the regime in which the emitters are far from saturation. Let us now consider a strong laser drive which can considerably modify the equations of motion of the emitter amplitudes, mainly leading to a power broadening effect responsible with a decrease of the overall reflectivity at higher incoming intensities. Let us first consider the instructing case of a single emitter, driven by an arbitrarily large external light source. Considering only a single polarization degree of freedom, the equations of motion for the coherence β and population inversion variable $\beta^z = \langle \sigma^\dagger \sigma - \sigma \sigma^\dagger \rangle$ are given by

$$\dot{\beta} = -\left(\frac{\Gamma_0}{2} - \mathrm{i}\Delta\right)\beta + \mathrm{i}\beta^z \eta, \tag{27a}$$

$$\dot{\beta}^z = -\Gamma_0(\beta^z + 1) - 4\eta\,\mathrm{Im}(\beta). \tag{27b}$$

Assuming steady-state, this yields a simple expression of the coherence

$$\beta = \frac{\mathrm{i}\eta(\Gamma_0/2 + \mathrm{i}\Delta)}{\Gamma_0^2/4 + \Delta^2}\beta^z = -\beta^{\mathrm{lin}}\beta^z, \tag{28}$$

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where we denote by β^{lin} the response in the linear regime, where β^z is close to -1 (low excitation regime). The population inversion is given by

$$\beta^z = -\frac{1}{1+\frac{2\eta^2}{\Gamma_0^2/4+\Delta^2}}, \tag{29}$$

showing that, with increasing saturation of the emitter, its coherence gets reduced. Assuming moderately high driving intensity, the first correction reads

$$\beta \approx \left[1-\frac{2\eta^2}{\Gamma_0^2/4+\Delta^2}\right]\beta^{\text{lin}}, \tag{30}$$

which is responsible for the Kerr effect where the polarizability of the emitter depends on the intensity of the applied light field. The overall effect on the emitter's response to the external stimulation is then a simple reduction in the overall reflectivity by the reduction in its coherence.

Let us now extend the analysis to the full two-dimensional emitter arrays case (also see Refs. [43,44] for related works on nonlinear effects in quantum emitter arrays). A full classical real-space simulation allows to identify the effect of higher driving power as an inhibitor for the overall reflectivity as seen in the progression from Fig. 7(a) to Fig. 7(d) corresponding to an increase of η from $\Gamma_0/20$ to Γ_0. Note that we assume a regime where the applied magnetic field shifts the y resonance very far from the x resonance such that a simplified two-level description fully captures the dynamics of the system. After performing a semiclassical approximation, where we factorize all correlation functions $\langle\sigma_i\sigma_j\rangle$ and $\langle\sigma_i\,\sigma_j^z\rangle$ if $i\neq j$, we see that similarly to Sec. 2.3 all emitters respond the same for an infinite array. This allows us to make a mean-field approach and identify $\beta_j=\beta$. The equations of motion become

$$\dot{\beta} = -\left(\frac{\Gamma_0}{2}-\mathrm{i}\Delta\right)\beta+\mathrm{i}\beta\beta^z\left(\tilde{\Omega}(0)-\mathrm{i}\frac{\tilde{\Gamma}(0)-\Gamma_0}{2}\right)+\mathrm{i}\eta\beta^z, \tag{31a}$$

$$\dot{\beta}^z = -\Gamma_0(\beta^z+1)-2(\tilde{\Gamma}(0)-\Gamma_0)|\beta|^2-4\eta\,\mathrm{Im}(\beta). \tag{31b}$$

We calculate the results of the equations above in steady-state to obtain an expression for β (and thus a renormalized reflectivity) and compare the predicted reflectivity with that of full numerics for real-space scattering. For a relatively moderate array size of 71×71 emitters, the exact numerics converge and agree with the reflectivity derived from the equations above, as illustrated in Fig. 7(e). The main observed effect is the reduction in reflectivity (increase in transmission) with increasing driving power, which can be simply explained by the reduction in the dipole strength predicted by Eq. (28) combined with a modification of the collective interactions due to the nonzero occupancy. Another effect that contributes to the total intensity emitted from the array is incoherent radiation due to spontaneous emission. We can however see in Figs. 7(b-d) that this effect is short range and small in comparison to the coherent part of the intensity and does thus not contribute significantly in the far-field. While this equation in principle supports two solutions (see Ref. [44]), we only find a single solution numerically. This is in complete agreement with the results in Ref. [44] for large lattice spacings such as $a=0.8\lambda$.

4.4. Beyond square lattice: honeycomb lattice

Let us now exemplify the extension of our formalism and result to more complicated lattice structures, in particular numerically simulating a two-dimensional honeycomb lattice (as illustrated in Fig. 8(a)). The underlying Bravais lattice of the honeycomb lattice is a triangular lattice with a two-atom unit cell. For non-Bravais lattices like the honeycomb lattice, an additional sublattice degree of freedom must be introduced, which we denote by ν. The modification of

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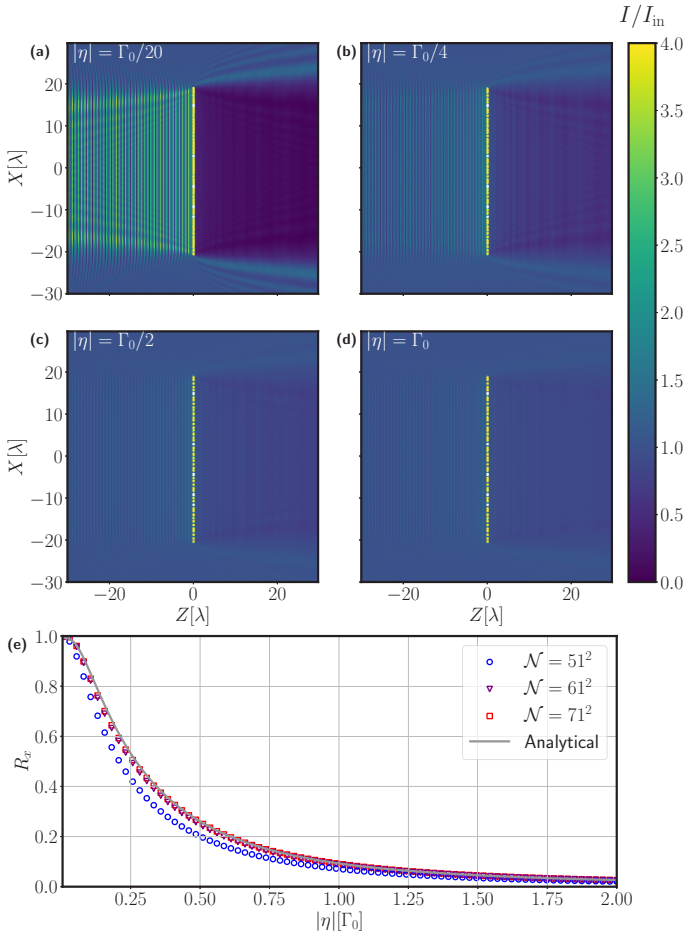


Fig. 7. Numerical simulations in the regime reaching beyond linear optical response, for a strong laser drive of the x-polarization component in the case where a y-polarizer is implemented. Since the drive has the same magnitude everywhere, $|\eta|$ is used as a measure for the nonlinearity. (a) $|\eta| = \Gamma_0/20$, (b) $|\eta| = \Gamma_0/4$, (c) $|\eta| = \Gamma_0/2$, and (d) $|\eta| = \Gamma_0$. The calculation is performed in real space for a square lattice with $a = 0.8\lambda$ also for $\mathcal{N} = 40^2 = 1600$ emitters. (e) Numerical calculation of the reflectivity of the array R_x for the x polarization for different array sizes ($\mathcal{N} = 51^2, 61^2, 71^2$). It is compared to the analytical expression (obtained from the steady-state solution of Eqs. (31)) as a function of the driving strength $|\eta|$.

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the theory is detailed in Appendix E and it mainly consists of a recalculation of the Green's tensor now including the additional degree of freedom. For a particular configuration chosen with a nearest-neighbor spacing of 0.9λ, the intensity pattern seen in Fig. 8(c) shows perfect reflection of the x component, while the y component is fully transmitted as seen in Fig. 8(d). The operating conditions were set such that $\Delta = -0.18\Gamma_0$ with respect to the natural transition frequency and a magnetic field of strength $\mu B_x = 5\Gamma_0$ is applied.

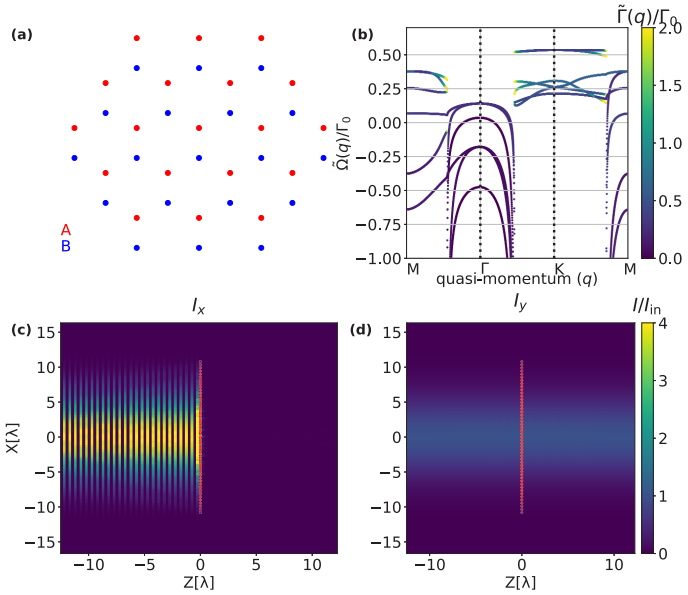


Fig. 8. Implementation of a polarizer on a honeycomb lattice with nearest-neighbor spacing 0.9λ. (a) The honeycomb lattice is described by a combination of two triangular (Bravais) sublattices labeled by A and B. (b) Band structure for such a lattice of three level systems (without magnetic field). (c) and (d) Intensities of the polarization components. The laser is detuned by $\Delta = -0.18\Gamma_0$ with respect to the natural transition frequency. A magnetic field of strength $\mu B_x = 5\Gamma_0$ is applied in x-direction.

5. Conclusions

We have investigated the applicability of subwavelength quantum emitter arrays as optical elements in the linear regime. To this end, we have highlighted the necessary steps to first derive the characteristics of the excitation modes on the two-dimensional array and then quantify the linear optical response via the polarizability tensor of the array. We then provided a general analytical expression for the transmission matrix which connects the polarization components of the outgoing, transmitted field to the ones of the incoming field. The subwavelength nature of the array, combined with the particular case of normal incident driving, allowed for a reduction to an exact two-by-two matrix formulation for this scattering problem. The possibility to design the array geometry, in combination with external control through an applied magnetic field allows for the implementation of various Jones matrices such as polarizers and phase retarders. We have also provided some numerical and analytical treatment of various effects that might be detrimental to the efficient operation of such metasurfaces, ranging from thermal motion to disorder and to optical nonlinearities induced by high intensity operation.

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Such subwavelength arrays might find other applications as, for example, bilayers of emitter arrays present an interesting platform for cavity quantum electrodynamics with extremely narrow and strongly frequency-dependent resonances [3,45]. Another aspect is that of imprinted spin patterns realizable via a spatially varying magnetic field. A possibility is that these spin patterns could be used to implement more generic optical elements such as perfect waveplates. Similar ideas have been put forward for classical metasurfaces [46].

A. Dyadic Green's function

The free-space photonic Green's tensor evaluated at the resonance of the emitters $\omega_0 = ck_0$ is given by

$$\mathbf{G}(\mathbf{R}) = \left(\mathbb{1} + \frac{1}{k_0^2}\nabla \otimes \nabla\right)\frac{e^{ik_0 R}}{4\pi R} - \frac{\mathbb{1}}{3k_0^2}\delta(\mathbf{R}), \tag{32}$$

where $\otimes$ denotes the dyadic product, $R = |\mathbf{R}|$ and the last term removes the divergence on the self interaction terms at $\mathbf{R} = 0$. This can be expressed more explicitly as

$$\mathbf{G}(\mathbf{R}) = \frac{e^{ik_0 R}}{4\pi k_0^2}\left[\left(\frac{k_0^2}{R} + \frac{ik_0}{R^2} - \frac{1}{R^3}\right)\mathbb{1} + \left(-\frac{k_0^2}{R} - \frac{3ik_0}{R^2} + \frac{3}{R^3}\right)\frac{\mathbf{R}\otimes\mathbf{R}}{R^2}\right] - \frac{\mathbb{1}}{3k_0^2}\delta(\mathbf{R}), \tag{33}$$

where $\mathbb{1}$ is a 3×3 identity matrix, the vector $\mathbf{R}$ has components X, Y and Z and the dyadic product is explicitly given by the following matrix

$$\mathbf{R}\otimes\mathbf{R} = \begin{pmatrix} X^2 & XY & XZ \\ YX & Y^2 & YZ \\ ZX & ZY & Z^2 \end{pmatrix}. \tag{34}$$

In the far field, only terms falling off as $1/R$ are contributing and the far field Green's tensor can be expressed in matrix form as

$$\mathbf{G}^{\text{far}}(\mathbf{R}) = \frac{e^{ik_0 R}}{4\pi R}\begin{pmatrix} 1 - X^2/R^2 & -XY & -XZ \\ -YX & 1 - Y^2/R^2 & -YZ \\ -ZX & -ZY & 1 - Z^2/R^2 \end{pmatrix}. \tag{35}$$

Notice that for the case where the only component is Z, the matrix can be reduced to a 2×2 diagonal matrix with $e^{ik_0 Z}/(4\pi Z)$ on the diagonal. Furthermore, for the scattering problem considered here, a Fourier decomposition of the Green's tensor with respect to the in-plane wave vector components $\mathbf{q} = (q_x, q_y)$ is useful. Decomposing also $\mathbf{R} = (\mathbf{r}_\parallel, Z)$ into lattice plane and out-of-plane coordinate, one can make use of the Weyl expansion [47]

$$\frac{e^{ik_0 R}}{R} = \frac{i}{2\pi}\int d\mathbf{q}\,\frac{1}{q_z} e^{i\mathbf{q}\cdot\mathbf{r}_\parallel} e^{iq_z|Z|}, \tag{36}$$

with $q_z = \sqrt{k_0^2 - q^2}$. The Green's tensor can now be written as

$$\mathbf{G}(\mathbf{R}) = \frac{i}{8\pi^2 k_0^2}\int d\mathbf{q}\,\frac{k_0^2\mathbb{1} - \mathbf{v}(\mathbf{q},Z)\otimes\mathbf{v}(\mathbf{q},Z)}{q_z} e^{i\mathbf{q}\cdot\mathbf{r}_\parallel} e^{iq_z|Z|}, \tag{37}$$

where we have introduced the function $\mathbf{v}(\mathbf{q},Z) = (q_x, q_y, \text{sgn}(Z)\sqrt{k_0^2 - q^2})$. Notice that the expansion above includes the divergent self-interaction (i.e., we have dropped the delta function term).

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B. Master equation formalism

The master equation describing the dynamics of the laser-driven quantum emitter array reads

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \mathrm{i}[\rho, \mathcal{H}] + \sum_{j,j',\alpha,\alpha'} \Gamma^{\alpha,\alpha'}(\mathbf{r}_{jj'}) \left[\sigma_{j,\alpha} \rho \sigma^{\dagger}_{j',\alpha'} - \frac{1}{2} \left\{ \sigma^{\dagger}_{j,\alpha} \sigma_{j',\alpha'}, \rho \right\} \right], \tag{38}$$

with the Hamiltonian

$$\mathcal{H} = -\sum_{j,\alpha} \Delta \sigma^{\dagger}_{j,\alpha} \sigma_{j,\alpha} + \sum_{j,\alpha} \left(\eta_{\alpha}(\mathbf{r}_j) \sigma^{\dagger}_{j,\alpha} + \eta^{*}_{\alpha}(\mathbf{r}_j) \sigma_{j,\alpha} \right) + \sum_{j,j',\alpha,\alpha'} \Omega^{\alpha,\alpha'}(\mathbf{r}_{jj'}) \sigma^{\dagger}_{j,\alpha} \sigma_{j',\alpha'} + \mathcal{H}_{\mathrm{B}}, \tag{39}$$

where $\eta_{\alpha}(\mathbf{r}_j) = \eta_{\alpha} \mathrm{e}^{\mathrm{i}\mathbf{k}_{\parallel} \cdot \mathbf{r}_j}$ is the generalized position-dependent Rabi frequency for oblique incidence with wave vector $\mathbf{k}_{\parallel}$ parallel to the array plane. $\mathcal{H}_{\mathrm{B}}$ is the Hamiltonian which describes the effect of a magnetic field. In the Cartesian basis which we mainly use this Hamiltonian reads

$$\mathcal{H}_{\mathrm{B}} = \mathrm{i} \varepsilon_{\alpha_1,\alpha_2,\alpha_3} \mu_{i,\alpha_1} B_{i,\alpha_1} |i, \alpha_2\rangle\langle i, \alpha_3|, \tag{40}$$

with Einstein sum convention where $\alpha_i = x, y, z$ and $\mu_{i,\alpha}$, $B_{i,\alpha}$ are the Cartesian components of the magnetic dipole moments and the magnetic field at emitter i, respectively. ε_{ijk} is the Levi-Civita symbol.

C. Fourier space expressions for far-field Green's tensor and decay rates

Let us now move to the Fourier domain. The dipole-scattered field can be expressed as an integral over all wave vectors in the Brillouin zone $\mathcal{B}$

$$\mathbf{E}^{(+)}_{\mathrm{dip}}(\mathbf{R}) = -\frac{3\pi\Gamma_0}{k_0 d} \int_{\mathbf{q} \in \mathcal{B}} \mathrm{d}\mathbf{q}\, \mathrm{e}^{\mathrm{i}\mathbf{q} \cdot \mathbf{r}_{\parallel}} \widetilde{\mathbf{G}}(\mathbf{q}; Z) \boldsymbol{\beta}_{\mathbf{q}}, \tag{41}$$

where $\widetilde{\mathbf{G}}(\mathbf{q}; Z) = \sum_{\mathbf{r}_{\parallel} \in \Lambda} \mathrm{e}^{-\mathrm{i}\mathbf{q} \cdot \mathbf{r}_{\parallel}} \mathbf{G}(\mathbf{R})$ is the (discrete) 2D lattice Fourier transform in the first two arguments of the Green's tensor. We can connect the discrete $\widetilde{\mathbf{G}}(\mathbf{q}; Z)$ and the continuous Fourier transform $\bar{\mathbf{G}}(\mathbf{q}; Z) = \frac{1}{(2\pi)^2} \int_{\mathbf{r}_{\parallel} \in \mathbb{R}^2} \mathrm{d}\mathbf{r}_{\parallel} \mathrm{e}^{-\mathrm{i}\mathbf{q} \cdot \mathbf{r}_{\parallel}} \mathbf{G}(\mathbf{R})$ by making use of the Poisson summation formula in two dimensions $\sum_{\mathbf{r}_{\parallel} \in \Lambda} f(\mathbf{r}_{\parallel}) = \frac{(2\pi)^2}{\mathcal{A}} \sum_{\mathbf{g} \in \Lambda^*} \bar{f}(\mathbf{g})$ where $\mathcal{A}$ is the area of the unit cell. To recast the lattice transform of the Green's tensor into a sum over reciprocal lattice vectors $\mathbf{g}$, we identify $f(\mathbf{r}_{\parallel}) = \mathrm{e}^{-\mathrm{i}\mathbf{q} \cdot \mathbf{r}_{\parallel}} \mathbf{G}(\mathbf{R})$ in the Poisson sum formula to obtain

$$\widetilde{\mathbf{G}}(\mathbf{q}; Z) = \frac{(2\pi)^2}{\mathcal{A}} \sum_{\mathbf{g} \in \Lambda^*} \bar{\mathbf{G}}(\mathbf{q} + \mathbf{g}; Z). \tag{42}$$

The expression for the continuous Fourier transform can be identified from the Weyl expansion Eq. (37)

$$\bar{\mathbf{G}}(\mathbf{q}; Z) = \frac{\mathrm{i}}{8\pi^2} \left(\mathbb{1} - \frac{\mathbf{v}(\mathbf{q}, Z) \otimes \mathbf{v}(\mathbf{q}, Z)}{k_0^2} \right) \frac{\mathrm{e}^{\mathrm{i}q_z |Z|}}{q_z}. \tag{43}$$

Instead of a sum over the real-space lattice, $\widetilde{\mathbf{G}}$ is now represented via a reciprocal lattice summation

$$\widetilde{\mathbf{G}}(\mathbf{q}; Z) = \frac{\mathrm{i}}{2\mathcal{A}} \sum_{\mathbf{g} \in \Lambda^*} \left(\mathbb{1} - \frac{\mathbf{v}(\mathbf{q} + \mathbf{g}, Z) \otimes \mathbf{v}(\mathbf{q} + \mathbf{g}, Z)}{k_0^2} \right) \frac{\mathrm{e}^{\mathrm{i}\sqrt{k_0^2 - (\mathbf{q}+\mathbf{g})^2} |Z|}}{\sqrt{k_0^2 - (\mathbf{q} + \mathbf{g})^2}} \tag{44}$$

Notably, the geometric prefactor in front of the plane wave ensures that the polarization of the outgoing plane wave is perpendicular to the propagation direction. It is thus a projector onto the plane orthogonal to $\mathbf{v}(\mathbf{q} + \mathbf{g}, Z)$.

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The fundamental argument for the directional character of the light scattered from subwavelength arrays is the fact that the exponential term $e^{i\sqrt{k_0^2-(\mathbf{q}+\mathbf{g})^2}|Z|}$ becomes evanescent for $\mathbf{g} \neq 0$ when $a<\lambda/2$ (this condition depends on the angle of incidence, with $a<\lambda/2$ for full oblique and $a<\lambda$ for normal incidence) so that only the zeroth summand in the reciprocal lattice sum remains

$$\widetilde{\mathbf{G}}^{\text{far}}(\mathbf{q};Z) \approx \frac{\mathrm{i}}{2\mathcal{A}}\frac{e^{\mathrm{i}q_z|Z|}}{q_z}\left(\mathbb{1}-\frac{\mathbf{v}(\mathbf{q},Z)\otimes\mathbf{v}(\mathbf{q},Z)}{k_0^2}\right) \tag{45}$$

For perpendicular incidence of the laser only the $\mathbf{q} = 0$ component will be selected and this simply becomes

$$\widetilde{\mathbf{G}}^{\text{far}}(0;Z) = \frac{\mathrm{i}}{2\mathcal{A}}\frac{e^{\mathrm{i}k_0|Z|}}{k_0}\begin{pmatrix}1&0&0\\0&1&0\\0&0&0\end{pmatrix} \tag{46}$$

The same argument as above can be used to deduce the decay rate for subwavelength arrays since the plane-wave term only contributes to the imaginary part if $\mathbf{g} = 0$ so that

$$\widetilde{\mathbf{\Gamma}}(\mathbf{q}) \approx \frac{3\Gamma_0}{4\pi}\left(\frac{\lambda^2}{\mathcal{A}}\right)\frac{k_0}{q_z}\left(\mathbb{1}-\frac{\mathbf{v}^+(\mathbf{q})\otimes\mathbf{v}^+(\mathbf{q})}{k_0^2}\right), \tag{47}$$

with $\mathbf{v}^+(\mathbf{q}) = \mathbf{v}(\mathbf{q},0^+) = (q_x, q_y, \sqrt{k_0^2-q^2})$. This can be used below to simplify the expression for the scattering matrix of the 2D array. Notice that the expression for the discrete Green's tensor does not include the delta function term. Therefore, it suffices for the analytical computation of the far field Green's tensor and of the decay rate matrix but it cannot be used to compute the dipole-dipole collective shifts. The reason for that is that the summation does not converge owing to the divergent self interaction. This can instead be dealt with more easily in real space by canceling any contributions from $\mathbf{r}_\parallel = 0$ or alternatively, via a fluctuation averaging procedure as in Ref. [27].

D. Explicit expression of scattering matrix

The matrix $\boldsymbol{\mathcal{M}}(\mathbf{q}) = -\Delta\mathbb{1} + \tilde{\mathbf{\Omega}}(\mathbf{q}) - \mathrm{i}\tilde{\mathbf{\Gamma}}(\mathbf{q})/2$, which appears in Sec. 2 when solving the linear system for the polarizations of the dipoles on the lattice is explicitly given by (without applied magnetic field)

$$\boldsymbol{\mathcal{M}}(\mathbf{q}) = -\Delta\mathbb{1} + \begin{pmatrix}\tilde{\Omega}^{xx}(\mathbf{q})-\mathrm{i}\tilde{\Gamma}^{xx}(\mathbf{q})/2 & \tilde{\Omega}^{xy}(\mathbf{q})-\mathrm{i}\tilde{\Gamma}^{xy}(\mathbf{q})/2 & 0\\ \tilde{\Omega}^{yx}(\mathbf{q})-\mathrm{i}\tilde{\Gamma}^{yx}(\mathbf{q})/2 & \tilde{\Omega}^{yy}(\mathbf{q})-\mathrm{i}\tilde{\Gamma}^{yy}(\mathbf{q})/2 & 0\\ 0 & 0 & \tilde{\Omega}^{zz}(\mathbf{q})-\mathrm{i}\tilde{\Gamma}^{zz}(\mathbf{q})/2\end{pmatrix}. \tag{48}$$

This matrix is is block-diagonal since there is no coupling to the z component for primitive lattice vectors in the xy plane. Defining the projector into the plane orthogonal to $\mathbf{v}$ as $\boldsymbol{\mathcal{P}}_{\mathbf{v}}$, the decay rate matrix for a subwavelength array in Fourier space may be approximated from Eq. (47) as $\tilde{\mathbf{\Gamma}}(\mathbf{q}) \approx \tilde{\Gamma}(\mathbf{q})\boldsymbol{\mathcal{P}}_{\mathbf{v}^+(\mathbf{q})}$. A magnetic field $\mathbf{B} = (B_x, B_y, B_z)$ introduces couplings among the Cartesian components with magnetic dipole moments $\boldsymbol{\mu} = (\mu_x, \mu_y, \mu_z)$ which adds to the

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above matrix $\boldsymbol{\mathcal{M}}(\mathbf{q})$ as

$$\mathbf{M}_{\mathrm{B}} = \begin{pmatrix} 0 & -\mathrm{i}B_z\mu_z & \mathrm{i}\mu_y B_y \\ \mathrm{i}B_z\mu_z & 0 & -\mathrm{i}\mu_x B_x \\ -\mathrm{i}\mu_y B_y & \mathrm{i}\mu_x B_x & 0 \end{pmatrix}. \tag{49}$$

Including the effect of the magnetic field, the $\mathbf{q}$ dependent scattering matrix for a subwavelength array may finally be expressed as

$$\boldsymbol{\mathcal{S}}(\mathbf{q}, Z) = \mathrm{i}\frac{\tilde{\Gamma}(\mathbf{q})}{2}\boldsymbol{\mathcal{P}}_{\mathbf{v}(\mathbf{q},Z)}\left[\begin{pmatrix} -\Delta+\tilde{\Omega}^{xx}(\mathbf{q}) & \tilde{\Omega}^{xy}(\mathbf{q})-\mathrm{i}B_z\mu_z & \mathrm{i}\mu_y B_y \\ \tilde{\Omega}^{yx}(\mathbf{q})+\mathrm{i}B_z\mu_z & -\Delta+\tilde{\Omega}^{yy}(\mathbf{q}) & -\mathrm{i}\mu_x B_x \\ -\mathrm{i}\mu_y B_y & \mathrm{i}\mu_x B_x & -\Delta+\tilde{\Omega}^{zz}(\mathbf{q}) \end{pmatrix} - \mathrm{i}\frac{\tilde{\Gamma}(\mathbf{q})}{2}\boldsymbol{\mathcal{P}}_{\mathbf{v}^+(\mathbf{q})}\right]^{-1}\boldsymbol{\mathcal{P}}_{\mathbf{v}^+(\mathbf{q})}. \tag{50}$$

This still depends on the sign of Z as for non-normal incidence, the reflected and transmitted waves have different propagation directions.

Generally, the matrix inversion in the equation above leads to a cumbersome expression for the scattering matrix. A simplified expression can be obtained by considering a square lattice at normal incidence $\mathbf{q} = 0$ which corresponds to the situation mostly considered in the main text of the manuscript. Additionally only considering a magnetic field along the x direction, $\mathbf{B} = B_x\hat{e}_x$, the scattering matrix $\boldsymbol{\mathcal{S}}(0) \equiv \boldsymbol{\mathcal{S}}$ becomes

$$\boldsymbol{\mathcal{S}} = C\begin{pmatrix} -\mu_x^2 B_x^2 + \left(\mathrm{i}\tilde{\Gamma}(0)/2 + \Delta - \tilde{\Omega}^{xx}(0)\right)\left(\Delta - \tilde{\Omega}^{zz}(0)\right) & \tilde{\Omega}^{xy}(0)\left(\Delta - \tilde{\Omega}^{zz}(0)\right) \\ \tilde{\Omega}^{xy}(0)\left(\Delta - \tilde{\Omega}^{zz}(0)\right) & \left(\mathrm{i}\tilde{\Gamma}(0)/2 + \Delta - \tilde{\Omega}^{xx}(0)\right)\left(\Delta - \tilde{\Omega}^{zz}(0)\right) \end{pmatrix}, \tag{51}$$

with

$$C = \frac{\mathrm{i}\tilde{\Gamma}(0)/2}{\mu_x^2 B_x^2\left(\mathrm{i}\tilde{\Gamma}(0)/2 + \Delta - \tilde{\Omega}^{xx}(0)\right) + \left[\left(\mathrm{i}\tilde{\Gamma}(0)/2 + \Delta - \tilde{\Omega}^{xx}(0)\right)^2 - \tilde{\Omega}^{xy}(0)^2\right]\left(\tilde{\Omega}^{zz}(0) - \Delta\right)}, \tag{52}$$

where we used that $\tilde{\Omega}^{xx}(0) = \tilde{\Omega}^{yy}(0)$ for the square lattice. In the limit of $B_x \to \infty$, this leads to the expression for the transmission matrix $\boldsymbol{\mathcal{T}} = \mathbb{1} + \boldsymbol{\mathcal{S}}$ in Eq. (22).

E. Generalization to non-Bravais lattices

For more complicated lattices that are non-Bravais, an additional sublattice degree of freedom must be introduced since the lattice Fourier transform is only well defined for a Bravais lattice [27]. Let us denote the sublattice degree of freedom by ν.

Indeed, first of all Eq. (41) is actually unchanged, but the Green's tensor now includes the sublattice degree of freedom. For the sublattice blocks, one can then more explicitly denote how the lattice Fourier transforms of the blocks differ

$$\widetilde{\mathbf{G}}_{\nu,\nu'}(\mathbf{q}; Z) = \sum_{\mathbf{r}_\parallel \in \Lambda} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{r}_\parallel}\mathbf{G}(\mathbf{r}_\parallel + \mathbf{b}_{\nu,\nu'} + \hat{e}_z Z), \tag{53}$$

where $\mathbf{b}_{\nu,\nu'} = \mathbf{b}_\nu - \mathbf{b}_{\nu'}$ for the sublattice vectors $\mathbf{b}_\nu$ which are also the locations of the emitters in the unit cell. For each sublattice block, the Poisson sum formula can now be leveraged to write

$$\widetilde{\mathbf{G}}_{\nu,\nu'}(\mathbf{q}; Z) = \frac{(2\pi)^2}{\mathcal{A}}\sum_{\mathbf{g}\in\Lambda^*} \mathrm{e}^{\mathrm{i}(\mathbf{q}+\mathbf{g})\cdot\mathbf{b}_{\nu,\nu'}}\tilde{\mathbf{G}}(\mathbf{q}+\mathbf{g}; Z). \tag{54}$$

This has some remarkable consequences. One point is the far-field (only taking the $\mathbf{g} = 0$ contribution)

$$\widetilde{\mathbf{G}}^{\mathrm{far}}_{\nu,\nu'}(\mathbf{q}; Z) \approx \frac{\mathrm{i}}{2\mathcal{A}}\frac{\mathrm{e}^{\mathrm{i}q_z|Z|}}{q_z}\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{b}_{\nu,\nu'}}\left(\mathbb{1} - \frac{\mathbf{v}(\mathbf{q}, Z)\otimes\mathbf{v}(\mathbf{q}, Z)}{k_0^2}\right) \tag{55}$$

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which simply implies that the plane waves emitted by the different sublattices interfere in a way that is determined by the spacing between the sublattices.

Furthermore, the matrix consisting of these sublattice phases is Hermitian such that the argument that only the $\mathbf{g} = 0$ component contributes to the decay rate of the band structure for $a<\lambda/2$ can also be applied for multiple sublattices

$$\widetilde{\mathbf{\Gamma}}_{\nu,\nu'}(\mathbf{q}) \approx \frac{3\Gamma_0}{4\pi}\left(\frac{\lambda^2}{\mathcal{A}}\right)\frac{k_0}{q_z}\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{b}_{\nu,\nu'}}\left(\mathbb{1} - \frac{\mathbf{v}^+(\mathbf{q})\otimes\mathbf{v}^+(\mathbf{q})}{k_0^2}\right). \tag{56}$$

It is particularly useful now to consider the special case of two sublattices. We can then choose the sublattice vectors as $\mathbf{b}_1 = 0$ and $\mathbf{b}_2 = \mathbf{b}$ so that the sublattice matrix becomes

$$\begin{pmatrix} 0 & \mathrm{e}^{\mathrm{i}\mathbf{q}\mathbf{b}} \\ \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{b}} & 0 \end{pmatrix}. \tag{57}$$

This matrix has eigenvalues $\xi_1 = 0$ and $\xi_2 = 2$. Since the $\mathbf{g} = 0$ term also contributes in a major way to the frequency component of the Green's function this implies that the diagonalization leads to a very flat, subradiant band and a more dispersive superradiant (with respect to the case without an additional sublattice) band corresponding respectively to the eigenvalues 0 and 2.

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2.5 P5: Purcell-modified Doppler cooling of quantum emitters inside optical cavities

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In this publication, my contributions were the following:

- Supervision and guidance on the development of all calculations and derived formulas.
- Contributions to rewriting and finalization of the manuscript (20%).

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Purcell-modified Doppler cooling of quantum emitters inside optical cavities

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Standard cavity cooling of atoms or dielectric particles is based on the action of dispersive optical forces in high-finesse cavities. We investigate here a complementary regime characterized by large cavity losses, resembling the standard Doppler cooling technique. For a single two-level emitter a modification of the cooling rate is obtained from the Purcell enhancement of spontaneous emission in the large cooperativity limit. This mechanism is aimed at cooling quantum emitters without closed transitions, which is the case for molecular systems, where the Purcell effect can mitigate the loss of population from the cooling cycle. We extend our analytical formulation to the many-particle case governed by small individual coupling, but exhibiting large collective coupling.

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I. INTRODUCTION

There are many ways to control the motion of atomic-sized objects via laser light and progress in cooling ions, and atoms have seen the emergence of techniques such as Doppler laser cooling, resolved sideband cooling, evaporative cooling, sub-Doppler cooling, and so on [1–4]. In general, these techniques make use of a cooling cycle between two electronic states where quick cycling of laser photons followed by many spontaneous emission events (at rate γ) removes kinetic energy into the electromagnetic bath. There are also alternatives which employ the enhanced coupling between a single photon and a single atom allowed by the use of optical cavities, i.e., within the cavity quantum electrodynamics (cQED) formalism [5–8]. Operation in a dispersive regime circumvents spontaneous emission and kinetic energy is removed via the loss of cavity photons (at rate κ) as proposed and discussed [9–12] and experimentally realized both for single atoms [13,14] as well as for ensembles [15,16].

Most of these techniques are not optimal for the cooling of molecules owing to their large number of vibrational and rotational sublevels where the population can migrate from the cooling cycle and thus reduce the cooling performance. In the context of cavity cooling, difficulties and mitigation solutions have been extensively discussed [17]. In other contexts, progress has been made in laser cooling of the center of mass of small molecules such as diatomics (CaF and SrF) [18–21], symmetric tops ($CaOCH_3$) [22], and asymmetric top molecules [23].

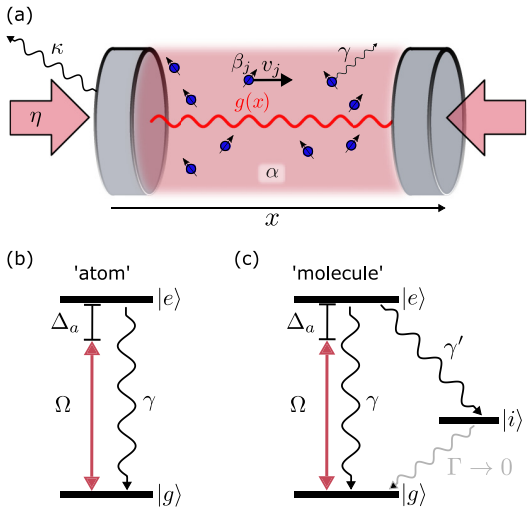


FIG. 1. (a) A variation of a standard one-dimensional Doppler cooling scheme for many emitters. Each emitter moves with some velocity v_j and the coupling between emitter coherence β_j and the cavity mode α is spatially dependent via $g(x_j)$, where g is the light-matter coupling constant. The cavity is driven with amplitude η. Spontaneous emission at rate γ and photon loss at rate κ are assumed. We consider two types of electronic level schemes for the emitters. (b) Electronic level scheme of a closed two-level system with energy eigenstates $|e\rangle$ (excited) and $|g\rangle$ (ground). Driving with amplitude Ω and detuning Δ_a is assumed. (c) Possible electronic level scheme mimicking a molecule with an additional level $|i\rangle$ to which population is lost from the cycle transition (with negligible repopulation rate Γ).

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We investigate here a hybrid scenario of Doppler-like cooling inside an optical cavity in the dissipative regime, where the rate of spontaneous emission of an atom or molecule is enhanced when operating in the Purcell regime of cQED, i.e., in the bad-cavity regime. This is inspired by experiments showing that the branching ratio of spontaneous emission in molecules can be strongly manipulated via optical cavities [24], albeit in solid-state environments where molecules are fixed in a host matrix. Extending this argument to molecules in a gas phase or in solvents can provide a mechanism to increase the cycling of photons and thus *close* the cooling cycle by reducing the rate of population loss into additional rotational or vibrational levels. We do not utilize dispersive optical forces as in standard cavity cooling, but simply employ the cavity as an additional dissipation channel for the emitters. This intuition is indeed validated for single quantum emitters, both with closed and nonclosed transitions in the regime where the cavity cooperativity is larger than unity. However, we find that extensions to many-particle systems, where the Purcell effect stems from a collective coupling to the cavity mode, indicates that individual loss of energy is not positively affected by collective properties. Our aim is solely in deriving a modified cooling or slowing down rate and to identify the mechanism via which the cavity can enhance it. The question of final temperature is not included in our formalism.

The paper is organized as follows. In Sec. II we proceed with computing analytical expressions for the cooling rates of quantum emitters with either closed or nonclosed transitions inside the one-dimensional geometry illustrated in Fig. 1. The results are compared to the standard situation of Doppler cooling in free space in the counterpropagating wave geometry. We identify the emitter-cavity cooperativity $\mathcal{C} = g^2/(\kappa\gamma)$ as the main tuning knob for speeding up the cooling process and maximizing the cooling time (for nonclosed transitions) with $\mathcal{C} \gg 1$. We then generalize in Sec. III to the many-particle case, where the single-particle cooperativity is small ($\mathcal{C} \ll 1$), but the collective cooperativity is large ($\mathcal{NC} \gg 1$). We derive analytical results for the cooling rate of each particle, which indicate that the collective Purcell regime with $\mathcal{NC} \gg 1$ does not positively affect the loss of kinetic energy at the individual particle level.

II. SINGLE-PARTICLE COOLING

Consider a one-dimensional scenario of a moving two-level system of mass m with an electronic transition between the ground state $|e\rangle$ and excited state $|g\rangle$ with frequency separation ω_0. We will first address the standard Doppler cooling scenario for a closed transition system in a standing wave. We refer to a closed transition system as one consisting of only two levels as in Fig. 1(b) where the excited state can undergo spontaneous emission only to the ground state. Next we consider the effect of placing the closed system within the confined electromagnetic volume of an optical cavity. We then depart from the closed system description by including an additional level in the electronic structure, which is exclusively populated via spontaneous emission from the excited state [see Fig. 1(c)]. We refer to the system as a nonclosed transition system. Again, we consider free space and cavity scenarios.

In the standard understanding of Doppler cooling, the condition of red-detuning $\Delta_a = \omega_0 - \omega_\ell > 0$ of the laser beam at frequency ω_ℓ with respect to the electronic transition is required. The cooling mechanism consists of the stimulated absorption of a photon below the resonance frequency, followed by spontaneous emission at the natural frequency. The energy difference then translates into a loss of kinetic energy and thus cooling. To derive a cooling rate, a semi-classical approach suffices, where an effective drag coefficient for the particle's momentum is derived that shows dependence on the driving power, detunings, and spontaneous emission rate. We start by reviewing such fundamental steps which we then expand to include the cavity scenario for both closed and nonclosed transition systems as depicted in Figs. 1(b) and 1(c).

The derivation is based on stating the master equation for the quantum emitter including motion from which we derive the equations of motion of the classical expectation values. Electronic transitions are described by the Pauli ladder operator $\hat{\sigma} = |g\rangle\langle e|$ and its Hermitian conjugate. The free Hamiltonian is

$$\hat{\mathcal{H}}_0 = \frac{\hat{p}^2}{2m} + \hbar\Delta_a \hat{\sigma}^\dagger\hat{\sigma}, \tag{1}$$

consisting of the kinetic energy operator and the two-level system Hamiltonian in a frame rotating with the laser frequency ω_ℓ, which we specify later. The spontaneous emission at rate γ is incorporated as a Lindblad superoperator

$$\mathcal{L}_{\rm em}[\hat{\rho}] = \gamma[2\hat{\sigma}\hat{\rho}\hat{\sigma}^\dagger - \hat{\sigma}^\dagger\hat{\sigma}\hat{\rho} - \hat{\rho}\sigma^\dagger\hat{\sigma}]. \tag{2}$$

The rate of spontaneous emission given by $\gamma = \omega_0^3 d_{eg}^2/(6\pi c^3\varepsilon_0)$ where d_{eg} is the transition dipole matrix element, ε_0 denotes the vacuum permittivity, and c is the speed of light in the vacuum. The dynamics of the system are then described by a master equation $i\dot{\hat{\rho}} = [\hat{\mathcal{H}}_0, \hat{\rho}]/\hbar + \mathcal{L}_{\rm em}[\hat{\rho}]$ for the system's density operator $\hat{\rho}$. The approach in terms of a Lindblad master equation as written in Eq. (2), neglects the effect of recoil in spontaneous emission. We will only consider operator averages and due to the isotropy of spontaneous emission, no momentum is imparted onto the emitter on average. However, the momentum imparted onto the emitter in spontaneous emission has nonzero variance, which results in a limit on the final temperature (Doppler temperature). We do not derive any modified temperature limits.

A. Free-space Doppler cooling of a closed transition system

Adding a classical laser drive with frequency ω_ℓ and Rabi frequency Ω with a standing wave spatial structure leads to a position-dependent Rabi frequency $\Omega(x) = \Omega\cos(k_\ell x)$. In a frame rotating at ω_ℓ, the time-independent Hamiltonian becomes

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \hbar\Omega(\hat{x})[\hat{\sigma} + \hat{\sigma}^\dagger]. \tag{3}$$

The dynamics of the expectation values of system operators such as $\beta = \langle\hat{\sigma}\rangle$, $p = \langle\hat{p}\rangle$, and $x = \langle\hat{x}\rangle$ can be deduced from

the master equation with $\hat{\mathcal{H}}$ as the total system Hamiltonian

$$\dot{\beta} = -(\gamma + \mathrm{i}\Delta_a)\beta - \mathrm{i}\Omega(x), \tag{4a}$$

$$\dot{p} = -\hbar\Omega'(x)[\beta + \beta^*], \tag{4b}$$

$$\dot{x} = p/m. \tag{4c}$$

We made the low-excitation approximation where $\langle\hat{\sigma}^\dagger\hat{\sigma} - \hat{\sigma}\hat{\sigma}^\dagger\rangle \approx -1$ and factorized quantum correlations between motional and internal degrees of freedom $\langle\hat{x}\hat{\sigma}\rangle = \langle\hat{x}\rangle\langle\hat{\sigma}\rangle$. To solve the equation of motion for the emitter coherence β we perform a Floquet expansion in the spatial harmonics of the driving field $\beta = \sum_{n=-\infty}^{\infty} b_n \mathrm{e}^{\mathrm{i}nk_\ell x}$, where the coefficients b_n are still time dependent. However, we assume that the expansion coefficients are stationary, which is a good approximation as long as the cooling rate is small compared to the rate of spontaneous emission γ. Inserting the expansion into the equation of motion Eq. (4a) gives only nonzero contributions for the harmonics of first order, i.e., b_n with $n = \pm 1$, which are not coupled in free space. We obtain the following set of equations:

$$b_n[\gamma + \mathrm{i}(\Delta_a + nk_\ell v)] = -\frac{\mathrm{i}\Omega}{2}(\delta_{n,+1} + \delta_{n,-1}), \tag{5}$$

where $v = \dot{x}$ is the instantaneous velocity of the emitter. The equations are solved by the following coefficients:

$$b_{\pm 1} = \frac{-\mathrm{i}\Omega}{2[\gamma + \mathrm{i}(\Delta_a \pm k_\ell v)]}. \tag{6}$$

For small Doppler shifts $k_\ell v \ll \Delta_a$, the coefficients may be approximated by

$$b_{\pm 1} \approx \frac{-\mathrm{i}\Omega}{2[\gamma + \mathrm{i}\Delta_a]} \pm \frac{-\Omega}{2[\gamma + \mathrm{i}\Delta_a]^2} k_\ell v \tag{7}$$

up to first order in $k_\ell v/\Delta_a$. The equation for the motion of the emitter momentum contains products of the gradient of the oscillating drive $\Omega'(x)$ and the oscillating emitter coherence $\beta(x)$. This leads to the occurrence of both constant terms and terms which oscillate at two times the fundamental spatial frequency of the standing wave. The constant term is a spatially independent force proportional to the emitter velocity (cooling force) and on timescales larger than half of the Doppler period $\pi/(k_\ell v)$ the oscillating terms average out, such that merely the cooling force remains. This results in an exponential decay of the emitter velocity $\dot{v} \approx -\xi_{\mathrm{fs}} v$. With the introduction of the recoil frequency $\omega_{\mathrm{rec}} = \hbar k_\ell^2/(2m)$, the cooling rate takes the following standard expression [25–29]:

$$\xi_{\mathrm{fs}} = \frac{4\Omega^2\omega_{\mathrm{rec}}\Delta_a\gamma}{\left[\gamma^2 + \Delta_a^2\right]^2}. \tag{8}$$

The validity of the analytical expression is illustrated in Fig. 2. The exponential cooling behavior is well captured in the regime where the Doppler shift is small compared to the emitter detuning. In the optimal regime, an additional effect of power broadening has to be taken into account limiting the applicable laser drive strength and an optimal detuning Δ_a close to the value of γ emerges. For smaller decay rates and some fixed $\Delta_a \gg \gamma$ the expression above instead shows a linear scaling with γ. This is the premise for using an optical cavity to enhance the rate of spontaneous emission and subsequently improve the cooling rate.

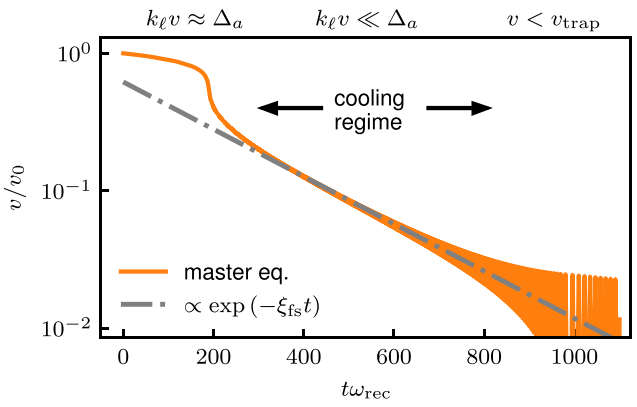


FIG. 2. Illustration of different cooling regimes obtained from numerical simulation of the master equation for a particle initially exhibiting a large Doppler shift. Within the regime of validity of the small shift approximation $k_\ell v \ll \Delta_a$, the exponential decay is well captured by the theoretical analysis. Finally, when the kinetic energy of the emitter is lower than the potential energy at a maximum of the standing wave ($v < v_{\mathrm{trap}}$), the emitter gets trapped around a potential minimum and starts oscillating around it. Parameters in units of γ: $\Omega = 1$, $\Delta_a = 10$, $\omega_{\mathrm{rec}} = 0.5$, $k_\ell v_0 = 18$.

B. Purcell-modified Doppler cooling of a closed transition system

Let us now assume that the two-level system is positioned inside an optical cavity and coupled to the spatially confined light field via the position-dependent light-matter coupling $g(x) = g\cos(k_c x)$, where k_c (corresponding frequency ω_c) is the wave vector of the cavity mode and g quantifies the maximum coupling at an antinode of the optical mode. For a two-level transition $g = d_{eg}\sqrt{\omega_c/(2\varepsilon_0 V)}$ where V is the optical mode volume. Furthermore, the cavity is driven with an amplitude η and frequency ω_ℓ. The description of the single-mode cavity is performed in terms of the bosonic annihilation operator $\hat{a}$ satisfying $[\hat{a}, \hat{a}^\dagger] = 1$. The time-independent Hamiltonian (in a frame rotating at ω_ℓ) is given by

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \hbar\Delta_c\hat{a}^\dagger\hat{a} + \mathrm{i}\hbar\eta[\hat{a} - \hat{a}^\dagger] + \hbar g(\hat{x})[\hat{\sigma}^\dagger\hat{a} + \hat{\sigma}\hat{a}^\dagger], \tag{9}$$

where $\Delta_c = \omega_c - \omega_\ell$ is the cavity detuning and the last two terms are the cavity drive and the light-matter coupling according to the Jaynes-Cummings model. Loss from the cavity at rate κ is described by the Lindblad superoperator

$$\mathcal{L}_{\mathrm{c}}[\hat{\rho}] = \kappa[2\hat{a}\hat{\rho}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}\hat{\rho} - \hat{\rho}\hat{a}^\dagger\hat{a}]. \tag{10}$$

We assume low excitation and factorizations of expectation values as in Eqs. (4). Additionally, we factorize light and matter expectation values $\langle\hat{a}\hat{\sigma}\rangle = \langle\hat{a}\rangle\langle\hat{\sigma}\rangle$, with the notation $\alpha = \langle\hat{a}\rangle$. Under these assumptions, we derive the following equations of motion:

$$\dot{\alpha} = -(\kappa + \mathrm{i}\Delta_c)\alpha - \mathrm{i}g(x)\beta - \eta, \tag{11a}$$

$$\dot{\beta} = -(\gamma + \mathrm{i}\Delta_a)\beta - \mathrm{i}g(x)\alpha, \tag{11b}$$

$$\dot{p} = -\hbar g'(x)(\beta\alpha^* + \beta^*\alpha), \tag{11c}$$

$$\dot{x} = p/m. \tag{11d}$$

The equations of motion in Eqs. (11) are a set of coupled nonlinear differential equations, such that already for a single particle a general analytical solution is nontrivial, if not impossible. However, from the physical problem considered (Doppler cooling in the Purcell regime) a clear hierarchy of timescales for the different degrees of freedom emerges, enabling an analytical treatment. The Purcell regime is characterized by large cavity losses $\kappa \gg (g, \gamma)$, such that the cavity field evolves on the shortest timescale. Furthermore, as we are interested in the Doppler cooling process, where the upper bound of the cooling rate is the recoil frequency ω_{rec}, the change in velocity corresponds to the longest timescale. We therefore start by eliminating the cavity field, proceed by determining the emitter coherence, and at last solve the motion of the emitter. Formal integration of the equation of motion for the cavity mode α to first order in g/κ gives (details in Appendix A 1)

$$\alpha = -\frac{\eta}{\kappa + \mathrm{i}\Delta_c} - \frac{\mathrm{i}g(x)\beta}{\kappa + \mathrm{i}\delta}, \tag{12}$$

with $\delta = \omega_0 - \omega_c$, the emitter-cavity detuning. We now see that the cavity field consists of two contributions, first the response to the direct drive and second the induced field of the emitter. The first term in Eq. (12) leads to the same dynamics as in the free space, i.e., a spatially dependent drive of the emitter coherence. The second term is therefore the crucial one, resulting in an effective cavity-mediated self-interaction of the emitter. The Floquet expansion of β now leads to a system of equations for the Floquet coefficients b_n of the form

$$\begin{aligned} &b_n[\gamma + \mathrm{i}(\Delta_a + \mathrm{i}nk_c v)] + \frac{g^2}{4\kappa}(b_{n+2} + b_{n-2} + 2b_n) \\ &\quad = -\frac{\mathrm{i}\Omega}{2}(\delta_{n,+1} + \delta_{n,-1}) \end{aligned} \tag{13}$$

for the cavity resonant with the atom $\delta = 0$ and the drive $\Omega = -g\eta/(\kappa + \mathrm{i}\Delta_c)$. The cavity couples all odd Floquet coefficients b_n since only the coefficients $b_{\pm 1}$ are directly driven and only coefficients with an index separated by ± 2 are coupled. The physical interpretation is the following. The cavity mode is driven with constant amplitude (in the rotating frame) and couples to the emitter via light-matter coupling with a standing wave spatial profile. Due to the nonzero velocity of the emitter any exchange of excitation between emitter and cavity results in a shift in frequency by the Doppler shift $\pm k_c v$. Elimination of the cavity field then results in an effective description where the emitter is directly driven with frequency components $\pm k_c v$ and an effective coupling between the frequency components separated by $\pm 2k_c v$ is obtained. The coupling corresponds to processes where a photon is emitted and then reabsorbed, such that the total Doppler shift either adds $\pm k_c v \pm k_c v = \pm 2k_c v$, resulting in the coupling between b_n and $b_{n\pm 2}$ or cancels $\pm k_c v \mp k_c v = 0$, resulting in a modification of b_n. The equations for the Floquet coefficients given by Eq. (13) can be cast into a matrix form with tridiagonal shape with constant sub and superdiagonal elements and nonconstant diagonal. In principle, the equations can be solved up to any order. However, we find that a reduction to a two-dimensional subspace involving only components $b_{\pm 1}$ suffices for $\mathcal{C}\gamma/\Delta_a \ll 1$ and allows for the derivation

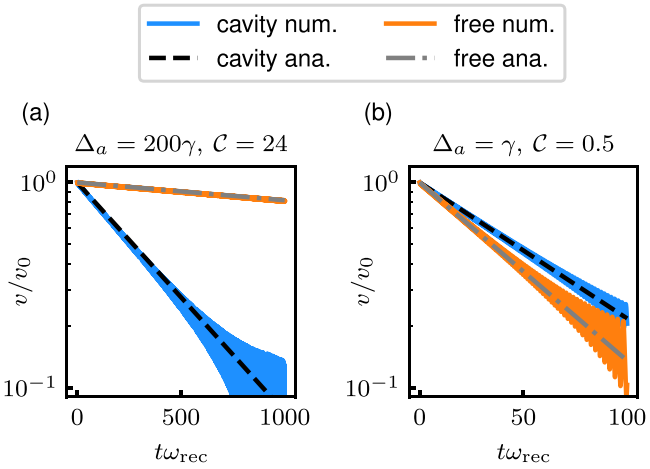


FIG. 3. Comparison between the numerical simulation of the emitter velocity obtained from the mean-field equations inside the cavity to free space, which confirms the scaling of the cooling rate in Eq. (15). (a) For large detuning $\Delta_a \gg \gamma(1 + \mathcal{C})$ the cooling rate ξ_c is enhanced compared to free space and scales linearly with the cooperativity. (b) For $\Delta_a = \gamma$ the cooling rate is reduced inside the cavity compared to free space. Parameters in units of γ for (a) $g = 155$, $\kappa = 1000$, $\Delta_a = \Delta_c = 200$, $\eta \approx 132$, $\omega_{\mathrm{rec}} = 2.5$, $k_c v_0 = 30$ and for (b) $g = \sqrt{500}$, $\kappa = 1000$, $\Delta_a = \Delta_c = 1$, $\eta \approx 6$, $\omega_{\mathrm{rec}} = 0.02$, $k_c v_0 = 0.15$. Note the large difference in ω_{rec} for the two cases.

of simple scaling laws of the cooling rate. An approach to include all b_n is sketched in Appendix A 1. We obtain the free-space case given by Eq. (5) from Eq. (13) by fixing the drive Ω and let $g^2/\kappa \to 0$.

Solving the reduced two-dimensional system leads to the following coefficients:

$$\begin{aligned} b_{\pm 1} = &-\frac{\mathrm{i}\Omega}{2}\frac{1}{\gamma(1 + \mathcal{C}/4) + \mathrm{i}(\Delta_a \pm k_c v)} \\ &\times \left[1 + \frac{g^2}{4\kappa}\sum_{\pm}\frac{1}{\gamma(1 + \mathcal{C}/4) + \mathrm{i}(\Delta_a \pm k_c v)}\right]^{-1}. \end{aligned} \tag{14}$$

Just as in free space we only keep terms in the equation of motion for the momentum which do not oscillate spatially, as the oscillating terms average to zero. Expanding Eq. (14) to first order in $k_c v/\Delta_a$ yields the cavity modified cooling rate

$$\xi_{\mathrm{c}} = \frac{4|\Omega|^2\omega_{\mathrm{rec}}\Delta_a\gamma(1 + \mathcal{C}/2)}{\left[\Delta_a^2 + \gamma^2(1 + \mathcal{C}/4)^2\right]\left[\Delta_a^2 + \gamma^2(1 + 3\mathcal{C}/4)^2\right]}. \tag{15}$$

For large detuning $\Delta_a \gg \gamma(1 + \mathcal{C})$ an expected linear increase in the cooling rate stemming from the Purcell-modified emission rate is obtained. We test this result against numerical simulation of the mean-field equations in Fig. 3(a) where an increase by a factor of $1 + \mathcal{C}/2$ in the cooling rate is observed. However, the improvement with $\mathcal{C}$ only holds in the regime $\Delta_a \gg \gamma(1 + \mathcal{C})$ which is suboptimal, but might be relevant for faster particles where the large Doppler shift requires higher detunings to allow for their capture, as $k_c v \gg \gamma$. In the regime $\Delta_a = \gamma$ the cooling rate decreases with increasing cooperativity, which is confirmed with numerics in Fig. 3(b) for small cooperativity where our analytical approach is valid.

C. Free-space Doppler cooling of a nonclosed transition system

We now consider a Λ-type three-level system, as displayed in Fig. 1(c) in free space. We assume that the drive couples solely to the transition between the ground state $|g\rangle$ and the excited state $|e\rangle$. Spontaneous emission, however, takes place between both excited state $|e\rangle$ and ground state $|g\rangle$ at rate γ and excited state $|e\rangle$ and intermediate state $|i\rangle$ at rate γ'. One could, in principle, assume an additional mechanism for population transfer from the intermediate state to the ground state at rate Γ. For molecules in a gas phase, this could correspond to population trapping in the rovibrational manifold and the value of Γ could be negligible (and we therefore neglect it in the following). This results in population trapping in the intermediate level and subsequently an effective loss of population from the cooling cycle. Since the intermediate state only couples via spontaneous emission from the excited state the Hamiltonian in Eq. (3) is unchanged and we merely include an additional Lindblad superoperator with collapse operator $\hat{\sigma}' = |i\rangle\langle e|$ at rate γ'. The corresponding mean-field equations including the populations of ground state n_g, excited state n_e, and intermediate state n_i read

$$\dot{\beta} = -(\gamma + \gamma' + \mathrm{i}\Delta_a)\beta - \mathrm{i}\Omega(x)[n_g - n_e], \tag{16a}$$

$$\dot{n}_g = 2\gamma n_e - \mathrm{i}\Omega(x)[\beta - \beta^*], \tag{16b}$$

$$\dot{n}_e = -2(\gamma + \gamma')n_e + \mathrm{i}\Omega(x)[\beta - \beta^*], \tag{16c}$$

$$\dot{n}_i = 2\gamma' n_e, \tag{16d}$$

$$\dot{p} = -\hbar\Omega'(x)[\beta + \beta^*], \tag{16e}$$

$$\dot{x} = p/m. \tag{16f}$$

In such a case, the system of equations are very similar to the ones for the closed transition system with the difference that the drive of the emitter coherence β has a term proportional to n_g for $n_e \ll n_g$. This simply suggests that the cooling rate for the nonclosed system is similar to the closed system case, with the distinction that it has an additional dependence on n_g such that it subsequently gets reduced to zero in time as population is lost to the intermediate state. We now solve Eqs. (16) under the assumption of low excitation $n_e \ll n_g$. Therefore, we can assume that the populations evolve much slower than the emitter coherence ($\dot{n}_g \ll \gamma + \gamma'$), such that we can directly solve Eq. (16a) in a similar fashion as already sketched out in the previous subsection. The Floquet coefficients of the emitter coherence now have a slow time dependence via the time-dependent ground-state population. Under the assumption of steady state for the excited-state population we obtain for the ground-state population

$$\dot{n}_g = -\frac{\gamma'\Omega^2}{\Delta_a^2 + \gamma_{\mathrm{tot}}^2} n_g = -\mu_{\mathrm{fs}} n_g, \tag{17}$$

where we define the total spontaneous decay rate $\gamma_{\mathrm{tot}} = \gamma + \gamma'$. The effective loss rate in Eq. (17) is simply the excitation probability times the rate of spontaneous emission into the intermediate state. The time-dependent ground-state population, which approaches 0 for $t \to \infty$, results in a time-dependent cooling rate of the form

$$\dot{v} = -\xi_{\mathrm{fs}} n_g(t) v = -\xi_{\mathrm{fs}} \mathrm{e}^{-\mu_{\mathrm{fs}} t} v, \tag{18}$$

with the solution

$$v(t) = v_0 \exp\left[\frac{\xi_{\mathrm{fs}}}{\mu_{\mathrm{fs}}}(\mathrm{e}^{-\mu_{\mathrm{fs}} t} - 1)\right], \tag{19}$$

where ξ_{fs} has the same form as in Eq. (8) but with γ replaced by γ_{tot}. For $t \to \infty$ when all population is lost to the intermediate state the final velocity is given by

$$v_{\mathrm{fs,final}} = v_0 \exp\left(-\frac{\xi_{\mathrm{fs}}}{\mu_{\mathrm{fs}}}\right) = v_0 \exp\left[-\frac{4\omega_{\mathrm{rec}}\gamma_{\mathrm{tot}}\Delta_a}{\gamma'(\Delta_a^2 + \gamma_{\mathrm{tot}}^2)}\right]. \tag{20}$$

The lowest final velocity is reached for $\Delta_a = \gamma_{\mathrm{tot}}$. In the regime $\Delta_a \gg \gamma_{\mathrm{tot}}$ the final velocity scales exponentially with the spontaneous decay rate γ_{tot}.

D. Purcell-modified Doppler cooling of a nonclosed transition system

We continue with the nonclosed transition system, now inside a cavity. The mean-field equations of motion derived from the Hamiltonian in Eq. (9) including populations and the spontaneous emission rates indicated in Fig. 1(c) read

$$\dot{\alpha} = -(\kappa + \mathrm{i}\Delta_c)\alpha - \mathrm{i}g(x)\beta - \eta, \tag{21a}$$

$$\dot{\beta} = -(\gamma + \gamma' + \mathrm{i}\Delta_a)\beta - \mathrm{i}g(x)\alpha[n_g - n_e], \tag{21b}$$

$$\dot{n}_g = 2\gamma n_e - \mathrm{i}g(x)[\beta\alpha^* - \beta^*\alpha], \tag{21c}$$

$$\dot{n}_e = -2(\gamma + \gamma')n_e + \mathrm{i}g(x)[\beta\alpha^* - \beta^*\alpha], \tag{21d}$$

$$\dot{n}_i = 2\gamma' n_e, \tag{21e}$$

$$\dot{p} = -\hbar g'(x)[\beta\alpha^* + \beta^*\alpha], \tag{21f}$$

$$\dot{x} = p/m. \tag{21g}$$

Again the equations of motion for the nonclosed system closely resemble the closed transition system, but with time-dependent populations. With the populations evolving much slower than the emitter coherence and the cavity, we can again utilize our solution for the closed transition system in terms of the Floquet coefficients which are now time dependent via the ground-state population. The population dynamics of the ground state are then dictated by the equation

$$\dot{n}_g = -\frac{\gamma'|\Omega|^2}{\gamma_{\mathrm{tot}}^2(1 + 3\mathcal{C}n_g/4)^2 + \Delta_a^2} n_g, \tag{22}$$

where we define the cooperativity as $\mathcal{C} = g^2/(\kappa\gamma_{\mathrm{tot}})$. Let us now distinguish two regimes: (i) $\gamma_{\mathrm{tot}}\mathcal{C}/(4\Delta_a) \ll 1$, in which case the reduced two-dimensional description of the Floquet coefficients suffices and analytical results are tractable and (ii) $\gamma_{\mathrm{tot}}\mathcal{C}/(4\Delta_a) \gtrsim 1$, in which case many Floquet coefficients have to be taken into account. An approach to include Floquet coefficients to arbitrary order is sketched in Appendix A 2. In regime (i), for sufficiently large Δ_a, Eq. (22) becomes equivalent to Eq. (17), such that the time evolution of the ground state is not modified by the cavity. This is confirmed with numerics in Fig. 4(b). Furthermore, the cooling rate scales linearly with the cooperativity in this regime. The reduction of ground-state population $n_g(t) = \exp(-\mu_{\mathrm{fs}} t)$ will then lead

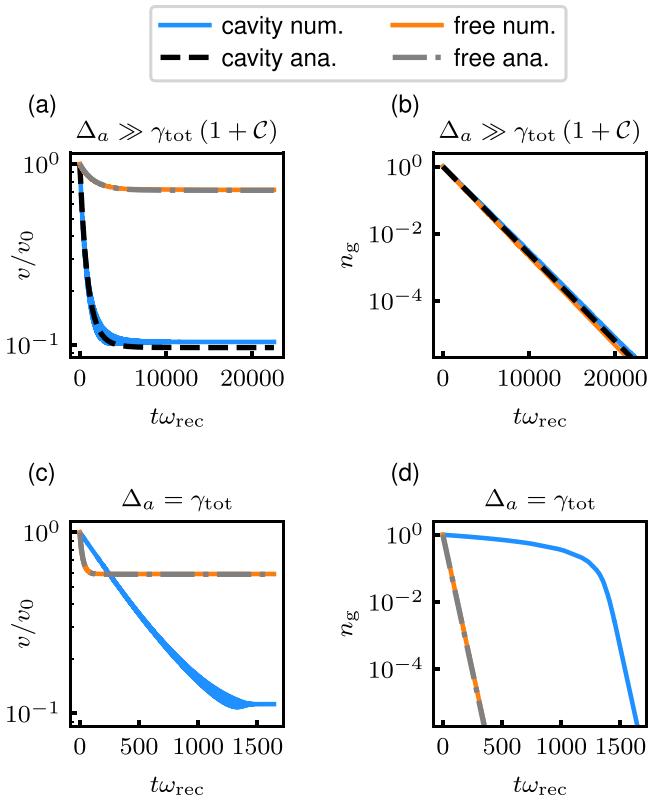


FIG. 4. Time evolution of (a) velocity and (b) ground-state population in regime (i) $\gamma_{\rm tot}\mathcal{C}/(4\Delta_a) \ll 1$ with $\Delta_a = 200\gamma_{\rm tot}$ and $\mathcal{C} = 24$. (a) Due to the Purcell-enhanced cooling rate in the strong detuned regime the final velocity is now reduced. (b) The loss of population to the intermediate state shows no Purcell modification in regime (i), as expected from Eq. (22). Time evolution of (c) velocity and (d) ground-state population in regime (ii) $\gamma_{\rm tot}\mathcal{C}/(4\Delta_a) \gg 1$ with $\Delta_a = \gamma_{\rm tot}$ and $\mathcal{C} = 24$. (c) The cooling rate inside the cavity is decreased compared to free space, but due to the reduced loss of population (d), the final velocity is still reduced. Numerical parameters for (a), (b) in units of $\gamma_{\rm tot}$: $\gamma = 0.85$, $g = 155$, $\kappa = 1000$, $\Delta_a = \Delta_c = 200$, $\eta \approx 132$, $\omega_{\rm rec} = 2.5$, $k_c v_0 = 30$ and for (c), (d): $\gamma = 0.85$, $g = 155$, $\kappa = 1000$, $\Delta_c = \Delta_a = 1$, $\eta \approx 0.9$, $\omega_{\rm rec} = 0.04$, $k_c v_0 = 0.15$. The performance in the different regimes is only similar due to the large difference in $\omega_{\rm rec}$.

to an exponential reduction in the cooling rate and an exponential reduction of the Purcell modification of the cooling rate with rate $2\mu_{\rm fs}$ [see Eq. (23)]. We can explicitly write the equation of motion for the velocity as

$$\dot{v} = -\xi_c(t)v = -\xi_{\rm fs}\left[n_g(t) + \frac{\mathcal{C}}{2}n_g^2(t)\right]v, \tag{23}$$

with the following solution:

$$v = v_0 \exp\left\{\frac{\xi_{\rm fs}}{\mu_{\rm fs}}\left[(e^{-\mu_{\rm fs}t} - 1) + \frac{\mathcal{C}}{4}(e^{-2\mu_{\rm fs}t} - 1)\right]\right\}. \tag{24}$$

We check the validity of Eq. (24) against the numerics in Fig. 4(a). The final velocity reached inside the cavity is then reduced due to the Purcell-enhanced cooling rate, while the loss of population is not modified. The performance of the Purcell-modified cooling mechanism compared to free space can then be quantified by the ratio of the final velocities when all population is lost to the intermediate state

$$\frac{v_{\rm c,final}}{v_{\rm fs,final}} = \exp\left(-\frac{\xi_{\rm fs}}{\mu_{\rm fs}}\frac{\mathcal{C}}{4}\right). \tag{25}$$

In the regime (ii) $\mathcal{C}\gamma_{\rm tot}/\Delta_a \gg 1$ we show only numerical results of the dynamics [see Figs. 4(c) and 4(d)], as the restriction to the Floquet coefficients $b_{\pm 1}$ is no longer valid. The loss of population to the intermediate state is now reduced by the Purcell effect inside the cavity, departing from the purely exponential decay [see Fig. 4(d)]. We see in Fig. 4(c) that the cooling rate inside the cavity is reduced compared to free space, as expected from the cooling rate for the closed transition system in Eq. (15). However, due to the reduction in population loss, the cooling time is increased and therefore a lower final velocity is reached, despite the reduced cooling rate. We have now identified two mechanisms by which the cavity can enhance the cooling process of a nonclosed transition system, i.e., increase the amount of removed kinetic energy. First, an increase in the cooling rate which appears in the regime $\Delta_a \gg \gamma_{\rm tot}(1+\mathcal{C})$, where the population dynamics are not modified by the cavity. Second, a decrease in the loss of population from the cooling cycle which appears most prominently in the regime $\Delta_a \approx \gamma_{\rm tot}$, where the cooling rate is reduced.

To understand how the effects of the cavity on cooling rate and population loss compete, we derive an analytical result for the final velocity, in regime (i) $\gamma_{\rm tot}\mathcal{C}/(4\Delta_a) \ll 1$, but now considering the Purcell modification of the dynamics of the ground-state population given by Eq. (22). The derivation, detailed in Appendix A 2, indicates that $v(t \to \infty) = v_0 \exp\{-\int_0^\infty \xi[n_g(t)]dt\}$, where the exponent is approximated to first order in $\mathcal{C}\gamma_{\rm tot}/\Delta_a$ by

$$\int_0^\infty \xi_{\rm c}[n_g(t)]dt \approx \frac{\xi_{\rm fs}}{\mu_{\rm fs}}\left[1 + \frac{\mathcal{C}\Delta_a^2}{4\left(\gamma_{\rm tot}^2 + \Delta_a^2\right)}\right]. \tag{26}$$

Therefore, already in regime (i) $\gamma_{\rm tot}\mathcal{C}/(4\Delta_a) \ll 1$ we see the onset of the behavior observed in Figs. 4(c) and 4(d), where the reduction of population loss at the expense of a reduced cooling rate leads to a reduction in the final velocity. This behavior differs from free space, where an increase in the rate of spontaneous emission is expected to reduce the final velocity for $\Delta_a = \gamma_{\rm tot}$ [see Eq. (20)]. Furthermore, Eq. (26) indicates that the lowest final velocity relative to free space is obtained for $\Delta_a = \gamma_{\rm tot}$, i.e., in the regime where the cooling rate is reduced. We perform numerical simulations of the final velocity beyond the validity of the analytical results (see Fig. 5), which show that the lowest final velocity is indeed obtained for $\Delta_a \approx \gamma_{\rm tot}$.

III. MANY-PARTICLE COOLING INSIDE OPTICAL CAVITIES

Let us now consider the case of $\mathcal{N}$ particles inside an optical cavity, where the single-particle cooperativity is small

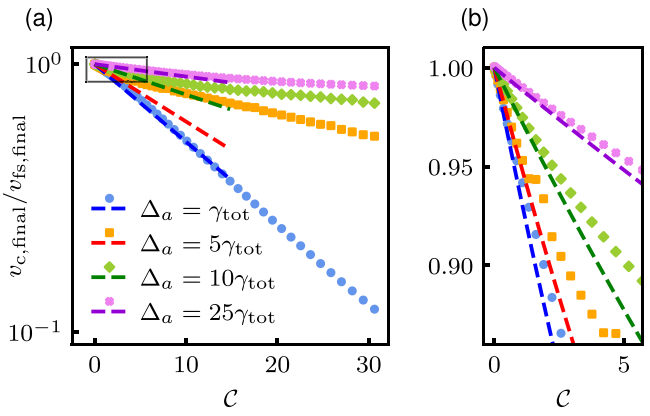


FIG. 5. (a) Performance of the cooling of a single nonclosed transition system inside the cavity compared to free space as a function of the cooperativity. The dashed lines correspond to the analytical scaling in Eq. (26) and the markers are obtained from numerical simulation of the mean-field equations. In (b) we show the rectangle marked in (a) for small cooperativity. For regime (i) $\mathcal{C}\gamma_{\mathrm{tot}}/\Delta_a \ll 1$ the scaling with the cooperativity given by Eq. (26) is confirmed. However, for large cooperativity the scaling of the final velocity with the cooperativity is generally lower than expected, but still, the final velocity remains reduced. Numerical parameters in units of γ_{tot}: $\gamma = 0.85, \kappa = 1000, \Delta_c = \Delta_a, \eta = \sqrt{0.01(\Delta_a^2 + \gamma_{\mathrm{tot}}^2)(\kappa^2 + \Delta_c^2)/g^2}$, $\omega_{\mathrm{rec}} = 0.04, k_c v_0 = 0.2\Delta_a$.

$\mathcal{C} \ll 1$ but the collective cooperativity is large $\mathcal{NC} \gg 1$. The aim is to elucidate whether the large collective cooperativity $\mathcal{CN}$ can influence the cooling dynamics of an individual emitter or whether it is solely the single-particle cooperativity $\mathcal{C}$ which is relevant for cooling.

A. Purcell-modified Doppler cooling of $\mathcal{N}$ closed transition systems

The total Hamiltonian for a set of $\mathcal{N}$ identical particles is the direct extension of the Hamiltonian of Eq. (9) where we now sum over the particle index $j = 1, \ldots, \mathcal{N}$. Similarly to the procedure in the previous section, we derive the set of coupled equations for the factorized expectation values in the low excitation regime

$$\dot{\alpha} = -(\kappa + \mathrm{i}\Delta_c)\alpha - \mathrm{i}\sum_{j=1}^{\mathcal{N}} g(x_j)\beta_j - \eta, \tag{27a}$$

$$\dot{\beta}_j = -(\gamma + \mathrm{i}\Delta_a)\beta_j - \mathrm{i}g(x_j)\alpha, \tag{27b}$$

$$\dot{p}_j = -\hbar g'(x_j)[\beta_j \alpha^* + \beta_j^* \alpha], \tag{27c}$$

$$\dot{x}_j = p_j/m. \tag{27d}$$

We proceed by performing a formal integration of the cavity mode in first order in g/κ to yield the $\mathcal{N}$ emitter equivalent of Eq. (12). In addition, each particle coherence is expanded in the harmonics of the cavity field $\beta_j = \sum_{n=-\infty}^{\infty} b_{j,n} \mathrm{e}^{\mathrm{i}nk_c x_j}$ (see Appendix B2). This now gives a system of equations where all Floquet coefficients $b_{j,n}$ are coupled, where we again truncate to $b_{j,\pm1}$ as in Sec. II B. The coupling is explicitly given by

$$\begin{pmatrix} a_{1,-} & 1 & \ldots & \ldots & 1 & 1 \\ 1 & \ddots & \ddots & & & 1 \\ \vdots & 1 & a_{\mathcal{N},-} & 1 & & \vdots \\ \vdots & & 1 & a_{1,+} & 1 & \vdots \\ 1 & & & \ddots & \ddots & 1 \\ 1 & 1 & \ldots & \ldots & 1 & a_{\mathcal{N},+} \end{pmatrix} \begin{pmatrix} b_{1,-} \\ \vdots \\ b_{\mathcal{N},-} \\ b_{1,+} \\ \vdots \\ b_{\mathcal{N},+} \end{pmatrix} = -\frac{2\mathrm{i}\kappa\Omega}{g^2} \begin{pmatrix} 1 \\ \vdots \\ 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \tag{28}$$

with $a_{j,\pm} = [\gamma + \mathrm{i}(\Delta_a \pm k_c v_j)]4\kappa/g^2 + 2$. The matrix can be inverted using the Sherman-Morrison formula [30], yielding the following coefficients:

$$b_{j,\pm1} = -\frac{\mathrm{i}\Omega}{2} \frac{1}{[\gamma(1 + \mathcal{C}/4) + \mathrm{i}(\Delta_a \pm k_c v_j)]} \times \left[1 + \frac{g^2}{4\kappa} \sum_{m,\pm}^{\mathcal{N}} \frac{1}{[\gamma(1 + \mathcal{C}/4) + \mathrm{i}(\Delta_a \pm k_c v_m)]}\right]^{-1}. \tag{29}$$

The Floquet coefficient $b_{j,\pm1}$ depends on all velocities v_m, which leads to a coupling of the equations of motion for all emitters. However, $b_{j,\pm}$ is an even function in all velocities $v_{m\neq j}$, whereas it has both even and odd parts in v_j. Therefore, a Taylor expansion up to first order in $k_c v_m/\Delta_a$ removes the dependency of $b_{j,\pm1}$ on all velocities v_m with $m \neq j$, resulting in a diagonal equations of motion for the velocities. The physical interpretation is illustrated in Fig. 6(c). Truncating the Floquet coefficients $b_{j,n}$ to $n = \pm1$ corresponds to the black rectangle in Fig. 6(c). Under this approximation a contribution to coefficient $b_{j,-1}$ is obtained from every coefficient $b_{i,\pm1}$, as indicated by the black arrows. This corresponds to processes where a photon is either directly absorbed by the emitter from the cavity, or the photon is absorbed by an emitter, reemitted into the cavity, and then absorbed again. As the Doppler shift from absorption and reemission in our approximation cancels, the coefficient $b_{j,-1}$ only depends on $-k_c v_j$. We can thus write

$$b_{j,\pm1} \approx b^{(0)} \pm k_c v_j b^{(1)}, \tag{30}$$

where we defer the explicit expression to Appendix B 1 Eq. (B4). With Eq. (30) we determine β_j and therefore also α. We obtain for α

$$\alpha = -\frac{\eta}{\kappa + \mathrm{i}\Delta_c} - \frac{\mathrm{i}g}{\kappa}\mathcal{N}b^{(0)}, \tag{31}$$

where we omitted terms which oscillate with the Doppler frequency. Collective effects appear both in $b_{j,\pm1}$, see Eq. (29), and in α, see Eq. (31). We obtain the cooling rate

$$\xi_c = \frac{4|\Omega|^2\omega_{\mathrm{rec}}\Delta_a\gamma(1 + \mathcal{C}/2)}{\left[\gamma^2(1 + \mathcal{C}/4)^2 + \Delta_a^2\right]\left\{\gamma^2[1 + \mathcal{C}(2\mathcal{N} + 1)/4]^2 + \Delta_a^2\right\}}. \tag{32}$$

The collective cooperativity $\mathcal{CN}$ only appears in the denominator of the cooling rate ξ_c, which implies worse cooling for increased collective cooperativity and hence particle number. The effects which lead to the form of ξ_c in Eq. (32) are schematically represented in Fig. 6. The cavity field drives

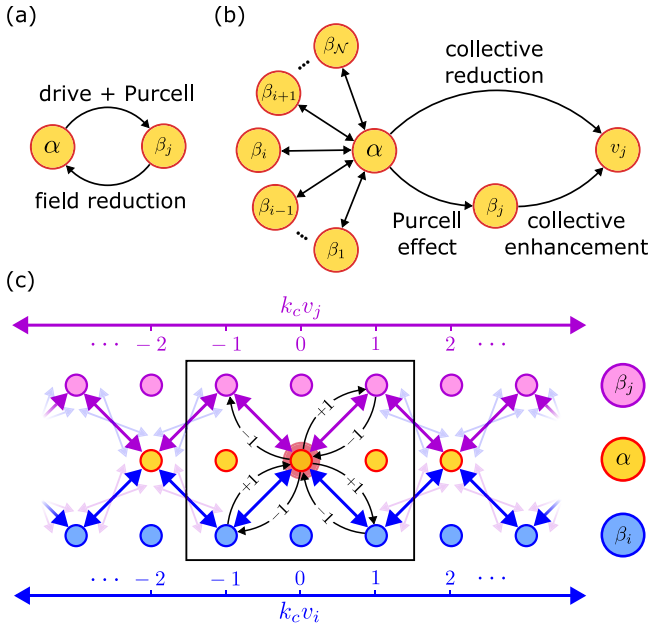


FIG. 6. (a) The cavity mode drives each emitter, i.e., induces a dipole moment and provides an additional decay channel (Purcell effect). The induced dipole moment of each emitter in turn reduces the amplitude of the cavity field. (b) The velocity of particle j evolves according to the interaction of its coherence with the cavity field [see Eq. (27c)]. The collective modification of the emitter coherence β_j leads to a collective enhancement of the emitter decay rate [see Eq. (29)]. However, due to the collective reduction of the cavity field by the emitters [see Eq. (31)], an increase in the amount of emitters always leads to a reduced cooling rate [see Eq. (32)]. (c) Illustration of the coupling between different Floquet coefficients, as described in the main text. On the single-particle level the exchange of a photon between the emitter and the cavity results in a frequency shift $\pm k_c v$, depending on whether the photon is absorbed from or emitted to the left or right. This is indicated by the solid coloured arrows. For many particles this results in a coupling between the Floquet coefficients of different particles. The black arrows indicate the contribution to $b_{j,-1}$ from $b_{j,+1}$, $b_{i,\pm 1}$. In the many-emitter case coupling to Floquet coefficients $b_{j,n}$ with $|n| > 1$, results in the generation of noninteger frequency components $k_c(nv_j + mv_i)$, as $v_i \neq v_j$, indicated by the translucent arrows.

each emitter coherence and provides an additional decay channel. In turn, the field generated by the emitters reduces the cavity field, such that the force component on particle j obtained from the interaction with the field generated by particle i with $i \neq j$ effectively corresponds to heating rather than cooling.

B. Purcell-modified Doppler cooling of $\mathcal{N}$ nonclosed transition systems

We now extend the results for $\mathcal{N}$ closed transition systems inside a cavity to $\mathcal{N}$ nonclosed transition systems and again derive the equations of motion for the factorized expectation values. As for the single nonclosed transition system we in-

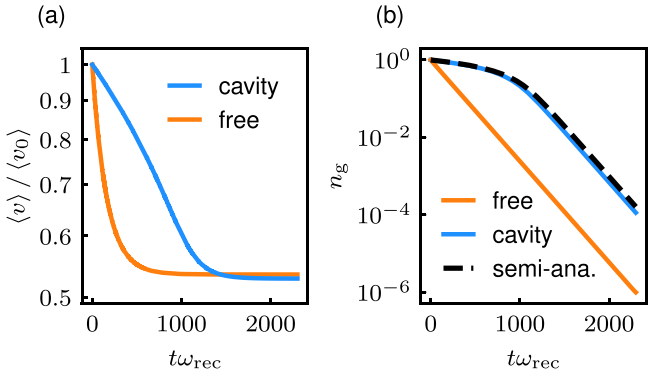


FIG. 7. (a) Numerical simulation of the mean-field equations for $\mathcal{N} = 400$ nonclosed transition emitters with $\mathcal{C} = 0.15$ and $\mathcal{CN} = 60$. The initial velocity distribution is Gaussian while the initial position distribution is uniform over $2\pi/k_c$. Despite the collective reduction of the cooling rate, as obtained from Eq. (32), the final velocity reached inside the cavity is almost identical to free space, as the cavity inhibits population migration from the cooling cycle. However, the reduction in the final velocity is only a single-particle effect $\propto \mathcal{C}$ with $\mathcal{C} \ll 1$, as derived in Eq. (35). The scale on the y axis is logarithmic. (b) Collective Purcell inhibition of population loss showing departure from the exponential dynamics in free space. The semi-analytical curve is a numerical simulation of Eq. (34). Numerical parameters normalized to $\gamma_{\rm tot}$: $\gamma = 0.7$, $g = 7.5$, $\eta = 50$, $\Delta_a = \Delta_c = 10$, $\kappa = 375$, $\omega_{\rm rec} = 0.5$, $k_c\langle v_0\rangle = 1.5$, $\sqrt{k_c^2(\langle v_0^2\rangle - \langle v_0\rangle^2)} = 0.1$.

clude the equations of motion for the populations, such that we obtain

$$\dot{\alpha} = -(\kappa + \mathrm{i}\Delta_c)\alpha - \mathrm{i}\sum_{m=1}^{\mathcal{N}} g(x_m)\beta_m - \eta, \tag{33a}$$

$$\dot{\beta}_j = -(\gamma + \gamma' + \mathrm{i}\Delta_a)\beta_j - \mathrm{i}g(x_j)\alpha[n_{j,g} - n_{j,e}], \tag{33b}$$

$$\dot{n}_{j,g} = 2\gamma n_{j,e} - \mathrm{i}g(x_j)[\beta_j\alpha^* - \beta_j^*\alpha], \tag{33c}$$

$$\dot{n}_{j,e} = -2(\gamma + \gamma')n_{j,e} + \mathrm{i}g(x_j)[\beta_j\alpha^* - \beta_j^*\alpha], \tag{33d}$$

$$\dot{n}_{j,i} = 2\gamma' n_{j,e}, \tag{33e}$$

$$\dot{p}_j = -\hbar g'(x_j)[\beta_j\alpha^* + \beta_j^*\alpha], \tag{33f}$$

$$\dot{x}_j = p_j/m. \tag{33g}$$

Again we follow the steps outlined in Sec. II D to derive a differential equation for the ground-state population

$$\dot{n}_g = -\frac{\gamma'|\Omega|^2}{\gamma_{\rm tot}^2[1 + (2\mathcal{N} + 1)\mathcal{C}n_g/4]^2 + \Delta_a^2} n_g, \tag{34}$$

where we dropped the particle index since the population transfer is position and velocity independent within our approximations and therefore identical for each particle. Numerical simulation of this equation shows agreement with the simulation of the full mean-field equations, as illustrated in Fig. 7(b). The reduced population loss, as already derived for a single particle in Eq. (22) shows now a dependence on the collective cooperativity $\mathcal{CN}$, instead of the single-particle cooperativity $\mathcal{C}$, i.e., it hints towards the possibility of a collective Purcell enhancement. A full analytical solution of the dynamics remains intractable. However, we can again find

an exact expression within our approximations for the final velocity, as already for the single nonclosed transition system (details in Appendix B 2). Here, we give the expression in leading order in the single-particle cooperativity

$$v_{\mathrm{c,final}} = \exp\left[-\frac{\xi_{\mathrm{fs}}}{\mu_{\mathrm{fs}}}\left(1+\frac{\mathcal{C}\Delta_a^2}{4\left(\Delta_a^2+\gamma_{\mathrm{tot}}^2\right)}\right)\right]. \tag{35}$$

The final velocity is independent of the number of emitters $\mathcal{N}$ since the collective effects in the cooling rate in Eq. (32) and population transfer in Eq. (34) cancel, such that only the single-particle effects remain. We confirm this with numerics in Fig. 7(a). Since we have small single-particle cooperativity $\mathcal{C} \ll 1$ the reduction of the final velocity inside the cavity compared to free space is insignificant.

IV. DISCUSSION AND CONCLUSION

We addressed the question of Purcell-modified Doppler cooling of quantum emitters, both with closed and nonclosed electronic transitions. The main effect, at the single-particle level, is the Purcell enhancement of spontaneous emission, which occurs when the cavity losses are high. This can lead to an improvement of cooling rates for both closed and nonclosed transition systems under far detuned conditions. In the regime of optimal cooling the cooling rate is not improved. However, for the nonclosed transition system, the Purcell effect leads to a reduction of population loss, which results in a lower final velocity when all population is lost to the intermediate state. At the level of many closed-transition systems, we show analytically how the cooling rate can be simply computed and find that the collective coupling does not lead to an enhancement of the cooling rate at the individual particle level, rather a collective decrease. For many nonclosed transition systems we show that the final velocity when all population is lost to the intermediate state is independent of the amount of emitters, i.e., shows no collective modification.

An extension of the one-dimensional treatment to three spatial dimensions can be envisioned using our equations by considering configurations of mirrors defining cavity modes in all three dimensions. Alternatively, one can simply reduce the kinetic energy along one direction and provide a thermalization mechanism which would lead to equipartition in all three dimensions.

We can also imagine scenarios where this extra effect of cooling via cavity Purcell enhancement of spontaneous emission could be implemented. For example, microfluidic devices could be integrated with optical cavities, and single molecule-photon coupling with cooperativity of the order of 50 could be reached as in Ref. [24]. For molecules moving in solutions, the cavity effect could enhance the cooling of their ballistic motion. Additionally, we will in the future address the reduction of temperature characterized by diffusive motion by supplementing our equations with classical Brownian noise terms based on a Wiener increment formalism.

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APPENDIX A: DOPPLER COOLING OF A SINGLE QUANTUM EMITTER

Let us sketch the procedure we follow to derive the cooling rate for closed and nonclosed transition quantum emitters inside an optical cavity.

1. Single closed transition system inside a cavity

Formal integration of the cavity mode amplitude expectation value from Eqs. (11), assuming free evolution of the emitter coherence β and linearized position $x = vt$, yields

$$\begin{aligned}\alpha &= -\int_0^t \mathrm{i}g[e^{-(\kappa+\mathrm{i}\Delta_c)(t-s)}\cos(k_c v s)\beta(t)e^{-(\gamma+\mathrm{i}\Delta_a)(s-t)} \\ &\qquad + \eta e^{-(\kappa+\mathrm{i}\Delta_c)(t-s)}]ds \\ &= -\mathrm{i}g\beta\sum_{\pm}\left[\frac{1}{2}\frac{e^{\pm\mathrm{i}k_c vt}}{\kappa-\gamma+\mathrm{i}(\delta\pm k_c v)} - \frac{1}{2}\frac{e^{-(\kappa-\gamma+\mathrm{i}(\delta\pm k_c v))t}}{\kappa-\gamma+\mathrm{i}(\delta\pm k_c v)}\right] \\ &\quad - \frac{\eta}{\kappa+\mathrm{i}\Delta_c} + \frac{\eta e^{-(\kappa+\mathrm{i}\Delta_c)t}}{\kappa+\mathrm{i}\Delta_c} \\ &\approx -\frac{\mathrm{i}g(x)\beta}{\kappa+\mathrm{i}\delta} - \frac{\eta}{\kappa+\mathrm{i}\Delta_c},\end{aligned} \tag{A1}$$

with $\delta = \Delta_c - \Delta_a$. Furthermore, we utilized the assumption that $\kappa \gg \gamma, k_c v$ and neglected the transient contributions due to large cavity loss in the Purcell regime. Inserting the final result of Eq. (A1) into the equation of motion for β with the cavity resonant to the emitter $\delta = 0$ and performing a temporal Fourier transform with linearised position $x = vt$ gives a discrete spectrum of the form

$$\begin{aligned}\mathrm{i}\omega\beta(\omega) &= -(\gamma+\mathrm{i}\Delta_a)\beta(\omega) \\ &\quad - \frac{g^2}{4\kappa}[\beta(\omega+2k_c v)+\beta(\omega-2k_c v)+2\beta(\omega)] \\ &\quad - \frac{\mathrm{i}\Omega}{2}[\delta(\omega-k_c v)+\delta(\omega+k_c v)].\end{aligned} \tag{A2}$$

Therefore, the emitter coherence contains only discrete frequencies and leads us to performing a Floquet expansion of the emitter coherence of the form

$$\beta = \sum_{n=-\infty}^{\infty} b_n e^{\mathrm{i}nk_c x}, \tag{A3}$$

which then gives an infinite set of coupled differential equations

$$\begin{aligned}\dot{b}_n + b_n[\gamma+\mathrm{i}(\Delta_a+nk_c v)] + \frac{g^2}{4\kappa}(b_{n+2}+b_{n-2}+2b_n) \\ = -\frac{\mathrm{i}\Omega}{2}(\delta_{n,+1}+\delta_{n,-1}).\end{aligned} \tag{A4}$$

We require the solution of β to derive the force acting on the particle. As the emitter velocity evolves much more slowly than the electronic degrees of freedom we may solve the differential equations for the Floquet coefficients b_n in the steady state $\dot{b}_n = 0$. In matrix notation the steady-state solution for the Floquet coefficients takes the form

$$\begin{pmatrix} \ddots & \ddots & \ddots & \ddots & \ddots & & & & & \\ & 0 & c & a_{-3} & c & 0 & & & & \\ & & 0 & c & a_{-1} & c & 0 & & & \\ & & & 0 & c & a_{+1} & c & 0 & & \\ & & & & 0 & c & a_{+3} & c & 0 & \\ & & & & & \ddots & \ddots & \ddots & \ddots & \ddots \end{pmatrix} \begin{pmatrix} \vdots \\ b_{-3} \\ b_{-1} \\ b_{+1} \\ b_{+3} \\ \vdots \end{pmatrix} = -\frac{i\Omega}{2} \begin{pmatrix} \vdots \\ 0 \\ 1 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \tag{A5}$$

with $a_n = [\gamma + i(\Delta_a + k_c n v)] + g^2/(2\kappa)$ and $c = g^2/(4\kappa)$. Neglecting couplings to harmonics of higher order ($b_{|n|>1} = 0$) reduces the problem to a 2×2 linear system with coupled coefficients $b_{\pm 1}$,

$$\begin{pmatrix} a_{-1} & c \\ c & a_{+1} \end{pmatrix} \begin{pmatrix} b_{-1} \\ b_{+1} \end{pmatrix} = -\frac{i\Omega}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{A6}$$

Inverting this matrix yields the solution

$$b_{\pm 1} = -\frac{i\Omega}{2[\gamma(1 + \mathcal{C}/4) + i(\Delta_a \pm k_c v)]} \left[1 + \sum_{\pm} \frac{g^2}{4\kappa} \frac{1}{[\gamma(1 + \mathcal{C}/4) + i(\Delta_a \pm k_c v)]} \right]^{-1}. \tag{A7}$$

Expansion to first order in $k_c v/\Delta_a$ gives

$$b_{\pm 1} \approx -\frac{i\Omega}{2[\gamma(1 + 3\mathcal{C}/4) + i\Delta_a]} \pm \frac{-\Omega}{2[\gamma(1 + \mathcal{C}/4) + i\Delta_a][\gamma(1 + 3\mathcal{C}/4) + i\Delta_a]} k_c v. \tag{A8}$$

However, one is not restricted to the approximation of two sidebands only, which holds for free space emitters but not when taking into account the interaction with the cavity for $\mathcal{C}\gamma/\Delta \gg 1$. We can cast the equations for the steady-state Floquet coefficients in the following form

$$(\boldsymbol{A} + i k_c v \boldsymbol{D}) \vec{b} = \vec{\Omega}, \tag{A9}$$

with $\boldsymbol{A}$ a symmetric tridiagonal Toeplitz matrix, $\boldsymbol{D}$ a diagonal matrix and $\vec{\Omega} = -i\Omega(\delta_{n,+1} + \delta_{n,-1})/2$ the drive of the spatial harmonics of first order. In matrix notation

$$\left[\begin{pmatrix} \ddots & \ddots & \ddots & \ddots & \ddots & & & & \\ & 0 & c & a & c & 0 & & & \\ & & 0 & c & a & c & 0 & & \\ & & & 0 & c & a & c & 0 & \\ & & & & 0 & c & a & c & 0 \\ & & & & & \ddots & \ddots & \ddots & \ddots & \ddots \end{pmatrix} + i k_c v \begin{pmatrix} \ddots & & & & & \\ & -3 & & & & \\ & & -1 & & & \\ & & & +1 & & \\ & & & & +3 & \\ & & & & & \ddots \end{pmatrix} \right] \begin{pmatrix} \vdots \\ b_{-3} \\ b_{-1} \\ b_{+1} \\ b_{+3} \\ \vdots \end{pmatrix} = -\frac{i\Omega}{2} \begin{pmatrix} \vdots \\ 0 \\ 1 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \tag{A10}$$

with $a = (\gamma + i\Delta_a) + g^2/(2\kappa)$ and $c = g^2/(4\kappa)$. As we are merely interested in a solution to linear order in $k_c v/\Delta_a$, which gives the cooling or friction-like force, we take a perturbative approach in the Doppler shift

$$\vec{b} = [\boldsymbol{A} + i k_c v \boldsymbol{D}]^{-1} \vec{\Omega} \approx [\boldsymbol{A^{-1}} - i k_c v \boldsymbol{A^{-1}} \boldsymbol{D} \boldsymbol{A^{-1}}] \vec{\Omega} = \vec{b}^{(0)} - i k_c v \vec{b}^{(1)}. \tag{A11}$$

The emitter coherence can then be written as

$$\beta = \sum_{n=-\infty}^{\infty} b_{2n+1} e^{i k_c (2n+1) x} = \sum_{n=0}^{\infty} \left\{ 2 b_{2n+1}^{(0)} \cos[(2n+1) k_c x] + 2 k_c v b_{2n+1}^{(1)} \sin[(2n+1) k_c x] \right\}. \tag{A12}$$

For sufficiently high harmonic order n the perturbative expansion in the Doppler shift breaks down since the perturbation diverges, i.e., $nk_c v/\Delta_a > 1$ for some n. However, from the formal integration of α to order g/κ we obtain the force

$$\begin{aligned}\dot{p} &= -\hbar g'(x)[\beta\alpha^* + \beta^*\alpha] \\ &= \hbar g k_c \sin(k_c x)\left[\beta\left(-\frac{\eta}{\kappa - \mathrm{i}\Delta_c} + \frac{\mathrm{i}g(x)}{\kappa - \mathrm{i}\delta}\beta^*\right)\right. \\ &\quad \left. + \beta^*\left(-\frac{\eta}{\kappa + \mathrm{i}\Delta_c} - \frac{\mathrm{i}g(x)}{\kappa + \mathrm{i}\delta}\beta\right)\right], \quad \text{(A13)}\end{aligned}$$

such that for $\delta = 0$ we obtain the spatially averaged force $\dot{p} \approx 2\hbar k_c^2 v \Im(\Omega^* b_{+1}^{(1)})$. So when the cavity is resonant with the emitter only coefficients $b_n^{(1)}$ with $n = \pm 1$ contribute with nonzero spatial average. This justifies the perturbative approach. The elements of the inverse of the tridiagonal Toeplitz operator $\boldsymbol{A}$ are given by [31]

$$\langle i|\boldsymbol{A}^{-1}|j\rangle = \frac{4\kappa}{g^2}\frac{\lambda^{|i-j|+1}}{\lambda^2 - 1}, \quad \text{(A14)}$$

with

$$\begin{aligned}\lambda &= (-a + \sqrt{a^2 - 4c^2})/(2c) \\ &= -1 - \frac{2\kappa(\gamma + \mathrm{i}\Delta_a)}{g^2}\left(1 - \sqrt{1 + \frac{g^2}{\kappa(\gamma + \mathrm{i}\Delta_a)}}\right). \quad \text{(A15)}\end{aligned}$$

The coefficients are given by

$$b_{2n+1}^{(0)} = -\frac{\mathrm{i}\Omega}{2}\frac{4\kappa}{g^2}\frac{\lambda^{n+1}}{\lambda - 1} \quad \text{with } n \in \mathbb{N}, \quad \text{(A16a)}$$

$$b_{+1}^{(1)} = -\frac{\mathrm{i}\Omega}{2}\left(\frac{4\kappa}{g^2}\right)^2\frac{\lambda^2(\lambda^2 + 1)}{(\lambda^2 - 1)^3}. \quad \text{(A16b)}$$

The Floquet coefficients $b_{2n+1}^{(0)}$ will be relevant for the population transfer in the nonclosed transition system and the coefficient $b_{+1}^{(1)}$ gives the cooling rate.

2. Single nonclosed transition system inside a cavity

The Floquet coefficients of first order to leading order in $k_c v/\Delta_a$ now have the following grounds-state dependency:

$$\begin{aligned}b_{\pm 1} \approx &-\frac{\mathrm{i}\Omega n_g}{2[\gamma(1 + 3\mathcal{C}n_g/4) + \mathrm{i}\Delta_a]} \\ &\pm\frac{-\Omega n_g}{2[\gamma(1 + \mathcal{C}n_g/4) + \mathrm{i}\Delta_a][\gamma(1 + 3\mathcal{C}n_g/4) + \mathrm{i}\Delta_a]}k_c v. \quad \text{(A17)}\end{aligned}$$

The differential equation for the ground state under steady-state assumption for the excited state $\dot{n}_e = 0$ is given by

$$\dot{n}_g = -\frac{|\Omega|^2\gamma' n_g}{\gamma_{\mathrm{tot}}^2(1 + n_g 3\mathcal{C}/4)^2 + \Delta_a^2}. \quad \text{(A18)}$$

This equation is separable and integrable, but not solvable for $n_g(t)$. We thus determine the final velocity when all population is lost to the intermediate state

$$v(t \to \infty) = v_0 \exp\left[-\int_0^\infty \xi(n_g(t))dt\right]. \quad \text{(A19a)}$$

We solve the integral by carrying out the integration over the ground-state population with $n_g(0) = 1$ and $n_g(t \to \infty) = 0$,

$$\begin{aligned}\int_0^\infty \xi_c(n_g(t))dt &= \int_1^0 \xi_c(n_g)\frac{dt}{dn_g}dn_g \\ &= \frac{2\hbar k_c^2 \Delta_a \gamma_{\mathrm{tot}}}{\gamma'\Delta_a^2}\int_0^1 \frac{(1 + \mathcal{C}/2n_g)}{\left[1 + \frac{\gamma_{\mathrm{tot}}^2}{\Delta_a^2}(1 + \mathcal{C}n_g/4)^2\right]}dn_g \\ &\approx \frac{4\omega_{\mathrm{rec.}}\Delta_a\gamma_{\mathrm{tot}}}{\gamma'(\gamma_{\mathrm{tot}}^2 + \Delta_a^2)}\left[1 + \frac{\mathcal{C}\Delta_a^2}{4(\Delta_a^2 + \gamma_{\mathrm{tot}}^2)}\right], \quad \text{(A20)}\end{aligned}$$

where the last step is a Taylor expansion in $\mathcal{C}\gamma_{\mathrm{tot}}/\Delta_a \ll 1$, which is already required for the cut-off of the Floquet expansion.

We can consider the population dynamics without restriction to the two-dimensional system of Floquet coefficients, i.e., consider the Floquet coefficients given by Eq. (A16). Elimination of the excited state $\dot{n}_e = 0$ yields

$$\dot{n}_g = -\frac{\gamma'}{\gamma_{\mathrm{tot}}}\mathrm{i}g(x)[\beta\alpha^* - \beta^*\alpha], \quad \text{(A21a)}$$

$$\dot{n}_i = \frac{\gamma'}{\gamma_{\mathrm{tot}}}\mathrm{i}g(x)[\beta\alpha^* - \beta^*\alpha], \quad \text{(A21b)}$$

where we insert the formal integration for the cavity mode to obtain

$$\dot{n}_g = \frac{\gamma'}{\gamma_{\mathrm{tot}}}\left[2\Im(\Omega^*(x)\beta) + \frac{2g^2(x)}{\kappa}|\beta|^2\right]. \quad \text{(A22)}$$

As β now contains all Floquet coefficients $2n + 1$ with $n \in \mathbb{N}$, calculating the second term in the drive $\propto g^2(x)|\beta|^2$ leads to geometric series. Once again invoking the previous argument that we can perform a spatial average to keep only constant terms

$$\begin{aligned}\langle g^2(x)|\beta|^2\rangle_x = \frac{g^2}{2}\sum_{m=0}^{\infty}&[b_{2m+1}^{(0)}b_{2m+1}^{(0)*}\delta_{m,0} + 2b_{2m+1}^{(0)}b_{2m+1}^{(0)*} \\ &+ b_{2m+3}^{(0)}b_{2m+1}^{(0)*} + b_{2m+1}^{(0)}b_{2m+3}^{(0)*}]. \quad \text{(A23)}\end{aligned}$$

Calculating the geometric series we obtain the differential equation for the ground state

$$\dot{n}_g = \frac{\gamma'}{\gamma_{\mathrm{tot}}}\frac{2\kappa|\Omega|^2}{g^2}\frac{|\lambda^2|(4 + \lambda + \lambda^*) + \lambda + \lambda^*}{|\lambda - 1|^2(1 - |\lambda|^2)}, \quad \text{(A24)}$$

where λ now has the following ground-state dependency:

$$\lambda = -1 - \frac{2\kappa(\gamma_{\mathrm{tot}} + \mathrm{i}\Delta_a)}{g^2 n_g}\left(1 - \sqrt{1 + \frac{g^2 n_g}{\kappa(\gamma_{\mathrm{tot}} + \mathrm{i}\Delta_a)}}\right). \quad \text{(A25)}$$

APPENDIX B: DOPPLER COOLING OF $\mathcal{N}$ QUANTUM EMITTERS

We now proceed with the treatment of an arbitrary number of emitters $\mathcal{N}$. As stated in the main text, we then assume that the single-particle cooperativity is small $\mathcal{C} \ll 1$, whereas the collective cooperativity $\mathcal{C}\mathcal{N} \gg 1$ is large.

1. $\mathcal{N}$ closed transition emitter inside a cavity

Formally integrating and inserting α into the equation of motion for β_j and expanding it in the Floquet coefficients of the cavity field

$$\beta_j = \sum_{n=-\infty}^{\infty} b_{j,n} e^{ink_c x_j}, \tag{B1}$$

leads to the following set of coupled equations for the steady-state Floquet coefficient $b_{j,n}$ for particle j of the nth-order harmonic

$$\begin{aligned}
&b_{j,n}[\gamma + i(\Delta_a + nk_c v_j)] \\
&= -\frac{i\Omega}{2}(\delta_{n,+1} + \delta_{n,-1}) - \frac{g^2}{4\kappa}\sum_{i=1}^{\mathcal{N}}\sum_{m=-\infty}^{\infty} b_{i,m}\left[e^{ik_c[(m+1)x_i-(n-1)x_j]} + e^{ik_c[(m+1)x_i-(n+1)x_j]} + e^{ik_c[(m-1)x_i-(n-1)x_j]} + e^{ik_c[(m-1)x_i-(n+1)x_j]}\right] \\
&\approx -\frac{i\Omega}{2}(\delta_{n,+1} + \delta_{n,-1}) \\
&\quad - \frac{g^2}{4\kappa}\sum_{i=1}^{\mathcal{N}}\sum_{m=-\infty}^{\infty} b_{i,m}\left[e^{ik_c[(m+1)v_i-(n-1)v_j]t} + e^{ik_c[(m+1)v_i-(n+1)v_j]t} + e^{ik_c[(m-1)v_i-(n-1)v_j]t} + e^{ik_c[(m-1)v_i-(n+1)v_j]t}\right] \\
&\approx -\frac{i\Omega}{2}(\delta_{n,+1} + \delta_{n,-1}) - \frac{g^2}{4\kappa}[b_{j,n-2} + b_{j,n+2} + 2b_{j,n}] - \frac{g^2}{4\kappa}\sum_{i\neq j}^{\mathcal{N}}[\delta_{n,1}(b_{i,1} + b_{i,-1}) + \delta_{n,-1}(b_{i,1} + b_{i,-1})].
\end{aligned} \tag{B2}$$

From numerical simulations we find that this holds in the relevant parameter regime, i.e., for small single-particle cooperativity $\mathcal{C} \ll 1$, such that Floquet coefficients $b_{j,n}$ with $|n| > 1$ can be neglected. Under the restriction to Floquet coefficients of first order $b_{j,\pm1}$ these equations may be cast into matrix form, as shown in Eq. (28), and inverted using the Sherman-Morrison formula. From this procedure we obtain the expression

$$b_{j,\pm1} = -\frac{i\Omega}{2}\frac{1}{\gamma(1+\mathcal{C}/4) + i(\Delta_a \pm k_c v_j)}\left[1 + \frac{g^2}{4\kappa}\sum_{m,\pm}^{\mathcal{N}}\frac{1}{[\gamma(1+\mathcal{C}/4) + i(\Delta_a \pm k_c v_m)]}\right]^{-1}. \tag{B3}$$

Expanding the coefficient $b_{j,\pm1}$ for particle j up to first order in all velocities $k_c v_i/\Delta_a$ gives

$$b_{j,\pm1} \approx -\frac{i\Omega}{2}\frac{1}{\gamma(1+\mathcal{C}(2\mathcal{N}+1)/4) + i\Delta_a} \pm \frac{-\Omega}{2}\frac{1}{[\gamma(1+\mathcal{C}/4) + i\Delta_a][\gamma(1+\mathcal{C}(2\mathcal{N}+1)/4) + i\Delta_a]}k_c v_j, \tag{B4}$$

which shows a collectively modified decay rate, similar to the single-particle case [see Eq. (A8)]. Inserting this solution into the steady-state solution for α gives

$$\alpha \approx -\frac{\eta}{\kappa + i\Delta_c} - \frac{ig\mathcal{N}b^{(0)}}{\kappa} = -\frac{\eta}{\kappa + i\Delta_c}\left[1 - \frac{g^2\mathcal{N}}{2\kappa}\frac{1}{\gamma(1+(2\mathcal{N}+1)\mathcal{C}/4) + i\Delta_a}\right], \tag{B5}$$

where we invoke the spatial averaging argument again, for $\mathcal{N}$ spatial variables x_j this time. The amplitude of the cavity field is now reduced.

2. $\mathcal{N}$ nonclosed transition systems inside a cavity

The final velocity reached inside the cavity can be calculated analogous to the single-particle case. The collective modifications cancel in the final velocity leaving only single-particle effects. Explicitly we obtain

$$\begin{aligned}
\int_0^{\infty}\xi_c(n_g(t))dt &= \int_1^0 \xi_c(n_g)\frac{dt}{dn_g}dn_g \\
&= \frac{8\hbar k_c^2\gamma_{\text{tot}}}{m\gamma'}\int_0^1 \frac{(1+n_g\mathcal{C}/2)\left[\Delta_a^2 + \gamma_{\text{tot}}^2(1+n_g(2\mathcal{N}+1)\mathcal{C}/4)^2\right]}{\left[\Delta_a^2 + \gamma_{\text{tot}}^2(1+n_g\mathcal{C}/4)^2\right]\left[\Delta_a^2 + \gamma_{\text{tot}}^2(1+n_g(2\mathcal{N}+1)\mathcal{C}/4)^2\right]}dn_g \\
&= \frac{4\omega_{\text{rec.}}\Delta_a\gamma_{\text{tot}}}{\gamma'\Delta_a^2}\int_0^1 \frac{(1+n_g\mathcal{C}/2)}{\left[1 + \frac{\gamma_{\text{tot}}^2}{\Delta_a^2}(1+n_g\mathcal{C}/4)^2\right]}dn_g,
\end{aligned} \tag{B6}$$

which is now equivalent to the single emitter case.

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